Density Bounds for Quadratic Forms under Product Measures

Abstract

Let $X \in \mathbb{R}^n$ have independent coordinates with Lebesgue densities bounded by one, and let B be an arbitrary real matrix. The quadratic form $X^T BX + b^T X + c$ depends on B only through its symmetric part $A = (B + B^T)/2$. We investigate density bounds under the hypothesis $s_3(A) \geq 1$. For every fixed dimension we obtain a bound of order $\sqrt{n}$ uniformly over A, b, and the coordinate laws. We prove dimension-free bounds for semidefinite forms, as well as bounds for diagonal forms, forms with suitable coordinate pivots, and several classes of indefinite forms satisfying quantitative geometric conditions. For product Gaussian scale mixtures satisfying an inverse-variance moment condition, the density bound is less than three for matrices of arbitrary rank and signature. Further estimates cover bounded-rank forms with unit-interval inputs and one side of a rank-three indefinite level set. We give a no-pivot reduction and integrability estimates for singular conditional densities. A dimension-free bound for general indefinite forms with arbitrary bounded coordinate densities remains open.

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1 Introduction

Let $X = (X_1, \dots, X_n)$ have independent coordinates with probability densities f_i satisfying $0 \le f_i \le 1$ almost everywhere. For a real $n \times n$ matrix B, put $A = (B + B^T)/2$ and consider

$$Q(X) = X^T B X + b^T X + c = X^T A X + b^T X + c, \tag{1}$$

where $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$. Write $s_j(A)$ for the singular values of the symmetric part in decreasing order; these are its eigenvalue moduli in decreasing order. The density norm below is the essential supremum. No moment, centering, identical-distribution, smoothness, bounded-support, connected-support, or log-concavity assumption is imposed unless stated in a particular theorem.

Conjecture 1.1 (Uniform three-eigenvalue density bound)**.** *There is a finite absolute constant C such that, for every dimension and every form (1) with $s_3(A) \ge 1$,*

$$\|p_Q\|_\infty \le C. \tag{2}$$

Equivalently, $\mathbb{P}(Q \in J) \le C|J|$ for every interval $J \subset \mathbb{R}$.

The spectral condition concerns A, not B. Indeed, a full-rank skew-symmetric matrix B in dimension four has $s_3(B) = 1$ after normalization, while $x^T B x = 0$ for every x. All subsequent theorems about symmetric matrices therefore apply to arbitrary real matrices through their symmetric parts.

The homogeneous case $b = c = 0$ is the motivating question. Arbitrary affine terms are included because conditioning on original coordinates creates them, often outside the range of the remaining quadratic matrix. Passing from the homogeneous question to an affine-uniform intermediate theorem is consequently a substantive requirement of many proofs. Absolute continuity itself is elementary when $A \neq 0$ (Lemma 3.1); the issue is a uniform density bound. A separate transversality argument gives a continuous density vanishing at infinity for each fixed matrix of rank at least three (Theorem 15.4); its constants may depend on the matrix and dimension.

The normalization is natural. If $s_3(A) \ge a > 0$ and the coordinate densities are bounded by a common $K > 0$, then scaling $Y = KX$ reduces the proposed bound to CK^2/a. The remaining eigenvalues may be arbitrarily small or arbitrarily large and may have either sign. Replacing A by a spectral truncation is not innocuous: the discarded quadratic part depends on the same coordinates.

1.1 Principal conclusions

The principal proved classes and their limitations are as follows. Bounds are stated at spectral scale one unless another scale is displayed.

Setting	**Conclusion**	**Scope**
Diagonal A	$\|p_Q\|_\infty \le \pi$	All affine terms; π is sharp in this class (Theorem 3.4).
Arbitrary real B	$\|p_Q\|_\infty \le \frac{2\sqrt{6\pi n}}{s_3(A)}$	$A = (B+B^T)/2$; every fixed dimension, all remaining spectrum and affine terms (Corollary 9.7).

Setting	Conclusion	Scope
Positive or negative semidefinite A	$\|p_Q\|_\infty \le 17100/s_3(A)$	Arbitrary rank, including tiny positive spectral tails and affine terms outside the range (Theorem 7.1).
Semidefinite, exact rank three	$\|p_Q\|_\infty \le 58/s_3(A)$	Arbitrary affine terms; $3e/s_3(A)$ suffices when the affine term is in the range.
One sign, separated nonzero spectrum	$\|p_Q\|_\infty \le 130/\theta$	Every nonzero eigenvalue has modulus at least θ; arbitrary rank at least three.
Indefinite with small absolute spectral diagonal	$\|p_Q\|_\infty < 35/s_3(A)$	Requires $\max_i(\|A\|)_{ii} \le s_3(A)/680$, where $\|A\| = (A^2)^{1/2}$. Small $\|a_{ii}\|$ alone is insufficient for this argument.
Physical indefinite 2×2 pair	Bound (7)	Only the determinant of the pair and $s_3(A)$ enter; no rank restriction.
Diffuse finite-rank part plus coordinate phases	Integrable Fourier decay	Geometric frame conditions are required; all coordinate phases are allowed (Theorem 4.6).
Cross-coordinate block with distributed volume	Dimension-free density	Quantitative volume and row-marginal conditions; these conditions are not automatic (Theorem 4.9).
Hub configurations	Several dimension-free bounds	The independent hub, transverse square, projected crossing, and law-regularity assumptions are stated separately in Section 13.
Gaussian scale mixtures	$\|p_Q\|_\infty < 3$ for the full form	Every rank and signature under the average component-peak condition (Theorem 6.1). A separate theorem permits adapted hubs.
Exact rank three, unit-interval laws	$\|p_Q\|_\infty < 372$	Every signature and affine term; bounded extra rank is allowed with a rank-dependent factor (Theorem 3.10).
Exact rank three, signature $(2, 1)$	$p_Q \le \pi\sqrt{2}$ on the ruled side	Affine term in the range only; the opposite side is not included (Theorem 3.12).

The dimension-free semidefinite theorem settles a broad portion of the question. For indefinite forms, the fixed-dimensional estimate above does not furnish a constant independent of n. The Gaussian-limit argument in Proposition 3.5 forces any universal constant to be at least $3\sqrt{2}$.

1.2 Proof methods

The proofs use three complementary kinds of information. Coordinate conditioning exploits scalar and two-variable changes of variables while preserving the input product law. Fourier arguments use geometric integral inequalities, decoupling, and the distribution of squared minors of a crossing block. Hub arguments retain the law of a distinguished coordinate when conditioning on that coordinate alone would create a nonuniform bound. Each method controls specific quantities; none permits arbitrary replacement of the joint law of spectral coordinates by a bounded joint density.

The no-pivot reduction in Section 9 isolates a class for which a dimension-free density bound is still needed. The integrability estimates in Section 8 control logarithmic and half-power singularities. Section 18 gives counterexamples to stronger estimates for general joint densities

and describes the open uniform case.

Classical inputs are cited at their points of use and discussed in Section 2.

1.3 Notation and measure-theoretic conventions

For a rectangular matrix M, $\|M\|$ and $\|M\|_F$ denote its operator and Frobenius norms. The elementary symmetric function of the eigenvalues of MM^T is denoted by $e_k(MM^T)$. By Cauchy–Binet, it is the sum of squared $k \times k$ minors. A principal coordinate block is always formed from original coordinates of X. The term *leverage* refers to a diagonal entry of an orthogonal projection, or to the analogous quadratic row norm relative to a positive Gram matrix. We set $\log_+ x = \max\{0, \log x\}$ for $x > 0$.

Our Fourier convention is $\widehat{f}(t) = \int e^{itx} f(x)\, dx$, so inverse Fourier transforms carry $(2\pi)^{-1}$ in one dimension. Conditional density formulas are interpreted through Tonelli against nonnegative output test functions. Integrable characteristic functions give bounded continuous density versions; an interval estimate alone gives an essentially bounded density without an additional continuity claim. Smooth approximation of the physical coordinate laws is used when a pointwise level-set calculation could otherwise encounter exceptional slices.

2 Relation to earlier work

The question considered here is an endpoint question with a uniformity requirement. If independent real random variables have densities bounded by K, and A has three eigenvalues of absolute value at least $a > 0$, the proposed conclusion is a bound of the form CK^2/a for the density of $X^{\mathsf{T}}AX + b^{\mathsf{T}}X + c$. The constant must be independent of the ambient dimension, the remaining eigenvalues, the eigenvectors, the linear term, and the supports and regularity of the coordinate densities. Several established theories address parts of this question. Their precise hypotheses matter when they are used in the proofs below.

2.1 Linear images, convolution, and geometric inequalities

The basic linear input is the theory of marginal densities of product measures. Rudelson and Vershynin (2015, Theorem 1.1) prove that an orthogonal projection onto a k-dimensional subspace has density bounded by $(CK)^k$. Livshyts et al. (2016, Theorem 1.1) obtain the sharper estimate

$$\|f_{PX}\|_\infty \le 2^{k/2} K^k,$$

as well as a refinement depending on the ambient dimension and unequal coordinate density bounds. In the one-dimensional case, the sharp constant follows from the convolution comparison of Rogozin (1987) and the cube slicing theorem of Ball (1986); the higher-codimension geometric background includes Ball (1989).

The integral arguments use the Brascamp–Lieb inequality (Brascamp and Lieb, 1976), its Gaussian extremization theory (Lieb, 1990), and its geometric formulation and transport proof (Barthe, 1998). For unit vectors u_j and nonnegative weights c_j satisfying $\sum_j c_j u_j u_j^{\mathsf{T}} = I_k$, the geometric inequality reads

$$\int_{\mathbb{R}^k} \prod_j h_j(\langle u_j, x\rangle)^{c_j}\, dx \le \prod_j \left(\int_{\mathbb{R}} h_j \right)^{c_j}.$$

These are established analytic tools. Gaussian extremization for this integral does not assert that Gaussian coordinate densities maximize every quadratic pushforward under an L^∞ constraint.

Likewise, a bound on a projected density does not give a uniform bound on the conditional density on each affine fiber: conditioning includes division by the marginal density, which can be arbitrarily small.

Where the proof partitions a frame with small individual vectors into nondegenerate groups, the relevant structural input is the partition consequence of the solution of the Kadison–Singer problem by Marcus et al. (2015). Applications must retain its upper bound on each vector norm and the resulting quantitative spectral bounds for the groups.

2.2 Fourier estimates for rough densities

An L^∞ bound on a probability density gives useful integral control of its characteristic function, but does not furnish a universal pointwise rate of decay at every frequency. The estimate in Rudelson and Vershynin (2015, Lemma 3.1) concerns the nonincreasing rearrangement of the modulus of the characteristic function. It is therefore suited to integral estimates and geometric Brascamp–Lieb arguments. Sharp Hausdorff–Young estimates provide another input (Beckner, 1975).

For linear combinations of independent variables, Friedland and Sodin (2007) use an analytic argument to bound Lévy concentration functions in terms of Diophantine properties of the coefficients, including a multidimensional extension. The arithmetic hypotheses and the linear map in that result differ from the quadratic pushforward considered here.

Bobkov et al. (2012, Theorems 1–2 and 8) give quantitative separation from modulus one using density bounds together with variance or quantile information, and give entropy versions of these estimates. The variance or quantile parameters are essential in those bounds. Related convolution concentration estimates are developed in Bobkov and Chistyakov (2015).

2.3 Bilinear forms and the role of independence

A particularly close result is the density estimate of Banks et al. (2024, Section 2). Although their Theorem 2.1 is titled "Density of Quadratic Forms," the expression there is bilinear in two independent random matrices. In the scalar case their Corollary 2.4 gives, for independent vectors X, Y with independent coordinates of density at most K,

$$\left\| f_{X^{\mathsf{T}}ZY+u^{\mathsf{T}}X+v^{\mathsf{T}}Y+w} \right\|_\infty \le CK^2 \min_{j\ge 2} \frac{1}{\sqrt{j}\left(\prod_{i=1}^{j}\sigma_i(Z)\right)^{1/j}}.$$

The minimum is over nonzero singular values. Conditioning reduces the proof to a linear density estimate and an integrable negative first moment in dimension at least two. This result explains why a well-conditioned crossing block with at least two directions is useful. It does not directly cover a rank-one interaction accompanied by a quadratic expression in the same light coordinates; the independence and rank needed by the displayed estimate are then absent.

2.4 The parabola extension estimate

For the Fourier convention

$$(Eh)(s,\tau) = \int_{\mathbb{R}} h(x) e^{i(sx^2+\tau x)}\, dx,$$

the bound

$$\int_{\mathbb{R}^2} |Eh(s,\tau)|^6 \, ds\, d\tau \le \frac{4\pi^3}{\sqrt{3}} \|h\|_2^6$$

is the sharp one-dimensional Schrödinger Strichartz inequality. The underlying restriction theory is due to Strichartz (1977); sharp constants and Gaussian extremizers are established in Foschi (2007); Hundertmark and Zharnitsky (2006). The normalization follows from the constant $12^{-1/12}$ in Foschi (2007, Theorem 1.1) and Plancherel. If h is a probability density bounded by one, then $\|h\|_2 \le 1$, giving the constant $4\pi^3/\sqrt{3}$ used below. An elementary coarea proof of this special case is included below.

2.5 General polynomial and Gaussian results

The polynomial anti-concentration inequality of Carbery and Wright (2001) gives, for a degree-d polynomial P under a log-concave probability measure, a bound of the form

$$\mu\{|P| \le \varepsilon\} \le Cd \left(\frac{\varepsilon}{\int |P| \, d\mu} \right)^{1/d}.$$

The degree-two exponent is $1/2$. Log-concavity and the normalization of the polynomial are part of that statement. Glazer and Mikulincer (2022) establish dimension-free covariance bounds for polynomials of product measures, and obtain sublevel and Fourier estimates under additional log-concavity hypotheses. Recent work treats further polynomial anti-concentration questions on L^p balls and symmetric measures (Glazer and Mikulincer, 2026), and regularity of nondegenerate polynomial pushforwards of log-concave measures (Kosov and Zhukova, 2026). These conclusions have different hypotheses or exponents from the linear-in-interval-length endpoint sought here.

For analytic maps, Glazer et al. (2026, Theorems 1.1–1.2) relate local L^p integrability of pushforward densities to singularity invariants. The value $\epsilon_\star = \infty$ means membership in every finite L^p; boundedness is a stronger property, treated for algebraic maps in their Corollary 6.2. In the real setting, Reiser (2018) proves continuity of pushforward densities of compactly supported smooth measures for flat maps with rational-singularity fibers. These statements concern specified map and input regularity and do not give a dimension-uniform constant for arbitrary bounded product densities.

For Gaussian input, rotation preserves independence and a quadratic polynomial can be diagonalized without leaving the input class. Density bounds for squared Gaussian norms and weighted chi-square variables are studied by Götze et al. (2019); Bobkov et al. (2020); signed, noncentral Gaussian quadratic forms are also treated in Choudhary and Kuchibhotla (2026). The elementary characteristic-function identity already gives

$$|\mathbb{E} e^{it(g^{\mathsf{T}} A g + b^{\mathsf{T}} g)}| \le \prod_i (1 + 4t^2 \lambda_i(A)^2)^{-1/4}.$$

Three nonzero eigenvalues make the right-hand side integrable at infinity. General product laws lack this rotational invariance, so the Gaussian computation does not settle their uniform density bound.

A related Gaussian quadratic estimate appears in Aizenman et al. (2017, Lemma 2.1): for any nonzero real symmetric matrix B, standard Gaussian vector g, fixed vector d, and scalar s,

$$\mathbb{P}\left\{ \frac{\|B(g+d)\|}{|(g+d)^{\mathsf{T}} B(g+d) - s|} \ge t \right\} \le \frac{C}{t}, \qquad t \ge 1,$$

with an absolute constant. Its proof uses Gaussian rotational invariance and a joint characteristic function of quadratic quantities, assumptions that do not hold for general bounded product densities.

A related regularity method of Herry et al. (2025) uses a Gaussian representation of independent identically distributed input laws and spectral quantities of a Hessian multiplied by a

random diagonal matrix. Their results combine inverse moments of the associated Dirichlet energy with spectral remainders and small influences. Its smooth representation and quantitative regularity hypotheses differ from the sole coordinate density bound here. In particular, these results do not yield a dimension-uniform estimate for arbitrary bounded coordinate laws and noncentral forms.

2.6 Scope of the present results

The semidefinite theorem below concerns bounded product densities, with an explicit constant controlled by the third spectral scale. The indefinite results require the additional geometric or analytic hypotheses in their statements. The unrestricted dimension-uniform bound remains a conjecture; the cited results do not establish it for arbitrary product laws.

3 Coordinate geometry and convex input laws

The estimates in this section use the original independent coordinates. An orthogonal change of variables in a geometric integral does not imply independence of the rotated random coordinates.

Lemma 3.1 (Absolute continuity)**.** *If $A \neq 0$ and the coordinates of X have densities, then $X^T AX + b^T X + c$ has an absolutely continuous distribution.*

Proof. If $a_{ii} \neq 0$, condition on all coordinates except X_i. The conditional polynomial is a nonconstant quadratic, and the inverse image of a Lebesgue null set under it is null: split the real line at its vertex and apply change of variables. If every diagonal entry vanishes, choose $a_{ij} \neq 0$. Conditional on X_{-i}, the coefficient of X_i is $2\sum_{j\neq i} a_{ij}X_j + b_i$. This is nonzero almost surely, since a nontrivial linear combination of independent absolutely continuous random variables is absolutely continuous. The conditional law is therefore absolutely continuous in this case as well. Fubini completes the argument. □

Lemma 3.2 (A scalar quadratic)**.** *Let X have density $f \leq 1$, and let $g(x) = a(x-m)^2 + \kappa$ with $a \neq 0$. On the side $(y-\kappa)/a > 0$,*

$$p_{g(X)}(y) = \frac{f(m+r) + f(m-r)}{2\sqrt{|a|\,|y-\kappa|}} \leq \frac{1}{\sqrt{|a|\,|y-\kappa|}}, \qquad r = \sqrt{|y-\kappa|/|a|}.$$

The density vanishes on the other side. Moreover, for every $u > 0$,

$$\sup_s \mathbb{P}(|g(X) - s| \leq u) \leq 2\sqrt{2u/|a|}, \qquad \mathbb{P}(|g(X) - s| \leq u) \leq \frac{4u}{\sqrt{|a|\max\{|s-\kappa|, u\}}}. \tag{3}$$

Proof. The density formula follows by change of variables on the two monotone half-lines. After translation and scaling, the admissible values of $|x-m|$ form an interval whose squared endpoints differ by at most $2u/|a|$. Its length is at most $\sqrt{2u/|a|}$, which proves the first bound after accounting for the two signs. If $s-\kappa$ lies on the occupied side and $|s-\kappa| > u$, rationalizing the difference of the two square roots gives the second estimate with constant 4. If $|s-\kappa| \leq u$, the first estimate implies it. If $s-\kappa$ is on the unoccupied side at distance greater than u, the probability is zero. □

Lemma 3.3 (A definite pair)**.** *Suppose that $W \in \mathbb{R}^2$ has a joint density bounded by K. If B is positive or negative definite and $D = |\det B|$, then every translated quadratic with quadratic part B has density bounded by $\pi K/\sqrt{D}$.*

Proof. Changing sign if necessary, write the eigenvalues as $\mu_1, \mu_2 > 0$. Complete the square. The parametrization

$$w = w_0 + \sqrt{\delta/\mu_1}\cos\theta\, e_1 + \sqrt{\delta/\mu_2}\sin\theta\, e_2, \qquad \delta > 0, \quad 0 \le \theta < 2\pi,$$

has Jacobian $1/(2\sqrt{D})$ in (δ, θ). Integrating the density on this ellipse proves the assertion. This lemma requires a joint density bound, but not independence. □

Theorem 3.4 (Diagonal forms and a sharp diagonal constant)**.** *If A is diagonal and $s_3(A) \ge s > 0$, then, for arbitrary affine terms,*

$$\|p_Q\|_\infty \le \pi/s.$$

For the normalization $s = 1$, the constant π is attained by $Q = X_1^2 + X_2^2 - X_3^2$ with independent uniform coordinates on $[-1/2, 1/2]$.

Proof. Among three diagonal coefficients of modulus at least s, two have the same sign. Condition on every other coordinate. The remaining two-variable form is definite, its determinant has modulus at least s^2, and its joint density is at most one. Apply Lemma 3.3 and average. For sharpness, the density of $X_1^2 + X_2^2$ equals π on $(0, 1/4)$, since the corresponding circles lie in the square. Averaging at X_3^2 gives density π at zero, in the continuous limiting sense. The uniform upper bound and dominated convergence justify this evaluation. □

Proposition 3.5 (Spectral threshold and a constant lower bound)**.** *One nonzero eigenvalue does not ensure bounded density, and two eigenvalues of opposite signs do not ensure it. Any universal constant in Conjecture 1.1 must satisfy $C \ge 3\sqrt{2}$.*

Proof. The square of a centered uniform variable has the inverse-square-root singularity of Lemma 3.2. For independent centered unit uniforms U, V, the density of $U^2 - V^2$ diverges logarithmically at zero; for $t > 0$ its defining integral contains

$$\int_t^{1/4} \frac{du}{\sqrt{u(u-t)}} \longrightarrow \infty.$$

For the lower bound on C, let $Z_{j,m}$, $j = 1, 2, 3$, be independent normalized sums of m centered unit uniforms. The three corresponding coefficient vectors in $\mathbb{R}^{3m}$ are orthonormal. Thus

$$Q_m = Z_{1,m}^2 + Z_{2,m}^2 - Z_{3,m}^2$$

is admissible with nonzero eigenvalues $1, 1, -1$. The central limit theorem and the continuous mapping theorem give weak convergence to $Q_\infty = G_1^2 + G_2^2 - G_3^2$, where the G_j are independent $N(0, 1/12)$ variables. If every Q_m has density at most C, the same interval bound passes to the weak limit, hence its density is also at most C. Here $G_1^2 + G_2^2$ is exponential of rate 6, and G_3^2 has density $\sqrt{6/\pi}\, t^{-1/2} e^{-6t}$ on $t > 0$. Their convolution at zero is

$$p_{Q_\infty}(0) = 6\sqrt{6/\pi} \int_0^\infty t^{-1/2} e^{-12t}\, dt = 3\sqrt{2}.$$

The density is continuous at zero. This argument uses weak convergence and interval domination, and does not require a local central limit theorem. It does not prove that $3\sqrt{2}$ is the sharp universal constant. □

3.1 The logarithmic envelope of an indefinite physical pair

Theorem 3.6 (Uniform hyperbolic envelope)**.** *Let W_1, W_2 be independent with densities bounded by one. Let $q(w) = (w - w_0)^T B(w - w_0) + \kappa$, where B has eigenvalues $\mu_+ > 0 > \mu_-$. Put*

$$D = |\det B|, \qquad \Lambda = \mu_+^{-1} + |\mu_-|^{-1}.$$

For almost every $s \neq \kappa$,

$$p_q(s) \leq \frac{1}{\sqrt{D}} \left(5.2 + \log_+ \frac{32}{\Lambda |s - \kappa|}\right) \leq \frac{1}{\sqrt{D}} \left(5.2 + \log_+ \frac{16\sqrt{D}}{|s - \kappa|}\right). \tag{4}$$

Consequently

$$\mathbb{P}(|q(W) - s| \leq u) \leq \frac{2u}{\sqrt{D}} \left(6.2 + \log_+ \frac{16\sqrt{D}}{u}\right). \tag{5}$$

Proof. Translation preserves the coordinate density caps. For $\delta = s - \kappa > 0$, the two branches of the level set are

$$w_\epsilon(t) = \epsilon\sqrt{\delta/\mu_+}\cosh t\, e_+ + \sqrt{\delta/|\mu_-|}\sinh t\, e_-, \quad \epsilon \in \{-1, 1\}, \quad t \in \mathbb{R}.$$

The Jacobian in (δ, t) equals $1/(2\sqrt{D})$. Thus

$$p_q(\kappa + \delta) = \frac{1}{2\sqrt{D}} \sum_{\epsilon = \pm 1} \int_{\mathbb{R}} f_1(w_{\epsilon,1}(t)) f_2(w_{\epsilon,2}(t))\, dt. \tag{6}$$

Write one branch as $ue^t + ve^{-t}$. Orthogonality of e_+, e_- gives $|u| = |v| = \rho_0 = \sqrt{\delta\Lambda}/2$. Choose a physical coordinate i with $|u_i| \geq \rho_0/\sqrt{2}$. For $t \geq T_1 = (3/4)\log 2$, the term $v_i e^{-t}$ has modulus at most $|u_i|e^t/2$. The coordinate $w_i(t)$ is monotone there and satisfies

$$|w_i'(t)| \geq \tfrac{1}{2}|u_i|e^t, \qquad \tfrac{1}{2}|u_i|e^t \leq |w_i(t)| \leq \tfrac{3}{2}|u_i|e^t, \qquad |w_i'(t)| \geq |w_i(t)|/3.$$

For $T = \max\{T_1, \log(2/|u_i|)\}$, put $m = |u_i|e^T/2 \geq 1$. Since $f_1 f_2 \leq f_i$ and $\int f_i = 1$, change of variables and the decreasing weight $1/|y|$ give

$$\int_T^\infty f_1(w_1(t)) f_2(w_2(t))\, dt \leq 3 \int_{|y| \geq m, \text{ one side}} \frac{f_i(y)}{|y|}\, dy \leq 3\log(1 + 1/m) \leq 3\log 2.$$

The penultimate inequality follows by filling the interval of length one closest to m; it also follows directly from layer cake. The negative tail is identical with v in place of u. Its cutoff T' and the first cutoff satisfy

$$T, T' \leq \tfrac{3}{4}\log 2 + \tfrac{1}{2}\log_+ \frac{32}{\delta\Lambda}.$$

On the middle interval, the integrand is at most one. Each branch integral is therefore at most

$$\tfrac{3}{2}\log 2 + 6\log 2 + \log_+ \frac{32}{\delta\Lambda} < 5.2 + \log_+ \frac{32}{\delta\Lambda}.$$

Substitute into (6). Negative δ exchanges the positive and negative eigenvalues. Since $\Lambda \geq 2/\sqrt{D}$, the second estimate in (4) follows. Finally, among intervals of length $2u$, the integral of $\log_+(K/|x|)$ is largest for the centered interval, and is at most $2u(1 + \log_+(K/u))$. Integration proves (5). All calculations may first be made against a nonnegative output test function, so the almost-everywhere formulas are legitimate under later conditioning. □

Theorem 3.7 (A determinant-only bound in arbitrary dimension)**.** *Suppose $s_3(A) \geq s > 0$ and A has an indefinite principal block B on two original coordinates. For $D = |\det B|$ and arbitrary affine terms,*

$$\|p_Q\|_\infty \leq D^{-1/2}\left(9.2 + \log_+ \frac{10000\sqrt{D}}{s}\right). \tag{7}$$

There is no rank bound and no restriction on the other entries of A.

Proof. Condition on the complementary physical coordinates, and complete the square in the pair. The conditional critical value is a quadratic R whose matrix is the Schur complement $S = A/B$. Lemma 9.1 gives $\|S\| \geq s$. Theorem 4.29 and layer cake imply, for every $K > 0$,

$$\sup_t \mathbb{E} \log_+ \frac{K}{|t - R|} \leq \int_0^\infty \min\{1, 5(Ke^{-u}/\|S\|)^{1/4}\}\, du \leq 4 + \log_+ \frac{625K}{\|S\|}.$$

Average (4) with $K = 16\sqrt{D}$. This proves (7) by Fubini. Notice that a small eigenvalue of the pair itself need not be controlled separately: its determinant is the only pair parameter in the conclusion. □

3.2 Convex superlevel sets and interval laws

Lemma 3.8 (The unit ball in three dimensions)**.** *Let Z be uniform on the unit ball of $\mathbb{R}^3$, and let q have invertible quadratic part M. Then*

$$\|p_{q(Z)}\|_\infty \leq C_B |\det M|^{-1/3}, \qquad C_B = \frac{3}{4\pi}\, 38.2.$$

Proof. Rotate the ball to diagonalize M, and complete squares in the polynomial. Let $|m_1|$ be the largest eigenvalue modulus and $D = |m_2 m_3| \leq m_1^2$. At a point (x_2, x_3) in the unit disc Δ, the first coordinate ranges along a chord of half-length h. The marginal density of (x_2, x_3) is $3h/(2\pi)$, while the conditional first coordinate is uniform on a segment of length $2h$. The chord factors cancel in Lemma 3.2, yielding

$$p_q(s) \leq \frac{3}{4\pi} |m_1|^{-1/2} \int_\Delta |s - R(x_2, x_3)|^{-1/2}\, dx_2 dx_3.$$

Layer cake bounds the integral by the integral over $v > 0$ of the area where $|s - R| < v^{-2}$. For a definite R, the area is bounded by $\min\{\pi, 2\pi v^{-2}/\sqrt{D}\}$, whose integral is $2\pi\sqrt{2}\, D^{-1/4}$. For an indefinite R, enclose Δ in $[-1, 1]^2$. The latter Lebesgue measure is four times a product probability measure with coordinate densities $1/2 \leq 1$. Theorem 3.6 gives the bound

$$D^{-1/4} \int_0^\infty \min\{\pi, 8v^{-2}(6.2 + \log_+(16v^2))\}\, dv.$$

Split this integral at $a = 28/5$. It is at most

$$\pi a + \frac{8}{a}(8.2 + \log 16 + 2 \log a) < 38.2.$$

Thus $p_q \leq C_B |m_1|^{-1/2} D^{-1/4}$. Since

$$|m_1|^{-1/2} D^{-1/4} = |\det M|^{-1/3} (D/m_1^2)^{1/12} \leq |\det M|^{-1/3},$$

the assertion follows. □

Theorem 3.9 (Three-dimensional convex superlevel estimate)**.** *Let $Z \in \mathbb{R}^3$ have density $g \le K$ whose positive superlevel sets are convex. For every quadratic polynomial with invertible quadratic part M,*

$$\|p_{q(Z)}\|_\infty \le 71.1\,|\det M|^{-1/3} K^{2/3}. \tag{8}$$

Proof. First consider Lebesgue measure on a convex body C. For an ellipsoid $E = c + TB_3$, Lemma 3.8 and $\det(T^T M T) = (\det T)^2 \det M$ imply that its pushforward density is at most

$$C_B \operatorname{vol}(B_3)^{2/3} |\det M|^{-1/3} \operatorname{vol}(E)^{1/3}.$$

John's ellipsoid theorem gives $E \subset C \subset 3E$ (with dilations about the ellipsoid center). Monotonicity of pushforward measures and $\operatorname{vol}(E) \le \operatorname{vol}(C)$ therefore bound the pushforward of C by

$$C_K |\det M|^{-1/3} \operatorname{vol}(C)^{1/3}, \quad C_K = 3(4\pi/3)^{2/3} C_B < 71.1.$$

The geometric input is John's theorem (John, 1948). Apply this to $C_t = \{g > t\}$ and use layer cake. Since $\int_0^K \operatorname{vol}(C_t)\,dt = 1$, Hölder gives

$$\int_0^K \operatorname{vol}(C_t)^{1/3}\,dt \le K^{2/3}.$$

This proves the result. Approximation by full-dimensional level sets handles degenerate or null superlevel sets. □

Theorem 3.10 (Bounded rank for unit-interval inputs)**.** *Let X be uniform on a product of unit intervals. Suppose*

$$A = R\Lambda R^T + R'\Lambda' R'^T,$$

where $[R\ R']$ has orthonormal columns, R has three columns, R' has k columns, and $\Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$ is invertible. Decompose $b = R\beta + R'\beta' + \gamma u$, with u a unit vector orthogonal to both ranges if $\gamma \neq 0$, and put $k' = k + \mathbf{1}_{\{\gamma \neq 0\}}$. Then

$$\|p_Q\|_\infty \le 142.2\,[\sqrt{2}(2\pi)^{1/3}]^{k'} |\lambda_1\lambda_2\lambda_3|^{-1/3}. \tag{9}$$

In particular, every exact rank-three form with $s_3(A) \ge 1$ and arbitrary affine term has density bounded by 372 for these inputs.

Proof. Write $Y = R^T X$ and $Y' = (R'^T X, \langle u, X\rangle)$, omitting the last coordinate if $\gamma = 0$. The joint density ρ of this orthogonal projection is log-concave by the Prékopa marginal theorem (Prékopa, 1973), and is bounded by $2^{(3+k')/2}$ by the product-projection theorem of Livshyts et al. (2016). The polynomial is $q(Y) + \varphi(Y')$, where q has quadratic part Λ. For almost every y' with marginal density $\rho_{Y'}(y') > 0$, the conditional density of Y is log-concave and bounded by $2^{(3+k')/2}/\rho_{Y'}(y')$. Theorem 3.9, followed by integration in y', gives

$$\|p_Q\|_\infty \le 71.1\,|\det \Lambda|^{-1/3} 2^{(3+k')/3} \int \rho_{Y'}(y')^{1/3}\,dy'.$$

Center Y' at its mean m. Hoeffding's bounded-variable exponential estimate (Hoeffding, 1963) and orthonormality give $\mathbb{E}e^{\langle \theta, Y'-m\rangle} \le e^{|\theta|^2/8}$. Introducing an independent standard Gaussian in $\mathbb{R}^{k'}$ yields $\mathbb{E}e^{|Y'-m|^2} \le 2^{k'/2}$. Hölder, applied after multiplication and division by $e^{|y'-m|^2/3}$, now gives

$$\int \rho_{Y'}^{1/3} \le \left(\int \rho_{Y'} e^{|y'-m|^2}\right)^{1/3} \left(\int e^{-|y'-m|^2/2}\,dy'\right)^{2/3} \le 2^{k'/6}(2\pi)^{k'/3}.$$

Combining the powers of two proves (9). The factor $\sqrt{2}(2\pi)^{1/3}$ is less than 2.61; when $k = 0$, one has $k' \le 1$ and the stated constant follows. □

Proposition 3.11 (Reductions in the class of coordinate laws)**.** *For fixed A, let $\mathcal{D}_A(f_1, \dots, f_n)$ denote the supremum of the density norm over all affine terms. This functional is translation invariant, convex in each coordinate law, and nonincreasing when one coordinate law is convolved with an arbitrary probability measure. The supremum over all densities $f_i \le 1$ equals the supremum over $f_i = \mathbf{1}_{E_i}$ with $|E_i| = 1$. A theorem uniform over a matrix class stable under diagonal congruences $A \mapsto DAD$, $D \ge I$, transfers from unit-interval laws to independent mixtures of widened unit-interval laws and their convolutions, with the same constant.*

Proof. Translation changes only the affine coefficients. For fixed output interval and fixed affine term, its probability is linear in each f_i separately. Taking suprema gives convexity. A convolution is a mixture of translations, giving the contraction assertion. To obtain the extremal-set statement, fix an output interval and all laws but one; its probability is $\int f_i(x) g_i(x)\,dx$ with $0 \le g_i \le 1$. The bathtub principle says its supremum over $0 \le f_i \le 1$, $\int f_i = 1$, is achieved or approximated arbitrarily well by the indicator of a measurable set of measure one. Replace the coordinates successively and then take the supremum over intervals and affine terms. No interval shape for E_i follows from this argument.

Finally, widening the ith interval by $r_i \ge 1$ replaces A by DAD, $D = \operatorname{diag}(r_i)$. The number of positive eigenvalues at least one cannot decrease: on a subspace where $x^T A x \ge |x|^2$, the transformed subspace $D^{-1}E$ satisfies $y^T DADy \ge |Dy|^2 \ge |y|^2$. Apply the same argument to $-A$ for negative eigenvalues. Rank is unchanged. The unit-interval theorem thus applies after widening; convexity and the convolution contraction finish the proof. □

The extremal-set argument above is an elementary special case of the more general extremal-law framework in Madiman et al. (2017). The convexity hypothesis in Theorem 3.9 and the restrictions on the input laws in Theorem 3.10 are substantive. They do not extend those results to arbitrary unions of separated intervals.

3.3 The ruled side of an exact rank-three form

Theorem 3.12 (A one-sheet bound)**.** *Let u_1, u_2, u_3 be orthonormal and $\lambda_1, \lambda_2, \mu \ge 1$. For*

$$Q = \lambda_1 \langle u_1, X\rangle^2 + \lambda_2 \langle u_2, X\rangle^2 - \mu \langle u_3, X\rangle^2,$$

the restriction of the law to $(0, \infty)$ has density bounded by $\pi\sqrt{2}$. Equivalently, for every interval $J \subset (0, \infty)$, $\mathbb{P}(Q \in J) \le \pi\sqrt{2}\,|J|$. An affine term in the range of the quadratic matrix is allowed after shifting the critical value. Sign reflection gives the corresponding half-line for signature $(1, 2)$.

Proof. First assume smooth compactly supported coordinate densities, bounded by one. Write $Y = (\langle u_1, X\rangle, \langle u_2, X\rangle, \langle u_3, X\rangle)$ and let ρ be its density. Put $D = \operatorname{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \sqrt{\mu})$ and $\Delta = \lambda_1\lambda_2\mu$. In the coordinates $z = Dy$, the quadratic is $z_1^2 + z_2^2 - z_3^2$. For $s > 0$, a ruling of its level set is

$$z(\theta, t) = \sqrt{s}\, e_\theta + t(e_\theta^\perp + e_3), \qquad e_\theta = (\cos\theta, \sin\theta, 0), \quad e_\theta^\perp = (-\sin\theta, \cos\theta, 0).$$

At fixed $t = z_3$, the planar radius is $\sqrt{s + t^2}$ and the planar polar angle differs from θ by $\arctan(t/\sqrt{s})$. Its angular Jacobian is one. Polar coarea, equivalently change of variables against an output test function, gives

$$p_Q(s) = \frac{1}{2\sqrt{\Delta}} \int_0^{2\pi} \int_{\mathbb{R}} \rho(D^{-1} z(\theta, t))\,dt\,d\theta.$$

The line velocity is $v_\theta = D^{-1}(e_\theta^\perp + e_3)$, with

$$|v_\theta|^2 = \frac{\sin^2\theta}{\lambda_1} + \frac{\cos^2\theta}{\lambda_2} + \frac{1}{\mu}.$$

The inner integral is $1/|v_\theta|$ times a two-dimensional orthogonal marginal density of Y. It is also a two-dimensional orthogonal projection density of the original product vector X, and is at most $2/|v_\theta|$ by Livshyts et al. (2016). In this smooth case, the marginal is continuous, so its essential bound is pointwise. Since

$$\Delta|v_\theta|^2 = \lambda_2\mu\sin^2\theta + \lambda_1\mu\cos^2\theta + \lambda_1\lambda_2 \geq 2,$$

the claimed bound follows by integration in θ.

To remove regularity, approximate each cap-one density in L^1 by smooth compactly supported probability densities with the same cap. Explicitly, truncate it, replace the missing mass by an indicator on a disjoint interval of the required length, and convolve with a nonnegative compactly supported mollifier. A diagonal choice of truncation and smoothing gives the approximation. The product densities converge in L^1 by the telescoping product estimate, and their pushforwards under Q converge in total variation. Pass the interval inequality to the limit for each fixed $J \subset (0,\infty)$. This step avoids evaluating an almost-everywhere bounded marginal on a potentially exceptional curve. If b belongs to the range of A, a translation in the original coordinates removes b and changes only the critical value; the coordinate caps are preserved. □

This theorem controls one side of the critical value. It supplies no estimate on the two-sheet side and does not accommodate arbitrary affine leakage into the kernel by itself.

4 Fourier estimates, product projections, and cross blocks

Throughout this section the coordinates of $X = (X_1, \dots, X_n)$ are independent, and their probability densities satisfy $0 \leq f_i \leq 1$. We use the Fourier transform

$$\widehat{h}(\xi) = \int_{\mathbb{R}^d} e^{i\langle\xi,x\rangle} h(x)\,dx, \qquad h(x) = (2\pi)^{-d}\int_{\mathbb{R}^d} e^{-i\langle\xi,x\rangle}\widehat{h}(\xi)\,d\xi.$$

Statements about essential suprema do not depend on the choice of a density representative. When the characteristic function is integrable, Fourier inversion supplies a continuous density vanishing at infinity.

4.1 Classical analytic inputs and an inversion formula

We use the geometric rank-one Brascamp–Lieb inequality (Brascamp and Lieb, 1976; Livshyts et al., 2016): if v_j are unit vectors in $\mathbb{R}^d$, $c_j > 0$, and $\sum_j c_j v_j v_j^{\mathsf{T}} = I_d$, then

$$\int_{\mathbb{R}^d}\prod_j H_j(\langle v_j, z\rangle)^{c_j}\,dz \leq \prod_j\left(\int_{\mathbb{R}} H_j\right)^{c_j}. \tag{10}$$

The functions H_j are arbitrary nonnegative measurable functions. We also use Lieb's Gaussian exhaustion theorem (Lieb, 1990). For nonzero row vectors b_j, positive exponents c_j with $\sum_j c_j = d$, the best constant in the corresponding rank-one Brascamp–Lieb inequality is

$$\sup_{\lambda_j>0}\left[\frac{\prod_j \lambda_j^{c_j}}{\det\left(\sum_j c_j\lambda_j b_j b_j^{\mathsf{T}}\right)}\right]^{1/2}. \tag{11}$$

The supremum may be infinite in general. The determinant estimates below show finiteness for the data considered here.

The sharp Hausdorff–Young and convolution inequalities (Beckner, 1975) are used with the following normalization:

$$\|\widehat{h}\|_{p'} \le (2\pi)^{1/p'} C_p \|h\|_p, \qquad C_p = p^{1/(2p)}(p')^{-1/(2p')}, \quad 1 \le p \le 2, \tag{12}$$

$$\|h_1 * h_2\|_r \le \frac{C_p C_q}{C_r} \|h_1\|_p \|h_2\|_q, \qquad \frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}. \tag{13}$$

In the convolution inequality $1 \le p, q, r \le \infty$, and each C_t is given by the same conjugate-exponent formula. Endpoint constants have their limiting values. In particular (12) applies to complex functions.

Lemma 4.1 (Integrable power envelopes)**.** *Let Y be real and suppose, for $a > 0$ and $b > 1$, that*

$$|\varphi_Y(t)| \le \min\{1, (a/|t|)^b\}.$$

Then Y has a continuous density vanishing at infinity and

$$\|p_Y\|_\infty \le \frac{ab}{\pi(b-1)}. \tag{14}$$

More generally, a bound $K|t|^{-b}$ can be used with $a = K^{1/b}$.

Proof. Splitting the integral at a gives

$$\int_{\mathbb{R}} |\varphi_Y(t)|\, dt \le 2a + 2a^b \int_a^\infty t^{-b}\, dt = \frac{2ab}{b-1}.$$

The inverse Fourier transform of φ_Y is continuous and vanishes at infinity, and is the density of its probability law. Its supremum is at most $(2\pi)^{-1}\|\varphi_Y\|_1$. □

4.2 Densities of projections and modulated scalar transforms

Lemma 4.2 (Product projection densities)**.** *Let U be an $n \times k$ real matrix with $U^{\mathsf{T}}U = I_k$, and put $c_i = \|U^{\mathsf{T}} e_i\|^2$. Then $U^{\mathsf{T}}X$ has a density satisfying the explicit estimate*

$$\|p_{U^{\mathsf{T}}X}\|_\infty \le \prod_{i : c_i < 1} (1 - c_i)^{-(1-c_i)/2} \le e^{k/2}. \tag{15}$$

The sharper classical product-marginal theorem (Livshyts et al., 2016) gives

$$\|p_{U^{\mathsf{T}}X}\|_\infty \le 2^{k/2}. \tag{16}$$

For any rank-k map $T : \mathbb{R}^n \to \mathbb{R}^k$, consequently,

$$\|p_{TX}\|_\infty \le \frac{2^{k/2}}{\sqrt{\det(TT^{\mathsf{T}})}} \le \frac{e^{k/2}}{\sqrt{\det(TT^{\mathsf{T}})}}. \tag{17}$$

Proof. To prove (15), complete U to an orthogonal matrix (U, W), with W having $n-k$ columns. Writing w_i^{T} for its rows and $b_i = \|w_i\|^2 = 1 - c_i$, Fubini's theorem gives a marginal density

$$p(y) = \int_{\mathbb{R}^{n-k}} \prod_i f_i((Uy)_i + \langle w_i, z\rangle)\, dz.$$

Terms with $b_i = 0$ are at most one and can be omitted. The other vectors $v_i = w_i/\sqrt{b_i}$ satisfy $\sum_i b_i v_i v_i^{\mathsf{T}} = I_{n-k}$. Apply (10) to

$$H_i(s) = f_i((Uy)_i + \sqrt{b_i}s)^{1/b_i}.$$

Because $b_i \le 1$, $f_i^{1/b_i} \le f_i$, so

$$\left(\int H_i\right)^{b_i} \le b_i^{-b_i/2}.$$

This proves the product bound uniformly in y. The entropy inequality

$$-(1-x)\log(1-x) \le x, \qquad 0 \le x \le 1, \tag{18}$$

with the endpoint interpreted continuously, shows that this product is at most $e^{\sum_i c_i/2} = e^{k/2}$. If $k = n$, the density is simply the orthogonal image of the product density and is at most one.

The product-marginal theorem of Livshyts et al. (2016, Theorem 1.1) gives, for $k < n$, the bound

$$\min\left\{\left(\frac{n}{n-k}\right)^{(n-k)/2}, 2^{k/2}\right\}.$$

Its hypotheses hold for the independent product densities with cap one considered here; see also Rudelson and Vershynin (2015) for a dimension-free projection theorem. Finally, write $T = (TT^{\mathsf{T}})^{1/2}U^{\mathsf{T}}$ and apply the change-of-variables formula to the invertible first factor. □

Lemma 4.3 (Modulated scalar Fourier estimate)**.** *Let f be a probability density with $0 \le f \le 1$, and let ω be measurable with $|\omega| \le 1$. For $h = f\omega$ and every real $p \ge 2$,*

$$\int_{\mathbb{R}} |\widehat{h}(\xi)|^p \, d\xi \le \frac{2\pi}{\sqrt{p}}\left(\frac{p}{p-1}\right)^{(p-1)/2} \le \frac{2\pi\sqrt{e}}{\sqrt{p}} \le \frac{4\pi}{\sqrt{p}}. \tag{19}$$

In particular the estimate is uniform in every measurable phase $\omega(x) = e^{iV(x)}$, and in translations of the Fourier variable.

Proof. Set $p' = p/(p-1)$. Since $|h| \le f \le 1$ and $\int f = 1$, we have $\|h\|_{p'} \le 1$. Apply (12) with input exponent p', then raise to the power p:

$$\|\widehat{h}\|_p^p \le 2\pi(p')^{p/(2p')}p^{-1/2} = 2\pi p^{-1/2}\left(1 + \frac{1}{p-1}\right)^{(p-1)/2}.$$

The inequality $\log(1+x) \le x$ proves the second bound, and $\sqrt{e} < 2$ proves the last one. Translation invariance of Lebesgue measure proves the assertion about Fourier shifts. □

Lemma 4.4 (Difference densities and Fourier powers)**.** *Let f be as in Lemma 4.3, put $\widetilde{f}(x) = f(-x)$ and $g = f * \widetilde{f}$. Then g is a probability density bounded by one, and, for $0 < c \le 1$,*

$$\|g\|_{1/c} \le 2^c c^{c/2}(1+c)^{-(1+c)/2}. \tag{20}$$

If $q \ge 1$ and $0 < d \le q/2$, then

$$\left\||\widehat{f}|^q\right\|_{1/d} \le (2\pi)^d q^{q/2-d}(q-d)^{-(q-d)/2}d^{d/2}. \tag{21}$$

Proof. The identity defining g gives $\int g = 1$ and $g(x) \le \|f\|_\infty \int \widetilde{f} \le 1$. Use (13) with both input exponents $2/(1+c)$ and output $1/c$. Their norms are at most one. Substitution in $C_{2/(1+c)}^2/C_{1/c}$ gives $2^c c^{c/2}(1+c)^{-(1+c)/2}$, proving (20). This remains valid at $c = 1$ by continuity. For the second estimate, the Fourier exponent q/d is at least two and its conjugate input exponent

is $q/(q-d)$. Apply (12) and raise to the power q. Since the input norm is at most one, the resulting constant is

$$(2\pi)^d \left(\frac{q}{q-d}\right)^{(q-d)/2} \left(\frac{q}{d}\right)^{-d/2},$$

which is (21). □

4.3 Fresnel linearization with separable phases

Lemma 4.5 (Regularized Fresnel formula)**.** *Let Λ be an invertible symmetric $r \times r$ real matrix, $t \neq 0$, and $\varepsilon > 0$. Then*

$$\begin{aligned} K_{\varepsilon,t}(\xi) &:= \int_{\mathbb{R}^r} e^{-\varepsilon|y|^2 + ity^{\mathsf{T}}\Lambda y - i\langle \xi, y\rangle}\, dy \\ &= \pi^{r/2} \det(\varepsilon I - it\Lambda)^{-1/2} \exp\left[-\frac{1}{4}\xi^{\mathsf{T}}(\varepsilon I - it\Lambda)^{-1}\xi\right], \end{aligned} \tag{22}$$

where the square root is the product of the branches having positive real part in the scalar Gaussian factors. Moreover,

$$|K_{\varepsilon,t}(\xi)| \le \pi^{r/2}|t|^{-r/2}|\det \Lambda|^{-1/2}. \tag{23}$$

If $g, \widehat{g} \in L^1(\mathbb{R}^r)$, it follows that

$$\left|\int_{\mathbb{R}^r} e^{ity^{\mathsf{T}}\Lambda y} g(y)\, dy\right| \le (2\pi)^{-r} \|\widehat{g}\|_1 \pi^{r/2}|t|^{-r/2}|\det \Lambda|^{-1/2}. \tag{24}$$

Proof. Diagonalize Λ orthogonally. Each one-dimensional integral is

$$\int_{\mathbb{R}} e^{-(\varepsilon - it\lambda)y^2 - i\eta y}\, dy = \sqrt{\pi}(\varepsilon - it\lambda)^{-1/2} e^{-\eta^2/[4(\varepsilon - it\lambda)]}.$$

The Gaussian identity for a positive real coefficient extends to the right half-plane by analytic continuation, with domination on compact subsets; this proves the displayed identity for every $\varepsilon > 0$. Multiplying the scalar factors proves (22). The real part of $(\varepsilon - it\lambda)^{-1}$ is $\varepsilon/(\varepsilon^2 + t^2\lambda^2) > 0$, so the exponential factors have modulus at most one. Also $|\varepsilon - it\lambda|^{-1/2} \le |t\lambda|^{-1/2}$. These observations prove (23).

Insert the inverse Fourier transform of g in the integral with the Gaussian cutoff. Fubini is justified by $\widehat{g} \in L^1$ and the integrable Gaussian. Bound its inner integral by (23). Finally let $\varepsilon \downarrow 0$; the left side converges by domination with $|g|$. This proves (24). □

Theorem 4.6 (Diffuse quadratic components and separable phases)**.** *Let $U^{\mathsf{T}}U = I_r$, let $c_i = \|U^{\mathsf{T}}e_i\|^2 \le 1/2$, and let Λ be invertible and symmetric. For arbitrary real measurable functions V_i, finite almost surely, define*

$$Y = (U^{\mathsf{T}}X)^{\mathsf{T}}\Lambda(U^{\mathsf{T}}X) + \sum_i V_i(X_i), \qquad B(U) = \prod_{i:c_i>0} (1-c_i)^{-(1-c_i)/2}.$$

Writing $\Delta = |\det \Lambda|$, for every $r \ge 1$,

$$|\varphi_Y(t)| \le \min\{1, B(U)\pi^{r/2}\Delta^{-1/2}|t|^{-r/2}\}, \qquad B(U) \le e^{r/2}. \tag{25}$$

If $r > 2$, then Y has a continuous density vanishing at infinity, and

$$\|p_Y\|_\infty \le \frac{r}{r-2} B(U)^{2/r}\Delta^{-1/r} \le \frac{er}{r-2}\Delta^{-1/r}. \tag{26}$$

The same conclusion holds if the indicated representation is available only for a principal block in the original coordinates of a quadratic form; all other blocks and couplings, as well as affine terms, may be arbitrary.

Proof. For fixed t, let $h_{i,t}(x) = f_i(x)e^{itV_i(x)}$. Push the complex product measure $\prod_i h_{i,t}(x_i)\,dx_i$ forward by $x \mapsto U^{\mathsf{T}}x$. Integrating over the orthogonal complement, as in Lemma 4.2, gives an L^1 density g_t with $\|g_t\|_1 \le 1$. Fubini gives

$$\widehat{g}_t(\xi) = \prod_i \widehat{h}_{i,t}(\langle U^{\mathsf{T}}e_i, \xi\rangle).$$

Rows with $c_i = 0$ contribute factors of modulus at most one. For the other rows set $v_i = U^{\mathsf{T}}e_i/\sqrt{c_i}$. Then $\sum_i c_i v_i v_i^{\mathsf{T}} = I_r$. The geometric Brascamp–Lieb inequality applied to $H_i(s) = |\widehat{h}_{i,t}(\sqrt{c_i}s)|^{1/c_i}$ yields

$$\|\widehat{g}_t\|_1 \le \prod_{i:c_i>0} c_i^{-c_i/2}\|\widehat{h}_{i,t}\|_{1/c_i}.$$

The output exponent $1/c_i$ is at least two. Sharp Hausdorff–Young with conjugate exponent $1/(1-c_i)$ gives

$$\|\widehat{h}_{i,t}\|_{1/c_i} \le (2\pi)^{c_i}(1-c_i)^{-(1-c_i)/2}c_i^{c_i/2}.$$

The c_i factors cancel, and $\sum_i c_i = r$, so

$$\|\widehat{g}_t\|_1 \le (2\pi)^r B(U). \tag{27}$$

The entropy inequality (18) gives $B(U) \le e^{r/2}$.

The characteristic function is exactly $\int e^{ity^{\mathsf{T}}\Lambda y}g_t(y)\,dy$. Lemma 4.5 and (27) prove (25); the bound one follows from the probability law of Y. Lemma 4.1, with $b = r/2$ and $a = \pi B(U)^{2/r}\Delta^{-1/r}$, proves (26).

For the principal-block statement, condition on the complementary original coordinates. The remaining coordinates keep their original independent distributions. The outside couplings induce only linear terms in them, and these are absorbed in the functions V_i. The bound (25) is uniform in all conditioning values, so averaging preserves it and its integrability. □

4.4 Decoupling and volume-sampling Brascamp–Lieb inequalities

Lemma 4.7 (A cross-block decoupling inequality)**.** *Let I, J be disjoint original coordinate sets, let M be a real $|I| \times |J|$ matrix, and let*

$$Y = F(X_I) + 2X_I^{\mathsf{T}}MX_J + G(X_J),$$

where F, G are arbitrary real measurable functions finite almost surely. If X'_J is an independent copy and $D = X_J - X'_J$, then

$$|\varphi_Y(t)|^2 \le \mathbb{E}\prod_{i\in I}|\widehat{f}_i(2t(MD)_i)|. \tag{28}$$

For every $q \ge 1$, consequently,

$$|\varphi_Y(t)|^{2q} \le \mathbb{E}\prod_{i\in I}|\widehat{f}_i(2t(MD)_i)|^q. \tag{29}$$

Each coordinate of D has its own difference density and these coordinates are independent.

Proof. Write the characteristic function by first integrating X_J. Cauchy–Schwarz in the probability law of X_I, after taking absolute values of the factor $e^{itF(X_I)}$, gives

$$|\varphi_Y(t)|^2 \le \mathbb{E}_{X_I}\left|\mathbb{E}_{X_J}e^{itG(X_J)+2itX_I^{\mathsf{T}}MX_J}\right|^2.$$

Expand the square using X'_J. Integration in X_I gives the product of the scalar transforms in (28), multiplied by $e^{it(G(X_J)-G(X'_J))}$. The triangle inequality proves (28). Jensen's inequality for the probability law of D proves (29). Independence and the difference-density description follow directly from the independent coordinate pairs. □

For a real $m \times d$ matrix M of rank at least k, write

$$Z_k = e_k(M^{\mathsf{T}}M) = \sum_{\substack{S\subseteq[m],\,T\subseteq[d]\\ |S|=|T|=k}} \det(M_{S,T})^2 > 0. \tag{30}$$

Here e_k is the elementary symmetric polynomial of the squared singular values. The equality follows from Cauchy–Binet. The squared-minor probability law and its marginals are

$$w_{S,T} = \frac{\det(M_{S,T})^2}{Z_k}, \qquad \alpha_i = \sum_{S\ni i,T} w_{S,T}, \qquad \beta_j = \sum_{S,T\ni j} w_{S,T}. \tag{31}$$

Thus $0 \le \alpha_i, \beta_j \le 1$ and $\sum_i \alpha_i = \sum_j \beta_j = k$.

Lemma 4.8 (Volume-sampling Fourier estimate)**.** *In the setting of Lemma 4.7, suppose $Z_k > 0$. Let $q \ge 1$ and assume*

$$\alpha_i \le q/2 \quad \textit{for every row } i. \tag{32}$$

Then, for $t \ne 0$,

$$|\varphi_Y(t)|^{2q} \le e^k q^{-k/2} \left(\frac{\pi}{|t|}\right)^k Z_k^{-1/2}, \tag{33}$$

$$|\varphi_Y(t)|^{2q} \le e^{k/2} q^{-k/2} \left(\frac{2\pi}{|t|}\right)^k Z_k^{-1/2}. \tag{34}$$

The first bound is stronger. Neither bound requires a condition on the column marginals.

Proof. Put $h_i(u) = |\widehat{f}_i(2tu)|^q$, and let g_j be the difference density of D_j. By (29) it suffices to bound

$$\int_{\mathbb{R}^d} \prod_i h_i((Mx)_i) \prod_j g_j(x_j)\, dx.$$

Use row maps $x \mapsto (Mx)_i$ with exponents α_i, and coordinate maps $x \mapsto x_j$ with exponents $1-\beta_j$. Their sum is d. Rows with $\alpha_i = 0$ can be dropped because $h_i \le 1$. Coordinates with $\beta_j = 1$ can be dropped because $g_j \le 1$. Every retained row map is nonzero: a zero row occurs in no nonzero minor.

We first bound the Brascamp–Lieb constant of this datum. Let $a_i > 0$ be Gaussian row parameters after absorbing the α_i factors, and let $b_j > 0$ be the similarly absorbed parameters of the active coordinate maps. Set $a_i = 0$ for omitted rows and $b_j = 0$ when $\beta_j = 1$. Expansion of a diagonal-plus-positive-semidefinite determinant, followed by Cauchy–Binet, gives

$$\begin{aligned}\det(D_b + M^{\mathsf{T}} D_a M) &= \sum_{T\subseteq[d]} \left(\prod_{j\notin T} b_j\right) \det\left((M^{\mathsf{T}} D_a M)_{T,T}\right) \\ &= \sum_{\substack{S\subseteq[m],\,T\subseteq[d]\\ |S|=|T|}} \det(M_{S,T})^2 \prod_{i\in S} a_i \prod_{j\notin T} b_j. \end{aligned} \tag{35}$$

This is a polynomial identity and is valid when some diagonal parameters vanish. Retain the terms with $|S| = |T| = k$. Weighted arithmetic–geometric mean with probabilities $w_{S,T}$ gives

$$\det(D_b + M^{\mathsf{T}} D_a M) \ge Z_k \prod_i a_i^{\alpha_i} \prod_j b_j^{1-\beta_j}. \tag{36}$$

Every column with $\beta_j = 1$ belongs to every T of positive weight, so its zero parameter never occurs in the product of an active summand. The same observation applies to omitted rows.

Thus all active products in the arithmetic–geometric mean step are positive, and no inverse of a zero parameter is used.

Substitute (36) in (11), restoring the absorbed exponents. The Brascamp–Lieb constant is at most

$$Z_k^{-1/2} \prod_i \alpha_i^{-\alpha_i/2} \prod_j (1-\beta_j)^{-(1-\beta_j)/2}. \tag{37}$$

Zero exponents in this formula have value one by continuity. Consequently the integral under consideration is at most this constant times $\prod_i \|h_i\|_{1/\alpha_i} \prod_j \|g_j\|_{1/(1-\beta_j)}$. The latter column norms are at most one.

The scalar estimate (21), including the change of variable $2tu$, gives

$$\|h_i\|_{1/\alpha_i} \le \left(\frac{\pi}{|t|}\right)^{\alpha_i} \left(\frac{\alpha_i}{q}\right)^{\alpha_i/2} \left(1-\frac{\alpha_i}{q}\right)^{-(q-\alpha_i)/2}. \tag{38}$$

Its use is legitimate precisely because $q/\alpha_i \ge 2$. After cancellation of the powers of α_i, the row entropy factor is at most $e^{k/2}$, by (18) applied to $x = \alpha_i/q$. The column entropy factor in (37) is also at most $e^{k/2}$. Since $\sum_i \alpha_i = k$, this proves (33).

Alternatively, use only the last bound in (19) to obtain

$$\|h_i\|_{1/\alpha_i} \le \left(\frac{2\pi}{|t|}\right)^{\alpha_i} \left(\frac{\alpha_i}{q}\right)^{\alpha_i/2}.$$

The same cancellation, followed by just the column entropy bound, proves (34). The stronger estimate follows also numerically from $e < 2\sqrt{e}$. □

Theorem 4.9 (General volume density criterion)**.** *Under the hypotheses of Lemma 4.8, suppose in addition that $k > 2q$. Then Y has a continuous density vanishing at infinity and*

$$\|p_Y\|_\infty \le \frac{ke}{(k-2q)\sqrt{q}}\, Z_k^{-1/(2k)}. \tag{39}$$

Using the second estimate of that lemma gives the also valid bound

$$\|p_Y\|_\infty \le \frac{2k\sqrt{e}}{(k-2q)\sqrt{q}}\, Z_k^{-1/(2k)}. \tag{40}$$

Both conclusions apply to a cross block $M = A_{I,J}$ in a larger quadratic form with arbitrary self blocks, outside couplings, and affine terms.

Proof. The first Fourier envelope has exponent $b = k/(2q) > 1$. Its crossover scale is $a = e\pi q^{-1/2} Z_k^{-1/(2k)}$. Lemma 4.1 proves (39). The second crossover scale is $2\pi\sqrt{e}\, q^{-1/2} Z_k^{-1/(2k)}$, giving (40). For a larger form, condition on the coordinates outside $I \cup J$. They change only the functions F, G and an additive constant. The Fourier envelopes remain uniform and can therefore be averaged. □

Corollary 4.10 (Three-volume and five-volume cases)**.** *The following explicit cases of Theorem 4.9 hold:*

1. *If $Z_3 > 0$ and $\max_i \alpha_i^{(3)} \le 1/2$, then*

$$\|p_Y\|_\infty \le 3e\, Z_3^{-1/6} \le 6\sqrt{e}\, Z_3^{-1/6}.$$

2. *If $Z_3 > 0$ and $\max_i \alpha_i^{(3)} \le h$ with $1/2 \le h < 3/4$, then*

$$\|p_Y\|_\infty \le \frac{3e}{(3-4h)\sqrt{2h}}\, Z_3^{-1/6} \le \frac{6\sqrt{e}}{(3-4h)\sqrt{2h}}\, Z_3^{-1/6}.$$

3. If $Z_5 > 0$, with no marginal hypothesis, then

$$\|p_Y\|_\infty \le \frac{5e}{\sqrt{2}}\, Z_5^{-1/10} \le 5\sqrt{2e}\, Z_5^{-1/10}.$$

These are respectively the three-volume, fractional three-volume, and unconditional five-volume criteria.

Proof. Take $(k,q) = (3,1), (3,2h), (5,2)$, respectively. In the last case every inclusion probability is at most one, so the gate $\alpha_i^{(5)} \le q/2$ is automatic. The strict inequality $k > 2q$ holds in all three cases. □

Remark 4.11 (Choice of a minor law). The canonical squared-minor probabilities can be replaced by any probability law $v_{S,T}$ supported on nonzero $k \times k$ minors. If its row marginals satisfy the same gate, the bound (39) acquires the factor

$$\exp\!\left(\frac{D(v\|w)}{2k}\right), \qquad D(v\|w) = \sum_{S,T} v_{S,T} \log \frac{v_{S,T}}{w_{S,T}}.$$

Indeed, apply weighted arithmetic–geometric mean to $\sum v_{S,T}[\det(M_{S,T})^2/v_{S,T}] \prod_{i\in S} a_i \prod_{j\notin T} b_j$. The determinant lower bound has $Z_k e^{-D(v\|w)}$ in place of Z_k. All subsequent steps are identical. This form allows the marginal constraint to be traded against an explicit entropy cost.

4.5 Exact marginal factors and entropy optimization

Theorem 4.12 (Exact weighted-volume factors)**.** *Let $w_{S,T}$ be any probability law on the nonzero $k \times k$ minors of a physical cross block M, and let α_i, β_j be its row and column inclusion probabilities. Set*

$$G(w) = \exp\left(\sum_{S,T} w_{S,T} \log \frac{\det(M_{S,T})^2}{w_{S,T}}\right).$$

For $q \ge 1$ with $\alpha_i \le q/2$, define

$$R_q(w) = \prod_i \left(1 - \frac{\alpha_i}{q}\right)^{-(q-\alpha_i)/2} \prod_j (1-\beta_j)^{-(1-\beta_j)/2}.$$

All zero-exponent factors have their continuous value one. Then, in the setting of Lemma 4.7,

$$|\varphi_Y(t)|^{2q} \le R_q(w) q^{-k/2} \pi^k G(w)^{-1/2} |t|^{-k}. \tag{41}$$

If $k > 2q$, it follows that

$$\|p_Y\|_\infty \le \frac{k}{(k-2q)\sqrt{q}}\, R_q(w)^{1/k} G(w)^{-1/(2k)}. \tag{42}$$

The density is continuous and vanishes at infinity. In particular, for $k = 3, q = 1$, writing $R = R_1$,

$$\|p_Y\|_\infty \le 3R(w)^{1/3} G(w)^{-1/6}, \qquad \|\varphi_Y\|_1 \le 6\pi R(w)^{1/3} G(w)^{-1/6}. \tag{43}$$

Proof. In the determinant expansion (35), apply arithmetic–geometric mean with the law w to obtain

$$\det(D_b + M^{\mathsf{T}} D_a M) \ge G(w) \prod_i a_i^{\alpha_i} \prod_j b_j^{1-\beta_j}.$$

The argument for saturated marginals in the proof of Lemma 4.8 applies without change. Thus its Brascamp–Lieb constant is (37) with $G(w)$ in place of Z_k. Use the exact scalar estimate (38) and leave the row and column factors unrelaxed. The powers of α_i cancel, and $\sum_i \alpha_i = k$; the result is (41). Its exponent is $k/(2q)$ and its crossover scale is $\pi q^{-1/2} R_q(w)^{1/k} G(w)^{-1/(2k)}$. Lemma 4.1 proves (42) and the last assertion. □

The entropy inequality gives $R_q(w) \le e^k$, recovering the coefficient in Lemma 4.8. For the canonical squared-minor law $p_{S,T} = \det(M_{S,T})^2/Z_k$, the identity

$$G(w) = Z_k e^{-D(w\|p)}$$

shows that p maximizes G without marginal constraints. It does not assert that p minimizes the whole expression in (42); the marginal factor depends on the law as well.

Proposition 4.13 (Balanced six-row and six-column blocks)**.** *If a law on 3×3 minors of a six-row, six-column physical cross block has every row and column marginal $1/2$, then*

$$\|p_Y\|_\infty \le 6G(w)^{-1/6}.$$

For example, let $u = 6^{-1/2}(1, \dots, 1)^{\mathsf{T}}$ and

$$M = sI_6 + (L - s)uu^{\mathsf{T}}, \qquad L > s > 0.$$

With arbitrary self functions on the two banks,

$$\|p_Y\|_\infty \le 6[10s^4(L^2 + s^2)]^{-1/6}. \tag{44}$$

This example need not satisfy the sufficient projection condition of Corollary 4.28.

Proof. Each of the twelve marginal factors equals $2^{1/4}$, so $R(w) = 8$. Formula (43) proves the first assertion. The displayed matrix has singular values L, s, s, s, s, s. Simultaneous coordinate-permutation symmetry makes its canonical row and column marginal vectors constant; since each sums to three, all entries are $1/2$. Moreover

$$Z_3 = \binom{5}{2} L^2 s^4 + \binom{5}{3} s^6 = 10s^4(L^2 + s^2).$$

The canonical law has $G = Z_3$, proving the bound. For the final assertion, any projection with all diagonal entries at most $1/2$ has rank at most three and is therefore proper. For a unit v in its orthogonal complement, $\|M^{\mathsf{T}} v\| \ge s$; hence $\|(I - P)M\|_{\mathrm{op}} \ge s$. Here $T_3 = 4s^2$, so the sufficient requirement $\|(I - P)M\| \le \sqrt{T_3}/15 = 2s/15$ fails. □

We next optimize the determinant entropy subject to a row marginal constraint. All optimization sets are finite dimensional, and all their coordinates refer to original rows. For a row k-set S, define

$$a_S = \det(M_S M_S^{\mathsf{T}}) = \sum_{|T|=k} \det(M_{S,T})^2, \qquad \mathcal{B} = \{S : a_S > 0\}, \qquad Z_k = \sum_{S \in \mathcal{B}} a_S.$$

For a probability law μ on $\mathcal{B}$, let

$$\alpha_i(\mu) = \sum_{S \ni i} \mu_S, \qquad G(\mu) = \exp\left(\sum_S \mu_S \log(a_S/\mu_S) \right).$$

Terms with $\mu_S = 0$ are interpreted by continuity.

Lemma 4.14 (Reduction to row entropy)**.** *For fixed row law μ, the lift*

$$w_{S,T} = \mu_S \frac{\det(M_{S,T})^2}{a_S} \tag{45}$$

has the same row marginals and maximizes $G(w)$ among all minor laws with row law μ. Its value is $G(\mu)$. If $p_S = a_S/Z_k$, then

$$G(\mu) = Z_k e^{-D(\mu\|p)}. \tag{46}$$

Proof. For a general lift let ν_S be the conditional column law given S, and let $\pi_S(T) = \det(M_{S,T})^2/a_S$. Expanding logarithms gives

$$\log G(w) = \sum_S \mu_S \log(a_S/\mu_S) - \sum_S \mu_S D(\nu_S\|\pi_S).$$

Relative entropy is nonnegative, as follows from $\log x \le x - 1$, with equality for identical laws. Thus the maximum is attained by $\nu_S = \pi_S$, which is (45). Expanding $D(\mu\|p)$ proves (46). □

Theorem 4.15 (Gibbs dual and concentration on a bounded row set)**.** *Assume $Z_k > 0$. For $0 < h < 1$, let*

$$\mathcal{F}_h = \{\mu : \alpha_i(\mu) \le h \text{ for every } i\}, \qquad G_h(M) = \max_{\mu\in\mathcal{F}_h} G(\mu),$$

with $G_h(M) = 0$ if $\mathcal{F}_h$ is empty. Then

$$G_h(M) = \inf_{\lambda_i\ge 0} \exp\left(h\sum_i \lambda_i\right) \sum_{S\in\mathcal{B}} a_S \exp\left(-\sum_{i\in S}\lambda_i\right). \tag{47}$$

If a finite vector λ attains the infimum, the corresponding maximizing law is

$$\mu_\lambda(S) = \frac{a_S e^{-\sum_{i\in S}\lambda_i}}{\sum_R a_R e^{-\sum_{i\in R}\lambda_i}}, \tag{48}$$

and $\lambda_i > 0$ implies $\alpha_i(\mu_\lambda) = h$. Regardless of dual attainment, there exists a set H of rows such that

$$|H| < k/h, \qquad e_k(M_{[m]\setminus H}^{\mathsf{T}} M_{[m]\setminus H}) \le G_h(M). \tag{49}$$

Proof. Let Δ be the probability simplex on $\mathcal{B}$, and define its continuous concave entropy function

$$F(\mu) = \sum_S \mu_S \log(a_S/\mu_S).$$

For $\lambda_i \ge 0$, put

$$\mathcal{L}(\mu,\lambda) = F(\mu) + \sum_i \lambda_i(h - \alpha_i(\mu)).$$

The Gibbs variational identity states

$$\max_{\mu\in\Delta} \mathcal{L}(\mu,\lambda) = h\sum_i \lambda_i + \log \sum_S a_S e^{-\sum_{i\in S}\lambda_i}. \tag{50}$$

Indeed, subtracting the right side from $\mathcal{L}(\mu,\lambda)$ gives $-D(\mu\|\mu_\lambda)$.

We justify the dual equality also when constraints are attained only on the boundary or are infeasible. Restrict first to $0 \le \lambda_i \le N$. The compact convex minimax theorem (Sion, 1958), applied to the continuous concave-affine function $\mathcal{L}$, gives

$$\min_{\lambda\in[0,N]^m} \max_{\mu\in\Delta} \mathcal{L}(\mu,\lambda) = \max_{\mu\in\Delta} \left[F(\mu) - N\sum_i (\alpha_i(\mu) - h)_+\right].$$

If $\mathcal{F}_h \neq \varnothing$, the right side decreases, as N increases, to $\max_{\mathcal{F}_h} F$. To see this explicitly, take maximizing laws along a sequence $N \to \infty$ and a convergent subsequence in the compact simplex. Since F is bounded and a feasible law gives a fixed finite lower bound, their constraint violations tend to zero. The limiting law is feasible. Dropping the nonpositive penalty and using continuity of F proves that the limiting value is at most the constrained maximum; the reverse inequality holds for every N. If $\mathcal{F}_h$ is empty, the continuous sum of positive constraint violations has a strictly positive minimum on Δ, so these maxima tend to $-\infty$. Taking $N \to \infty$ and using (50) proves the logarithm of (47), including the value $-\infty$ in the infeasible case.

If the dual minimizer is finite, differentiation of its logarithmic objective gives $h - \alpha_i(\mu_\lambda)$. Optimality on the nonnegative orthant therefore implies $\alpha_i(\mu_\lambda) \leq h$, and equality if $\lambda_i > 0$. Substitution in $\mathcal{L}$ shows that this Gibbs law attains the primal maximum.

It remains to prove (49). If $G_h = Z_k$, take $H = \varnothing$. Otherwise choose, for arbitrarily small $\varepsilon > 0$, a nonnegative vector λ whose dual objective is at most $G_h + \varepsilon < Z_k$. Then $\Lambda = \sum_i \lambda_i > 0$. Put $\omega_i = \lambda_i/\Lambda$ and

$$H = \{i : \omega_i > h/k\}.$$

Since $\sum_i \omega_i = 1$, necessarily $|H| < k/h$. For every k-set S avoiding H, $\sum_{i \in S} \lambda_i \leq h\Lambda$. The corresponding term in the dual objective is therefore at least a_S. Thus

$$\sum_{S : S \cap H = \varnothing} a_S \leq G_h + \varepsilon.$$

The left side is the residual k-volume in (49). There are only finitely many possible sets H; along a sequence $\varepsilon \downarrow 0$, one repeats infinitely often. Passing to that subsequence proves the desired bound, including when $G_h = 0$. □

Corollary 4.16 (A three-volume or bounded-row alternative)**.** *Let $1/2 \leq h < 3/4$ and $g > 0$. For a physical cross block M with $Z_3 > 0$, either*

$$\|p_Y\|_\infty \leq \frac{3e}{(3 - 4h)\sqrt{2h}}\, g^{-1/6},$$

with a continuous density vanishing at infinity, or there is a row set H with $|H| < 3/h$ for which the residual third volume is less than g. In particular $h = 3/5$ requires at most four rows in the second alternative.

Proof. If $G_h \geq g$, compactness gives an optimizing feasible law. Lift it by Lemma 4.14 and apply Theorem 4.12 with $q = 2h$ and $R_q \leq e^3$. If $G_h < g$, use (49). At $h = 3/5$, the strict bound $|H| < 5$ is equivalent to $|H| \leq 4$. □

Proposition 4.17 (A single active row constraint)**.** *Fix a distinguished row i_0, and write*

$$Z_1 = \sum_{S \ni i_0} a_S, \qquad Z_0 = \sum_{S \not\ni i_0} a_S.$$

Suppose $Z_0, Z_1 > 0$, $p = Z_1/(Z_0 + Z_1) > h$, and the law

$$\mu(S) = \begin{cases} h\, a_S/Z_1, & i_0 \in S, \\ (1-h) a_S/Z_0, & i_0 \notin S \end{cases}$$

satisfies all other row constraints. Then

$$G_h(M) = \left(\frac{Z_1}{h}\right)^h \left(\frac{Z_0}{1-h}\right)^{1-h} = (Z_0 + Z_1) e^{-D_{\mathrm{Ber}}(h\|p)}. \tag{51}$$

A dual optimizer has all multipliers zero except

$$e^{-\lambda_{i_0}} = \frac{hZ_0}{(1-h)Z_1}.$$

Proof. If a row law gives total mass x to sets containing i_0, the entropy chain rule bounds its logarithmic value by

$$x\log(Z_1/x)+(1-x)\log(Z_0/(1-x)),$$

with equality exactly when the conditional laws in the two classes are canonical. The derivative of this expression is $\log[Z_1(1-x)/(Z_0x)]$, which is positive for $x<p$. Under $x\le h<p$, its maximum is therefore attained at h. The displayed law attains this value and, by assumption, is feasible for every other constraint. This proves (51). Substitution in (48) verifies the stated multiplier and complementary slackness. □

4.6 Regularized determinants and ridge leverage

Lemma 4.18 (A graph inequality with a free regularizer)**.** *Let R be a real $m\times k$ matrix with columns r_j. Let $D_1,\dots,D_m$ be independent differences of independent copies of variables with densities bounded by one, and let $f_1,\dots,f_k$ be further probability densities bounded by one. Fix $\alpha>0$, $a\in\mathbb{R}$, and $1\le q\le 2$. Put*

$$H=\alpha I_m+a^2RR^{\mathsf T},\qquad c_i=\alpha e_i^{\mathsf T}H^{-1}e_i,\qquad d_j=a^2r_j^{\mathsf T}H^{-1}r_j,\qquad \theta=\sum_j d_j=\sum_i(1-c_i).$$

If $d_j\le q/2$ for every j, then, with $L_q=\pi e\sqrt{2/q}$,

$$\mathbb{E}\prod_j|\widehat{f}_j(a\langle r_j,D\rangle)|^q\le\det(I_m+a^2RR^{\mathsf T}/\alpha)^{-1/2}(L_q/\sqrt{\alpha})^{\theta}. \tag{52}$$

Proof. For $a=0$, both the expectation and the determinant are one, and $\theta=0$. Assume $a\neq0$. The vectors

$$u_i=\sqrt{\alpha}H^{-1/2}e_i,\qquad v_j=aH^{-1/2}r_j$$

satisfy $\sum_i u_iu_i^{\mathsf T}+\sum_j v_jv_j^{\mathsf T}=I_m$. Their squared lengths are c_i,d_j, respectively; thus $0<c_i\le1$, $0\le d_j\le1$. The trace identity gives the displayed expression for θ. Zero v_j may be omitted.

Let g_i be the density of D_i, and put $h_j=|\widehat{f}_j|^q$. Change variables $x=H^{-1/2}z$ in the expectation integral and normalize the nonzero vectors u_i,v_j to unit vectors. The geometric Brascamp–Lieb inequality gives

$$\det(H)^{-1/2}\prod_i\left[\alpha^{c_i/2}c_i^{-c_i/2}\|g_i\|_{1/c_i}\right]\prod_j\left[d_j^{-d_j/2}\|h_j\|_{1/d_j}\right]. \tag{53}$$

For example, the argument of g_i is $\sqrt{c_i/\alpha}$ times the relevant unit linear form, which accounts for its factor $\alpha^{c_i/2}c_i^{-c_i/2}$.

Define

$$G(c)=2^c(1+c)^{-(1+c)/2},\qquad H_q(d)=q^{q/2-d}(q-d)^{-(q-d)/2}.$$

Lemma 4.4 cancels the powers of c_i,d_j in (53), leaving

$$\det(H)^{-1/2}\alpha^{(m-\theta)/2}(2\pi)^{\theta}\prod_i G(c_i)\prod_j H_q(d_j). \tag{54}$$

The logarithms of these scalar functions are concave:

$$(\log G)''(c)=-\frac{1}{2(1+c)},\qquad(\log H_q)''(d)=-\frac{1}{2(q-d)}.$$

The tangent to $\log G$ at $c=1$, and the tangent to $\log H_q$ at $d=0$, therefore give

$$\log G(c)\le\frac{1-\log2}{2}(1-c),\qquad\log H_q(d)\le\frac{1-\log q}{2}d.$$

Their endpoint values are $G(1) = H_q(0) = 1$. The product in (54) is consequently at most $\exp\{[1 - \frac{1}{2}\log(2q)]\theta\}$. Factor α^m from $\det H$. The remaining scalar multiplier is

$$\left(\frac{2\pi e}{\sqrt{2q\alpha}}\right)^{\theta} = (L_q/\sqrt{\alpha})^{\theta},$$

which proves (52). □

Theorem 4.19 (Regularized cross-block determinant)**.** *Let Y have the form in Lemma 4.7, with cross block B. Let $1 \le q \le 2$. Suppose that either $q = 2$ or the projection $P_{\mathrm{range}(B)}$, acting on the row-index space, satisfies*

$$\max_i (P_{\mathrm{range}(B)})_{ii} \le q/2.$$

Then

$$|\varphi_Y(t)| \le \det\left(I + \frac{2qt^2}{\pi^2 e^2} BB^{\mathsf{T}}\right)^{-1/(4q)}. \tag{55}$$

In particular, without any leverage hypothesis,

$$|\varphi_Y(t)| \le \det\left(I + \frac{4t^2}{\pi^2 e^2} BB^{\mathsf{T}}\right)^{-1/8}. \tag{56}$$

If B has at least $r > 2q$ singular values at least $\sigma > 0$, the indicated hypothesis implies a continuous density with

$$\|p_Y\|_\infty \le \frac{e\sqrt{\pi}}{2\sqrt{2q}\,\sigma} \frac{\Gamma(r/(4q) - 1/2)}{\Gamma(r/(4q))} \tag{57}$$

$$\le \frac{e}{\sqrt{2q}\,\sigma} \frac{r}{r - 2q}. \tag{58}$$

All these statements allow arbitrary functions in the two self blocks, and conditioning on additional original coordinates.

Proof. Apply (29) and then Lemma 4.18 with $R = B^{\mathsf{T}}$, $a = 2t$, and $\alpha = L_q^2$. The graph weights on the Fourier factors are the diagonal entries of

$$(2t)^2 B(L_q^2 I + (2t)^2 B^{\mathsf{T}} B)^{-1} B^{\mathsf{T}}.$$

This matrix is bounded between zero and $P_{\mathrm{range}(B)}$, as is seen by singular-value decomposition. Its diagonal entries are always at most one, so $q = 2$ needs no assumption. The multiplier in (52) is one for this choice of α. Taking the $2q$-th root and using equality of the nonzero spectra of BB^{T} and $B^{\mathsf{T}}B$ proves (55).

Discarding determinant factors other than the r indicated ones gives

$$|\varphi_Y(t)| \le \left(1 + \frac{2q\sigma^2 t^2}{\pi^2 e^2}\right)^{-r/(4q)}.$$

The integral

$$\int_{\mathbb{R}} (1 + x^2)^{-a}\, dx = \sqrt{\pi}\, \frac{\Gamma(a - 1/2)}{\Gamma(a)}, \qquad a > 1/2,$$

follows from $u = x^2$ and the beta integral. Fourier inversion now proves (57). Bounding the same envelope by $\min\{1, (\pi e/(\sqrt{2q}\sigma|t|))^{r/(2q)}\}$ and applying Lemma 4.1 proves the simpler bound. The uniformity in self functions and outside coordinates is inherited from Lemma 4.7. □

Theorem 4.20 (Ridge-leverage criterion)**.** *For a real cross block B and $\gamma > 0$, define*

$$K_\gamma = BB^{\mathsf{T}}(\gamma^2 I + BB^{\mathsf{T}})^{-1}, \quad \theta_\gamma = \operatorname{tr} K_\gamma, \quad D_\gamma = \det(I + BB^{\mathsf{T}}/\gamma^2).$$

Suppose, for some $1 \le q \le 2$, that

$$\max_i (K_\gamma)_{ii} \le q/2, \qquad \theta_\gamma > 2q. \tag{59}$$

Then Y in Lemma 4.7 has a continuous density vanishing at infinity and

$$\|p_Y\|_\infty \le \frac{e}{\sqrt{2q}\,\gamma} \frac{\theta_\gamma}{\theta_\gamma - 2q} D_\gamma^{-1/(2\theta_\gamma)} \tag{60}$$

$$\le \frac{\sqrt{e}}{\sqrt{2q}\,\gamma} \frac{\theta_\gamma}{\theta_\gamma - 2q}. \tag{61}$$

The two coordinate sets may be interchanged to use the ridge diagonal on the other side.

Proof. For $t \neq 0$, use the same graph argument as in Theorem 4.19, but choose $\alpha = 4\gamma^2 t^2$. The graph weights d_i are now exactly $(K_\gamma)_{ii}$, independent of t; their sum is θ_γ. The first condition in (59) makes the scalar Fourier exponents admissible. Keeping the multiplier in (52), and taking the $2q$-th root, gives

$$|\varphi_Y(t)| \le D_\gamma^{-1/(4q)} \left(\frac{L_q}{2\gamma|t|} \right)^{\theta_\gamma/(2q)}.$$

The power is greater than one. Its crossover scale is

$$a = \frac{L_q}{2\gamma} D_\gamma^{-1/(2\theta_\gamma)}.$$

Lemma 4.1 and $L_q/(2\pi) = e/\sqrt{2q}$ prove (60). If $x_j = s_j(B)^2/\gamma^2$, then

$$\log D_\gamma = \sum_j \log(1 + x_j) \ge \sum_j \frac{x_j}{1 + x_j} = \theta_\gamma,$$

because the difference between the summands has derivative $x/(1+x)^2$ and vanishes at zero. Therefore $D_\gamma^{-1/(2\theta_\gamma)} \le e^{-1/2}$, which gives (61). □

Corollary 4.21 (Three cross singular values)**.** *Suppose $s_3(B) \ge \sigma > 0$, set $\gamma = \sigma/\sqrt{6}$, and assume $\max_i (K_\gamma)_{ii} \le 1/2$. Then*

$$\|p_Y\|_\infty \le \frac{9\sqrt{3e}}{2\sigma}. \tag{62}$$

Proof. The first three singular values give $\theta_\gamma \ge 3/(1 + 1/6) = 18/7$. Apply (61) with $q = 1$. The function $\theta/(\theta - 2)$ decreases for $\theta > 2$, and its value at $18/7$ is $9/2$. □

Lemma 4.22 (Stability of ridge leverage)**.** *Let $B = B_0 + E$, $\|E\|_{\mathrm{op}} \le \eta$, and let P be the orthogonal projection onto* $\operatorname{range}(B_0)$*. For every $\gamma > 0$, writing $\ell_i = P_{ii}$,*

$$(K_\gamma)_{ii} \le \left(\sqrt{\ell_i} + \frac{\eta}{\gamma} \sqrt{1 - \ell_i} \right)^2. \tag{63}$$

In particular, if $\ell_i \le \ell$ for all i, the weaker bound $(\sqrt{\ell} + \eta/\gamma)^2$ holds.

Proof. The identity

$$K_\gamma = B(\gamma^2 I + B^\mathsf{T} B)^{-1} B^\mathsf{T}$$

shows $0 \le K_\gamma \le I$. For $v \perp \operatorname{range}(B_0)$, $B_0^\mathsf{T} v = 0$; hence

$$v^\mathsf{T} K_\gamma v \le \gamma^{-2} \|B^\mathsf{T} v\|^2 \le (\eta/\gamma)^2 \|v\|^2.$$

Decompose $e_i = Pe_i + (I - P)e_i$ and use the triangle inequality after applying $K_\gamma^{1/2}$. Its norm on the first component is at most $\sqrt{\ell_i}$, and the preceding bound on the second is at most $(\eta/\gamma)\sqrt{1 - \ell_i}$. □

4.7 A spectral-window consequence

We use the two-part frame partition consequence of Marcus et al. (2015, Corollary 1.5): if $\sum_i u_i u_i^\mathsf{T} = I_r$ and $\|u_i\|^2 \le \varepsilon$, the indices admit a partition I, J such that both partial frame operators have norm at most $(1/\sqrt{2} + \sqrt{\varepsilon})^2$. The following corollary applies this partition to the range of a quadratic matrix.

Corollary 4.23 (Diffuse full range)**.** *Let A be symmetric, $s_3(A) \ge s > 0$, and let P_A be the projection onto its full range. If $\max_i (P_A)_{ii} \le 1/32$, then, for arbitrary affine terms and arbitrary additional separable measurable phases,*

$$\|p_{X^\intercal AX + \sum_i V_i(X_i)}\|_\infty \le \frac{96e}{7\sqrt{2}\, s}. \tag{64}$$

The density is continuous and vanishes at infinity.

Proof. Write $A = U\Lambda U^\mathsf{T}$, where U has orthonormal columns spanning its full range. The rows of U form a tight frame with squared lengths at most $1/32$. The cited partition theorem gives

$$\frac{7}{32} I \le M_I := U_I^\mathsf{T} U_I,\ M_J := U_J^\mathsf{T} U_J \le \frac{25}{32} I.$$

Indeed $(1/\sqrt{2} + \sqrt{1/32})^2 = 25/32$, and the two operators sum to I. The nonzero singular values of $B = A_{I,J} = U_I \Lambda U_J^\mathsf{T}$ are those of $M_I^{1/2} \Lambda M_J^{1/2}$, since $U_I M_I^{-1/2}$ and $U_J M_J^{-1/2}$ have orthonormal columns. Multiplication by the two square roots gives $s_3(B) \ge (7/32)s$. Both restricted frames have full column rank, and Λ is invertible, so $\operatorname{range}(B) = \operatorname{range}(U_I)$. Its projection has diagonal

$$u_i^\mathsf{T} M_I^{-1} u_i \le \frac{1/32}{7/32} = \frac{1}{7} < \frac{1}{2}.$$

Apply (58) with $q = 1$, $r = 3$, and $\sigma = 7s/32$. □

Theorem 4.24 (A diffuse spectral window)**.** *Let A be symmetric with $s_3(A) \ge s > 0$, and let P be its spectral projection onto the eigenspaces with $|\lambda| \ge s/64$. If*

$$\max_i P_{ii} \le 1/32, \tag{65}$$

then $X^\mathsf{T} AX + \sum_i V_i(X_i)$, for arbitrary real measurable V_i finite almost surely, has a continuous density vanishing at infinity and

$$\|p\|_\infty \le \frac{288\sqrt{3e}}{13s} < \frac{64}{s}. \tag{66}$$

The number, signs, eigenvectors, and nonzero sizes of the smaller spectral directions are unrestricted.

Proof. Write $A = A_H + E$, where $A_H = U\Lambda_H U^\mathsf{T}$ is the indicated spectral truncation. Then $\|E\|_{\mathrm{op}} \le \eta = s/64$. Apply the same frame partition as in Corollary 4.23. For $B_0 = (A_H)_{I,J}$, its proof gives

$$s_3(B_0) \ge \frac{7s}{32}, \qquad \max_i (P_{\mathrm{range}(B_0)})_{ii} \le \frac{1}{7}.$$

The actual cross block is $B = B_0 + E_{I,J}$, and $\|E_{I,J}\|_{\mathrm{op}} \le \eta$. The singular-value perturbation inequality gives

$$s_3(B) \ge \frac{7s}{32} - \frac{s}{64} = \sigma := \frac{13s}{64}.$$

For completeness, that inequality follows by restricting to the top-three right-singular subspace of B_0, where $\|Bx\| \ge (\frac{7s}{32} - \eta)\|x\|$, and applying the singular-value min–max principle.

Set $\gamma = \sigma/\sqrt{6}$. Lemma 4.22 yields

$$\max_i (K_\gamma)_{ii} \le \left(\frac{1}{\sqrt{7}} + \frac{\sqrt{6}}{13}\right)^2 < \frac{1}{2}.$$

For example $1/\sqrt{7} < 2/5$ and $\sqrt{6}/13 < 1/5$, so the square is less than $9/25$. Corollary 4.21 proves

$$\|p\|_\infty \le \frac{9\sqrt{3e}}{2\sigma} = \frac{288\sqrt{3e}}{13s}.$$

All small spectral directions have remained in the actual cross block B; none was removed from the random quadratic expression. All self terms are permitted by decoupling. This proves the claim including arbitrary separable perturbations. □

4.8 Stability of volume marginals and distributed tails

Proposition 4.25 (Spectral formula for volume marginals)**.** *Let M have rank at least k, with singular-value decomposition $M = U\,\mathrm{diag}(s_j)V^\mathsf{T}$. Put $\lambda_j = s_j^2$, $Z_k = e_k(\lambda)$, and*

$$p_j = \frac{\lambda_j e_{k-1}(\lambda_{\neq j})}{Z_k}, \qquad W_k = U\,\mathrm{diag}(p_j)U^\mathsf{T}.$$

Then the row marginals in (31) satisfy $(W_k)_{ii} = \alpha_i$, and

$$0 \le W_k \le I, \qquad W_k \le \frac{e_{k-1}(\lambda)}{Z_k} MM^\mathsf{T}. \tag{67}$$

Writing $T_k = \sum_{j\ge k} \lambda_j > 0$, we also have

$$W_k \le \frac{k}{T_k} MM^\mathsf{T} \le \frac{k}{s_k(M)^2} MM^\mathsf{T}. \tag{68}$$

Proof. Multiply row i of M by $\sqrt{1+t}$. Each squared k-minor containing that row is multiplied by $1+t$, while the other minors are unchanged. Thus the coefficient of t in the new Z_k is $Z_k\alpha_i$. Let $C = MM^\mathsf{T}$. Using Sylvester's determinant identity and the rank-one determinant formula, the generating polynomial of this row scaling is

$$\det(I + zC)\left[1 + tz\, e_i^\mathsf{T}(I + zC)^{-1}Ce_i\right].$$

Its coefficient of tz^k, after diagonalizing C, is $\sum_j U_{ij}^2 \lambda_j e_{k-1}(\lambda_{\neq j})$. Comparison proves $(W_k)_{ii} = \alpha_i$. This polynomial argument also covers repeated singular values and zero modes.

The decomposition

$$Z_k = \lambda_j e_{k-1}(\lambda_{\neq j}) + e_k(\lambda_{\neq j})$$

gives $0 \le p_j \le 1$, while $e_{k-1}(\lambda_{\neq j}) \le e_{k-1}(\lambda)$. These observations prove (67). Finally,

$$kZ_k = \sum_{|S|=k-1} \Big(\prod_{i\in S} \lambda_i\Big) \sum_{j\notin S} \lambda_j.$$

Removing any $k-1$ indices removes at most the sum of the largest $k-1$ squared singular values. Hence every inner sum is at least T_k, and $kZ_k \ge T_k e_{k-1}(\lambda)$. This proves (68). □

Lemma 4.26 (Projection control of volume marginals)**.** *Let P be any orthogonal projection on the row-index space, $\ell_i = P_{ii}$, and $\varepsilon = \|(I-P)M\|_{\mathrm{op}}$. With the notation of Proposition 4.25,*

$$\sqrt{\alpha_i} \le \sqrt{\ell_i} + \sqrt{\frac{k}{T_k}}\,\varepsilon\sqrt{1-\ell_i}. \tag{69}$$

In particular for $k=3$, $\kappa = s_3(M) > 0$,

$$\sqrt{\alpha_i} \le \sqrt{\ell_i} + \frac{\sqrt{3}\,\varepsilon}{\kappa}\sqrt{1-\ell_i}. \tag{70}$$

Proof. Use the triangle inequality after applying $W_k^{1/2}$ to $e_i = Pe_i + (I-P)e_i$. The bound $W_k \le I$ controls the first component by $\sqrt{\ell_i}$. The first inequality in (68) controls the second by

$$\sqrt{k/T_k}\,\|M^{\mathsf{T}}(I-P)e_i\| \le \sqrt{k/T_k}\,\varepsilon\sqrt{1-\ell_i}.$$

Since $\|W_k^{1/2}e_i\| = \sqrt{\alpha_i}$, this proves the first assertion. The bound $T_3 \ge \kappa^2$ proves the second. □

Theorem 4.27 (A cross block near a diffuse range)**.** *Suppose a physical cross block of a quadratic form satisfies*

$$M = M_0 + E, \qquad s_3(M_0) \ge a > 0, \qquad \max_i (P_{\mathrm{range}(M_0)})_{ii} \le \tfrac{1}{2}, \qquad \|E\|_{\mathrm{op}} \le a/16.$$

Then, with arbitrary self blocks, outside couplings, and affine terms, the quadratic form has a continuous density vanishing at infinity and

$$\|p\|_\infty \le \frac{24\sqrt{3}\,e}{5a} < \frac{23}{a}. \tag{71}$$

No rank bound on either M_0 or E is required.

Proof. Let $P = P_{\mathrm{range}(M_0)}$. Then $\varepsilon = \|(I-P)M\| \le a/16$, and singular-value perturbation gives $\kappa = s_3(M) \ge 15a/16$. Lemma 4.26 gives

$$\sqrt{\alpha_i} \le \sqrt{\ell_i} + \frac{\sqrt{3}}{15}\sqrt{1-\ell_i}.$$

For $0 \le \ell \le 1/2$ and $0 \le d \le 1$, the function $\sqrt{\ell} + d\sqrt{1-\ell}$ is increasing in ℓ. Consequently

$$\alpha_i \le \frac{(1+\sqrt{3}/15)^2}{2} < \frac{2}{3}.$$

For a rational check, replacing $\sqrt{3}$ by 2 gives the still smaller-than-2/3 bound 289/450. Theorem 4.9 with $k=3$, $q=4/3$ therefore gives

$$\|p\|_\infty \le \frac{9\sqrt{3}\,e}{2}\,Z_3^{-1/6}.$$

Since $Z_3 \ge s_1(M)^2 s_2(M)^2 s_3(M)^2 \ge \kappa^6$, the claimed constant follows from $\kappa \ge 15a/16$. □

Corollary 4.28 (Distributed spectral tails)**.** *Let M be a physical cross block of rank at least three and put*

$$T_3 = \sum_{j\ge 3} s_j(M)^2.$$

Suppose a projection P on its row-index space satisfies

$$\max_i P_{ii} \le \tfrac{1}{2}, \qquad \|(I-P)M\|_{\mathrm{op}} \le \sqrt{T_3}/15.$$

Then

$$\begin{aligned} \|p\|_\infty &\le \frac{9\sqrt{3}\,e}{2}\, e_3(M^{\mathsf T}M)^{-1/6} \\ &\le \frac{9\sqrt{3}\,e}{2}[s_1(M)^2 s_2(M)^2 T_3]^{-1/6}. \end{aligned} \tag{72}$$

If $s_2(M) \ge a > 0$ and $T_3 \ge b^2 > 0$, the latter bound is $(9\sqrt{3}\,e/2)a^{-2/3}b^{-1/3}$. In particular, when P projects onto the top two left singular directions and its diagonal is at most $1/2$, the condition $T_3/s_3(M)^2 \ge 225$ suffices.

Proof. Use (69) with $k = 3$. The same marginal estimate as in Theorem 4.27 gives $\alpha_i < 2/3$. The fractional three-volume theorem proves the first bound. The terms of e_3 containing the first two indices sum to $s_1^2 s_2^2 T_3$, proving the second. For the indicated projection, $\|(I-P)M\|_{\mathrm{op}} = s_3(M)$. □

4.9 Small balls for an arbitrary nonzero quadratic form

Lemma 4.29 (Quarter-power concentration)**.** *Let $S \ne 0$ be any real symmetric matrix and let*

$$R = X^{\mathsf T}SX + b^{\mathsf T}X + c.$$

For every $\rho > 0$,

$$\sup_w \mathbb{P}(|R-w| < \rho) \le \min\left\{1, 5\left(\frac{\rho}{\|S\|_{\mathrm{op}}}\right)^{1/4}\right\}. \tag{73}$$

In particular R has no atoms. The same concentration estimate holds with a closed interval.

Proof. Put $L = \|S\|_{\mathrm{op}} > 0$ and $m = \max_i |S_{ii}|$. First suppose $m \ge L/2$, and condition on all coordinates except one attaining m. A real quadratic $ax^2 + dx + e$ with $a \ne 0$ has a preimage of any interval of length 2ρ of total length at most $2\sqrt{2\rho/|a|}$. Indeed, complete the square and note that for an interval $[u, u+\delta]$ the set where $(x-x_0)^2$ belongs to that interval has length at most $2\sqrt{\delta}$: on the nonnegative half-line this is $2(\sqrt{u+\delta} - \sqrt{u})$, with the evident truncation when $u < 0$. The density cap therefore bounds the conditional probability by $4\sqrt{\rho/L}$. For $\rho \le L$, this is at most $4(\rho/L)^{1/4}$; for $\rho > L$, the bound one suffices.

Now suppose $m < L/2$. Let $S_0 = S - \mathrm{diag}(S)$, so $\|S_0\|_{\mathrm{op}} \ge L/2$. For independent uniform signs on the diagonal of a matrix E, define $C_E = (S_0 - ES_0E)/2$. The zero diagonal of S_0 implies $\mathbb{E}C_E = S_0/2$. Convexity of the operator norm shows that some sign choice satisfies $\|C_E\|_{\mathrm{op}} \ge \|S_0\|_{\mathrm{op}}/2 \ge L/4$. For the cut into positive and negative signs,

$$C_E = \begin{pmatrix} 0 & B \\ B^{\mathsf T} & 0 \end{pmatrix}, \qquad \|B\|_{\mathrm{op}} = \|C_E\|_{\mathrm{op}} \ge L/4.$$

This is a cut in the original coordinates. The unrestricted determinant bound (56), retaining only the largest singular value of B, gives

$$|\varphi_R(t)| \le \left(1 + \frac{4\|B\|_{\mathrm{op}}^2 t^2}{\pi^2 e^2}\right)^{-1/8} \le \left(\frac{2\pi e}{L|t|}\right)^{1/4}. \tag{74}$$

The affine terms are allowed throughout the decoupling argument.

For the concentration constant, let $v > 0$. Then $\mathbf{1}_{\{|x|<\rho\}} \le e^v e^{-vx^2/\rho^2}$. Gaussian Fourier inversion and Fubini give

$$\mathbb{P}(|R - w| < \rho) \le \frac{e^v \rho}{2\sqrt{\pi v}} \int_{\mathbb{R}} e^{-\rho^2 t^2/(4v)} |\varphi_R(t)|\, dt.$$

For $0 < \alpha < 1$,

$$\int_{\mathbb{R}} e^{-at^2} |t|^{-\alpha}\, dt = a^{(\alpha-1)/2}\Gamma((1-\alpha)/2),$$

by the substitution $u = at^2$. Use (74), $\alpha = 1/4$, and $v = 1/8$. The coefficient of $(\rho/L)^{1/4}$ is

$$2^{3/8}\pi^{-1/4}e^{3/8}\Gamma(3/8) < 5.$$

For an elementary verification, $\Gamma(3/8) \le 8/3 + e^{-1} < 73/24$, obtained by splitting its defining integral at one. Also $2^{3/8} < 4/3$, $\pi^{-1/4} < 4/5$, and $e^{3/8} < 3/2$. The product of these four upper bounds is $73/15 < 5$. This proves (73). Letting $\rho \downarrow 0$ proves absence of atoms. The closed-interval version follows by applying the result with $\rho + \varepsilon$ and then letting $\varepsilon \downarrow 0$. □

Corollary 4.30 (Logarithmic and fractional potentials)**.** *In the setting of Lemma 4.29, let $L = \|S\|_{\mathrm{op}}$. For every $a > 0$, put $M = 5(a/L)^{1/4}$. Uniformly in w,*

$$\mathbb{E}\log_+ \frac{a}{|R - w|} \le \begin{cases} 4M, & M \le 1, \\ 4 + 4\log M, & M \ge 1, \end{cases} \tag{75}$$
$$\le 20(a/L)^{1/4}.$$

For every $0 < \beta < 1/4$, moreover,

$$\sup_w \mathbb{E}|R - w|^{-\beta} \le \frac{5^{4\beta}}{1 - 4\beta} L^{-\beta}. \tag{76}$$

Proof. Tonelli's theorem and the quarter-power bound give

$$\mathbb{E}\log_+ \frac{a}{|R-w|} = \int_0^\infty \mathbb{P}(|R - w| < ae^{-u})\, du \le \int_0^\infty \min\{1, Me^{-u/4}\}\, du.$$

If $M \le 1$, this integral is $4M$. If $M > 1$, split at $u = 4\log M$ to obtain $4 + 4\log M$. Both cases are at most $4M$. For the negative moment use

$$\mathbb{E}|R - w|^{-\beta} = \beta \int_0^\infty r^{-\beta-1}\mathbb{P}(|R - w| < r)\, dr.$$

Split at $r_0 = L/5^4$. The probability is at most $(r/r_0)^{1/4}$ for $r < r_0$ and at most one otherwise. The two integrals sum to $r_0^{-\beta}/(1 - 4\beta)$, giving (76). □

The spectral-window, volume, and ridge criteria above are quantitative sufficient hypotheses. A lower bound for the third singular value of the full quadratic matrix alone does not verify their coordinate cut or marginal conditions. Their usefulness is that the estimates retain arbitrary self interactions and, where specified, all additional spectral directions without any assumption that a remainder is independent of the principal part.

5 Fractional potentials of rank-two forms

An indefinite rank-two form can have an unbounded density, but its half-power potential admits a uniform determinant bound, including for affine terms outside the range of its quadratic matrix. Such terms arise under conditioning on the original coordinates.

Lemma 5.1 (A rank-one affine small-ball estimate)**.** *Let $|u| = 1$, $\lambda \neq 0$, and let*

$$R = \lambda\langle u, X\rangle^2 + b^{\mathsf{T}}X + d.$$

For every $w \in \mathbb{R}$ and $\rho > 0$,

$$\mathbb{P}\{|R - w| < \rho\} \leq \min\{1, 8\sqrt{\rho/|\lambda|}\}.$$

Proof. If $u_i^2 \geq 1/2$ for some original coordinate, conditioning on the others leaves a quadratic of leading coefficient λu_i^2. Its inverse image of an interval of length 2ρ has total length at most $2\sqrt{2\rho/|\lambda u_i^2|} \leq 4\sqrt{\rho/|\lambda|}$.

Otherwise Theorem 4.6, with one spectral direction and the given affine phase, gives

$$|\phi_R(t)| \leq \sqrt{e\pi}\,|\lambda t|^{-1/2}.$$

For $v > 0$, Gaussian Fourier inversion and $\mathbf{1}_{\{|z|<\rho\}} \leq e^v e^{-vz^2/\rho^2}$ imply

$$\mathbb{P}\{|R - w| < \rho\} \leq \frac{e^v\rho}{2\sqrt{\pi v}} \int_{\mathbb{R}} e^{-\rho^2 t^2/(4v)}|\phi_R(t)|\,dt \leq \frac{e^v\sqrt{e}\,\Gamma(1/4)}{\sqrt{2}\,v^{1/4}}\sqrt{\rho/|\lambda|}.$$

Take $v = 1/4$. The resulting constant $e^{3/4}\Gamma(1/4)$ is less than 8. An elementary upper bound sufficient to verify this numerical inequality is

$$\Gamma(1/4) \leq 4 - \frac{4}{5} + \frac{2}{9} - \frac{2}{39} + \frac{1}{102} + e^{-1}.$$

Indeed, on $[0, 1]$ use the degree-four Taylor upper bound for e^{-x}, and on $[1, \infty)$ use $x^{-3/4} \leq 1$. Together with $e^{-1} < 3/8$ and $e^{3/4} < 17/8$, the displayed expression gives a product less than 8. □

Define an even function on $\mathbb{R}$ by

$$\Psi(a) = \begin{cases} 1 + \log(1/|a|), & 0 < |a| \leq 1, \\ |a|^{-1/4}, & |a| \geq 1, \\ +\infty, & a = 0. \end{cases}$$

Lemma 5.2 (A translated quadratic kernel)**.** *For $a \neq 0$ and every probability density $f \leq 1$,*

$$\int_{\mathbb{R}} f(x)|a - x^2|^{-1/2}\,dx \leq 5\Psi(a).$$

More generally, a probability density bounded by K satisfies

$$\int_{\mathbb{R}} f(x)|a - x^2|^{-1/2}\,dx \leq 5K\Psi(K^2a).$$

Proof. Let $m_a(s)$ be the measure of $\{x : |a - x^2|^{-1/2} > s\}$. The integral is at most $\int_0^\infty \min\{1, m_a(s)\}\, ds$. For $a > 0$, put $\varepsilon = s^{-2}$. If $\varepsilon \geq a$, then $m_a(s) = 2\sqrt{a+\varepsilon} \leq 2\sqrt{2}/s$. If $\varepsilon < a$, then

$$m_a(s) = 2(\sqrt{a+\varepsilon} - \sqrt{a-\varepsilon}) \leq \frac{4}{\sqrt{a}\, s^2}.$$

For $0 < a \leq 1$, split the integral at 1 and $a^{-1/2}$. It is at most $5 + \sqrt{2}\log(1/a) \leq 5\Psi(a)$. For $a \geq 1$, splitting at $2a^{-1/4}$ gives at most $4a^{-1/4}$.

If $a < 0$, then $m_a(s) \leq 2/s$, and it vanishes when $s \geq |a|^{-1/2}$. For $-1 \leq a < 0$, split at 1 to obtain $1 + \log(1/|a|)$. For $a \leq -1$, the integrand kernel itself is at most $|a|^{-1/2} \leq |a|^{-1/4}$. This proves the first assertion. For the second, substitute $y = Kx$; the rescaled density is bounded by one and the kernel gains the factor K. □

Lemma 5.3 (Averaging the kernel)**.** *If $\mathbb{P}\{|W| < \rho\} \leq L\sqrt{\rho}$ for all $\rho > 0$, then*

$$\mathbb{E}\Psi(W) \leq 4\sqrt{L}.$$

Proof. The hypothesis excludes an atom at zero. For $0 < s < 1$, the event $\Psi(W) > s$ implies $|W| < s^{-4}$, so its probability is at most Ls^{-2}. For $s \geq 1$, it is at most $Le^{(1-s)/2}$. Since $\sup_{s\geq 1} s^2 e^{(1-s)/2} = 16e^{-3/2} < 4$, both ranges are bounded by $\min\{1, 4L/s^2\}$. Its integral over $s > 0$ is $4\sqrt{L}$. □

Theorem 5.4 (Rank-two half-power potential)**.** *Let $U^{\mathsf{T}}U = I_2$, let Λ be an invertible real symmetric 2-by-2 matrix, and put $D = |\det \Lambda|$. For every original-coordinate affine term,*

$$R = (U^{\mathsf{T}}X)^{\mathsf{T}}\Lambda(U^{\mathsf{T}}X) + b^{\mathsf{T}}X + c_0,$$

one has

$$\sup_w \mathbb{E}|R - w|^{-1/2} \leq 40\, 2^{3/4} D^{-1/4} < 68 D^{-1/4}. \tag{77}$$

Proof. Suppose first that every row leverage of U is at most $1/2$. Theorem 4.6 gives

$$|\phi_R(t)| \leq \min\{1, K/|t|\}, \qquad K = e\pi D^{-1/2}.$$

The Gaussian representation

$$|z|^{-1/2} = \frac{1}{\Gamma(1/4)} \int_0^\infty v^{-3/4} e^{-vz^2}\, dv$$

and Gaussian Fourier inversion give

$$\mathbb{E}|R - w|^{-1/2} \leq (2\pi)^{-1/2} \int_{\mathbb{R}} |t|^{-1/2} |\phi_R(t)|\, dt.$$

To justify the order of integration, first write the Gaussian formula at each $v > 0$, take absolute values, and then apply Tonelli to the nonnegative integrand. The integral in v is $\sqrt{2}\,\Gamma(1/4)|t|^{-1/2}$ before its fixed prefactor, giving the stated coefficient. Splitting the last integral at $|t| = K$ gives

$$\mathbb{E}|R - w|^{-1/2} \leq 4\sqrt{2e}\, D^{-1/4}.$$

Otherwise an original coordinate i has leverage $h > 1/2$. Rotate only the two-dimensional spectral frame so that its row is $(\sqrt{h}, 0)$, and write

$$Y_1 = \sqrt{h}\, X_i + T, \qquad Y_2 = Z, \qquad \Lambda = \begin{pmatrix} \beta & \gamma \\ \gamma & \delta \end{pmatrix}.$$

The variables T, Z depend on X_{-i}, and Z is a unit linear form in those original coordinates. Put $k = h^{-1/2}$, $d = b_i/\sqrt{h}$, and $L_0 = b_{-i}^{\mathsf{T}} X_{-i} - dT + c_0$. Then

$$R = \beta Y_1^2 + 2\gamma Y_1 Z + \delta Z^2 + dY_1 + L_0.$$

Conditional on X_{-i}, the density of Y_1 is bounded by k; its translation is unrestricted.

If $\beta \neq 0$, complete the square:

$$R = \beta\left(Y_1 + \frac{\gamma Z}{\beta} + \frac{d}{2\beta}\right)^2 + R_0, \qquad R_0 = \frac{\det\Lambda}{\beta} Z^2 - \frac{\gamma d}{\beta} Z - \frac{d^2}{4\beta} + L_0.$$

This is a rank-one quadratic in the remaining original coordinates plus an arbitrary affine term. Lemma 5.2, conditionally, yields

$$\mathbb{E}[|R - w|^{-1/2} \mid X_{-i}] \leq 5k|\beta|^{-1/2}\Psi(W), \qquad W = k^2(w - R_0)/\beta.$$

The magnitude of the rank-one coefficient of W is $\lambda_0 = k^2 D/\beta^2$. Lemmas 5.1 and 5.3 therefore give

$$\mathbb{E}|R - w|^{-1/2} \leq 20\sqrt{8}\, k|\beta|^{-1/2}\lambda_0^{-1/4} = 20\sqrt{8}\,\sqrt{k}\, D^{-1/4} \leq 40\, 2^{3/4} D^{-1/4}.$$

All powers of β cancel.

If $\beta = 0$, then $D = \gamma^2 > 0$, and the conditional polynomial is affine in X_i, of slope $\sqrt{h}(2\gamma Z + d)$. For any probability density bounded by H, layer cake gives

$$\int f(x)|x - a|^{-1/2}\, dx \leq 2\sqrt{2H}.$$

Apply this first to X_i, then to Z, whose density is at most $\sqrt{2}$ by Lemma 4.2. The result is

$$\mathbb{E}|R - w|^{-1/2} \leq 4\, 2^{3/4} h^{-1/4} D^{-1/4} \leq 8 D^{-1/4}.$$

The exceptional value $2\gamma Z + d = 0$ has probability zero. These cases exhaust the possibilities. □

Lemma 5.5 (A conditional two-coordinate potential)**.** *Suppose Z has density at most H, and the conditional density of S given Z is at most K. For an invertible real symmetric matrix L, and arbitrary deterministic affine terms,*

$$\sup_w \mathbb{E}\left|(S, Z)L(S, Z)^{\mathsf{T}} + \alpha S + \beta Z + \gamma - w\right|^{-1/2} \leq 10\sqrt{15}\sqrt{KH}\,|\det L|^{-1/4}.$$

Proof. An invertible deterministic quadratic matrix permits removing its affine term by translating S, Z; both density caps are preserved. Write $L = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$. For $a \neq 0$, complete the square in S. Lemma 5.2 reduces the conditional bound to

$$5K|a|^{-1/2}\Psi(A + \epsilon\delta_0 Z^2), \qquad \delta_0 = K^2|\det L|/a^2, \quad \epsilon \in \{-1, 1\}.$$

The elementary sublevel estimate $|\{z : |z^2 + A| < \rho\}| \leq 4\sqrt{\rho}$ shows that the function $v(z) = \Psi(A + \epsilon\delta_0 z^2)$ has weak L^2 norm at most $\sqrt{15}\delta_0^{-1/4}$. Indeed, for levels $0 < s < 1$, $s^2|\{v > s\}| \leq 4\delta_0^{-1/2}$. For $s \geq 1$, the bound is $4\delta_0^{-1/2} s^2 e^{(1-s)/2} \leq 64 e^{-3/2}\delta_0^{-1/2} < 15\delta_0^{-1/2}$. For any probability density $g \leq H$, layer cake gives

$$\int gv \leq \int_0^\infty \min\{1, H\|v\|_{2,\infty}^2/s^2\}\, ds = 2\sqrt{H}\,\|v\|_{2,\infty}.$$

Substitution cancels a and proves the asserted constant. For $a = 0$, condition first on Z and use the affine negative-half-moment estimate from the last proof; it gives the smaller constant $4\sqrt{2}\sqrt{KH}\,|\det L|^{-1/4}$. □

Corollary 5.6 (Affine terms in the range)**.** *In Theorem 5.4, if $b \in \mathrm{range}(U)$, then*

$$\sup_w \mathbb{E}|R - w|^{-1/2} \le 10\sqrt{30}\, D^{-1/4}.$$

Proof. A deterministic translation of the original coordinates removes the affine term. In the diffuse case the bound $4\sqrt{2e}D^{-1/4}$ obtained above suffices. In the heavy-coordinate notation of that proof, the conditional density of Y_1 given Z is at most $h^{-1/2}$: average the conditional bound given all X_{-i}. The marginal density of Z is at most $\sqrt{2}$. Lemma 5.5 gives

$$10\sqrt{15}(2/h)^{1/4}D^{-1/4} \le 10\sqrt{30}\, D^{-1/4}.$$

□

Theorem 5.7 (An exact-rank-three Schur estimate)**.** *Suppose an original-coordinate principal block has representation*

$$A_{II} = VMV^{\mathsf{T}}, \qquad V^{\mathsf{T}}V = I_3, \qquad \det M \neq 0.$$

Choose $i \in I$ with $a = A_{ii} \neq 0$, and write $h = |V^{\mathsf{T}}e_i|^2$. Then the full ambient quadratic form, with arbitrary other blocks and affine terms, satisfies

$$\|p_Q\|_\infty \le 68(h|a|\,|\det M|)^{-1/4}.$$

For a full exact-rank-three matrix with all nonzero eigenvalues of modulus at least one, this implies $\|p_Q\|_\infty \le 68/\sqrt{|a|}$.

Proof. Condition first on the original coordinates outside I. They create only an arbitrary affine term in the independent coordinates of I. Completing the square in X_i gives a remainder with Schur matrix

$$S = A_{I\setminus\{i\},I\setminus\{i\}} - a^{-1}A_{I\setminus\{i\},i}A_{i,I\setminus\{i\}}.$$

The square kernel gives

$$p_Q(t) \le |a|^{-1/2}\mathbb{E}|t - R|^{-1/2}.$$

Put $v = V^{\mathsf{T}}e_i$, choose a 3-by-2 orthonormal frame T for $v^\perp$, and let V_- be V with row i removed. Then $W = V_-T$ satisfies

$$W^{\mathsf{T}}W = T^{\mathsf{T}}(I - vv^{\mathsf{T}})T = I_2.$$

The symmetric matrix $H = M - Mvv^{\mathsf{T}}M/a$ has kernel $\mathrm{span}(v)$. Writing $N = T^{\mathsf{T}}HT$ gives $S = WNW^{\mathsf{T}}$. In a spectral-space basis with $v = \sqrt{h}\,e_1$, write

$$M = \begin{pmatrix} \alpha & q^{\mathsf{T}} \\ q & G \end{pmatrix}, \qquad a = h\alpha.$$

Then $N = G - qq^{\mathsf{T}}/\alpha$, up to an orthogonal change of its two-dimensional basis, and

$$\det N = \det M/\alpha = h\det M/a.$$

Theorem 5.4, with arbitrary induced affine terms allowed, proves the first bound. In the normalized full-rank-three case, $|\det M| \ge \|M\|$ and $|a| \le h\|M\|$; hence $h|a||\det M| \ge |a|^2$, proving the last assertion. □

6 Arbitrary quadratic forms under Gaussian mixture inputs

The Gaussian determinant formula and weighted arithmetic–geometric mean yield a dimension-free estimate for product laws with controlled Gaussian-mixture component peaks. The result applies to quadratic forms of arbitrary rank and signature, provided the hypothesis holds for every coordinate. For related Gaussian quadratic-form and regularity estimates, see Aizenman et al. (2017); Götze et al. (2019); Herry et al. (2025).

Theorem 6.1 (Gaussian mixtures with controlled average component peaks)**.** *Let*

$$X_i = M_i + \sqrt{T_i}\, G_i, \qquad 1 \le i \le n,$$

where the pairs (M_i, T_i) are independent across coordinates, $0 < T_i < \infty$ almost surely, and the G_i are independent standard Gaussian variables independent of all pairs. Suppose

$$\mathbb{E}T_i^{-1/2} \le \sqrt{2\pi} \qquad (1 \le i \le n). \tag{78}$$

Let A be any real symmetric matrix, let b, c be arbitrary, and let $3 \le k \le \operatorname{rank} A$. Put

$$Z_k = e_k(A^2) = \sum_{|I|=|J|=k} \det(A_{IJ})^2 > 0, \qquad I_k = \int_0^\infty (1+u^{2k})^{-1/4}\, du.$$

Then $Q = X^T AX + b^T X + c$ has a bounded continuous density satisfying

$$\|p_Q\|_\infty \le I_k Z_k^{-1/(2k)}, \qquad I_k = \frac{1}{2k}\mathrm{B}\Big(\frac{1}{2k}, \frac{1}{4} - \frac{1}{2k}\Big) < \frac{k}{k-2}. \tag{79}$$

In particular, $s_3(A) \ge 1$ implies $\|p_Q\|_\infty < 3$ in this input class.

Proof. Condition first on all the mixture variables (M_i, T_i). With $D = \operatorname{diag}(T_1, \dots, T_n)$ and $M = (M_1, \dots, M_n)$, write

$$Q = G^T BG + \ell^T G + c', \qquad B = D^{1/2} A D^{1/2}, \quad \ell = D^{1/2}(2AM + b).$$

Orthogonally diagonalizing B is legitimate for the standard Gaussian vector G. If its eigenvalues are μ_j and the rotated linear coefficients are β_j, direct one-dimensional Gaussian integration gives

$$\begin{aligned} |\varphi_{Q|M,D}(t)| &= \prod_j (1+4t^2\mu_j^2)^{-1/4} \exp\left[-\sum_j \frac{t^2\beta_j^2}{2(1+4t^2\mu_j^2)}\right] \\ &\le \det(I + 4t^2 B^2)^{-1/4}. \end{aligned}$$

Terms with $\mu_j = 0$ obey the same formula and contribute only a Gaussian linear factor. Expanding the determinant in elementary symmetric functions yields

$$|\varphi_{Q|M,D}(t)| \le [1 + (2t)^{2k} W_k]^{-1/4}, \qquad W_k = e_k(B^2) > 0.$$

This bound is integrable because $k > 2$. Fourier inversion and the substitution $u = 2tW_k^{1/(2k)}$ give

$$\|p_{Q|M,D}\|_\infty \le \frac{1}{2\pi}\int_{\mathbb{R}} [1 + (2t)^{2k} W_k]^{-1/4}\, dt = \frac{I_k}{2\pi} W_k^{-1/(2k)}. \tag{80}$$

It remains to average the random spectral factor. For $T_I = \prod_{i \in I} T_i$, Cauchy–Binet gives the exact identity

$$W_k = \sum_{|I|=|J|=k} \det(A_{IJ})^2 T_I T_J. \tag{81}$$

The sum is over ordered pairs of coordinate subsets; intersections are allowed. On the nonzero minors, define probability weights

$$w_{IJ} = \frac{\det(A_{IJ})^2}{Z_k}, \qquad \alpha_i = \sum_{I,J} w_{IJ}(\mathbf{1}_{\{i\in I\}} + \mathbf{1}_{\{i\in J\}}).$$

Then $0 \le \alpha_i \le 2$ and $\sum_i \alpha_i = 2k$. Weighted arithmetic–geometric mean in (81) gives

$$W_k \ge Z_k \prod_i T_i^{\alpha_i}, \qquad W_k^{-1/(2k)} \le Z_k^{-1/(2k)} \prod_i T_i^{-\alpha_i/(2k)}.$$

Independence across coordinates and concavity of $s \mapsto s^{\alpha_i/k}$ give

$$\begin{aligned}
\mathbb{E}W_k^{-1/(2k)} &\le Z_k^{-1/(2k)} \prod_i \mathbb{E}(T_i^{-1/2})^{\alpha_i/k} \\
&\le Z_k^{-1/(2k)} \prod_i (\mathbb{E}T_i^{-1/2})^{\alpha_i/k} \\
&\le 2\pi Z_k^{-1/(2k)}.
\end{aligned}$$

Here $\alpha_i/k \le 2/k < 1$, and zero exponents are omitted. This estimate also proves that the expectation of the conditional Fourier L^1 bound is finite. Hence $\int |\varphi_Q(t)|\,dt \le \mathbb{E}\int |\varphi_{Q|M,D}(t)|\,dt < \infty$ by conditional expectation, the triangle inequality, and Tonelli. Fourier inversion therefore supplies a continuous density, and averaging (80) proves (79).

The beta-integral formula follows from $v = u^{2k}$. Finally,

$$I_k < \int_0^1 1\,du + \int_1^\infty u^{-k/2}\,du = 1 + \frac{2}{k-2} = \frac{k}{k-2}.$$

If $s_3(A) \ge 1$, then $Z_3 \ge s_1(A)^2 s_2(A)^2 s_3(A)^2 \ge 1$, which gives the final assertion. □

Corollary 6.2 (Centered scale mixtures)**.** *Conjecture 1.1 holds, with a constant smaller than* 3*, for all independent coordinate laws that are Gaussian scale mixtures about deterministic coordinate centers and have densities bounded by one. There is no lower bound on the component variances and no moment assumption on the coordinate variables themselves.*

Proof. If $M_i = m_i$ is deterministic, the mixture density is

$$f_i(x) = \mathbb{E}\left[(2\pi T_i)^{-1/2} e^{-(x-m_i)^2/(2T_i)}\right].$$

It is maximized at m_i, and its essential supremum is $(2\pi)^{-1/2}\mathbb{E}T_i^{-1/2}$, also when this is infinite. The latter assertion follows by monotone convergence as $x \to m_i$ along either side. Thus the density cap is equivalent to (78), and Theorem 6.1 applies. □

Remark 6.3 (Scope of the mixture condition). For random component centers, (78) still implies a cap-one marginal density, but the converse can fail. Narrow components placed far apart can have a bounded final mixture density while their average component peak is large. The theorem therefore does not extend to all bounded densities by an arbitrary Gaussian-mixture approximation. Independent centered Laplace laws, after scaling to the stated cap, are examples within the centered scale-mixture class. The result also differs from the adapted-hub theorem: here the entire input vector satisfies the mixture hypothesis, whereas the adapted theorem allows a conditional hub law outside that class.

Remark 6.4 (Unequal component-peak bounds). The same proof retains the factor

$$\frac{I_k}{2\pi} Z_k^{-1/(2k)} \prod_i (\mathbb{E} T_i^{-1/2})^{\alpha_i/k}.$$

In particular, a common bound $\mathbb{E} T_i^{-1/2} \le \sqrt{2\pi} K$ gives $I_k K^2 Z_k^{-1/(2k)}$. The powers sum to two, as required by quadratic scaling. The weighted-minor argument allows the conditional Gaussian eigenvectors to depend on the mixture variables.

7 Semidefinite forms and spectral spread

Throughout this section, $X = (X_1, \dots, X_n)$ has independent coordinates with Lebesgue densities $0 \le f_i \le 1$. We write

$$Q = X^T A X + b^T X + c, \qquad \varphi_Q(t) = \mathbb{E} e^{itQ},$$

where A is real and symmetric. The vector b and scalar c are arbitrary. Density bounds refer to the essential supremum. Eigenvalues of a positive semidefinite matrix are numbered in decreasing order, including zeros.

Theorem 7.1 (Semidefinite forms). *If A is positive semidefinite and $\lambda_3(A) > 0$, then Q has a density satisfying*

$$\|p_Q\|_\infty \le \frac{17100}{\lambda_3(A)}. \tag{82}$$

The same conclusion holds for negative semidefinite A, with $\lambda_3(-A)$ in the denominator. The bound is independent of dimension and rank, and allows a linear term outside the range of A.

Remark 7.2 (Density at the lower endpoint). For the homogeneous form $Q = X^T A X$ with $A \succeq 0$ and $\lambda_3(A) > 0$, only three positive eigendirections are needed to show that the density vanishes *at zero*. Let u_1, u_2, u_3 be orthonormal eigenvectors with positive eigenvalues $\lambda_1, \lambda_2, \lambda_3$, and put $Y_j = \langle u_j, X \rangle$. Lemma 4.2 bounds the joint density of (Y_1, Y_2, Y_3) by $2^{3/2}$. Positivity of the discarded spectral terms gives

$$Q \ge \sum_{j=1}^{3} \lambda_j Y_j^2, \qquad \mathbb{P}(Q \le t) \le \frac{8\sqrt{2}\,\pi}{3\sqrt{\lambda_1\lambda_2\lambda_3}} t^{3/2} \quad (t \ge 0).$$

Thus the distribution function has right derivative zero at zero; the continuous density furnished by Theorem 15.4 satisfies $p_Q(0) = 0$. The comparison concerns the lower tail of a nonnegative homogeneous form. It does not identify the law of Q after spectral truncation or prove a density bound at positive levels, and it does not apply at zero when an affine term moves zero away from the lower endpoint.

7.1 Concentration and conditioning

For a real random variable Z, define

$$L_Z(u) = \sup_s \mathbb{P}(|Z - s| \le u), \qquad \alpha_1(Z) = \sup_{u>0} \frac{L_Z(u)}{\sqrt{u}}, \qquad \alpha_2(Z) = \sup_{0<u\le 1} \frac{L_Z(u)}{u(1 + \log(1/u))}.$$

Lemma 7.3 (Fourier concentration). *For every $u > 0$,*

$$L_Z(u) \le \frac{u}{2 \sin^2 1} \int_{|t| \le 2/u} |\varphi_Z(t)|\, dt. \tag{83}$$

If $|\varphi_Z(t)| \le B|t|^{-1/2}$ for $t \ne 0$, then $\alpha_1(Z) \le 2\sqrt{2} B / \sin^2 1$. If $|\varphi_Z(t)| \le \min(1, B/|t|)$ and $B \ge 2$, then $\alpha_2(Z) \le B / \sin^2 1$.

Proof. The function $K(x) = (\sin x/x)^2$, with $K(0) = 1$, is at least $\sin^2 1$ on $[-1, 1]$. Its Fourier transform, with kernel e^{-itx}, is $\pi(1 - |t|/2)_+$. Fourier inversion and Fubini's theorem therefore give

$$\mathbb{P}(|Z - s| \le u) \le \frac{\mathbb{E}K((Z-s)/u)}{\sin^2 1} \le \frac{u}{2\sin^2 1} \int_{|t|\le 2/u} |\varphi_Z(t)|\, dt.$$

Integrating $B|t|^{-1/2}$ proves the first consequence. For $0 < u \le \min(1, 2/B)$, integration of $\min(1, B/|t|)$ gives

$$L_Z(u) \le \frac{Bu}{\sin^2 1}\left(1 + \log\frac{2}{Bu}\right) \le \frac{Bu}{\sin^2 1}(1 + \log(1/u)).$$

For $2/B < u \le 1$, the last inequality follows from $L_Z(u) \le 1$ and $Bu > 2$. □

Lemma 7.4 (Square-root profiles)**.** *Suppose that, for every nonnegative Borel function h,*

$$\mathbb{E}h(Q) \le K\,\mathbb{E}\int_{\mathbb{R}} h(y)|y - \xi|^{-1/2}\, dy. \tag{84}$$

Then Q has a density. Moreover:

1. *$\alpha_1(Q) \le 4K$;*
2. *if $\alpha_1(\xi) \le K_1$, then $\alpha_2(Q) \le 4\sqrt{2}K\max(K_1, 1)$;*
3. *if $\alpha_2(\xi) \le K_2$ and $K_2 \ge 1$, then*

$$\|p_Q\|_\infty \le 3K\sqrt{K_2(3 + \log K_2)}. \tag{85}$$

Proof. Tonelli's theorem shows that the right side of (84) vanishes when h is the indicator of a Lebesgue null set. Thus Q has a density, with $p_Q(y) \le K\mathbb{E}|y - \xi|^{-1/2}$ almost everywhere. The integral of $|y - \xi|^{-1/2}$ over an interval of length $2u$ is at most $4\sqrt{u}$, proving the first assertion.

For any $d \in \mathbb{R}$,

$$\int_{-u}^{u} |y + d|^{-1/2}\, dy \le 4\sqrt{2}\, u \max(|d|, u)^{-1/2}. \tag{86}$$

For $|d| \le 2u$ this follows from the preceding integral bound. For $|d| > 2u$, use $|y + d| \ge |d|/2$. When $0 < u \le 1$, layer cake gives

$$\begin{aligned}\mathbb{E}\max(|s - \xi|, u)^{-1/2} &= \int_0^{u^{-1/2}} \mathbb{P}(|s - \xi| < t^{-2})\, dt \\ &\le 1 + K_1 \int_1^{u^{-1/2}} \frac{dt}{t} \\ &\le \max(K_1, 1)(1 + \log(1/u)).\end{aligned}$$

Combine this with (84) and (86).

For the last assertion, put $V = \sqrt{K_2(3 + \log K_2)} \ge 1$. For every s,

$$\begin{aligned}\mathbb{E}|s - \xi|^{-1/2} &\le V + K_2 \int_V^\infty t^{-2}(1 + 2\log t)\, dt \\ &= V + K_2(3 + 2\log V)/V.\end{aligned}$$

Writing $a = 3 + \log K_2 \ge 3$ gives $3 + 2\log V = a + \log a \le 2a$. The last display is at most $3V$, proving the claim. □

Lemma 7.5 (Positive pivots)**.** *Suppose A is positive semidefinite and $a = a_{ii} > 0$. Completing the square gives $Q = a(X_i - m(X_{-i}))^2 + R(X_{-i})$, with m affine and R quadratic. The matrix of R is the positive semidefinite Schur complement*

$$S = A_{-i,-i} - a^{-1} A_{-i,i} A_{i,-i}.$$

The profile (84) *holds with $K = a^{-1/2}$ and $\xi = R$. For $k \geq 2$,*

$$\lambda_{k-1}(S) \geq \lambda_k(A). \tag{87}$$

If A has rank r, then S has rank $r - 1$.

Proof. For fixed X_{-i}, the density of $a(X_i - m)^2$ at $w > 0$ is

$$\frac{f_i(m + \sqrt{w/a}) + f_i(m - \sqrt{w/a})}{2\sqrt{aw}} \leq (aw)^{-1/2}.$$

Integration and averaging prove the profile. No independence between m and R is needed. Writing $A = \left(\begin{smallmatrix} a & v^T \\ v & D \end{smallmatrix}\right)$, the identity

$$(x, y)^T A (x, y) = a(x + a^{-1} v^T y)^2 + y^T S y$$

proves positivity of S and the rank assertion. A k-dimensional subspace where the form of A is at least $\lambda_k(A)$ times the squared norm intersects $ax + v^T y = 0$ in dimension at least $k-1$. Projection onto y is injective on this intersection, and on its image $y^T S y \geq \lambda_k(A)(x^2 + |y|^2) \geq \lambda_k(A)|y|^2$. The variational principle proves (87). □

7.2 The decoupling normalization

For a $p \times q$ matrix M of rank at least k, set

$$Z_k = e_k(M^T M) = \sum_{\substack{S \subseteq [p], T \subseteq [q] \\ |S| = |T| = k}} \det(M_{S,T})^2, \qquad w_{S,T} = \det(M_{S,T})^2 / Z_k.$$

Let $\alpha_i = \sum_{S \ni i, T} w_{S,T}$ and $\beta_j = \sum_{S, T \ni j} w_{S,T}$ be the row and column inclusion probabilities. If $M = A_{I,J}$ for disjoint coordinate sets I, J and $\alpha_i \leq 1/2$ for every $i \in I$, Lemma 4.8 gives

$$|\varphi_Q(t)|^2 \leq e^{k/2} (2\pi)^k |t|^{-k} Z_k^{-1/2}. \tag{88}$$

Arbitrary quadratic terms inside I and J, and arbitrary linear terms, are allowed. For $k = 3$, Fourier inversion gives

$$\|p_Q\|_\infty \leq 6\sqrt{e}\, Z_3^{-1/6}. \tag{89}$$

Indeed, with $B = (2\pi)^{3/2} e^{3/4} Z_3^{-1/4}$,

$$(2\pi)^{-1} \int_{\mathbb{R}} \min(1, B|t|^{-3/2})\, dt = 3B^{2/3}/\pi = 6\sqrt{e}\, Z_3^{-1/6}.$$

The analytic input in (88) is the integrated Fourier estimate of Lemma 4.3 and the Gaussian Brascamp–Lieb theorem (Brascamp and Lieb, 1976; Lieb, 1990). No pointwise decay assumption on an individual characteristic function is used.

7.3 A partition in Gram coordinates

For positive semidefinite A of rank r, choose

$$A = BB^T, \qquad B^TB = \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_r) > 0.$$

Write b_i^T for row i of B, so that $|b_i|^2 = a_{ii}$. For independent equiprobable signs ε_i, define

$$I = \{i : \varepsilon_i = 1\}, \quad J = \{i : \varepsilon_i = -1\}, \quad N = \tfrac{1}{2}\sum_i \varepsilon_i b_i b_i^T.$$

Then $H_I = B_I^T B_I = \Lambda/2 + N$ and $H_J = B_J^T B_J = \Lambda/2 - N$.

Lemma 7.6 (Partition identity)**.** *For every $g \in \mathbb{R}^r$,*

$$\langle B_I g, A_{I,J} B_J g\rangle = |\Lambda g|^2/4 - |Ng|^2, \tag{90}$$

$$|B_I g|\,|B_J g| \le g^T \Lambda g/2. \tag{91}$$

If $K \subseteq [r]$ has cardinality k, $\lambda_m \ge \theta > 0$ for $m \in K$, and $T_K = \sum_{m\in K} |Ne_m|^2/\lambda_m^2 \le 1/8$, then $s_k(A_{I,J}) \ge \theta/4$. If $\max_i a_{ii} \le d$, then

$$\mathbb{E}T_K \le kd/(4\theta), \qquad \mathbb{P}(T_K > 1/8) \le 2kd/\theta.$$

Proof. The scalar product is $g^T H_I H_J g$. Expanding $H_I H_J$ gives $\Lambda^2/4 - N^2 + (N\Lambda - \Lambda N)/2$, whose last term is skew symmetric. This proves (90). Equation (91) follows from $H_I + H_J = \Lambda$. For g in the span G of $\{e_m : m \in K\}$, Cauchy–Schwarz gives $|Ng| \le |\Lambda g| T_K^{1/2}$. Thus the scalar product is at least $|\Lambda g|^2/8$. It is positive for $g \ne 0$, so B_J is injective on G. For $x = B_J g \ne 0$,

$$\frac{|A_{I,J}x|}{|x|} \ge \frac{\langle B_I g, A_{I,J} B_J g\rangle}{|B_I g|\,|B_J g|} \ge \frac{|\Lambda g|^2}{4 g^T \Lambda g} \ge \theta/4.$$

The singular-value variational principle on the k-dimensional subspace $B_J G$ proves the assertion. Finally,

$$\mathbb{E}|Ne_m|^2 = \tfrac{1}{4}\sum_i b_i(m)^2 |b_i|^2 \le \tfrac{d}{4}\sum_i b_i(m)^2 = d\lambda_m/4.$$

Sum over m and apply Markov's inequality. □

7.4 Volume marginals and ridge leverage

Lemma 7.7 (Volume marginals)**.** *Let $G = MM^T$ have eigenvalues $\sigma_1 \ge \sigma_2 \ge \cdots \ge 0$ and an orthonormal eigenbasis (u_m). If $\operatorname{rank} M \ge k$, then*

$$\alpha_i = \sum_m p_m u_m(i)^2, \qquad p_m = \frac{\sigma_m e_{k-1}(\sigma_{-m})}{e_k(\sigma)}, \tag{92}$$

where $e_0 = 1$ and σ_{-m} is the list with entry m removed. Moreover $p_m \le \min(1, k\sigma_m/\sigma_k)$. If $0 < \kappa \le \sigma_k/k$ and $m_i = M^T e_i$, then

$$\alpha_i \le 2 m_i^T (M^T M + \kappa I)^{-1} m_i. \tag{93}$$

Proof. By Cauchy–Binet, the numerator of α_i is $\sum_{S\ni i, |S|=k} \det G_{S,S}$. Differentiate $\det(I + tD(z)G)$ with respect to z_i at $z = (1, \dots, 1)$, where $D(z) = \operatorname{diag}(z)$, and take the coefficient of t^k. The principal-minor expansion gives this numerator, whereas spectral decomposition gives the coefficient in

$$\sum_m u_m(i)^2\, t\sigma_m \prod_{\ell\ne m}(1 + t\sigma_\ell).$$

This proves (92). The identity $e_k(\sigma) = \sigma_m e_{k-1}(\sigma_{-m}) + e_k(\sigma_{-m})$ gives $p_m \le 1$. For $m \le k$ it also gives $p_m \le k\sigma_m/\sigma_k$. Otherwise the list $\mu = \sigma_{-m}$ contains k entries at least σ_k, and

$$ke_k(\mu) = \sum_{|S|=k-1} \left(\prod_{j\in S} \mu_j\right) \sum_{j\notin S} \mu_j \ge \sigma_k e_{k-1}(\mu).$$

Indeed, each $(k-1)$-element set omits one of those k entries. This proves the remaining bound for p_m. The inequality $\min(1,x) \le 2x/(1+x)$ now yields $p_m \le 2\sigma_m/(\sigma_m+\kappa)$. The spectral decomposition of $M(M^TM+\kappa I)^{-1}M^T$ proves (93). □

Lemma 7.8 (Gram comparison)**.** *For $M = A_{I,J} = B_I B_J^T$, $i \in I$, and $\kappa > 0$,*

$$m_i^T (M^TM + \kappa I)^{-1} m_i \le b_i^T (H_I + \kappa\Lambda^{-1})^{-1} b_i. \tag{94}$$

Proof. The identity $(B_J H_I B_J^T + \kappa I)B_J = B_J(H_I H_J + \kappa I)$ gives

$$m_i^T (M^TM + \kappa I)^{-1} m_i = b_i^T H_J (H_I H_J + \kappa I)^{-1} b_i.$$

If $H_J > 0$, the middle matrix is $(H_I + \kappa H_J^{-1})^{-1}$. Since $H_J \le \Lambda$, order reversal under inversion gives (94). If H_J is singular, replace it in this matrix identity by $H_J + \varepsilon I \le \Lambda + \varepsilon I$ and let $\varepsilon \downarrow 0$. Continuity at a positive semidefinite matrix T follows from the expression $T^{1/2}(T^{1/2} H_I T^{1/2} + \kappa I)^{-1} T^{1/2}$. □

Proposition 7.9 (A semidefinite partition)**.** *For $k \in \{1,2,3\}$, let $\theta = \lambda_k(A) > 0$, with A positive semidefinite, and put $\gamma_1 = 128$, $\gamma_2 = 300$, $\gamma_3 = 640$. If $\max_i a_{ii} \le \theta/\gamma_k$, some partition $[n] = I \sqcup J$ satisfies*

$$s_k(A_{I,J}) \ge \theta/4, \qquad \max_{i\in I} \alpha_i \le (64k/\theta) \max_i a_{ii} \le 1/2.$$

Uniformly in the linear and constant terms, this implies

$$k = 1: \quad |\varphi_Q(t)| \le 2e^{1/4}\sqrt{2\pi}\,(\theta|t|)^{-1/2}, \tag{95}$$

$$k = 2: \quad |\varphi_Q(t)| \le 8\pi\sqrt{e}\,(\theta|t|)^{-1}, \tag{96}$$

$$k = 3: \quad \|p_Q\|_\infty \le 24\sqrt{e}/\theta. \tag{97}$$

Proof. Put $d = \max_i a_{ii}$. Split the Gram-coordinate space into its first k coordinates, denoted by the subscript b, and the remaining coordinates, denoted by s. These subscripts denote blocks, not the linear coefficient of Q. Set

$$\kappa = \theta^2/(16k), \qquad \rho = \kappa/\theta = \theta/(16k), \qquad \ell_i = |\Lambda_b^{-1/2} b_i^b|^2.$$

Then $\sum_i \ell_i = k$ and $\ell_i \le d/\theta$. Consider the events

$$\begin{aligned}
\mathcal{S}: &\quad \sum_{m=1}^{k} |Ne_m|^2/\lambda_m^2 \le 1/8,\\
\mathcal{E}_1: &\quad \|\Lambda_b^{-1/2} N_{bb} \Lambda_b^{-1/2}\|_F^2 \le 1/64,\\
\mathcal{E}_2: &\quad (2/\rho)\|N_{sb}\Lambda_b^{-1/2}\|_F^2 \le 1/8.
\end{aligned}$$

An empty s block contributes zero. On $\mathcal{S}$, Lemma 7.6 gives $s_k(M) \ge \theta/4$, so $\kappa \le s_k(M)^2/k$. Lemmas 7.7 and 7.8 give $\alpha_i \le 2b_i^T(H_I + \kappa\Lambda^{-1})^{-1} b_i$.

For $T > 0$, completing a square gives the block inequality

$$\begin{pmatrix} P & R \\ R^T & T \end{pmatrix} \ge \begin{pmatrix} P - 2RT^{-1}R^T & 0 \\ 0 & T/2 \end{pmatrix}. \tag{98}$$

The quadratic form of the difference is $2|T^{-1/2}R^Tx + T^{1/2}y/2|^2$. As $\Lambda_s \le \theta I$ and $(H_I)_{ss} \ge 0$, (98), with $T = \rho I$, yields

$$H_I + \kappa\Lambda^{-1} \ge (\Lambda_b/2 + N_{bb} - (2/\rho)N_{bs}N_{sb}) \oplus (\rho/2)I_s.$$

On $\mathcal{E}_1$, $N_{bb} \ge -\Lambda_b/8$. On $\mathcal{E}_2$, conjugating by $\Lambda_b^{-1/2}$ and bounding operator norm by trace gives $(2/\rho)N_{bs}N_{sb} \le \Lambda_b/8$. Therefore $H_I + \kappa\Lambda^{-1} \ge \Lambda_b/4 \oplus (\rho/2)I_s$ and

$$\alpha_i \le 8\ell_i + (4/\rho)|b_i^s|^2 \le (64k/\theta)|b_i|^2 = (64k/\theta)a_{ii}.$$

It remains to show that these events can hold simultaneously. Lemma 7.6 gives $\mathbb{P}(\mathcal{S}^c) \le 2kd/\theta$. Independence and centering of the signs also give

$$\mathbb{E}\|\Lambda_b^{-1/2}N_{bb}\Lambda_b^{-1/2}\|_F^2 = \tfrac{1}{4}\sum_i \ell_i^2 \le kd/(4\theta),$$
$$\mathbb{E}\|N_{sb}\Lambda_b^{-1/2}\|_F^2 = \tfrac{1}{4}\sum_i |b_i^s|^2\ell_i \le kd/4.$$

Markov's inequality gives failure bounds $16kd/\theta$ and $64k^2d/\theta$ for $\mathcal{E}_1$ and $\mathcal{E}_2$. The total is at most

$$(18k + 64k^2)/\gamma_k = \begin{cases} 82/128, & k = 1, \\ 292/300, & k = 2, \\ 630/640, & k = 3, \end{cases}$$

which is strictly below one. The marginal bound is at most $1/2$ for each stated γ_k. Since $Z_k \ge \prod_{j=1}^k s_j(M)^2 \ge (\theta/4)^{2k}$, (88)–(89) prove (95)–(97). □

7.5 Completion of the semidefinite proof

Proposition 7.10 (The first two levels)**.** *For positive semidefinite A and arbitrary b, c:*

1. *if $\lambda_1(A) \ge 1$, then $\alpha_1(Q) \le K_1 := 4\sqrt{128} < 45.26$;*
2. *if $\lambda_2(A) \ge 1$, then $\alpha_2(Q) \le K_2 := 4\sqrt{600}\,K_1 < 4435$.*

Proof. If some $a_{ii} \ge 1/128$, Lemmas 7.5 and 7.4(1) give the first bound. Otherwise Proposition 7.9, with $k = 1$, and Lemma 7.3 give

$$\alpha_1(Q) \le (2\sqrt{2}/\sin^2 1)\, 2e^{1/4}\sqrt{2\pi} < 25.72 < K_1.$$

For the second assertion, a coefficient $a_{ii} \ge 1/300$ leaves a positive semidefinite Schur complement with an eigenvalue at least one. Apply the first assertion to its quadratic form, which is the random center in the profile. Lemma 7.4(2) gives $\alpha_2(Q) \le 4\sqrt{2}\sqrt{300}\,K_1 = K_2$. If there is no such coefficient, (96) and Lemma 7.3 give $\alpha_2(Q) \le 8\pi\sqrt{e}/\sin^2 1 < 58.53 < K_2$. □

Proof of Theorem 7.1. Suppose first that $\lambda_3(A) \ge 1$. A coefficient $a_{ii} \ge 1/640$ leaves a positive semidefinite Schur complement with at least two eigenvalues at least one. Apply Proposition 7.10(2) to its quadratic form and Lemma 7.4(3) to obtain

$$\|p_Q\|_\infty \le 3\sqrt{640}\sqrt{K_2(3 + \log K_2)} < 17100.$$

Here $K_2 = 4\sqrt{600}\,4\sqrt{128}$ is used without rounding. If every diagonal coefficient is below $1/640$, Proposition 7.9, with $k = 3$, instead gives $\|p_Q\|_\infty \le 24\sqrt{e}$. For $\lambda_3(A) = \theta > 0$, apply this argument to Q/θ and use density scaling. Applying the result to $-Q$ proves the negative semidefinite case. □

7.6 Indefinite forms with a small absolute spectral diagonal

For a real symmetric A, write $|A| = (A^2)^{1/2}$ and let $|\lambda|_{(1)}(A) \geq |\lambda|_{(2)}(A) \geq \cdots$ be its eigenvalue moduli. In particular $(|A|)_{ii} \geq |a_{ii}|$.

Theorem 7.11 (Absolute spectral spread)**.** *Let $k \in \{1,2,3\}$, set $\theta = |\lambda|_{(k)}(A) > 0$, and put $\gamma'_1 = 130$, $\gamma'_2 = 360$, $\gamma'_3 = 680$. If*

$$\max_i (|A|)_{ii} \leq \theta/\gamma'_k, \tag{99}$$

some partition $[n] = I \sqcup J$ satisfies $s_k(A_{I,J}) \geq 23\theta/80$ and $\max_{i\in I} \alpha_i \leq 1/2$. Consequently, for every b, c,

$$|\varphi_Q(t)| \leq e^{k/4}(2\pi)^{k/2}(23\theta|t|/80)^{-k/2}. \tag{100}$$

For $k = 3$, Q has a density and

$$\|p_Q\|_\infty \leq \frac{480\sqrt{e}}{23\theta} < \frac{35}{\theta}. \tag{101}$$

Proof. On the range of A, factor

$$A = B\Sigma B^T, \qquad B^T B = \Lambda > 0, \qquad \Sigma = \operatorname{diag}(\pm 1),$$

where the entries of Λ are the nonzero eigenvalue moduli in decreasing order. Thus $BB^T = |A|$ and $|b_i|^2 = (|A|)_{ii} \leq d := \max_i (|A|)_{ii}$. Define the random partition and N as before. Then $H_I = \Lambda/2 + N$, $H_J = \Lambda/2 - N$, and $M = A_{I,J} = B_I \Sigma B_J^T$. Let G be the first k Gram coordinates. Decompose $g \in G$ as $g = g_+ + g_-$ according to the signs in Σ. Direct multiplication gives

$$\langle B_I\Sigma g, M B_J g\rangle = |\Lambda g|^2/4 - g_-^T \Lambda N g_+ - g_+^T \Lambda N g_- - (\Sigma N \Sigma g)^T N g, \tag{102}$$

$$|B_I\Sigma g|^2 + |B_J g|^2 = g^T \Lambda g - 4 g_+^T N g_-. \tag{103}$$

Suppose

$$T := \sum_{m=1}^{k} |Ne_m|^2/\lambda_m^2 \leq \eta^2, \qquad \|\Lambda_b^{-1/2} N_{bb} \Lambda_b^{-1/2}\|_F \leq \eta', \tag{104}$$

where b denotes the first k Gram coordinates. Then $|Ng| \leq \eta|\Lambda g|$ for $g \in G$. The sum of the two middle terms in (102) has modulus at most $\eta|\Lambda g|^2$, by applying this bound separately to g_+ and g_- and then using $2ab \leq a^2 + b^2$. The last term has modulus at most $2\eta^2|\Lambda g|^2$ (the smaller coefficient η^2 would also suffice). Since $|g_+^T N g_-| \leq \eta' |\Lambda^{1/2} g_+|\,|\Lambda^{1/2} g_-|$, (103) gives

$$|B_I\Sigma g|\,|B_J g| \leq (1/2 + \eta') g^T \Lambda g.$$

If $1/4 - \eta - 2\eta^2 > 0$, positivity of the scalar product makes B_J injective on G. Testing against $B_I\Sigma g$ and applying the singular-value variational principle gives

$$s_k(M) \geq \theta \, \frac{1/4 - \eta - 2\eta^2}{1/2 + \eta'}. \tag{105}$$

Take $\eta = 1/16$ and $\eta' = 1/8$; the last ratio is $q = 23/80$.

The ridge comparison carries over to the signed factorization. The calculation of Lemma 7.8, with $\Sigma H_I \Sigma$ and Σb_i, gives

$$\begin{aligned} m_i^T (M^T M + \kappa I)^{-1} m_i &\leq (\Sigma b_i)^T (\Sigma H_I \Sigma + \kappa \Lambda^{-1})^{-1} (\Sigma b_i) \\ &= b_i^T (H_I + \kappa \Lambda^{-1})^{-1} b_i. \end{aligned}$$

Here Σ commutes with Λ; the same limiting argument handles a singular H_J. Set $\kappa = q^2\theta^2/k$ and $\rho = \kappa/\theta = q^2\theta/k$, and additionally require

$$(2/\rho)\|N_{sb}\Lambda_b^{-1/2}\|_F^2 \le 1/8.$$

On these events, $\kappa \le s_k(M)^2/k$, and the block comparison in Proposition 7.9 gives $H_I + \kappa\Lambda^{-1} \ge \Lambda_b/4 \oplus (\rho/2)I_s$. Thus

$$\alpha_i \le 8\ell_i + (4/\rho)|b_i^s|^2 \le \frac{4k}{q^2\theta}(|A|)_{ii}.$$

All second moments used in Proposition 7.9 depend only on $|b_i|^2 \le d$ and are unchanged. Markov's inequality bounds the failure probabilities of the three events by $64kd/\theta$, $16kd/\theta$, and $4k^2d/(q^2\theta)$. Their sum is at most

$$\frac{80k + (25600/529)k^2}{\gamma'_k} < 1, \qquad k = 1, 2, 3.$$

The three ratios are less than 0.988, 0.983, and 0.994. Also $4k/(q^2\gamma'_k) < 1/2$ in every case. A good partition therefore exists. Since $Z_k \ge (q\theta)^{2k}$, (88)–(89) prove the theorem. □

Remark 7.12. Condition (99) concerns the diagonal of $|A|$, not merely that of A. The positive and negative spectral contributions to a_{ii} can cancel for an indefinite matrix. Such cancellation is absent from $(|A|)_{ii}$. Thus Theorem 7.11 does not settle the unrestricted indefinite case.

7.7 A smaller constant for a separated nonzero spectrum

The following argument gives a smaller constant when every nonzero eigenvalue, rather than just the third, is bounded away from zero.

Lemma 7.13 (A spread spectral frame)**.** *Let R be an $n \times r$ matrix with $R^TR = I_r$ and let Λ be a nonsingular real diagonal $r \times r$ matrix. If every row r_i^T of R satisfies $|r_i|^2 \le 1/2$, then for $Q = X^TR\Lambda R^TX + b^TX + c$,*

$$|\varphi_Q(t)| \le (4\pi)^{r/2}|t|^{-r/2}|\det\Lambda|^{-1/2}. \tag{106}$$

Proof. Put $c_i = |r_i|^2$ and $u_i = r_i/\sqrt{c_i}$ when $c_i > 0$. Then $\sum_i c_i u_i u_i^T = I_r$ and $\sum_i c_i = r$. The geometric Brascamp–Lieb inequality (Brascamp and Lieb, 1976; Lieb, 1990) applied to $F_i(s) = |\varphi_{X_i}(\sqrt{c_i}s + tb_i)|^{1/c_i}$ gives

$$\begin{aligned}\int_{\mathbb{R}^r}\prod_i|\varphi_{X_i}(\langle r_i,u\rangle + tb_i)|\,du &\le \prod_{i:c_i>0}\left(\int_{\mathbb{R}}F_i(s)\,ds\right)^{c_i}\\ &\le (4\pi)^r.\end{aligned}$$

The last inequality is Lemma 4.3 with exponent $1/c_i \ge 2$, after changing variables. Rows with $c_i = 0$ contribute factors of modulus at most one.

For a justified quadratic Fourier inversion, put $Y = R^TX$ and regularize the quadratic exponential as

$$e^{itY^T\Lambda Y - \varepsilon|Y|^2}, \qquad \varepsilon > 0.$$

The Fourier transform with kernel e^{-iu^Ty} of this complex Gaussian has modulus at most $\pi^{r/2}|t|^{-r/2}|\det\Lambda|^{-1/2}$. Fourier inversion and independence therefore bound the regularized expectation by

$$(4\pi|t|)^{-r/2}|\det\Lambda|^{-1/2}\int_{\mathbb{R}^r}\prod_i|\varphi_{X_i}(\langle r_i,u\rangle + tb_i)|\,du.$$

For $\varepsilon > 0$ the Gaussian transform is integrable, which justifies the initial use of Fubini. The preceding Brascamp–Lieb bound supplies an integrable majorant for the product after multiplication by the uniformly bounded Gaussian Fourier factor. Let $\varepsilon \downarrow 0$ to prove (106). □

Theorem 7.14 (Separated nonzero semidefinite spectrum)**.** *Suppose A is positive semidefinite of rank $r \geq 3$ and every nonzero eigenvalue is at least $\theta > 0$. Then*

$$\|p_Q\|_\infty \leq 130/\theta \tag{107}$$

for all b, c. The same assertion holds if all nonzero eigenvalues are at most $-\theta$.

Proof. Scale so that $\theta = 1$, and write $A = R\Lambda R^T$ with $R^T R = I_r$ and $\Lambda \geq I_r$. Then $|r_i|^2 \leq a_{ii}$. Use a positive pivot if some $a_{ii} \geq 1/2$. Otherwise Lemma 7.13 applies, and

$$|\varphi_Q(t)| \leq \min(1, (4\pi/|t|)^{r/2}) \leq \min(1, (4\pi/|t|)^{j/2}) \quad (r \geq j).$$

For rank at least one, a pivot gives $\alpha_1(Q) \leq 4\sqrt{2} < 15$. Without a pivot, Lemma 7.3 gives $\alpha_1(Q) \leq 2\sqrt{2}\sqrt{4\pi}/\sin^2 1 < 15$. Thus $\alpha_1 \leq 15$ throughout this spectral class.

For rank at least two, a pivot leaves a Schur complement of rank one less with all nonzero eigenvalues at least one. The first level and Lemma 7.4(2) give $\alpha_2(Q) \leq 4\sqrt{2}\sqrt{2} \cdot 15 = 120$. Without a pivot, Lemma 7.3 gives $\alpha_2(Q) \leq 4\pi/\sin^2 1 < 18$.

For rank at least three, a pivot and the second level give

$$\|p_Q\|_\infty \leq 3\sqrt{2}\sqrt{120(3 + \log 120)} < 130.$$

Without a pivot, integrating $\min(1, (4\pi/|t|)^{3/2})$ gives $\|p_Q\|_\infty \leq 12$. Scaling and reflection complete the proof. □

7.8 Sharper bounds in definite rank three

Theorem 7.15 (Exact definite rank three)**.** *Suppose A has rank three and its nonzero eigenvalues are all at least $\theta > 0$, or all at most $-\theta$. Then, for every $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$,*

$$\|p_{X^TAX+b^TX+c}\|_\infty \leq 58/\theta. \tag{108}$$

If in addition $b \in \operatorname{range}(A)$, the stronger bound

$$\|p_{X^TAX+b^TX+c}\|_\infty \leq 3e/\theta \tag{109}$$

holds.

Proof. Reflection and scaling reduce the assertion to $A \geq 0$, $\operatorname{rank}(A) = 3$, and $\lambda_3(A) \geq 1$. Write $A = R\Lambda R^T$ with $R^T R = I_3$ and $\Lambda \geq I_3$. If every row r_i^T of R satisfies $|r_i|^2 \leq 1/2$, Theorem 4.6, with the individual linear terms as separable phases, gives

$$\|p_Q\|_\infty \leq 3e\,(\det \Lambda)^{-1/3} \leq 3e.$$

This proves both assertions in this case.

Otherwise choose i with $|r_i|^2 > 1/2$. Then $a = a_{ii} \geq |r_i|^2 > 1/2$. Lemma 7.5 writes $Q = a(X_i - m(X_{-i}))^2 + T(X_{-i})$. The matrix of T has rank two, with both nonzero eigenvalues at least one; its linear term may be arbitrary. Theorem 9.12 therefore gives $\alpha_2(T) \leq 46$. The conditional square density yields

$$\|p_Q\|_\infty \leq \sqrt{2} \sup_s \mathbb{E}|s - T|^{-1/2}.$$

For any $V \geq 1$, the layer-cake calculation used in Lemma 7.4 gives

$$\mathbb{E}|s-T|^{-1/2} \leq V + \frac{46(3+2\log V)}{V}.$$

Taking $V = 20$ proves (108), since

$$\sqrt{2}\left(20 + \frac{46(3+2\log 20)}{20}\right) < 57.531 < 58.$$

Now suppose $b \in \operatorname{range}(A)$. Choose z with $2Az = b$ and replace X by $X + z$, absorbing the resulting constant into c. Translation preserves independence and all coordinate density caps, so it is enough to treat $b = 0$. In the heavy-row case the Schur complement is a positive semidefinite matrix of rank two with nonzero eigenvalues $\mu_1, \mu_2 \geq 1$. Thus

$$T = Y^T \operatorname{diag}(\mu_1, \mu_2) Y + c', \qquad Y = U^T X_{-i}, \quad U^T U = I_2.$$

Lemma 4.2 bounds the joint density of Y by 2. Lemma 3.3 consequently gives $\|p_T\|_\infty \leq 2\pi$. Keep the support restriction in the square density formula:

$$p_Q(s) \leq a^{-1/2}\mathbb{E}\big[(s-T)^{-1/2}\mathbf{1}_{\{T<s\}}\big] \quad \text{for almost every } s.$$

For any real random variable with density bounded by M, layer cake gives

$$\mathbb{E}\big[(s-T)^{-1/2}\mathbf{1}_{\{T<s\}}\big] = \int_0^\infty \mathbb{P}(s - t^{-2} < T < s)\, dt \leq \int_0^\infty \min(1, Mt^{-2})\, dt = 2\sqrt{M}.$$

It follows that $\|p_Q\|_\infty \leq 4\sqrt{\pi} < 3e$. Together with the diffuse case, this proves (109). Restoring the scale θ completes the proof. □

8 The first two levels without a coordinate pivot

Let $X = (X_1, \dots, X_n)$ have independent coordinates with densities bounded by one, and write

$$Q = X^{\mathsf{T}} M X + b^{\mathsf{T}} X + c_0, \qquad \varphi_Q(t) = \mathbb{E}e^{itQ}.$$

Here M is real symmetric, and all estimates are uniform in b, c_0. The definition of a c-pivot is given in Section 9. In particular, absence of a c-pivot, with $0 < c \leq 1$, implies

$$|m_{ii}| < c, \qquad m_{ij}^2 < c + c^2 \leq 2c \quad (i \neq j). \tag{110}$$

Set

$$O = M - \operatorname{diag} M, \qquad r_i^2 = \sum_{j \neq i} m_{ij}^2, \qquad L_Q(u) = \sup_s \mathbb{P}(|Q-s| \leq u).$$

The truncated half-power potential is

$$\mathcal{H}(Q) = \sup_s \mathbb{E}\big[|Q-s|^{-1/2};\ |Q-s| \leq 1\big].$$

Theorem 8.1 (One large eigenvalue)**.** *Suppose that M has no c-pivot, where $c \leq 2 \cdot 10^{-6}$, and $\|M\| \geq 1$. Then*

$$L_Q(u) \leq 25\sqrt{u} \qquad (u > 0).$$

In particular,

$$\sup_s \mathbb{E}\log_+ \frac{1}{|Q-s|} \leq 2 + 2\log 25 < 8.44.$$

Theorem 8.2 (Two large eigenvalues)**.** *Put $c_* = 10^{-15}$. If M has no c_*-pivot and has at least two eigenvalues of modulus at least one, then*

$$\mathcal{H}(Q) \le 10^6.$$

The second theorem is a negative-moment estimate. Its statement does not require an estimate of order $u(1+\log(1/u))$ for $L_Q(u)$. The small pivot threshold is used to make the separate cases below quantitative.

8.1 Fourier concentration and volume marginals

Put $\kappa_0 = 4\sin^2(1/2) > 0.91939$. The Fejér kernel identity

$$\left(\frac{\sin(x/2)}{x/2}\right)^2 = \int_{-1}^{1} (1-|\tau|)e^{i\tau x}\,d\tau$$

gives, for any real random variable Z,

$$L_Z(u) \le \kappa_0^{-1} u \int_{|t|\le 1/u} |\varphi_Z(t)|\,dt. \tag{111}$$

Indeed, the kernel is at least κ_0 on $[-1,1]$; take its expectation at $(Z-s)/u$, use Fubini, and then take absolute values. In particular,

$$|\varphi_Z(t)| \le B|t|^{-\beta}, \quad 0<\beta<1 \quad\Longrightarrow\quad L_Z(u) \le \frac{2B}{\kappa_0(1-\beta)} u^{\beta}. \tag{112}$$

For disjoint coordinate sets I, J, let $A = M_{I,J}$. Write

$$Z_k(A) = \sum_{|S|=|T|=k} \det(A_{S,T})^2 = e_k(A^{\mathsf{T}}A), \qquad \alpha_i^{(k)} = \frac{\sum_{S\ni i,T} \det(A_{S,T})^2}{Z_k(A)}.$$

The sums are over row and column subsets of A of the indicated cardinalities. Lemma 4.8, with $1 \le q \le 2$, gives

$$|\varphi_Q(t)| \le \left[e^{k/2}(2\pi)^k q^{-k/2} Z_k(A)^{-1/2}\right]^{1/(2q)} |t|^{-k/(2q)}, \qquad \max_i \alpha_i^{(k)} \le q/2. \tag{113}$$

This estimate allows arbitrary quadratic terms within I and J. Other coordinates may be conditioned on, and the orientation of A may be reversed. These facts follow from the decoupling step in that lemma; the resulting bounds are uniform in the conditioned affine terms.

Lemma 8.3 (First-volume estimate)**.** *Suppose $A = M_{I,J} \neq 0$, and let $1/2 \le h \le 1$. If every row satisfies $|A_{i,\cdot}|^2 \le h\|A\|_F^2$, then*

$$|\varphi_Q(t)| \le \left[2\pi e^{1/2}(2h)^{-1/2}\|A\|_F^{-1}\right]^{1/(4h)} |t|^{-1/(4h)}. \tag{114}$$

The same assertion holds with columns in place of rows. When $h = 1$ there is no additional marginal hypothesis.

Proof. For $k=1$, the minors are the entries of A, so that $Z_1(A) = \|A\|_F^2$ and $\alpha_i^{(1)} = |A_{i,\cdot}|^2/\|A\|_F^2$. Apply (113) with $q = 2h$. □

Lemma 8.4 (Choice of orientation)**.** *Let $Z = \|A\|_F^2 > 0$ and $1/2 < h < 1$. If $A_{ij}^2 \le (2h-1)Z$ for every entry, then either every row share is at most h, or every column share is at most h.*

Proof. If a row i and a column j both had shares greater than h, then

$$Z \geq |A_{i,\cdot}|^2 + |A_{\cdot,j}|^2 - A_{ij}^2 > 2hZ - A_{ij}^2,$$

contrary to the entry hypothesis. □

Combining these two lemmas with (112) at $h = 3/4$ gives the following useful consequence:

$$\|M_{I,J}\|_F \geq \rho, \qquad m_{ij}^2 \leq \tfrac{1}{2}\|M_{I,J}\|_F^2 \quad \Longrightarrow \quad L_Q(u) \leq K(\rho)u^{1/3},$$
$$K(\rho) = \frac{3}{\kappa_0}\left(\frac{2\pi\sqrt{2e/3}}{\rho}\right)^{1/3} < 3.264\left(\frac{8.46}{\rho}\right)^{1/3}. \tag{115}$$

8.2 Proof at the first level

Proof of Theorem 8.1. Let $F^2 = \|O\|_F^2$. The triangle inequality gives $F \geq \|O\| \geq 1 - c$. Put $\varepsilon = 1/100$ and call an index heavy if $r_i^2 > \varepsilon F^2$. There are fewer than $1/\varepsilon$ heavy indices. Place all of them in J, and place each remaining index independently in I or J, with equal probabilities. Set $Z = \|M_{I,J}\|_F^2$.

The squared entries between heavy indices have total at most $2c/\varepsilon^2$, by (110). Every other off-diagonal squared entry contributes to $\mathbb{E}Z$ with probability at least $1/4$, with the two orientations counted in F^2. Consequently,

$$\mathbb{E}Z \geq \frac{1}{4}(F^2 - 2c/\varepsilon^2) \geq \frac{19}{80}F^2.$$

The last step uses $c \leq \varepsilon^2 F^2/40$, which follows from $c \leq 2 \cdot 10^{-6}$ and $F \geq 1 - c$.

Changing the coin at a light index i changes Z by at most r_i^2. Conditional variance for one fair coin, followed by variance tensorization, gives

$$\operatorname{Var} Z \leq \frac{1}{4}\sum_{i\ mlight} r_i^4 \leq \frac{\varepsilon}{4}F^4.$$

Chebyshev's inequality therefore gives

$$\mathbb{P}\{Z < \tfrac{1}{2}\mathbb{E}Z\} \leq \frac{\varepsilon F^4}{(\mathbb{E}Z)^2} \leq \frac{6400\varepsilon}{361} < 0.18.$$

Choose a realization with $Z \geq \mathbb{E}Z/2 > F^2/9$. All its rows are light, so

$$\frac{|M_{i,J}|^2}{Z} \leq 9\varepsilon = 0.09 < 1/2.$$

Lemma 8.3, with $h = 1/2$, now yields

$$|\varphi_Q(t)| \leq \left(\frac{6\pi\sqrt{e}}{F}\right)^{1/2}|t|^{-1/2} < 5.6|t|^{-1/2}.$$

Equation (112) gives $L_Q(u) \leq (22.4/\kappa_0)\sqrt{u} < 25\sqrt{u}$. Finally, layer cake gives

$$\mathbb{E}\log_+\frac{1}{|Q-s|} = \int_0^\infty \mathbb{P}(|Q-s| < e^{-v})\,dv \leq \int_0^\infty \min(1, 25e^{-v/2})\,dv = 2 + 2\log 25.$$

□

With only $\max_i |m_{ii}| \leq 1/2$ and $\|M\| \geq 1$, a fair partition satisfies $\mathbb{E}\|M_{I,J}\|_F^2 = \|O\|_F^2/4 \geq 1/16$. The unrestricted $h = 1$ case gives a fourth-root estimate; Lemma 4.29 provides such an estimate for every nonzero quadratic matrix. In particular it already gives a uniform logarithmic potential. The argument above upgrades the concentration exponent when the principal pivots are small.

8.3 A discriminant estimate

Fix a coordinate h, and write $X' = X_{-h}$ and

$$a = m_{hh}, \qquad w = M_{-h,h}, \qquad L = w^{\mathsf{T}} X' + b_h/2, \qquad Q = aX_h^2 + 2LX_h + R'(X').$$

For a target level s, set

$$\Delta_s = L^2 + a(s - R'). \tag{116}$$

Its quadratic matrix is

$$N^{(h)} = ww^{\mathsf{T}} - aM_{-h,-h}, \qquad N^{(h)}_{ij} = m_{hi}m_{hj} - m_{hh}m_{ij}. \tag{117}$$

The diagonal entries are negatives of the principal two-coordinate minors through h.

Lemma 8.5 (Conditional sublevel bound)**.** *For every $u > 0$,*

$$\mathbb{P}(|Q - s| \le u \mid X') \le \min\left\{1, \frac{2\sqrt{2}\sqrt{u}}{\sqrt{|a|}}, \frac{4u}{\sqrt{|\Delta_s|}}\right\}. \tag{118}$$

When $a = 0$, omit the middle expression.

Proof. If $a \neq 0$, complete the square:

$$Q = a(X_h + L/a)^2 + \xi, \qquad \xi = R' - L^2/a, \qquad |a|\,|s - \xi| = |\Delta_s|.$$

Apply the one-dimensional quadratic sublevel estimate in Lemma 3.2. If $a = 0$, the conditional polynomial is affine with slope $2L$ and has sublevel probability at most $\min(1, u/|L|)$. Since $\Delta_s = L^2$, this implies the stated weaker bound, including the case $L = 0$. □

Lemma 8.6 (Integrated discriminant bound)**.** *With the middle term omitted for $a = 0$, define*

$$I(d, a) = \int_1^\infty \min\left\{1, \frac{2\sqrt{2}}{v\sqrt{|a|}}, \frac{4}{v^2\sqrt{|d|}}\right\} dv.$$

Then

$$I(d, a) \le 4|d|^{-1/4}, \tag{119}$$

$$\mathbb{E}[|Q - s|^{-1/2};\ |Q - s| \le 1] \le 1 + \mathbb{E}I(\Delta_s, a). \tag{120}$$

If $0 < |a| \le 1$, one also has

$$I(d, a) \le \frac{2\sqrt{2}}{\sqrt{|a|}}\left(2 + \frac{1}{2}\log_+ \frac{a^2}{4|d|}\right). \tag{121}$$

Proof. Put $D = 4|d|^{-1/2}$. Dropping the middle term, the integral is at most D if $D \le 1$, and at most $2\sqrt{D}$ if $D > 1$, by splitting at $\sqrt{D}$. Both bounds imply (119). For (121), put $A_0 = 2\sqrt{2}|a|^{-1/2} \ge 1$ and $V_0 = D/A_0$. If $V_0 \le A_0$, integrate 1 to A_0 and D/v^2 afterwards; the result is at most $2A_0$. If $V_0 > A_0$, integrate 1 to A_0, A_0/v to V_0, and D/v^2 afterwards. The result is at most $A_0(2 + \log(V_0/A_0))$. Since $V_0/A_0 = |a|/(2\sqrt{|d|})$, this proves the claim. Finally, layer cake and Lemma 8.5 give

$$\mathbb{E}[|Q - s|^{-1/2};\ |Q - s| \le 1] \le 1 + \int_1^\infty \mathbb{P}(|Q - s| < v^{-2})\, dv \le 1 + \mathbb{E}I(\Delta_s, a).$$

□

Theorem 8.7 (A coordinate discriminant)**.** *Let $0 < \nu \le 1$. Suppose M has no c-pivot, where $c \le \nu/9$, and some coordinate h satisfies $\|N^{(h)}_{\mathrm{off}}\|_F \ge \nu$. Then*

$$\mathcal{H}(Q) \le 5 + 40\left(\frac{16.92}{\nu}\right)^{1/3}. \tag{122}$$

No spectral hypothesis is needed for this assertion.

Proof. By (110) and $c \le 1/9$,

$$|N^{(h)}_{ii}| < c, \qquad |N^{(h)}_{ij}| \le 2c + c\sqrt{2c} \le 3c.$$

A random partition of the coordinates other than h yields a realization satisfying

$$\|N^{(h)}_{I,J}\|_F^2 \ge \nu^2/4, \qquad |N^{(h)}_{ij}|^2 \le 9c^2 \le \nu^2/9 \le \tfrac{1}{2}\|N^{(h)}_{I,J}\|_F^2.$$

Apply (115) to Δ_s, whose affine terms are arbitrary. Uniformly in s and in a second target z,

$$\mathbb{P}(|\Delta_s - z| \le r) \le K_\Delta r^{1/3}, \qquad K_\Delta = K(\nu/2) < 3.264(16.92/\nu)^{1/3}.$$

In particular Δ_s has no atom at zero, and layer cake gives

$$\mathbb{E}|\Delta_s|^{-1/4} \le 1 + \int_1^\infty K_\Delta v^{-4/3}\, dv = 1 + 3K_\Delta.$$

Equations (119) and (120) now give $\mathcal{H}(Q) \le 5 + 12K_\Delta$, which implies (122). □

Lemma 8.8 (Two tests for a discriminant)**.** *If M has no c-pivot, then every unit eigenpair (μ, v) and every coordinate h satisfy*

$$\|N^{(h)}_{\mathrm{off}}\|_F \ge |\mu|\big(|\mu| v(h)^2 - |m_{hh}|\big) - c, \tag{123}$$

$$\|N^{(h)}_{\mathrm{off}}\|_F \ge r_h^2 - |m_{hh}|\,\|M\| - c. \tag{124}$$

Proof. Extend $N^{(h)}$ by a zero hth row and column. Then $N^{(h)} = m_h m_h^{\mathsf{T}} - m_{hh}M$, where $m_h = Me_h$. Its diagonal norm is at most c, so

$$\|N^{(h)}_{\mathrm{off}}\|_F \ge \|N^{(h)}\| - c.$$

The Rayleigh quotient on v is $\mu^2 v(h)^2 - m_{hh}\mu$, proving (123). Also $\|N^{(h)}\| \ge |m_h|^2 - |m_{hh}|\|M\|$, and $|m_h|^2 \ge r_h^2$, proving (124). □

8.4 Two-volume marginals and random partitions

Lemma 8.9 (Second-volume marginals)**.** *Let A have rank at least two, with singular values $s_1 \ge s_2 \ge \cdots$ and left singular vectors u_m. Put $G = AA^{\mathsf{T}}$, $\lambda_m = s_m^2$ and $Z_2 = e_2(G)$. Then*

$$\alpha_i^{(2)} = \frac{G_{ii}\,\mathrm{tr}\,G - (G^2)_{ii}}{e_2(G)} = \sum_m p_m u_m(i)^2, \quad p_m = \frac{\lambda_m \sum_{\ell \ne m}\lambda_\ell}{e_2(\lambda)} \le 1. \tag{125}$$

Consequently,

$$\alpha_i^{(2)} \le 2|A_{i,\cdot}|^2/s_2^2, \tag{126}$$

$$\alpha_i^{(2)} \le u_1(i)^2 + |(A - s_1u_1v_1^{\mathsf{T}})_{i,\cdot}|^2\left(\frac{1}{T} + \frac{1}{S}\right), \tag{127}$$

$$S = s_1^2, \qquad T = \sum_{m\ge 2} s_m^2.$$

There are identical formulas for the column marginals.

Proof. Cauchy–Binet gives $\sum_T \det(A_{S,T})^2 = \det G_{S,S}$. Summing the two-coordinate minors that contain i gives

$$\sum_{j\neq i}(G_{ii}G_{jj} - G_{ij}^2) = G_{ii}\,\mathrm{tr}\,G - (G^2)_{ii}.$$

Spectral decomposition gives (125). The bound $p_m \le 1$ follows because its numerator is the sum of the terms of $e_2(\lambda)$ that contain λ_m. Moreover,

$$e_2(\lambda) = \tfrac{1}{2}\sum_m \lambda_m \sum_{\ell\neq m}\lambda_\ell \ge \tfrac{1}{2}\lambda_2 \sum_m \lambda_m,$$

which proves (126). Since $e_2 \ge ST$, for $m \ge 2$ one has $p_m \le \lambda_m(S+T)/(ST)$. Separating the term $m = 1$ proves (127). □

Let $(\mu_1, v_1), (\mu_2, v_2)$ be orthonormal eigenpairs of M, write $V = [v_1\ v_2]$, and put $\Lambda = \mathrm{diag}(\mu_1, \mu_2)$. For independent fair signs ϵ_i, put

$$E = \mathrm{diag}(\epsilon_i), \quad I = \{i : \epsilon_i = 1\}, \quad J = I^c, \quad \Sigma = V^{\mathsf{T}}EV, \quad B = V_I^{\mathsf{T}} M_{I,J} V_J.$$

Writing $D = \mathrm{diag}\,M$, direct expansion gives

$$4B = \Lambda + [\Sigma, \Lambda] - V^{\mathsf{T}}DV - C, \qquad C_{kl} = \sum_{i\neq j}\epsilon_i\epsilon_j v_k(i) m_{ij} v_l(j). \tag{128}$$

Indeed $4B = V^{\mathsf{T}}(I+E)M(I-E)V$, and $MV = V\Lambda$. If $M_3 = M - V\Lambda V^{\mathsf{T}}$, the alternative expansion is

$$4B = \Lambda + [\Sigma, \Lambda] - \Sigma\Lambda\Sigma - V^{\mathsf{T}}\,\mathrm{diag}(M_3)V - C^{(3)}, \tag{129}$$

where $C^{(3)}$ is defined as C with M_3 in place of M. This follows from $M_3 V = 0$.

The elementary second-moment identities used below are

$$\mathbb{E}\Sigma_{kl}^2 = \sum_i v_k(i)^2 v_l(i)^2, \tag{130}$$

$$\mathbb{E}(C_{kl}^{(H)})^2 \le 2\sum_{i\neq j} v_k(i)^2 H_{ij}^2 v_l(j)^2 \le 2\max_i v_k(i)^2 \|H\|^2. \tag{131}$$

For the second identity only equal or reversed index pairs survive the sign average; bound each cross product by half the sum of its two squares. Finally,

$$[\Sigma, \Lambda] = \begin{pmatrix} 0 & s \\ -s & 0 \end{pmatrix}, \qquad s = (\mu_2 - \mu_1)\Sigma_{12},$$

Thus the matrix $P = \left(\begin{smallmatrix} \mu_1 & s \\ -s & \mu_2 \end{smallmatrix}\right)$ has determinant $\mu_1\mu_2 + s^2$. Since $\|V_I\|, \|V_J\| \le 1$, the singular values satisfy $s_k(M_{I,J}) \ge s_k(B)$ for $k = 1, 2$.

Lemma 8.10 (Comparable eigenvalues and light rows)**.** *Suppose*

$$|m_{ii}| \le c, \quad 1 \le |\mu_2| \le |\mu_1| \le K, \quad r_i \le \rho, \quad c \le \frac{1}{8(K+2)}, \quad \rho \le \frac{3}{64(K+2)}.$$

With probability at least 11/16*, a fair partition has*

$$s_2(M_{I,J}) \ge \sigma_A := \frac{3}{32(K+2)}, \qquad \max_i \alpha_i^{(2)} \le 1/2.$$

Consequently,

$$|\varphi_Q(t)| \le 2\pi\sqrt{e}\,\sigma_A^{-1}|t|^{-1}. \tag{132}$$

Proof. The eigenvector equation gives

$$|v_k(i)| \le \frac{r_i}{|\mu_k - m_{ii}|} \le \frac{\rho}{1-c} \qquad (k = 1, 2).$$

Equation (131), using the off-diagonal row bound before the last inequality, gives

$$\mathbb{E}\|C\|_F^2 \le \frac{8\rho^4}{(1-c)^2} \le 16\rho^4.$$

Also $\mathbb{E}\Sigma_{12}^2 \le \rho^2/(1-c)^2$. Except on a set of probability at most $1/16 + 1/4 = 5/16$, both

$$|\Sigma_{12}| \le 4\rho/(1-c), \qquad \|C\|_F \le 8\rho^2$$

hold. On this event,

$$|s| \le 8K\rho/(1-c) \le 1/2, \quad |\det P| \ge 3/4, \quad \|P\| \le K + 1/2.$$

Thus $s_2(P) \ge 3/[4(K+2)]$. Using (128),

$$s_2(4B) \ge \frac{3}{4(K+2)} - c - 8\rho^2 \ge \frac{3}{8(K+2)}.$$

Division by four gives the claimed singular-value lower bound. Every row of $M_{I,J}$ has norm at most ρ, and hence (126) gives $\alpha_i^{(2)} \le 2\rho^2/\sigma_A^2 \le 1/2$. Finally $Z_2 \ge s_2^4 \ge \sigma_A^4$, so (113) with $k = 2, q = 1$ proves (132). □

Lemma 8.11 (A dominant eigenvalue)**.** *Order the eigenvalues of M by decreasing modulus, and suppose*

$$|\mu_2| = 1, \quad K = |\mu_1| \ge 10^4, \quad |m_{ii}| \le c, \quad |m_{ij}| \le \sqrt{2c}, \quad c \le 10^{-6}.$$

Assume, for all i, that $Kv_1(i)^2 \le \eta$ and $v_2(i)^2 \le \eta$, where $\eta \le 6 \cdot 10^{-5}$. Some partition and orientation satisfy

$$Z_2(M_{I,J}) \ge 1, \qquad \max_i \alpha_i^{(2)} \le 0.6.$$

In particular,

$$|\varphi_Q(t)| \le 6.51|t|^{-5/6}, \qquad L_Q(u) \le 85u^{5/6}, \qquad \mathcal{H}(Q) < 129.$$

Proof. We first find a partition with two singular directions. Since $\|M_3\| \le 1$, equations (130)–(131) show that, with probability at least 53/64, all four bounds

$$\begin{gathered} |\Sigma_{11}|, |\Sigma_{12}| \le 8\sqrt{\eta/K}, \qquad |\Sigma_{22}| \le 8\sqrt{\eta}, \\ \|C^{(3)}\|_F \le 8\sqrt{\eta} \end{gathered} \tag{133}$$

hold. The failure probabilities for the three entries of Σ are at most 1/64 each. The bound $\mathbb{E}\|C^{(3)}\|_F^2 \le 4\eta/K + 4\eta \le 8\eta$ gives an additional failure probability at most 8/64.

On this event,

$$\|\Sigma\Lambda\Sigma\|_F \le K(\Sigma_{11}^2 + \Sigma_{12}^2) + \Sigma_{12}^2 + \Sigma_{22}^2 \le 256\eta.$$

Moreover,

$$\|V^{\mathsf{T}} \operatorname{diag}(M_3) V\| \le c + 2\eta, \qquad s^2 \le 64\eta K(1 + 1/K)^2 \le 65\eta K.$$

It follows that $|\det P| \ge 0.996K$, $\|P\| \le 1.001K$, and $s_2(P) \ge 0.995$. The remaining terms in (129) have norm at most

$$256\eta + c + 2\eta + 8\sqrt{\eta} < 0.078.$$

Hence, with $A = M_{I,J}$,

$$s_2(A) \geq 0.229, \qquad s_1(A) \geq 0.249K. \tag{134}$$

Next write

$$A = \mu_1 x y^{\mathsf{T}} + F, \qquad x = (v_1)_I, \quad y = (v_1)_J, \quad \|F\| \leq 1.$$

Equation (133) implies $|x|^2, |y|^2 \geq 0.4996$. Put $\sigma_0 = K|x||y|$, so that $0.4996K \leq \sigma_0 \leq K/2$. Absorbing the sign of μ_1 into one normalized vector, write $\widehat{x} = x/|x|$, $\widehat{y} = \pm y/|y|$. Then

$$AA^{\mathsf{T}} = \sigma_0^2 \widehat{x}\widehat{x}^{\mathsf{T}} + P', \qquad \|P'\| \leq 2\sigma_0 + 1.$$

If u is a top left singular vector of A, projecting its eigenvector equation perpendicular to $\widehat{x}$ gives

$$\sin \angle(u, \widehat{x}) \leq \frac{2\sigma_0 + 1}{(\sigma_0 - 1)^2} \leq \frac{2.01}{\sigma_0}.$$

Here $s_1(A) \geq \sigma_0 - 1$, and $\sigma_0 \geq 4996$. The identical bound holds for the top right singular vector v relative to $\widehat{y}$.

Let $A_0 = A - s_1 u v^{\mathsf{T}}$. For each applicable coordinate,

$$|u_i|, |v_j| \leq R_K := \sqrt{\frac{\eta}{0.4996K}} + \frac{2.01}{0.4996K}, \qquad R_K^2 < 3 \cdot 10^{-7}.$$

Using $A_0 = A(I - vv^{\mathsf{T}})$ and its transposed version gives

$$|(A_0)_{i,\cdot}|, |(A_0)_{\cdot,j}| \leq 1 + 2.01\sqrt{\frac{\eta}{0.4996K}} < 1.001.$$

For the entries, $s_1 \leq \sigma_0 + 1 \leq K/2 + 1$ yields

$$|(A_0)_{ij}| \leq \sqrt{2c} + (K/2 + 1)R_K^2 < 0.003.$$

The last bound follows by substituting $c \leq 10^{-6}$, $\eta \leq 6 \cdot 10^{-5}$ and $K \geq 10^4$; after expansion, $(K/2+1)R_K^2$ is decreasing in K. By (134), $T = \|A_0\|_F^2 \geq 0.229^2$. Since $0.003^2 < 0.02 \cdot 0.229^2$, Lemma 8.4 with $h = 0.51$ gives an orientation in which every residual row has squared norm at most $0.51T$.

Equation (127) therefore gives

$$\alpha_i^{(2)} \leq 3 \cdot 10^{-7} + 0.51 + \frac{1.001^2}{(0.249K)^2} < 0.6.$$

Also $Z_2 \geq s_1^2 s_2^2 \geq (0.249K)^2\, 0.229^2 > 1$. Apply (113) with $k = 2, q = 6/5$:

$$|\varphi_Q(t)| \leq \left[\frac{5e(2\pi)^2}{6}\right]^{5/12} |t|^{-5/6} < 6.51|t|^{-5/6}.$$

Equation (112) gives $L_Q(u) \leq 85u^{5/6}$. Finally,

$$\mathcal{H}(Q) \leq 1 + \int_1^\infty 85v^{-5/3}\, dv = 128.5 < 129.$$

□

8.5 Completion of the second-level estimate

Proof of Theorem 8.2. Order the eigenvalues by modulus and let $r = |\mu_2| \geq 1$. Scaling by r preserves the absence of a c_*-pivot. More precisely,

$$\mathcal{H}(Q) \leq r^{-1/2}\mathcal{H}(Q/r),$$

because the event $|Q - s| \leq 1$ becomes $|Q/r - s/r| \leq 1/r \leq 1$. It is therefore enough to assume $|\mu_2| = 1$ and put $K = |\mu_1| = \|M\|$.

Suppose first that $K \leq 10^4$, and put

$$\rho_A = \frac{3}{64 \cdot 10002}, \qquad \nu_A = \rho_A^2/2.$$

If some $r_h > \rho_A$, equation (124) gives

$$\|N_{\mathrm{off}}^{(h)}\|_F \geq \rho_A^2 - 10^{-15}(10^4 + 1) \geq \nu_A.$$

Here $\nu_A > 1.098 \cdot 10^{-11}$ and $10^{-15} < \nu_A/9$. Theorem 8.7 consequently gives

$$\mathcal{H}(Q) \leq 5 + 40(16.92/\nu_A)^{1/3} < 4.7 \cdot 10^5.$$

If all $r_i \leq \rho_A$, apply Lemma 8.10 with $K = 10^4$. It gives $|\varphi_Q(t)| \leq \min(1, B_0/|t|)$, where

$$B_0 = 2\pi\sqrt{e}\,\frac{32 \cdot 10002}{3} < 1.106 \cdot 10^6.$$

For $u \leq 1/B_0$, equation (111) implies

$$L_Q(u) \leq \frac{2B_0 u}{\kappa_0}\left(1 + \log\frac{1}{B_0 u}\right).$$

For larger u use $L_Q(u) \leq 1$. Splitting the half-power integral at $v = \sqrt{B_0}$ yields

$$\begin{aligned}\mathcal{H}(Q) &\leq \sqrt{B_0} + \frac{2B_0}{\kappa_0}\int_{\sqrt{B_0}}^{\infty} v^{-2}\left(1 + 2\log\frac{v}{\sqrt{B_0}}\right)dv \\ &= \left(1 + \frac{6}{\kappa_0}\right)\sqrt{B_0} < 8000.\end{aligned}$$

Now suppose $K > 10^4$ and put $\nu_B = 5 \cdot 10^{-5}$. If some h and $k \in \{1, 2\}$ satisfy

$$|\mu_k|(|\mu_k| v_k(h)^2 - c_*) \geq \nu_B + 2c_*,$$

equation (123) gives $\|N_{\mathrm{off}}^{(h)}\|_F \geq \nu_B$. Theorem 8.7 gives $\mathcal{H}(Q) < 3000$. Otherwise, for every i,

$$K v_1(i)^2 < c_* + \frac{\nu_B + 2c_*}{K} \leq 6 \cdot 10^{-5}, \qquad v_2(i)^2 < \nu_B + 3c_* \leq 6 \cdot 10^{-5}.$$

The entry bounds (110) permit an application of Lemma 8.11, which gives $\mathcal{H}(Q) < 129$. Every case is bounded by 10^6, as asserted. □

Theorems 8.1 and 8.2 provide the first two integrability estimates used in the coordinate-pivot reduction. Lemma 9.1 preserves the required number of large eigenvalues at each pivot. The reduction still requires a third-level estimate for matrices without a fixed small pivot. The estimates above allow all affine terms produced by conditioning.

9 Spectral reductions and finite coordinate cores

Throughout this section $X = (X_1, \dots, X_n)$ has independent coordinates with probability densities bounded by one. Write

$$Q_A(X) = X^{\mathsf{T}} A X + b^{\mathsf{T}} X + c_0, \qquad N_A(t) = \#\{j : |\lambda_j(A)| \geq t\}.$$

All assertions are uniform in the affine terms. For $0 < c \leq 1$, a *c-pivot* is either a diagonal entry of modulus at least c, or a principal 2-by-2 submatrix whose determinant has modulus at least c. If there is no c-pivot, then

$$|a_{ii}| < c, \qquad |a_{ij}|^2 < c + c^2 \leq 2c. \tag{135}$$

9.1 Schur complements and logarithmic potentials

Lemma 9.1 (Spectral count under a coordinate pivot)**.** *Let B be an invertible principal submatrix of a real symmetric matrix A, of order k, and let $S = A/B$ be its Schur complement. For every $t > 0$,*

$$N_S(t) \geq N_A(t) - k.$$

Proof. First suppose A is invertible. Then S is invertible and S^{-1} is the complementary principal submatrix of A^{-1}. Interlacing bounds the number of eigenvalues of S^{-1} in (t^{-1}, ∞) by the corresponding number for A^{-1}. It gives the analogous bound in $(-\infty, -t^{-1})$. Consequently, at most $n - N_A(t)$ of the $n - k$ eigenvalues of S^{-1} have modulus greater than t^{-1}. This is the asserted inequality. For singular A, apply the argument to $A + \varepsilon I$ along values for which this matrix is invertible, at threshold $t - |\varepsilon|$. The pivot remains invertible for sufficiently small ε; its Schur complement converges to S. Continuity of the ordered eigenvalues gives the result. □

Define

$$\mathcal{L}(Y) = \sup_s \mathbb{E} \log_+ \frac{1}{|Y - s|}, \qquad \mathcal{H}(Y) = \sup_s \mathbb{E}\left[|Y - s|^{-1/2} \mathbf{1}_{\{|Y-s| \leq 1\}} \right].$$

Lemma 9.2 (A global logarithmic bound)**.** *If $A \neq 0$, then*

$$\mathcal{L}(Q_A) \leq 4 + \log_+ \frac{5^4}{\|A\|}.$$

In particular, $N_A(1) \geq 1$ implies $\mathcal{L}(Q_A) \leq L_0 := 4 + 4 \log 5$.

Proof. The quarter-power estimate of Lemma 4.29 and the layer-cake formula give

$$\mathbb{E} \log_+ \frac{1}{|Q_A - s|} \leq \int_0^\infty \min\{1, 5\|A\|^{-1/4} e^{-u/4}\}\, du.$$

The integral is $4 + 4 \log K$ when $K = 5\|A\|^{-1/4} \geq 1$, and $4K \leq 4$ otherwise. □

Lemma 9.3 (Half-power integrability at a pivot)**.** *If $N_A(1) \geq 2$ and A has a c-pivot, then*

$$\mathcal{H}(Q_A) \leq 44 c^{-1/2}.$$

Proof. Suppose first that $a = a_{hh}$ satisfies $|a| \geq c$. Completing the square in the original coordinate X_h writes $Q_A = a(X_h - m)^2 + K$, where m is affine in the remaining coordinates

and the quadratic matrix of K is A/a. Lemma 9.1 gives $N_{A/a}(1) \geq 1$. The conditional density estimate of Lemma 3.2 therefore yields

$$\mathcal{H}(Q_A) \leq |a|^{-1/2} \sup_s \mathbb{E}J(s-K), \qquad J(d) = \int_{-1}^{1} |x|^{-1/2}|x-d|^{-1/2}\,dx.$$

For $d \neq 0$,

$$J(d) \leq 8 + 3\log_+ \frac{1}{|d|}. \tag{136}$$

Indeed, on $|x| \leq 2|d|$ substitute $x = |d|u$. The resulting integral is at most

$$\int_{-2}^{2} |u|^{-1/2}|u-1|^{-1/2}\,du = 2\operatorname{arsinh}\sqrt{2} + \pi + 2\operatorname{arcosh}\sqrt{2} < 8.$$

On $2|d| < |x| \leq 1$, the integrand is at most $\sqrt{2}/|x|$, giving the remaining term in (136). Lemma 9.2 bounds this case by $c^{-1/2}(8 + 3L_0) < 40c^{-1/2}$.

Next suppose that a principal two-coordinate block B is definite with determinant $D \geq c$. Its conditional density is at most $\pi D^{-1/2}$; integrating $|x-s|^{-1/2}\mathbf{1}_{\{|x-s|\leq 1\}}$ gives $\mathcal{H}(Q_A) \leq 4\pi c^{-1/2}$.

Finally suppose $\det B = -D$, $D \geq c$. Lemma 3.6 bounds the conditional density by

$$D^{-1/2}\left(5.2 + \log_+ \frac{16\sqrt{D}}{|y-K|}\right).$$

For every real a,

$$\int_{-1}^{1} |x|^{-1/2} \log_+ \frac{1}{|x-a|}\,dx \leq 8.$$

To see this without a regularity assumption, apply layer cake to both factors. Their superlevel sets are intervals, and the overlap of two intervals of prescribed lengths is maximal when their centres coincide. The integral is therefore at most its value at $a = 0$, namely $2\int_0^1 x^{-1/2}\log(1/x)\,dx = 8$. It follows that

$$\mathcal{H}(Q_A) \leq D^{-1/2}\left(28.8 + 4\log_+(16\sqrt{D})\right) \leq c^{-1/2}(28.8 + 4\log 16) + 4/e < 44c^{-1/2}.$$

All conditionings are on original independent coordinates. □

9.2 Gaussian padding and reduction to small pivots

Theorem 9.4 (Reduction to the three-eigenvalue no-pivot case)**.** *Fix $0 < c \leq 1$. Suppose that*

$$\|p_{Q_A}\|_\infty \leq C_3 \quad \text{whenever } N_A(1) \geq 3 \text{ and } A \text{ has no } c\text{-pivot.} \tag{137}$$

Then the bounded-density conjecture holds for all real symmetric matrices with $N_A(1) \geq 3$. One admissible bound is

$$\max\left\{C_3,\ c^{-1/2}(1+G_2),\ 20c^{-1/2}\right\}, \qquad G_2 = \max\{2\sqrt{2\pi e}\,C_3,\ 44c^{-1/2}\}.$$

The converse implication is immediate.

Proof. First consider a matrix M with $N_M(1) \geq 2$ and no c-pivot. Choose $N > 1/c$, put $u = N^{-1/2}(1,\dots,1)$, and let $G_1,\dots,G_N$ be independent standard Gaussian variables, independent of X. The variable $Z = \langle u, G\rangle$ is standard Gaussian. The matrices $M \oplus uu^{\mathsf{T}}$ and $M \oplus (-uu^{\mathsf{T}})$ have at least three eigenvalues of modulus at least one and no c-pivot: all new diagonal entries have modulus $1/N < c$, all two-coordinate minors inside the new block vanish, and a mixed

minor is the product of two diagonal entries of modulus less than c. The original Gaussian coordinate densities are bounded by one. Thus (137) applies to $Q_M \pm Z^2$.

The density of Z^2 is

$$k(t) = \frac{e^{-t/2}}{\sqrt{2\pi t}} \mathbf{1}_{\{t>0\}}.$$

For $0 < |t| \le 1$,

$$k(t) + k(-t) \ge (2\pi e)^{-1/2} |t|^{-1/2}.$$

Adding the two convolution bounds gives $\mathcal{H}(Q_M) \le 2\sqrt{2\pi e}\, C_3$. The estimate holds for every centre s, rather than only almost every centre: the symmetrized kernel $(2\pi|t|)^{-1/2} e^{-|t|/2}$, with value $+\infty$ at zero, is lower semicontinuous. Its convolution has the same property, so an almost-everywhere upper bound extends everywhere. Lemma 4.29 also ensures that Q_M has no atoms. Together with Lemma 9.3, this proves

$$N_M(1) \ge 2 \quad \Longrightarrow \quad \mathcal{H}(Q_M) \le G_2 \tag{138}$$

for every matrix M.

Now let $N_A(1) \ge 3$. The case with no pivot is the hypothesis. At a one-coordinate pivot a, the square kernel and Lemma 9.1 give

$$p_{Q_A}(s) \le |a|^{-1/2} \mathbb{E}|s - K|^{-1/2} \le c^{-1/2}(1 + G_2),$$

using (138) for the Schur complement. A definite two-coordinate pivot gives $\pi c^{-1/2}$. At an indefinite pivot of determinant $-D$, the centre has at least one large eigenvalue, so Lemmas 3.6 and 9.2 give

$$p_{Q_A}(s) \le D^{-1/2}\bigl(5.2 + \log_+(16\sqrt{D}) + L_0\bigr) \le 20 c^{-1/2}.$$

This exhausts the cases. □

Remark 9.5. The reduction assumes (137) and requires neither regularity of the original coordinate densities nor moment bounds.

9.3 Finite-dimensional bounds and projected gradients

Theorem 9.6 (Slicing by a projected gradient)**.** *Let E be a three-dimensional invariant subspace of A on which every eigenvalue has modulus at least $\sigma > 0$. Put*

$$\ell_i = |P_E e_i|^2, \qquad \mathcal{R}(E) = \sum_i \sqrt{\ell_i}.$$

Then

$$\|p_{Q_A}\|_\infty \le \frac{2\sqrt{2\pi}}{\sigma} \mathcal{R}(E) \le \frac{2\sqrt{6\pi n}}{\sigma}. \tag{139}$$

For $n = 3$, the bound improves to $6\sqrt{\pi}/\sigma$. All eigenvalues outside E are unrestricted.

Proof. Write $v(x) = \nabla Q_A(x)$ and $g(x) = P_E v(x)$. There is a unique $z_0 \in E$ with $2(A|_E)z_0 + P_E b = 0$, and hence

$$g(x) = 2(A|_E)(P_E x - z_0), \qquad |g(x)| \ge 2\sigma |P_E x - z_0|.$$

The set $g = 0$ has codimension three and has probability zero. The identities $v^{\mathsf{T}} g = |g|^2$ and $g \in E$ imply

$$1 \le \sum_i \frac{|\partial_i Q_A|\, |g_i|}{|g|^2}, \qquad |g_i| \le \sqrt{\ell_i}\, |g|. \tag{140}$$

The absolute values in this inequality are necessary; individual terms of $v^{\mathsf{T}}g$ need not be nonnegative.

Let ψ be a nonnegative bounded compactly supported Borel function. Multiply (140) by $\psi(Q_A(x))\prod_j f_j(x_j)$, integrate, and apply Tonelli. For the i-th summand, fix all coordinates except $x_i = t$. On the monotonicity intervals of the resulting nonconstant quadratic or linear polynomial, one-dimensional change of variables gives

$$\int \psi(Q_A) f_i(t) \frac{|\partial_i Q_A|\,|g_i|}{|g|^2}\,dt = \int \psi(s) \sum_{\substack{t: Q_A(t,x_{-i})=s \\ \partial_i Q_A \neq 0}} f_i(t) \frac{|g_i(t,x_{-i})|}{|g(t,x_{-i})|^2}\,ds.$$

Critical values form a null set. If the fibre polynomial is constant, the left side is zero and the fibre contributes zero.

Choose a two-dimensional subspace $E_i \subset E \cap e_i^{\perp}$. It exists because $\dim E = 3$. The projection $P_{E_i}x$ is independent of the coordinate $t = x_i$, and

$$|g(x)| \geq 2\sigma |P_{E_i}x - P_{E_i}z_0|.$$

There are at most two noncritical roots, and $f_i \leq 1$. The root sum is consequently at most

$$\frac{\sqrt{\ell_i}}{\sigma |P_{E_i}x_{-i} - P_{E_i}z_0|}.$$

The vector $P_{E_i}X$ is an orthonormal two-dimensional projection of the remaining original product coordinates. Its density is at most 2, by Livshyts et al. (2016, Theorem 1.1). For a two-dimensional probability density bounded by M, layer cake gives

$$\mathbb{E}|Y - z|^{-1} \leq \int_0^{\infty} \min\{1, \pi M/u^2\}\,du = 2\sqrt{\pi M}.$$

The i-th summand is therefore bounded by $(2\sqrt{2\pi}/\sigma)\sqrt{\ell_i}\int\psi$. Summing proves domination of the output law by the first constant in (139) times Lebesgue measure. This establishes absolute continuity as well as the essential density bound, without a pointwise density assumption on a level set.

Finally $\sum_i \ell_i = 3$, so $\mathcal{R}(E) \leq \sqrt{3n}$. When $n = 3$, E_i is the full original coordinate complement, whose density is at most one; also $\mathcal{R}(E) = 3$. This gives $6\sqrt{\pi}/\sigma$. □

Corollary 9.7 (The finite-dimensional spectral problem)**.** *If A is n-by-n and $N_A(\lambda) \geq 3$, then*

$$\|p_{Q_A}\|_{\infty} \leq 2\sqrt{6\pi n}/\lambda.$$

Proof. Choose E to be spanned by three corresponding eigenvectors and apply Theorem 9.6. □

Corollary 9.8 (A finite coordinate core)**.** *Suppose a principal submatrix A_{KK}, with $|K| \leq N$, has three eigenvalues of modulus at least $\lambda > 0$. Then*

$$\|p_{Q_A}\|_{\infty} \leq 2\sqrt{6\pi N}/\lambda,$$

irrespective of the ambient dimension and all other blocks of A.

Proof. Condition on X_{K^c}. The remaining variables retain their independent densities, the quadratic matrix is A_{KK}, and the conditioned values affect only its affine terms. Apply Corollary 9.7 and average. □

Theorem 9.9 (A localized spectral core with a dense perturbation)**.** *Let H contain $m \geq 3$ original coordinates, and suppose*

$$A_{HH} = U_H \Lambda U_H^{\mathsf{T}} + T_H, \qquad U^{\mathsf{T}} U = I_3, \qquad U_H^{\mathsf{T}} U_H \geq a I_3,$$

where Λ is symmetric, $s_{\min}(\Lambda) \geq s > 0$, and $\|T_H\| \leq \varepsilon < as$. Then every choice of the other blocks of A satisfies

$$\|p_{Q_A}\|_\infty \leq \frac{2\sqrt{6\pi m}}{as - \varepsilon}.$$

Proof. Put $G = U_H^{\mathsf{T}} U_H$ and $V = U_H G^{-1/2}$. The columns of V are orthonormal. On their span, the rank-three matrix $U_H \Lambda U_H^{\mathsf{T}}$ is represented by $K = G^{1/2} \Lambda G^{1/2}$, for which

$$|Kz| \geq \sqrt{a}\, |\Lambda G^{1/2} z| \geq as|z|.$$

For $x \in \operatorname{range}(V)$, the triangle inequality gives $|A_{HH} x| \geq (as - \varepsilon)|x|$. The variational characterization of singular values yields $s_3(A_{HH}) \geq as - \varepsilon$. Now apply Corollary 9.8. □

The perturbation in Theorem 9.9 is retained in the actual physical matrix. Its small operator norm certifies the spectrum on the core; the proof makes no assertion that $\varepsilon|X|^2$ is small.

Proposition 9.10 (Small balls of a projected gradient)**.** *Let $u_1, \dots, u_r$ be orthonormal eigenvectors of A, with nonzero eigenvalues $\lambda_1, \dots, \lambda_r$, and put $U = (u_1, \dots, u_r)$. The vector $Y = U^{\mathsf{T}} \nabla Q_A(X)$ has density at most*

$$K = \frac{2^{-r/2}}{\prod_{j=1}^{r} |\lambda_j|}.$$

Writing v_r for the volume of the unit ball in $\mathbb{R}^r$, for every $z \in \mathbb{R}^r$, $a > 0$, and $0 < p < r$,

$$\mathbb{P}\{|Y - z| \leq a\} \leq \min\{1, K v_r a^r\}, \qquad \mathbb{E}|Y - z|^{-p} \leq \frac{r}{r - p} (K v_r)^{p/r}.$$

Proof. Since

$$U^{\mathsf{T}} \nabla Q_A = 2 \operatorname{diag}(\lambda_1, \dots, \lambda_r) U^{\mathsf{T}} X + U^{\mathsf{T}} b,$$

the projection-density bound of Lemma 4.2 and change of variables give K. Integrating this density over a ball gives the first estimate. The second follows by integrating the first estimate in the layer-cake identity $\mathbb{E}|Y - z|^{-p} = \int_0^\infty \mathbb{P}\{|Y - z| < t^{-1/p}\}\, dt$. □

The projected-gradient estimate controls neighbourhoods of critical points. It does not, by itself, bound the density of Q_A: the level-set integral also contains the product density, which is allowed to be rough.

9.4 An affine estimate for exact rank two

Lemma 9.11 (A square with an arbitrary linear term)**.** *For $|u| = 1$, $\mu \neq 0$, and arbitrary b, c_0,*

$$\sup_s \mathbb{P}\{|\mu \langle u, X \rangle^2 + b^{\mathsf{T}} X + c_0 - s| \leq a\} \leq 8\sqrt{a/|\mu|}.$$

Proof. This is Lemma 5.1; its Gaussian smoothing argument includes arbitrary original-coordinate affine terms. The closed-interval version follows by decreasing the radius to a from above. □

Theorem 9.12 (Exact rank two)**.** *Suppose $A = RMR^{\mathsf{T}}$, where $R^{\mathsf{T}}R = I_2$, M is real symmetric, and $|\det M| \geq 1$. Then, for $0 < a \leq 1$,*

$$\sup_s \mathbb{P}\{|Q_A - s| \leq a\} \leq 32\sqrt{2}\, a(1 + \log(1/a)) \leq 46a(1 + \log(1/a)).$$

Moreover, Theorem 5.4 gives the stronger fractional-potential bound $\sup_s \mathbb{E}|Q_A - s|^{-1/2} < 68$.

Proof. Let $\rho_i = R^{\mathsf{T}}e_i$. If every $|\rho_i|^2 \leq 1/2$, Fresnel linearization in the two spectral directions and the geometric Brascamp–Lieb estimate give $|\phi_{Q_A}(t)| \leq 4\pi/(|t|\sqrt{|\det M|})$. The Fejér kernel $(\sin(x/2)/(x/2))^2$ is greater than $9/10$ on $[-1, 1]$. Its triangular Fourier representation yields

$$\mathbb{P}\{|Y - s| \leq a\} \leq \frac{10}{9} a \int_{|t| \leq a^{-1}} |\phi_Y(t)|\, dt.$$

Using $|\phi| \leq 1$ on $|t| \leq 1$ gives the asserted bound with constant less than 28.

Otherwise choose h with $l = |\rho_h|^2 > 1/2$. Rotate the two-dimensional frame so that $\rho_h = \sqrt{l}\, e_2$, and write

$$M = \begin{pmatrix} s & q \\ q & a_0 \end{pmatrix}.$$

The first column r_1 of R, restricted to h^c, is a unit vector; the second restricted column r_2 is orthogonal to it. As a quadratic in $x = X_h$, write $Q_A = ax^2 + 2Lx + T$, and define $\Delta = L^2 + a(s_0 - T)$, for a target level s_0. Its quadratic matrix is

$$cc^{\mathsf{T}} - aA_{h^c,h^c} = -l \det(M)\, r_1 r_1^{\mathsf{T}}, \qquad c = A_{h^c,h} = \sqrt{l}(qr_1 + a_0 r_2), \quad a = la_0.$$

This identity follows by expanding both sides; the coefficients of $r_2 r_2^{\mathsf{T}}$ and $r_1 r_2^{\mathsf{T}} + r_2 r_1^{\mathsf{T}}$ cancel. The other terms of Δ are affine. Lemma 9.11 thus gives $\mathbb{P}\{|\Delta - t| \leq u\} \leq 8\sqrt{2}\sqrt{u}$.

The one-variable quadratic sublevel estimate, including the linear case $a = 0$, bounds the conditional probability by $\min\{1, 4a_s/\sqrt{|\Delta|}\}$, where a_s is the small-ball radius. Hence

$$\mathbb{P}\{|Q_A - s_0| \leq a_s\} \leq \int_0^1 \min\{1, 32\sqrt{2}\, a_s/\theta\}\, d\theta \leq 32\sqrt{2}\, a_s(1 + \log(1/a_s)).$$

The potential estimate is Theorem 5.4, since $|\det M| \geq 1$. □

10 Off-diagonal rank and quantitative skeletons

For a real symmetric matrix A, let

$$\rho(A) = \max_{S \cap T = \varnothing} \operatorname{rank} A_{ST}, \qquad \operatorname{mr}(A) = \min_{D \text{ diagonal}} \operatorname{rank}(A - D).$$

The sets S, T in this definition need not have the same size.

Theorem 10.1 (Exact completion by a diagonal)**.** *For every real symmetric A,*

$$\operatorname{mr}(A) \leq 3\rho(A).$$

For every positive integer r, examples satisfy $\rho(A) = r$ and $\operatorname{mr}(A) = 2r$. If $\rho(A) \leq 1$, then $\operatorname{mr}(A) \leq 2$.

Proof. Put $r = \rho(A)$. The case $r = 0$ means that A is diagonal. Otherwise choose disjoint r-sets S, T with A_{ST} invertible, and put $C = A_{ST}^{-1}$, $K = S \cup T$, $L = K^c$. For $l \in L$, write $x_l = A_{lT}$, $y_l = A_{lS}$, and collect these rows in matrices X, Y. Every off-diagonal $(r+1)$-minor vanishes; adjoining distinct $l, m \in L$ to the selected minor gives

$$a_{lm} = x_l C y_m^{\mathsf{T}}.$$

Symmetry therefore implies that the span of the vectors $(y_l, x_l) \in \mathbb{R}^{2r}$ is isotropic for the nondegenerate alternating form

$$\omega((y, x), (y', x')) = xC(y')^{\mathsf{T}} - x'Cy^{\mathsf{T}}.$$

An isotropic subspace has dimension at most r: it is contained in its orthogonal complement for this nondegenerate form, and the two dimensions sum to $2r$. The matrix $P = XCY^{\mathsf{T}}$ is symmetric because its off-diagonal entries agree with those of A_{LL}. Choose $D_K = 0$ and $D_L = \operatorname{diag}(A_{LL} - P)$. Each L-row of $A - D$ is a fixed linear image of (y_l, x_l), so these rows span at most r dimensions. The $2r$ rows in K add at most $2r$, proving $\operatorname{mr}(A) \le 3r$.

A star with centre h, zero leaf–leaf entries and at least two nonzero edges has $\rho = 1$ and $\operatorname{mr} = 2$. Its off-diagonal matrix already has rank two. A symmetric rank-one completion would have the form $\pm vv^{\mathsf{T}}$; two nonzero centre–leaf entries would then force a nonzero entry between those leaves. Direct sums of r such stars have the stated parameters, since both quantities add over direct sums.

It remains to prove the improved upper bound for $\rho \le 1$. Consider the support graph of the off-diagonal entries. For four distinct indices the hypothesis reads $a_{ik}a_{jl} = a_{il}a_{jk}$. Two disjoint edges consequently force all six edges on their four endpoints. A vertex adjacent to a clique of order at least three is adjacent to every vertex of that clique: pair its incident edge with edges between the other clique vertices and use the same identity. Thus, if disjoint edges occur, all nonzero edges lie in a single complete graph with at least four vertices. On this graph the numbers

$$c_i = \frac{a_{ij}a_{ik}}{a_{jk}}$$

are independent of the distinct choices $j, k \ne i$. The four-index identities give $c_i c_j = a_{ij}^2 > 0$; all c_i have a common sign. The edge signs obey the same identities, so choices of signs for $v_i = \pm\sqrt{|c_i|}$ give $a_{ij} = \pm v_i v_j$ with a common overall sign. This supplies a rank-one diagonal completion. If there are no disjoint edges, the graph is a star, a triangle, or a subgraph of one. A star has the rank-two completion already described. For a triangle, subtracting any eigenvalue from the three diagonal entries makes the corresponding block singular, giving rank at most two. □

For $r \ge 1$, define

$$V_r(A) = \sum_{\substack{|S|=|T|=r \\ S\cap T=\varnothing}} \det(A_{ST})^2, \qquad V_0(A) = 1.$$

The sum is over ordered pairs of subsets; each subset is listed in its natural order.

Lemma 10.2 (Volume-sampled skeleton inequality)**.** *Suppose $V_r(A) > 0$. Sample disjoint r-sets S, T with probability $\det(A_{ST})^2/V_r(A)$, omitting zero weights, and put*

$$P^{ST} = A_{\cdot T} A_{ST}^{-1} A_{S\cdot}.$$

Then

$$\mathbb{E}_{S,T} \sum_{\substack{l,m\notin S\cup T \\ l\ne m}} (A - P^{ST})_{lm}^2 \le (r+1)^2 \frac{V_{r+1}(A)}{V_r(A)}. \tag{141}$$

Proof. For every sampled nonsingular seed and every allowed l, m, the bordered determinant formula gives

$$\det(A_{S+l,T+m})^2 = \det(A_{ST})^2 (A - P^{ST})^2_{lm}.$$

Sum over the seeds and l, m. A disjoint pair of $(r+1)$-sets is counted at most $(r+1)^2$ times, according to which row and column are distinguished. Some of these deletions may leave a singular seed and are then absent from the sum. Division by $V_r(A)$ proves (141). □

Remark 10.3 (Why equality is not automatic). Take

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad r = 1.$$

Then $V_1 = V_2 = 4$. Every nonzero seed has squared off-diagonal residual sum 2, so the left side of (141) is 2, whereas the right side is 4. Singular seeds may have nonsingular extensions. Thus the valid general statement is an inequality, which suffices for the following consequence.

Theorem 10.4 (A robust low-rank completion)**.** *Suppose $V_r(A) > 0$ and $V_{r+1}(A) \le \kappa V_r(A)$. There is a decomposition*

$$A = D + R + E, \qquad \operatorname{rank} R \le 4r, \qquad \|E\|_{\mathrm{F}} \le 3(r+1)\sqrt{\kappa},$$

where D is diagonal and E is symmetric, has zero diagonal, and vanishes on $2r$ selected rows and columns.

Proof. Choose a seed from Lemma 10.2 for which the off-diagonal residual on $L = (S \cup T)^c$ has squared Frobenius norm at most $(r+1)^2\kappa$. Write

$$P = A_{LT} A_{ST}^{-1} A_{SL}, \qquad F = \operatorname{offdiag}(A_{LL} - P).$$

Since A_{LL} is symmetric, $\|P - P^{\mathsf{T}}\|_{\mathrm{F}} \le 2\|F\|_{\mathrm{F}}$. Thus $P_{\mathrm{sym}} = (P + P^{\mathsf{T}})/2$ satisfies $\|P_{\mathrm{sym}} - P\|_{\mathrm{F}} \le \|F\|_{\mathrm{F}}$. Let P_r retain the r largest eigenvalues in modulus of P_{sym}. The best rank-r approximation property and $\operatorname{rank} P \le r$ imply

$$\|P_{\mathrm{sym}} - P_r\|_{\mathrm{F}} \le \|F\|_{\mathrm{F}}.$$

All columns of P_r lie in $\operatorname{col}[A_{LS}, A_{LT}]$, a space of dimension at most $2r$, because this contains the range of P_{sym}.

With $K = S \cup T$, set

$$R = \begin{pmatrix} A_{KK} & A_{KL} \\ A_{LK} & P_r \end{pmatrix}, \quad E_{LL} = \operatorname{offdiag}(A_{LL} - P_r), \quad E_{K,\cdot} = E_{\cdot,K} = 0,$$

and let $D = A - R - E$. The L-row block of R has rank at most $2r$, and its K-row block adds at most $2r$, so $\operatorname{rank} R \le 4r$. The matrix D is diagonal and

$$\|E\|_{\mathrm{F}} \le \|F\|_{\mathrm{F}} + \|P - P_r\|_{\mathrm{F}} \le 3\|F\|_{\mathrm{F}} \le 3(r+1)\sqrt{\kappa}.$$

□

Corollary 10.5. *If every coordinate partition I, J satisfies $\sum_{j \ge r+1} s_j(A_{IJ})^2 \le \delta^2$, and $V_r(A) > 0$, then Theorem 10.4 applies with $\kappa = 4\delta^2$. In particular $\|E\|_{\mathrm{F}} \le 6(r+1)\delta$.*

Proof. For nonincreasing nonnegative numbers x_j,

$$e_{r+1}(x) \le \left(\sum_{j \ge r+1} x_j \right) e_r(x).$$

Assign each $(r+1)$-set to its largest index to verify this inequality term by term. For a random fair coordinate partition, Cauchy–Binet gives $\mathbb{E} e_r(A_{IJ} A_{IJ}^{\mathsf{T}}) = 4^{-r} V_r(A)$. Average the preceding inequality with $x_j = s_j(A_{IJ})^2$. □

The conclusion of Corollary 10.5 uses a bound on the entire squared singular-value tail. A bound only on s_{r+1} yields a dimension factor through that tail and is not the same hypothesis.

11 Grouped projections and projected crossing blocks

The following arguments retain arbitrary phases inside each coordinate group. The projected coordinates need not be independent. This distinction is useful when many original coordinates share a small number of coupling directions.

Lemma 11.1 (A phase-weighted projection)**.** *Let a random vector W have density $\rho \le M$, and let Z be any random variable on the same probability space with $|Z| \le 1$. The complex measure $\mathbb{E}[Z; W \in dw]$ has a density $a(w)$ satisfying*

$$|a(w)| \le \rho(w), \qquad \|a\|_1 \le 1, \qquad \|a\|_2^2 \le M.$$

Proof. Disintegration gives $a(w) = \rho(w) \mathbb{E}[Z \mid W = w]$ almost everywhere. Its modulus is at most ρ, and $\int |a|^2 \le \int \rho^2 \le M \int \rho = M$. □

Theorem 11.2 (An exact-rank crossing block)**.** *Let X and Y be independent random vectors and $B = U \Sigma V^{\mathsf{T}}$ have exact rank r, where U, V have orthonormal columns and $\Sigma = \operatorname{diag}(s_1, \dots, s_r) > 0$. Suppose $U^{\mathsf{T}} X$ and $V^{\mathsf{T}} Y$ have densities bounded by M_X, M_Y. For arbitrary measurable real functions F, G, put*

$$Q = F(X) + 2X^{\mathsf{T}} B Y + G(Y).$$

Then, for $t \ne 0$,

$$|\phi_Q(t)| \le \min \left\{ 1, \left(\frac{\pi}{|t|} \right)^{r/2} \frac{\sqrt{M_X M_Y}}{\sqrt{s_1 \cdots s_r}} \right\}.$$

If $r > 2$, its density satisfies

$$\|p_Q\|_\infty \le \frac{r}{r-2} \left(\frac{M_X M_Y}{s_1 \cdots s_r} \right)^{1/r}.$$

For product inputs with coordinate densities bounded by one, one may take $M_X, M_Y \le 2^{r/2}$.

Proof. Push $e^{itF(X)} d\mathbb{P}_X$ onto $U^{\mathsf{T}} X$, obtaining a complex density a_t, and similarly obtain b_t from $e^{itG(Y)} d\mathbb{P}_Y$. Lemma 11.1 gives $\|a_t\|_2^2 \le M_X$ and $\|b_t\|_2^2 \le M_Y$. Independence and Fubini give

$$\phi_Q(t) = \int_{\mathbb{R}^r} \widehat{a}_t(2t\Sigma z) b_t(z)\, dz.$$

By Plancherel,

$$\int |\widehat{a}_t(2t\Sigma z)|^2\, dz = \left(\frac{\pi}{|t|} \right)^r (s_1 \cdots s_r)^{-1} \|a_t\|_2^2.$$

Cauchy–Schwarz proves the characteristic-function bound. For $\alpha > 1$,

$$\frac{1}{2\pi}\int_{\mathbb{R}} \min\{1, K|t|^{-\alpha}\}\,dt = \frac{\alpha}{\pi(\alpha-1)}K^{1/\alpha}.$$

Take $\alpha = r/2$. The last assertion is the projection bound of Livshyts et al. (2016). □

We next allow the light crossing matrix to have arbitrary rank. For a matrix C with rows c_i, let

$$Z_\ell(C) = e_\ell(CC^{\mathsf{T}}) = \sum_{|S|=\ell} \det(C_S C_S^{\mathsf{T}}).$$

When $Z_\ell(C) > 0$, its ℓ-volume row marginals are

$$\alpha_i = \sum_{S \ni i} \frac{\det(C_S C_S^{\mathsf{T}})}{Z_\ell(C)}.$$

They satisfy $\sum_i \alpha_i = \ell$.

Lemma 11.3 (A weighted-subset Fourier estimate)**.** *Let $D \in \mathbb{R}^m$ have independent coordinate densities bounded by one. Let M be a k-by-m matrix of row rank k, with product of singular values S_M. Write Π for projection onto its row space. For a matrix A_0 with rows a_i, put $C = A_0(I - \Pi)$. Suppose $Z_\ell(C) > 0$ and every ℓ-volume row marginal is at most $1/2$. Let ψ_i be characteristic functions of scalar probability densities bounded by one, and let κ be any nonnegative probability density on $\mathbb{R}^k$. Then*

$$\mathbb{E}\left[\kappa(MD)\prod_i |\psi_i(2t\langle a_i, D\rangle)|\right] \le \frac{2^{(k+\ell)/2}(2\pi)^\ell e^{\ell/2}}{S_M\sqrt{Z_\ell(C)}\,|t|^\ell}. \tag{142}$$

When $\ell = 1$, the factor $e^{\ell/2}$ may be omitted.

Proof. Set

$$w_S = \frac{\det(C_S C_S^{\mathsf{T}})}{Z_\ell(C)}.$$

Discard factors with $\alpha_i = 0$, which only increases the left side. Weighted Hölder for the finite measure $\kappa(Md)d\mathbb{P}_D(d)$ bounds the left side by $\prod_{S:w_S>0} I_S^{w_S}$, where

$$I_S = \mathbb{E}\left[\kappa(MD)\prod_{i\in S}|\psi_i(2t\langle a_i, D\rangle)|^{1/\alpha_i}\right].$$

Indeed, the exponent of each factor after taking this weighted product is $\sum_{S\ni i} w_S/\alpha_i = 1$.

For $i \in S$, write $a_i = c_i + \lambda_i M$. The linear image $(MD, (\langle c_i, D\rangle)_{i\in S})$ has density bounded by

$$\frac{2^{(k+\ell)/2}}{S_M\sqrt{\det(C_S C_S^{\mathsf{T}})}}.$$

To verify the determinant, the rows of C are perpendicular to the rows of M, so the Gram determinant of the combined map is $S_M^2\det(C_S C_S^{\mathsf{T}})$. Factor this map through an orthonormal projection and apply Lemma 4.2. For each fixed $x = MD$, shift the remaining coordinates by $\lambda_i x$, integrate κ, and use Lemma 4.3 at exponent $1/\alpha_i \ge 2$. The result is

$$I_S \le \frac{2^{(k+\ell)/2}(2\pi)^\ell}{S_M|t|^\ell}\,\frac{\prod_{i\in S}\sqrt{\alpha_i}}{\sqrt{\det(C_S C_S^{\mathsf{T}})}}.$$

The weighted product of the last factors is

$$Z_\ell(C)^{-1/2}\exp\left\{\frac{1}{2}\left(H(w)+\sum_i \alpha_i\log\alpha_i\right)\right\},$$

where $H(w)=-\sum_S w_S\log w_S$. Entropy subadditivity for the indicator coordinates of the random subset S gives

$$H(w)\le\sum_i\left[-\alpha_i\log\alpha_i-(1-\alpha_i)\log(1-\alpha_i)\right].$$

This also follows directly from nonnegativity of relative entropy against the independent Bernoulli distribution with parameters α_i. Since $-(1-a)\log(1-a)\le a$ and $\sum_i\alpha_i=\ell$, the exponential is at most $e^{\ell/2}$. For $\ell=1$, $w_{\{i\}}=\alpha_i$, so the entropy expression cancels exactly. □

Theorem 11.4 (Grouped hubs with an arbitrary-rank crossing remainder)**.** *Let X_H and X' be independent, with X' having independent coordinate densities bounded by one. Let U have k orthonormal columns, and suppose $W=U^{\mathsf T}X_H$ has density at most M_H. Consider*

$$Q=F(X_H)+2\langle W,MX'+\beta\rangle+R(X'),$$

where F is arbitrary measurable, M has row rank k, and R is an arbitrary quadratic polynomial with matrix A_R. Split the light coordinates into I,J, assume M_J has row rank k, and write S for the product of its singular values. Define

$$C=(A_R)_{IJ}(I-\Pi_{\mathrm{row}(M_J)}).$$

Suppose $Z_\ell(C)>0$ and its ℓ-volume row marginals are at most $1/2$. Put $d=k+\ell$. For $t\neq0$,

$$|\phi_Q(t)|\le\min\{1,K|t|^{-d/2}\},\quad K=\frac{M_H^{1/2}2^{(k+3\ell)/4}\pi^{d/2}e^{\ell/4}}{S^{1/2}Z_\ell(C)^{1/4}}.$$

If $d>2$, then

$$\|p_Q\|_\infty\le\frac{d}{d-2}2^{(k+3\ell)/(2d)}e^{\ell/(2d)}\frac{M_H^{1/d}}{S^{1/d}Z_\ell(C)^{1/(2d)}}.\tag{143}$$

For $\ell=1$, omit the exponential factors.

Proof. Push $e^{itF(X_H)}d\mathbb{P}_H$ onto W, obtaining a complex density a_t with $\|a_t\|_2^2\le M_H$, and put $h_t(l)=\widehat{a}_t(2t(l+\beta))$. Plancherel gives

$$\int|h_t(l)|^2\,dl\le M_H(\pi/|t|)^k.$$

Let $\mu_t(l)$ be the density of the complex measure $\mathbb{E}[e^{itR(X')};MX'\in dl]$. The ordinary projection density of MX' is bounded, so Lemma 11.1 implies $\mu_t\in L^2$. Independence yields

$$\phi_Q(t)=\int h_t(l)\mu_t(l)\,dl,\qquad|\phi_Q(t)|^2\le M_H(\pi/|t|)^k\|\mu_t\|_2^2.$$

To bound the last norm, take a nonnegative even compactly supported approximate identity k_ε on $\mathbb{R}^k$ and write

$$m_\varepsilon(l)=\mathbb{E}[e^{itR(X')}k_\varepsilon(l-MX')].$$

Conditional Cauchy–Schwarz in X_I, followed by integration in l and an independent copy X_J^*, gives

$$\|m_\varepsilon\|_2^2\le\mathbb{E}\left[e^{it(R(X_I,X_J)-R(X_I,X_J^*))}\kappa_\varepsilon(M_JD_J)\right],$$

where $D_J = X_J - X_J^*$ and $\kappa_\varepsilon = k_\varepsilon * k_\varepsilon$ is a nonnegative probability density. Every term purely in I cancels. Terms purely in J give a phase of modulus one, while the cross terms give $2\langle X_I, (A_R)_{IJ} D_J\rangle$. Averaging over X_I and taking absolute values bounds the last expression by

$$\mathbb{E}\left[\kappa_\varepsilon(M_J D_J)\prod_{i\in I}|\widehat{f}_i(2t\langle (A_R)_{iJ}, D_J\rangle)|\right].$$

The coordinates of D_J are independent with densities $f_j * \widetilde{f}_j \le 1$. Apply Lemma 11.3, uniformly in ε, and pass to the limit by L^2 approximate-identity convergence. This yields

$$\|\mu_t\|_2^2 \le \frac{2^{(k+\ell)/2}(2\pi)^\ell e^{\ell/2}}{S\sqrt{Z_\ell(C)}\,|t|^\ell}.$$

Combining with the hub Plancherel estimate proves the characteristic-function bound. The integration formula in Theorem 11.2 gives (143). □

Corollary 11.5 (One hub and a spread second crossing volume)**.** *For one original hub, let*

$$Q = F(X_h) + 2X_h(\langle w, X'\rangle + \beta) + R(X').$$

For a split I, J, suppose $w_J \neq 0$, and let $C = (A_R)_{IJ}(I - \Pi_{w_J})$. If $Z_2(C) > 0$ and every second-volume row marginal is at most $1/2$, then

$$\|p_Q\|_\infty \le 3\,2^{7/6}e^{1/3}(|w_J|^2 Z_2(C))^{-1/6} < 9.4(|w_J|^2 Z_2(C))^{-1/6}.$$

There is no restriction on $\operatorname{rank} C$.

Proof. Use Theorem 11.4 with $k = 1, \ell = 2$, and $M_H = 1$. □

Corollary 11.6 (Many coordinates with two exact hub directions)**.** *Suppose a hub–light cross block has exact rank two and write it as UM, with $U^{\mathsf{T}}U = I_2$. Every internal hub quadratic, and even every measurable internal hub function, is allowed. If a light split has* $\operatorname{rank} M_J = 2$*, singular-value product S, and projected crossing matrix C with*

$$F_0 = \|C\|_{\mathrm{F}} > 0, \qquad \|c_i\|^2 \le F_0^2/2,$$

then product hub and light laws satisfy

$$\|p_Q\|_\infty \le 3\,2^{7/6}(SF_0)^{-1/3} < 6.74(SF_0)^{-1/3}.$$

Proof. The projection $U^{\mathsf{T}}X_H$ has density at most two. Apply Theorem 11.4 with $k = 2, \ell = 1$, $M_H = 2$, and $Z_1(C) = F_0^2$. □

12 Concentrated directions and uniform bounds

Lemma 12.1 (Few concentrated directions in exact finite rank)**.** *Let $A = RMR^{\mathsf{T}}$, with $R^{\mathsf{T}}R = I_r$ and every eigenvalue of M of modulus at least one. Assume A has no c-pivot. Let H be any set of original coordinates and*

$$G_H = R_H^{\mathsf{T}}R_H, \qquad \tau = |H|\sqrt{2c}.$$

If the eigenvalues of G_H are $g_1 \ge \cdots \ge g_r$, then $g_k \le \tau$ whenever $k > r/2$. If M is definite, every row leverage $|R^{\mathsf{T}}e_i|^2$ is less than c.

Proof. By (135), $\|A_{HH}\| \le \|A_{HH}\|_{\mathrm{F}} \le \tau$. Rotate the frame so that G_H is diagonal. If $g_k > \tau$, the compression of M to its first k frame coordinates has norm at most $\tau/g_k < 1$. Indeed, writing $R_H = VG_H^{1/2}$ on its support expresses A_{HH} as an isometric copy of $G_H^{1/2} M G_H^{1/2}$, and division by the first k eigenvalues of $G_H^{1/2}$ gives this norm bound. Interlacing then forces M to have at least k eigenvalues greater than -1, hence at least 1, and at least k eigenvalues less than 1, hence at most -1. These are disjoint sets, so $2k \le r$. For definite M, its eigenvalue hypothesis gives $|a_{ii}| \ge |R^{\mathsf{T}} e_i|^2$; use the no-pivot diagonal bound. □

For example, if $H = \{i : |R^{\mathsf{T}} e_i|^2 \ge \delta\}$, then $|H| \le r/\delta$. In exact rank three, Lemma 12.1 permits only one strongly concentrated frame direction when c is small relative to δ. It makes no assertion about a decomposition with an arbitrary additional diagonal: that diagonal may cancel the diagonal entries of the low-rank component.

Proposition 12.2 (A normalized exact-rank-three transition)**.** *Let w, v, p be orthonormal vectors on the light coordinates, write $L = \langle w, X' \rangle$, $V = \langle v, X' \rangle$, $P = \langle p, X' \rangle$, and put*

$$Q_\varepsilon = 2X_h L + V^2 + 2\varepsilon LP.$$

For every real ε, its quadratic matrix has exact rank three and nonzero eigenvalues

$$\sqrt{1+\varepsilon^2}, \qquad -\sqrt{1+\varepsilon^2}, \qquad 1.$$

For every fixed $c > 0$ and $0 \le \varepsilon \le 1$, there are arbitrarily large spread frames for which this matrix has no c-pivot. If the light frame entries all have modulus $m^{-1/2}$, then every principal submatrix of order at most N has third singular value at most N/m.

Proof. In the orthonormal basis (e_h, w, v, p), the matrix is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & \varepsilon \\ 0 & 0 & 1 & 0 \\ 0 & \varepsilon & 0 & 0 \end{pmatrix}.$$

On the plane spanned by w and $(e_h + \varepsilon p)/\sqrt{1+\varepsilon^2}$, it is $\left(\begin{smallmatrix} 0 & \sqrt{1+\varepsilon^2} \\ \frac{}{\sqrt{1+\varepsilon^2}} & 0 \end{smallmatrix}\right)$. The v-direction has eigenvalue one and the remaining orthogonal direction has eigenvalue zero.

Take three orthogonal Walsh columns on m coordinates, normalized to entries of modulus $m^{-1/2}$, with $m > 6/c$. The hub diagonal is zero, every hub–leaf minor has modulus $1/m < c$, and light-block entries have modulus at most $3/m$. Their diagonals are less than c and their principal two-coordinate determinants have modulus at most $18/m^2 < c$, after enlarging m if necessary. This proves the no-pivot assertion.

For any coordinate set K, the principal matrix is the sum of a matrix of rank at most two,

$$\left((e_h + \varepsilon p) w^{\mathsf{T}} + w(e_h + \varepsilon p)^{\mathsf{T}}\right)_{KK},$$

and $v_K v_K^{\mathsf{T}}$, of norm $|v_K|^2 \le |K|/m$. The singular-value approximation inequality gives $s_3(A_{KK}) \le |K|/m$. □

The family in Proposition 12.2 has a third singular value of one for every ε. The endpoint $\varepsilon = 0$ and every fixed nonzero ε are covered by results in Section 13, but the available bounds for arbitrary rough laws need not stay bounded as ε tends to zero. Corollary 13.11 gives a much smaller loss than the direct leakage estimate. Theorem 11.4 also retains a dependence on this parameter: in the one-hub split its projected crossing matrix is

$$C = v_I (v_J^{\perp})^{\mathsf{T}} + \varepsilon w_I (p_J^{\perp})^{\mathsf{T}},$$

so $Z_2(C) \le 4\varepsilon^2$ for $0 \le \varepsilon \le 1$. The finite-core theorem is likewise insufficient uniformly, by the last assertion of Proposition 12.2. These are limitations of the respective estimates, not counterexamples to a bounded density.

Proposition 12.3 (Two hubs do not suffice with an arbitrary remainder)**.** *For every $0 < c \le 1$, there is a real symmetric matrix with no c-pivot, exactly two eigenvalues of modulus at least one, and a two-row hub–light block whose singular values are both one, such that the corresponding quadratic form under independent standard Gaussian coordinates has unbounded density.*

Proof. Choose $0 < b < c/2$, put $B = \operatorname{diag}(b, -b)$, and let M have two orthonormal, sufficiently spread rows. Set

$$A = \begin{pmatrix} B & M \\ M^{\mathsf{T}} & M^{\mathsf{T}} B^{-1} M \end{pmatrix}.$$

Writing $L = MX'$,

$$Q_A = (X_H + B^{-1}L)^{\mathsf{T}} B (X_H + B^{-1}L).$$

The matrix has rank two. On the two mutually orthogonal hub–partner planes its nonzero eigenvalues are $b + b^{-1}$ and $-b - b^{-1}$. By taking the two rows of M with entries of modulus at most η, where $\eta < c/2$ and $2\eta^2/b < c/2$, every entry of A has modulus less than $c/2$. Thus every diagonal entry is smaller than c and every principal two-coordinate determinant has modulus less than $c^2/2 < c$.

For Gaussian inputs, $Y = X_H + B^{-1}L$ is a nondegenerate Gaussian vector whose density is bounded below by some $\gamma > 0$ on a neighbourhood of the origin. Use the change of variables $u = \sqrt{b}(Y_1 - Y_2)$, $v = \sqrt{b}(Y_1 + Y_2)$. Then $Q_A = uv$ and the Jacobian is $2b$. Choose $r > 0$ so that the square $|u|, |v| \le r$ maps into that neighbourhood. For $0 < a \le r^2$,

$$\mathbb{P}\{|Q_A| \le a\} \ge \frac{\gamma}{2b} \, |\{(u, v) : |u|, |v| \le r, \ |uv| \le a\}| = \frac{2\gamma}{b} a(1 + \log(r^2/a)).$$

The ratio to $2a$ is unbounded as $a \downarrow 0$. □

Proposition 12.3 has only two large eigenvalues. It shows why a two-hub bound must use additional information about the remainder, such as that in Theorem 11.4; it does not refute the three-eigenvalue conjecture.

The structural statements above do not imply that every no-pivot matrix admits a split for which the quantities in (143) are uniformly nondegenerate. In particular, the grouped hub block must have exact rank; discarding the smaller singular directions does not cover a nearly rank-two block. A dimension-independent bound requires uniform control of these limiting configurations and of the one-hub transition above.

13 Quadratic forms with hub coordinates

Let $X_0, X_1, \dots, X_m$ be independent real random variables with probability densities bounded by one, and write $X' = (X_1, \dots, X_m)$. For orthonormal vectors $w, v \in \mathbb{R}^m$ put $L = \langle w, X' \rangle$ and $V = \langle v, X' \rangle$. The basic family in this section is

$$Q = aX_0^2 + 2X_0(L + \beta) + \lambda V^2 + 2L\langle q, X' \rangle + \sum_{i=1}^{m} \phi_i(X_i), \qquad \phi_i(x) = d_i x^2 + b_i x + c_i, \quad (144)$$

where $\lambda \neq 0$. Unless a stronger condition is stated, assume

$$w_i^2 + v_i^2 \le \tfrac{1}{2} \qquad (1 \le i \le m). \quad (145)$$

The coefficients of the one-coordinate polynomials are unrestricted. We use the Fourier convention $\widehat{\mu}(t) = \int e^{itx}\,\mu(dx)$, and all logarithms are natural.

The one-coordinate term X_0 is a hub: its coupling to the remaining coordinates is carried by the single linear statistic L. The estimates below distinguish the independent hub law from the dependence between L, V, and the other linear statistics of X'.

13.1 Projection and Fourier estimates

We use the following consequence of the density bound for projections of product measures (Livshyts et al., 2016; Rudelson and Vershynin, 2015): if T is an $r \times m$ matrix of rank r, then

$$\|p_{TX'}\|_\infty \le \frac{2^{r/2}}{\sqrt{\det(TT^T)}}. \tag{146}$$

The bound for an orthonormal projection is $2^{r/2}$. No independence between the projected coordinates is asserted.

Lemma 13.1 (Modulated frame integral)**.** *Let $f \le 1$ be a probability density and let g be a complex function with $|g| \le f$. For $p \ge 2$,*

$$\int_{\mathbb{R}} |\widehat{g}(t)|^p\,dt \le \frac{4\pi}{\sqrt{p}}. \tag{147}$$

Let $e_1, \dots, e_r$ be orthonormal vectors in $\mathbb{R}^m$, set $u_i = (e_{1i}, \dots, e_{ri})$, and assume $|u_i|^2 \le 1/2$. For complex functions g_i dominated by the coordinate densities and arbitrary real shifts s_i,

$$\int_{\mathbb{R}^r} \prod_i |\widehat{g}_i(\langle u_i, z\rangle + s_i)|\,dz \le (4\pi)^r. \tag{148}$$

Proof. For an integer $k \ge 1$, Plancherel and convolution give

$$\int |\widehat{g}|^{2k} = 2\pi (g * \widetilde{g})^{*k}(0) \le 2\pi (f * \widetilde{f})^{*k}(0), \qquad \widetilde{g}(x) = \overline{g(-x)}.$$

The density on the right is that of a sum of $2k$ independent variables whose densities are bounded by one. Applying (146) to this one-dimensional linear map gives a bound $1/\sqrt{k}$ for that density. Since $|\widehat{g}| \le 1$, choose $k = \lfloor p/2 \rfloor$. For $p \ge 4$, $k \ge p/4$, and for $2 \le p < 4$ Plancherel gives the required bound directly.

Set $c_i = |u_i|^2$ and omit the zero rows. The vectors $u_i/\sqrt{c_i}$ with weights c_i form geometric Brascamp–Lieb data, since $\sum_i u_i u_i^T = I_r$. The geometric Brascamp–Lieb inequality (Brascamp and Lieb, 1976; Ball, 1986; Barthe, 1998), including arbitrary translations of the one-dimensional arguments, bounds the integral in (148) by

$$\prod_i \left(c_i^{-1/2} \int |\widehat{g}_i(t)|^{1/c_i}\,dt \right)^{c_i} \le (4\pi)^{\sum_i c_i} = (4\pi)^r.$$

Here $1/c_i \ge 2$, so (147) applies. □

For $\ell \in \mathbb{R}$ define the frozen polynomial

$$Y_\ell = \lambda V^2 + 2\ell\langle q, X'\rangle + \sum_i \phi_i(X_i).$$

For each fixed ℓ_0, let $\mu_{\ell_0,\ell}$ denote the unnormalized slice of the joint law of (L, Y_{ℓ_0}) at $L = \ell$. Its total mass is $\rho_L(\ell)$, not one.

Lemma 13.2 (Frozen slices)**.** *For every fixed ℓ_0, for almost every ℓ, and every $t \neq 0$,*

$$|\widehat{\mu}_{\ell_0,\ell}(t)| \le \min\left\{\rho_L(\ell), \frac{4\sqrt{\pi}}{\sqrt{|t\lambda|}}\right\}. \tag{149}$$

In particular, for every interval I,

$$\mu_{\ell_0,\ell}(I) \le 2\pi^2\sqrt{\frac{|I|}{|\lambda|}}. \tag{150}$$

Proof. Write $Y_{\ell_0} = \lambda V^2 + \sum_i (d_i X_i^2 + c_i' X_i) + c'$. For fixed $t \neq 0$ the Fourier transform in L of the complex measure $\mathbb{E}[e^{itY_{\ell_0}}; L \in \cdot]$ is

$$\Psi(z,t) = \mathbb{E}e^{izL + itY_{\ell_0}}.$$

The Fresnel identity, first with Gaussian damping, gives

$$|\Psi(z,t)| \le (4\pi|t\lambda|)^{-1/2} \int \prod_i |\widehat{g}_i(zw_i + uv_i + tc_i')|\, du, \qquad g_i(x) = f_i(x)e^{itd_i x^2}.$$

Lemma 13.1 bounds the integral in (z,u) by $(4\pi)^2$. Fourier inversion in z therefore gives the second term in (149). The first follows from the total mass of the slice. Disintegration first identifies this transform for almost every pair (ℓ,t); continuity in t identifies it for every t on a common full-measure set of ℓ.

If I has center y_0 and length $2b$, the function

$$K(y) = \left(\frac{\sin(\pi(y-y_0)/(4b))}{\pi(y-y_0)/(4b)}\right)^2$$

satisfies $K \ge (8/\pi^2)\mathbf{1}_I$; its Fourier transform is supported on $|t| \le \pi/(2b)$ and has modulus at most $4b$. Parseval and (149) give

$$\mu(I) \le \frac{\pi^2}{8}\frac{4b}{2\pi}\int_{|t|\le\pi/(2b)} \frac{4\sqrt{\pi}}{\sqrt{|t\lambda|}}\, dt = 2\pi^2\sqrt{\frac{2b}{|\lambda|}}.$$

All the Fourier manipulations follow first with a positive Gaussian damping; the displayed integrable bounds justify its removal. □

Remark 13.3 (Freezing and exceptional slices)*.* An estimate valid for every fixed ℓ_0 and almost every ℓ cannot be substituted on the diagonal $\ell_0 = \ell$ without justification. We use the following discretization whenever the slice parameter varies. Partition the real line into cells C_j of width h, choose $\ell_j \in C_j$, and replace Y_L by Y_{ℓ_j} on $\{L \in C_j\}$. Then $Y_{\ell_j} - Y_L = 2(\ell_j - L)\langle q, X'\rangle \to 0$ almost surely. For a bounded kernel $G(\ell, y)$ continuous off the null set $\ell = -\beta$, dominated convergence gives

$$\mathbb{E}G(L, Y_L) = \lim_{h\downarrow 0} \sum_j \int_{C_j} \int G(\ell_j, y)\, \mu_{\ell_j,\ell}(dy)\, d\ell. \tag{151}$$

For each partition only countably many frozen parameters occur, so the slice bounds apply simultaneously. The analogous construction with rectangular cells applies to several hub partners. Riemann sums used below can first be restricted to bounded sets; tails are controlled by the total mass. This also justifies the formulas for arbitrary unbounded coordinate laws.

13.2 Real-variable estimates for one hub

Lemma 13.4 (A profile criterion)**.** *Assume $a = 0$. Suppose that for every fixed ℓ_0, almost every ℓ, and every Borel set S,*

$$\mu_{\ell_0,\ell}(S) \le K|S|^\theta, \qquad 0 < \theta \le 1.$$

Then

$$\|p_Q\|_\infty \le \frac{1}{2} + \frac{2^\theta K}{\theta}. \tag{152}$$

More generally, a concave nondecreasing profile ω with $\omega(0) = 0$ gives $\|p_Q\|_\infty \le \frac{1}{2} + \int_0^2 \omega(u)\,du/u$ whenever that integral is finite.

Proof. Fix a level s and a window radius $\delta > 0$. With $t = \ell + \beta$ put

$$G(\ell, y) = \int f_0(x)\mathbf{1}_{\{|2tx+y-s|\le\delta\}}\,dx.$$

For $t \neq 0$, $0 \le G \le \delta/|t|$, and $\int G(\ell, y)\,dy = 2\delta$. If $M = \delta/|t|$, layer cake and Jensen's inequality give

$$\int G\,d\mu \le K\int_0^M |\{G > u\}|^\theta\,du \le KM(2\delta/M)^\theta = K\delta\,2^\theta|t|^{\theta-1}.$$

For $|t| \ge 1$ use $G \le \delta$ and the slice mass $\rho_L(\ell)$, whose integral is one. Apply (151), integrate the displayed bound over $|t| < 1$, divide by 2δ, and let $\delta \downarrow 0$. This proves (152). The same layer-cake argument gives $M\omega(2\delta/M)$ for a concave profile and yields the final assertion. The resulting uniform interval bounds imply absolute continuity and the stated essential density bound. □

Theorem 13.5 (An interval-supported hub)**.** *Let $a = 0$ and suppose X_0 is supported in an interval of length D. Then every form (144) satisfies*

$$\|p_Q\|_\infty \le \frac{3}{2}(4\sqrt{2}\pi^2)^{2/3}(D/|\lambda|)^{1/3} < 22(D/|\lambda|)^{1/3}. \tag{153}$$

Proof. Let J be the supporting interval and replace f_0 by $\mathbf{1}_J$. For $t = \ell + \beta \neq 0$ the resulting window kernel is

$$G(\ell, y) = \frac{|(s - y + [-\delta, \delta]) \cap 2tJ|}{2|t|}.$$

Three estimates follow by Fubini and Lemma 13.2:

$$\int G\,d\mu \le (\delta/|t|)\rho_L(\ell),$$
$$\int G\,d\mu \le (\delta/|t|)2\pi^2\sqrt{(2|t|D + 2\delta)/|\lambda|},$$
$$\int G\,d\mu \le 2\pi^2 D\sqrt{2\delta/|\lambda|}.$$

Use the first on $|t| \ge T$ and the minimum of the other two on $|t| < T$, with the freezing procedure of Remark 13.3. After division by 2δ, the range $|t| < \delta$ tends to zero. On $\delta \le |t| < T$, dominated convergence applies with an integrable multiple of $|t|^{-1/2}$. It follows that, uniformly in s,

$$\limsup_{\delta\downarrow 0} \frac{\mathbb{P}(|Q - s| \le \delta)}{2\delta} \le \frac{1}{2T} + 4\sqrt{2}\pi^2\sqrt{DT/|\lambda|}.$$

Choose $T = (4\sqrt{2}\pi^2)^{-2/3}(D/|\lambda|)^{-1/3}$. □

The next result needs no spread condition. Its only use of the product law is the two-dimensional projection bound.

Theorem 13.6 (A transverse square with an independent hub)**.** *Let w, v be orthonormal and let v_c, F_c be arbitrary measurable real functions. Set*

$$Q = 2X_0(L+\beta) + \lambda(V - v_c(L))^2 + F_c(L), \qquad \mu = |\lambda| > 0.$$

Then

$$\|p_Q\|_\infty \le \begin{cases} 2\sqrt{2} + \sqrt{2}\log(1/3 + 4\sqrt{2}/\mu), & 0 < \mu \le 6\sqrt{2}, \\ (192/\mu)^{1/3}, & \mu \ge 6\sqrt{2}. \end{cases} \tag{154}$$

Proof. For fixed s, integrating out X_0 gives

$$p_Q(s) = \int \rho_{L,V}(\ell, v)\, \frac{f_0((s - \lambda(v - v_c(\ell))^2 - F_c(\ell))/(2t))}{2|t|}\, d\ell\, dv, \qquad t = \ell + \beta.$$

Denote the fraction by $K(\ell, v)$. It is at most $1/(2|t|)$. Changing variables on the two branches of the square and using $f_0 \le 1$, $\int f_0 = 1$, gives

$$\int K(\ell, v)\, dv \le \sqrt{2}(\mu|t|)^{-1/2}.$$

Indeed the remaining integral is a one-sided negative half moment of f_0, bounded by $\int_0^1 u^{-1/2}\, du = 2$. Since $\rho_{L,V} \le 2$ and $\rho_L \le \sqrt{2}$, the contribution at t is bounded by $g(t)/(2|t|)$ for some nonnegative function satisfying

$$\int g \le 1, \qquad g(t) \le M(t) := \min\{\sqrt{2}, 4\sqrt{2}\mu^{-1/2}\sqrt{|t|}\}.$$

The weight $1/(2|t|)$ decreases with $|t|$. Thus its maximal integral under these constraints is obtained by taking $g = M$ on $[-\tau, \tau]$, where $\int_{-\tau}^{\tau} M = 1$. The envelope switches at $t_* = \mu/16$, and its mass on $[-t_*, t_*]$ is $\sqrt{2}\mu/12$. If this mass is at most one, the inner contribution is $2\sqrt{2}$ and the outer contribution is $\sqrt{2}\log(\tau/t_*)$, with $\tau/t_* = 1/3 + 4\sqrt{2}/\mu$. Otherwise $\tau = (3\sqrt{\mu}/(16\sqrt{2}))^{2/3}$, and the integral is $8\sqrt{2}\mu^{-1/2}\sqrt{\tau} = (192/\mu)^{1/3}$. The argument can equivalently be applied to interval probabilities, so no prior existence of p_Q is required. □

13.3 A nonzero transverse cross term

Write

$$q = q_w w + q_v v + \varepsilon e, \qquad e \perp w, v, \quad |e| = 1, \quad \varepsilon > 0,$$

and set $P = \langle e, X' \rangle$.

Theorem 13.7 (Nonzero leakage)**.** *Assume $w_i^2 + v_i^2 + e_i^2 \le 1/2$ for every light coordinate. For (144), including an arbitrary hub diagonal a,*

$$|\mathbb{E}e^{itQ}| \le 4\sqrt{2}\pi^{3/2}(\varepsilon|\lambda|)^{-1/2}|t|^{-3/2}, \tag{155}$$

$$\|p_Q\|_\infty \le 3\, 2^{5/3}(\varepsilon|\lambda|)^{-1/3}. \tag{156}$$

Proof. Complete the square in V: $V' = V + (q_v/\lambda)L$ and $\sigma = 2q_w - q_v^2/\lambda$. After moving $-2\varepsilon\beta P$ into the linear terms, write

$$Q = aX_0^2 + 2X_0(L+\beta) + \sigma L^2 + 2\varepsilon(L+\beta)P + W, \quad W = \lambda V'^2 - 2\varepsilon\beta P + \sum_i \phi_i(X_i).$$

Let $g(\omega) = \int f_0(x) e^{ita x^2 + i\omega x}\, dx$ and

$$H(\ell, p) = g(2t(\ell + \beta)) e^{it\sigma\ell^2 + 2it\varepsilon(\ell+\beta)p}.$$

Let $n(\ell, p)$ be the density of $\mathbb{E}[e^{itW}; (L, P) \in \cdot]$. Its Fourier transform is integrable by the Fresnel identity and the three-dimensional frame integral. Fourier transformation of H in p pins $\ell + \beta = -\gamma/(2t\varepsilon)$; consequently

$$|\widehat{H}(z, \gamma)| = \frac{\pi}{|t|\varepsilon} |g(-\gamma/\varepsilon)|.$$

This identity is interpreted with Gaussian damping, or as a tempered distribution followed by the indicated integrable pairing. Put $G(\gamma) = \int |\widehat{n}(z, \gamma)|\, dz$. Fresnel linearization in V' and translation of the z variable give

$$\|G\|_\infty \le (4\pi|t\lambda|)^{-1/2} (4\pi)^2, \qquad \|G\|_1 \le (4\pi|t\lambda|)^{-1/2} (4\pi)^3.$$

Thus $\|G\|_2 \le (4\pi|t\lambda|)^{-1/2} (4\pi)^{5/2}$. Plancherel gives $\int |g(\gamma/\varepsilon)|^2\, d\gamma \le 2\pi\varepsilon$. Parseval and Cauchy–Schwarz yield

$$|\mathbb{E} e^{itQ}| \le \frac{1}{4\pi|t|\varepsilon} (2\pi\varepsilon)^{1/2} (4\pi|t\lambda|)^{-1/2} (4\pi)^{5/2},$$

which is (155). Finally, $\int_{\mathbb{R}} \min(1, B|t|^{-3/2})\, dt = 6B^{2/3}$. Fourier inversion gives (156). □

Example 13.8 (A frozen-hub obstruction)**.** The nonzero-leakage estimate cannot be made uniform by bounding every conditional density obtained on fixing $X_0 = x$. Fix $x = r > 0$, let the light vector have law $X' = -(r/\varepsilon)e + G$, where the coordinates of G are independent standard Gaussians, and take no other terms. Then the conditional quadratic form is

$$\lambda \langle v, G\rangle^2 + 2\varepsilon \langle w, G\rangle \langle e, G\rangle.$$

For $\lambda > 0$, let Z_1, Z_2, Z_3 be its three independent standard Gaussian coordinates. The density of $2Z_1Z_2$ is positive and continuous on any compact interval avoiding zero. In the convolution formula for $\lambda Z_3^2 + 2\varepsilon Z_1 Z_2$ at zero, restrict λZ_3^2 to $[\varepsilon, 2\varepsilon]$. The square density is bounded below by a constant times $(\lambda u)^{-1/2}$ there when $\varepsilon \le \lambda$; the second density is bounded below by a constant times ε^{-1}. Integration gives a lower bound $c(\lambda\varepsilon)^{-1/2}$. All input densities are bounded by one. This is an obstruction to uniform frozen-hub estimates, not a counterexample to the full quadratic form after averaging the independent hub.

13.4 Near-linear small-ball bounds and symmetric-window recovery

Theorem 13.9 (Double-logarithmic small balls)**.** *Assume $a = 0$ and (145). No restriction is imposed on q beyond (144). For every $s \in \mathbb{R}$ and $\delta > 0$,*

$$\frac{\mathbb{P}(|Q - s| \le \delta)}{2\delta} \le A_\delta := \frac{1}{2} + 2\sqrt{2} + \sqrt{2} \log_+(2B_\delta^2), \tag{157}$$

where

$$B_\delta = \frac{4 + \sqrt{4/\pi + 32 \log_+(|\lambda|/\delta)}}{2\sqrt{|\lambda|}}. \tag{158}$$

Proof. Use the window kernel in the proof of Lemma 13.4. For fixed $t = \ell + \beta \ne 0$, it is $G(\ell, \cdot) = \mathbf{1}_{[-\delta,\delta]} * g$, where $g(u) = f_0((s-u)/(2t))/(2|t|)$. Hence

$$\widehat{G}(\eta) = \frac{2\sin(\eta\delta)}{\eta} \widehat{g}(\eta), \qquad \int |\widehat{g}|^2 \le \frac{\pi}{|t|}.$$

For every applicable frozen slice, Lemma 13.2 gives

$$\int_{|\eta|\le 1/\delta} |\widehat{\mu}(\eta)|^2\, d\eta \le \frac{4 + 32\pi \log_+(|\lambda|/\delta)}{|\lambda|}.$$

To see this, use $|\widehat{\mu}|^2 \le 2$ below $1/|\lambda|$ and $|\widehat{\mu}|^2 \le 16\pi/(|\eta\lambda|)$ above it. Parseval and Cauchy–Schwarz on the regions below and above $1/\delta$ now give

$$\begin{aligned}\int G\, d\mu &\le \frac{1}{2\pi}\left(2\delta \int_{|\eta|\le 1/\delta} |\widehat{g}||\widehat{\mu}| + \int_{|\eta|>1/\delta} \frac{2}{|\eta|}|\widehat{g}||\widehat{\mu}|\right) \\ &\le \delta(|\lambda t|)^{-1/2}\left(\sqrt{4/\pi + 32\log_+(|\lambda|/\delta)} + 4\right).\end{aligned}$$

The high-frequency calculation uses $\int_{|\eta|>1/\delta} |\eta|^{-3}\, d\eta = \delta^2$. The alternative bound is $\int G\, d\mu \le \sqrt{2}\delta/|t|$. For $|t| \ge 1$ retain the slice mass and obtain a contribution at most δ. Formula (151) therefore implies

$$\frac{\mathbb{P}(|Q - s| \le \delta)}{2\delta} \le \frac{1}{2} + \int_{|t|<1} \min\left\{\frac{\sqrt{2}}{2|t|}, B_\delta |t|^{-1/2}\right\} dt.$$

If $2B_\delta^2 \ge 1$, split at $|t| = 1/(2B_\delta^2)$; the integral is $2\sqrt{2} + \sqrt{2}\log(2B_\delta^2)$. Otherwise it equals $4B_\delta \le 2\sqrt{2}$. □

Lemma 13.10 (Recovery from a symmetric window)**.** *Suppose a real random variable Z satisfies $|\mathbb{E}e^{itZ}| \le B|t|^{-3/2}$ for $t \ne 0$. Then it has a continuous bounded density and, for every $h > 0$,*

$$\|p_Z\|_\infty \le \sup_s \frac{\mathbb{P}(|Z - s| \le h)}{2h} + \frac{2\sqrt{2\pi}}{3\pi} B\sqrt{h}. \tag{159}$$

Proof. The integrable characteristic function gives the density by Fourier inversion. Let $U_h = \mathbf{1}_{[-h,h]}/(2h)$. Its Fourier transform is $\sin(ht)/(ht)$, which is at most one. Consequently

$$|p_Z(s) - (p_Z * U_h)(s)| \le \frac{B\sqrt{h}}{\pi}\int_0^\infty u^{-3/2}\left(1 - \frac{\sin u}{u}\right) du.$$

Since $\sin u/u = \int_0^1 \cos(ru)\, dr$, Tonelli and an integration by parts give

$$\begin{aligned}\int_0^\infty u^{-3/2}\left(1 - \frac{\sin u}{u}\right) du &= \int_0^1 \int_0^\infty u^{-3/2}(1 - \cos(ru))\, du\, dr \\ &= \sqrt{2\pi}\int_0^1 \sqrt{r}\, dr = \frac{2}{3}\sqrt{2\pi}.\end{aligned}$$

For completeness, the inner integral at $r = 1$ is $2\int_0^\infty u^{-1/2}\sin u\, du = \sqrt{2\pi}$; this identity follows by exponential damping and the Gamma integral. Finally $(p_Z * U_h)(s) = \mathbb{P}(|Z - s| \le h)/(2h)$. □

Corollary 13.11 (Improved double-logarithmic density estimate)**.** *Under the assumptions of Theorem 13.7, with $a = 0$,*

$$\|p_Q\|_\infty \le A_h + \frac{16\pi}{3}\sqrt{\frac{h}{\varepsilon|\lambda|}} \qquad (h > 0). \tag{160}$$

In particular,

$$\|p_Q\|_\infty \le \frac{1}{2} + 2\sqrt{2} + \sqrt{2}\log_+(2B_\varepsilon^2) + \frac{16\pi}{3\sqrt{|\lambda|}}. \tag{161}$$

For fixed $|\lambda| \ge 1$, this is $O(1 + \log\log(1/\varepsilon))$ as $\varepsilon \downarrow 0$.

Proof. Apply Lemma 13.10 with the constant in (155), and then Theorem 13.9. The recovery constant simplifies to $(2\sqrt{2\pi}/(3\pi))4\sqrt{2}\pi^{3/2} = 16\pi/3$. □

Remark 13.12. The estimates above hold for each positive ε. Their constants are not bounded as $\varepsilon \downarrow 0$. Thus they do not establish a uniform three-eigenvalue theorem. Even though the family at exactly zero leakage is bounded in several important normal forms, continuity for fixed input laws does not supply uniformity over all bounded product densities.

13.5 Localization and regularity of the hub law

Proposition 13.13 (A localized leakage direction)**.** *Consider*

$$Q = aX_0^2 + 2X_0(L+\beta) + \lambda V^2 + 2\varepsilon LP + \sum_i \phi_i(X_i).$$

Suppose the coefficient vector of P is supported on T, let $I = T^{\mathrm{c}}$, and put $G_I = \sum_{i\in I}(w_i, v_i)^T(w_i, v_i)$. If G_I is invertible and $(w_i, v_i)G_I^{-1}(w_i, v_i)^T \le 1/2$ on I, then

$$|\mathbb{E}e^{itQ}| \le 4\pi^{3/2}(|\lambda| \det G_I)^{-1/2}|t|^{-3/2}, \qquad \|p_Q\|_\infty \le 3\,2^{4/3}(|\lambda| \det G_I)^{-1/3}.$$

The estimate is uniform in ε, a, and all other displayed coefficients.

Proof. For fixed t, integrating the hub and using $f_0 \le 1$ bounds the characteristic function by $(2|t|)^{-1}\int |\widehat{n}(z)|\,dz$, where $\widehat{n}(z)$ is the light expectation with the additional phase zL. Fresnel linearization of λV^2 reduces its integral to $(4\pi|t\lambda|)^{-1/2}J$, where

$$J = \iint \left|\mathbb{E}e^{izL+iuV+2it\varepsilon LP+it\sum_i \phi_i(X_i)}\right| dz\,du.$$

Disintegrate the complex light measure over $P = p$. On that slice the cross term translates z by $2t\varepsilon p$. The triangle inequality followed by this translation bounds J by the averaged conditional frame integral. The coordinates in I are independent of those carrying P; the latter push forward to a complex measure of total variation at most one. Therefore

$$J \le \iint \prod_{i\in I} |\widehat{g}_i(zw_i + uv_i + tb_i)|\,dz\,du \le (4\pi)^2(\det G_I)^{-1/2}.$$

The last inequality follows by the change of variables associated with G_I and Lemma 13.1. Multiplication of the three factors proves the characteristic-function bound; its integral proves the density estimate. □

Proposition 13.14 (A square on independent coordinates)**.** *Let I be a set of light coordinates on which w vanishes and let X_R denote the others. Suppose*

$$Q = 2X_0(L+\beta) + \lambda V^2 + 2Vg(X_R) + F(X_R), \qquad \omega = |v_I| > 0,$$

where g, F are arbitrary measurable functions. Then

$$\|p_Q\|_\infty \le \frac{1}{2} + \frac{8\sqrt{2}}{\omega\sqrt{|\lambda|}}.$$

Proof. Write $V = V_I + V_R$. The variable V_I is independent of X_R and has density at most $\sqrt{2}/\omega$ by (146). Given X_R, the remainder is $\lambda(V_I + c_1)^2 + c_2$. For every Borel set S,

$$|\{u : \lambda u^2 \in S\}| \le 2\sqrt{|S|/|\lambda|},$$

which follows by placing a set of fixed measure at the origin in the one-sided density of the square. Thus the conditional probability is at most $2\sqrt{2}\omega^{-1}\sqrt{|S|/|\lambda|}$. Since L is a function of X_R and $\rho_L \le \sqrt{2}$, its unnormalized slices have profile $4\omega^{-1}\sqrt{|S|/|\lambda|}$. Lemma 13.4 finishes the proof. □

Proposition 13.15 (A block carrying only the hub partner)**.** *Assume $a = 0$. Suppose a set I of light coordinates satisfies $v_i = d_i = 0$, $q_i = \kappa w_i$, and $b_i = \mu w_i$ on I. Put $\omega = |w_I| > 0$. Then*

$$\|p_Q\|_\infty \le \frac{1}{2} + \frac{\sqrt{2}}{\omega}\left[\frac{7}{2}\Big(2 + \log(1 + 4\pi^4/|\lambda|)\Big) + \frac{1}{2}\log(1 + 4|\kappa|)\right].$$

If $\kappa = 0$, the sharper bound is

$$\|p_Q\|_\infty \le \frac{1}{2} + \frac{\sqrt{2}}{\omega}\Big(2 + \log(1 + 2\pi^4/|\lambda|)\Big).$$

Proof. Let $R = I^c$, $L_I = \langle w_I, X_I\rangle$ and $L_R = L - L_I$. Collecting terms gives $Q = 2X_0 t + 2\kappa L^2 + 2LP + W$, where $t = L + \beta$ and P, W depend only on X_R. The value of the hub at level s has the form

$$x_* = \frac{\alpha}{2t} + c_0 - \kappa t, \quad \alpha = s - 2\kappa\beta^2 + 2\beta P - W, \quad c_0 = 2\kappa\beta - P.$$

Conditional on X_R, the variable t has density at most $\sqrt{2}/\omega$. On $|t| \ge 1$ its contribution to p_Q is at most $1/2$. Thus the remaining contribution is bounded by

$$\frac{\sqrt{2}}{\omega}\mathbb{E}_{X_R}\int_{0<|t|<1} \widetilde{f}_0\Big(\frac{\alpha}{2t} - \kappa t\Big)\frac{dt}{2|t|}, \qquad \widetilde{f}_0(x) = f_0(x + c_0).$$

For $\kappa = 0$, the substitution $x = \alpha/(2t)$ and the bathtub principle bound the inner integral by $\log(1 + 1/|\alpha|)$. For $\kappa \neq 0$, set $\sigma = 2\kappa$ and $\tau = \min(1, \sqrt{|\alpha|/(2|\sigma|)})$. For $0 < |t| < \tau$, the map $x(t) = \alpha/(2t) - \sigma t/2$ is monotone on each half-line, its two images are disjoint, $|x(t)| \ge |\alpha|/(4\tau)$, and $dt/|t| \le 3\,|dx|/|x|$. The bathtub principle bounds this part by $3\log(1 + 2\tau/|\alpha|)$; the range $\tau \le |t| < 1$ contributes at most $\log(1/\tau)$. Together these are at most

$$\frac{7}{2}\log(1 + 2/|\alpha|) + \frac{1}{2}\log(1 + 2|\sigma|).$$

The variable α is a quadratic polynomial on X_R consisting of $-\lambda V^2$ and one-coordinate terms: the L-cross terms have already been put into P. One-dimensional Fresnel linearization and Lemma 13.1 give $|\mathbb{E}e^{it\alpha}| \le 2\sqrt{\pi/|t\lambda|}$. The kernel in Lemma 13.2 therefore gives $\mathbb{P}(|\alpha - u| \le r) \le \pi^2\sqrt{2r/|\lambda|}$. For $c = 2\pi^2/\sqrt{|\lambda|}$, layer cake implies

$$\mathbb{E}\log(1 + 2/|\alpha|) \le \int_0^\infty \min\{1, c(e^u - 1)^{-1/2}\}\,du \le \log(1 + c^2) + 2.$$

Indeed, above $u = \log(1 + c^2)$ the integral equals $2c\arctan(1/c) < 2$. Replacing 2 by 1 in the logarithm gives $2 + \log(1 + 2\pi^4/|\lambda|)$, proving both assertions. □

Proposition 13.16 (A hub density of bounded variation)**.** *Assume $a = 0$ and let the total variation of f_0 be $V_0 < \infty$. For $A = 4\sqrt{\pi/|\lambda|}$,*

$$|\mathbb{E}e^{itQ}| \le V_0|t|^{-3/2}\left[A\Big(1 + \log_+(2|t|^{3/2}/V_0)\Big) + \frac{1}{2}\right]. \tag{162}$$

In particular, for fixed $|\lambda| \ge 1$, $\|p_Q\|_\infty \le CV_0^{2/3}$ with an absolute constant C, uniformly in every light diagonal, linear term, and cross term.

Proof. Integration by parts against the distributional derivative of f_0 gives $|\widehat{f_0}(z)| \le \min(1, V_0/|z|)$. Freeze L and apply (149). With $u = L + \beta$,

$$|\mathbb{E}e^{itQ}| \le \int \min\{1, V_0/(2|tu|)\}\min\{\rho_L(u - \beta), A|t|^{-1/2}\}\,du.$$

On $|u| \le T = |t|^{1/2}$, use the second entry in the second minimum. The integral of the first minimum is at most $(V_0/|t|)(1 + \log_+(2|t|T/V_0))$. On $|u| > T$, retain ρ_L and use its unit mass, obtaining $V_0/(2|t|^{3/2})$. This proves (162). Substitute $t = V_0^{2/3}u$ into the integral of the minimum of one and the right side. The resulting integral is finite and uniformly bounded for $A \le 4\sqrt{\pi}$, proving the density estimate. No numerical quadrature is needed for the existence of C. □

13.6 Bounded light supports and a fixed transverse projection

Proposition 13.17 (A maximal-density criterion)**.** *Let E be a fixed k-dimensional subspace orthogonal to w, v, and write Z for the orthonormal coordinates of the projection of X' onto E. Let*

$$Q = 2X_0(L + \beta) + \lambda V^2 + 2Vg(L, Z) + F(L, Z),$$

where g, F are measurable. Put

$$M_k = \operatorname*{ess\,sup}_{\ell} \int_{\mathbb{R}^k} \operatorname*{ess\,sup}_{u} \rho_{L,V,Z}(\ell, u, z)\, dz.$$

If $M_k < \infty$, then

$$\|p_Q\|_\infty \le \frac{1}{2} + 8M_k|\lambda|^{-1/2}.$$

This result needs no leverage condition.

Proof. On the slice $L = \ell$ complete the square in V: $\lambda(V - V_0(z))^2 + F_0(z)$. The slab $|V - V_0(z)| \le r$ has slice mass at most $2rM_k$. Outside the slab, the two inverse branches each have derivative at most $(2|\lambda|r)^{-1}$ in the output variable, so the restricted output density is at most $M_k/(|\lambda|r)$. For every Borel set S, choose $r = \sqrt{|S|/(2|\lambda|)}$ to obtain the profile $2\sqrt{2}M_k\sqrt{|S|/|\lambda|}$. Lemma 13.4 gives the result; freezing is unnecessary here because the remainder is already a function of (L, V, Z). □

Proposition 13.18 (Bounded light supports)**.** *Suppose every light coordinate is supported in an interval of length at most K. Then the quantity in Proposition 13.17 satisfies*

$$M_k \le 2^{(k+2)/2}K^k \int_{\mathbb{R}^k} e^{-2(|y|-k-2)_+^2}\, dy.$$

Moreover, for (144) with $a = d_i = 0$, arbitrary q and arbitrary linear terms,

$$\|p_Q\|_\infty \le \frac{1}{2} + 8\sqrt{2}\left[2^{3/2}\left(6 + \sqrt{\pi/2}\right)\frac{K}{|\lambda|}\right]^{1/2} < \frac{1}{2} + 52\sqrt{K/|\lambda|}. \tag{163}$$

The hub law is arbitrary subject to its density cap.

Proof. Tilt the product law by $e^{\langle\theta,Z\rangle}$. Each tilted coordinate density is at most $e^{|\theta|K}$ times its original cap, because the tilting factor varies by at most that amount on its supporting interval. The $(k+2)$-dimensional projection-density bound for the tilted product law is therefore $2^{(k+2)/2}e^{(k+2)|\theta|K}$. Independence and the bounded-variable estimate of Hoeffding (1963) give

$$\mathbb{E}e^{\langle\theta,Z-\mathbb{E}Z\rangle} \le e^{|\theta|^2K^2/8}.$$

Undoing the tilt and optimizing θ in the direction $z - \mathbb{E}Z$ yields

$$\rho_{L,V,Z}(\ell, u, z) \le 2^{(k+2)/2}\exp\left[-2\frac{(|z - \mathbb{E}Z| - (k+2)K)_+^2}{K^2}\right].$$

The optimization uses $|\theta| = 4(|z - \mathbb{E}Z| - (k+2)K)_+/K^2$. Integration over z proves the first assertion.

For the second assertion freeze L as in Remark 13.3. With no light diagonal, every frozen remainder has the form $\lambda(V - V_0)^2 + \gamma B + c$, where B is a unit linear projection orthogonal to w, v; its direction may depend on the frozen parameter. The preceding tilt bound is uniform in that direction and gives

$$\int \sup_u \rho_{L,V,B}(\ell, u, b)\, db \le M := 2^{3/2}\left(6 + \sqrt{\pi/2}\right) K.$$

The slab $|V - V_0| \le r$ has slice mass at most $4r$, using $\rho_{L,V} \le 2$. Off the slab the output density is at most $M/(|\lambda| r)$. Optimizing $4r + M|S|/(|\lambda| r)$ gives the Borel profile $4\sqrt{M|S|/|\lambda|}$. Lemma 13.4 proves (163). If $\gamma = 0$, the direct square profile is at least as strong, since $K \ge 1$. □

Corollary 13.19 (Window-mixture light laws)**.** *Suppose each light law is of the form $\mathcal{U}[0,1] * \nu_i$. For (144) with $a = d_i = 0$, every independent cap-one hub law satisfies*

$$\|p_Q\|_\infty \le \frac{1}{2} + 52|\lambda|^{-1/2}.$$

Proof. Represent $X_i = U_i + Z_i$, with independent uniform U_i on $[0,1]$, and condition on the shifts Z_i. The light coordinates are now supported on unit intervals. The shifts change only the linear coefficients and the constant of the quadratic form, which are unrestricted in Proposition 13.18. Apply that proposition and average. □

13.7 Several hub coordinates

Theorem 13.20 (Three or more hubs)**.** *Let $X_H \in \mathbb{R}^k$ and X' be independent, with $\int f_H^2 \le 1$, and suppose $L = MX' \in \mathbb{R}^k$ has density bounded by D. For any real measurable F, R and $\beta \in \mathbb{R}^k$,*

$$Q = F(X_H) + 2\langle X_H, L + \beta\rangle + R(X')$$

satisfies

$$|\mathbb{E}e^{itQ}| \le \min\{1, D^{1/2}(\pi/|t|)^{k/2}\}.$$

If $k > 2$, then

$$\|p_Q\|_\infty \le \frac{k}{k-2} D^{1/k}. \tag{164}$$

For independent cap-one light coordinates and singular values $s_1, \dots, s_k > 0$ of M, this is at most $\frac{k}{k-2}\sqrt{2}(s_1 \cdots s_k)^{-1/k}$.

Proof. Let $h_t(\ell) = \int f_H(x) e^{itF(x) + 2it\langle x, \ell + \beta\rangle}\, dx$. Plancherel gives

$$\int_{\mathbb{R}^k} |h_t(\ell)|^2\, d\ell = (\pi/|t|)^k \int f_H^2 \le (\pi/|t|)^k.$$

After conditioning on X',

$$|\mathbb{E}e^{itQ}| \le \mathbb{E}|h_t(L)| \le \|\rho_L\|_2 \|h_t\|_2 \le D^{1/2}(\pi/|t|)^{k/2}.$$

Integrating the minimum with one gives (164). The final assertion follows from (146). □

Theorem 13.21 (Two hubs and a transverse square)**.** *Let $X_H \in \mathbb{R}^2$ have density $f_H \le 1$ and be independent of the cap-one product vector X'. Let M have rank two, singular values s_1, s_2, and orthonormal row-space basis e_1, e_2. Suppose $v \perp e_1, e_2$, $|v| = 1$, and $e_{1i}^2 + e_{2i}^2 + v_i^2 \le 1/2$.*

Put $L = MX'$ and $V = \langle v, X'\rangle$. For every real symmetric 2×2 matrix B, vectors q_1, q_2, and coordinate quadratics,

$$Q = X_H^T B X_H + 2\langle X_H, L + \beta\rangle + \lambda V^2 + 2\sum_{j=1}^{2} L_j \langle q_j, X'\rangle + \sum_i \phi_i(X_i)$$

has

$$\|p_Q\|_\infty \le 5\, 2^{6/5} (s_1 s_2)^{-2/5} |\lambda|^{-1/5} < 11.5 (s_1 s_2)^{-2/5} |\lambda|^{-1/5}.$$

Proof. Freeze the two partners on rectangular cells. Plancherel gives

$$\int |h_t(\ell)|^2 \, d^2\ell \le \pi^2/t^2$$

for the hub factor. Fresnel linearization of the transverse square and the three-dimensional frame bound give for the frozen slice

$$|\widehat{\mu}_{\ell_0,\ell}(t)| \le \min\left\{ \rho_L(\ell), \frac{8\sqrt{\pi}}{s_1 s_2 \sqrt{|t\lambda|}} \right\}.$$

The factor $(s_1 s_2)^{-1}$ is the Jacobian converting the frequency of L to that of its orthonormal row-space coordinates. For each fixed t, Cauchy–Schwarz on the cells and passage to the Riemann-sum limit give

$$|\mathbb{E} e^{itQ}| \le K |t|^{-5/4}, \quad K = \pi \left(\frac{8\sqrt{\pi}}{s_1 s_2} \right)^{1/2} |\lambda|^{-1/4}.$$

Indeed, the squared slice factor integrates to at most its displayed uniform bound times $\int \rho_L = 1$. The characteristic function is also at most one. Its integral is $10K^{4/5}$, and division by 2π gives the stated constant. □

Theorem 13.22 (Two hubs with an interval slice profile)**.** *Let $X_H \in \mathbb{R}^2$ have density at most one, be independent of X', and let $L = MX'$ have a density. Let Y_ℓ be a family of real functions of X' such that $Y_{\ell'} \to Y_\ell$ pointwise as $\ell' \to \ell$. Suppose its frozen slices satisfy*

$$\mu_{\ell_0,\ell}(I) \le K |I|^b \quad (0 < b \le 1)$$

for every fixed ℓ_0, almost every ℓ, and every interval I. Then, for every real symmetric B and $\beta \in \mathbb{R}^2$,

$$Q = X_H^T B X_H + 2\langle X_H, L + \beta\rangle + Y_L$$

has

$$\|p_Q\|_\infty \le \frac{1}{\pi} + \frac{\sqrt{10K}}{1 - 2^{-b/2}}.$$

Proof. First restrict the light measure to $\{\rho_L(L) \le N\}$. The resulting subprobability measure has bounded partner density, and all the assumed slice inequalities persist with the same constants. The bounds below are independent of N; monotone convergence at the end removes the restriction. Use rectangular cells C_j and frozen values ℓ_j. Define the finite positive measure

$$\nu_h = \sum_j \int_{C_j} \mu_{\ell_j,\ell} * \widetilde{\mu}_{\ell_j,\ell} \, d^2\ell.$$

It has nonnegative Fourier transform $\widehat{\nu}_h(t) = \sum_j \int_{C_j} |\widehat{\mu}_{\ell_j,\ell}(t)|^2 d^2\ell$ and satisfies $\nu_h([-r, r]) \le K(2r)^b$, since each slice has mass $\rho_L(\ell)$ and $\int \rho_L = 1$.

For any positive measure ν with nonnegative transform and $\nu([-r,r]) \le K' r^b$, the Fejér kernel gives

$$\int_{-T}^{T} \widehat{\nu}(t)\,dt \le 4T \int \left(\frac{\sin(Tx)}{Tx}\right)^2 \nu(dx) \le 20K'T^{1-b}.$$

For the last inequality split at $|x| = 1/T$ and into the dyadic shells $2^j/T < |x| \le 2^{j+1}/T$; use $\min(1,(Tx)^{-2})$ and $1 + 2^b/(1-2^{b-2}) \le 5$.

Let $\Omega_A = \{A \le |t| \le 2A\}$. Cauchy–Schwarz in the partners and then in t, followed by Fatou and the freezing limit, give

$$\int_{\Omega_A} |\mathbb{E}e^{itQ}|\,dt \le \left(\int_{\Omega_A}\int |h_t(\ell)|^2 d^2\ell\,dt\right)^{1/2} \liminf_{h\downarrow 0}\left(\int_{\Omega_A} \widehat{\nu}_h(t)dt\right)^{1/2}.$$

This can first be done on finitely many cells and then exhausted; all slice-mass estimates remain unchanged. The first factor is at most $\pi A^{-1/2}$ by hub Plancherel, and the second is at most $(20\cdot 2^b K(2A)^{1-b})^{1/2}$. Thus the integral is at most $\pi\sqrt{40K}A^{-b/2}$. Sum over $A = 2^j$, $j \ge 0$, add the bound two on $|t| < 1$, and divide by 2π. □

Proposition 13.23 (A projected crossing remainder)**.** *Let $k = 1$ or 2, let $X_H \in \mathbb{R}^k$ have density at most one, and let X' be an independent cap-one product vector. Consider*

$$Q = X_H^T B X_H + 2\langle X_H, MX' + \beta\rangle + R(X'), \qquad R(x) = x^T A_R x + b^T x + c.$$

Split the light coordinates as $I \cup J$. Suppose M_J has rank k, with product S of its singular values, and put

$$C = (A_R)_{I,J}(I - \Pi), \qquad F = \|C\|_F > 0,$$

where Π projects onto the row space of M_J. If every row c_i of C satisfies $|c_i|^2 \le F^2/2$, then

$$|\mathbb{E}e^{itQ}| \le 2^{(k+3)/4}\pi^{(k+1)/2}(SF)^{-1/2}|t|^{-(k+1)/2}. \tag{165}$$

For $k = 2$ this gives

$$\|p_Q\|_\infty \le 3\,2^{5/6}(SF)^{-1/3} < 5.35(SF)^{-1/3}.$$

For $k = 1$, $S = |w_J|$, it gives

$$\mathbb{P}(|Q - s| \le \delta) \le \frac{\pi^2\delta}{\sqrt{SF}}\left(1 + \log_+\frac{\sqrt{SF}}{4\delta}\right).$$

Proof. Let $\mu_t(\ell)$ be the density of $\mathbb{E}[e^{itR(X')}; MX' \in d\ell]$. The hub factor and Plancherel imply

$$|\mathbb{E}e^{itQ}|^2 \le (\pi/|t|)^k \int |\mu_t(\ell)|^2 d\ell.$$

Convolve μ_t with an even nonnegative normalized approximate identity k_h. Conditional Cauchy–Schwarz in X_I, an independent copy X_J^*, and integration over ℓ yield

$$\int |\mu_t * k_h|^2 \le \mathbb{E}\left[\kappa_h(M_J D_J)\prod_{i\in I}|\widehat{f}_i(2t\langle a_i, D_J\rangle)|\right],$$

where $D_J = X_J - X_J^*$, $\kappa_h = k_h * k_h$, and a_i is the ith row of $(A_R)_{I,J}$. Terms of R internal to I cancel; terms internal to J contribute only a phase of modulus one.

Put $\theta_i = |c_i|^2/F^2$ and discard the zero rows, whose factors are at most one. Hölder with weights θ_i bounds the last expectation by

$$\prod_i \left(\mathbb{E}\left[\kappa_h(M_J D_J)|\widehat{f}_i(2t\langle a_i, D_J\rangle)|^{1/\theta_i}\right]\right)^{\theta_i}.$$

The coordinates of D_J are independent with densities bounded by one. Since $c_i \perp \mathrm{row}(M_J)$, the density of $(M_J D_J, \langle c_i, D_J\rangle)$ is at most $2^{(k+1)/2}/(S|c_i|)$. Write $a_i = c_i + \alpha_i M_J$ and translate the last variable for each value of $M_J D_J$. Applying (147) gives a bound

$$\frac{2^{(k+1)/2}}{S|c_i|}\frac{4\pi\sqrt{\theta_i}}{2|t|} = \frac{2^{(k+3)/2}\pi}{SF|t|}$$

for every factor. It is independent of h. Pass to the L^2 approximate-identity limit and combine with the hub factor to obtain (165). For $k = 2$, integrate its minimum with one. For $k = 1$, put $K_1 = 2\pi/\sqrt{SF}$ and apply the kernel used in Lemma 13.2 to $\min(1, K_1/|t|)$. The integral over $|t| \le \pi/(2\delta)$ is at most $2K_1(1 + \log_+(\pi/(2K_1\delta)))$, which gives the assertion. □

Remark 13.24. The projected-crossing estimate permits arbitrary terms internal to either light block. Its extension to several volume directions and arbitrary-rank crossing matrices is given in Theorem 11.4; exact rank of the crossing matrix is not needed there. The nondegeneracy and volume-marginal assumptions remain essential hypotheses of that theorem.

Corollary 13.25 (A logarithmic bound with a hub diagonal)**.** *Suppose Theorem 13.7 and the $k = 1$ case of Proposition 13.23 both apply. Then*

$$\|p_Q\|_\infty \le \frac{2}{\sqrt{SF}}\left(3 + \log_+ \frac{4(SF)^{3/2}}{\varepsilon|\lambda|}\right).$$

Proof. The characteristic function is bounded by $\min(1, K_1|t|^{-1}, K_2|t|^{-3/2})$, where $K_1 = 2\pi/\sqrt{SF}$ and $K_2 = 4\sqrt{2}\pi^{3/2}/\sqrt{\varepsilon|\lambda|}$. Splitting the positive half-line at K_1 and $(K_2/K_1)^2$ gives an integral at most $K_1[3 + \log_+(K_2^2/K_1^3)]$. The same bound holds when the latter split precedes the former by using just the first and third entries of the minimum. Since $K_2^2/K_1^3 = 4(SF)^{3/2}/(\varepsilon|\lambda|)$, division of the two-sided integral by 2π proves the claim. □

Proposition 13.26 (A spread linear hub)**.** *Let X_H and X' be independent cap-one product vectors, let $U = \langle c, X_H\rangle$, and assume $c_h^2 \le |c|^2/2$ for every hub coordinate. For*

$$Q = 2U(L + \beta) + \sum_{h\in H}\phi_h(X_h) + \lambda V^2 + 2L\langle q, X'\rangle + \sum_i \phi_i(X_i),$$

with (145)*,*

$$|\mathbb{E}e^{itQ}| \le 8\pi^{3/2}|c|^{-1}|\lambda|^{-1/2}|t|^{-3/2}, \qquad \|p_Q\|_\infty \le 12(|c|^2|\lambda|)^{-1/3}.$$

Proof. The conditional hub expectation is

$$H_t(u) = \prod_{h\in H}\widehat{g}_h(2tuc_h + tb_h), \qquad g_h(x) = f_h(x)e^{itd_h x^2}.$$

Hölder with exponents $p_h = |c|^2/c_h^2 \ge 2$ and (147) gives $\int |H_t(u)|\,du \le 2\pi/(|t||c|)$. Freeze L and pair this factor with the uniform slice-transform bound $4\sqrt{\pi}/\sqrt{|t\lambda|}$ from Lemma 13.2. Their product is the claimed characteristic-function bound. Integrating the minimum with one gives the density bound. □

13.8 An adapted-hub variational formulation

The entangled case admits a variational formulation in terms of a restricted conditional law. Its dependence structure is absent from arbitrary bounded two-dimensional densities.

Proposition 13.27 (Adapted-hub variational reduction)**.** *Let $a = 0$ and $q = q_0 + \varepsilon e$, where $q_0 \in \operatorname{span}\{w, v\}$ and $P = \langle e, X' \rangle$. Absorb $-2\varepsilon\beta P$ into the light linear terms and write*

$$Q = 2Y(L + \beta) + R_0(X'), \qquad Y = X_0 + \varepsilon P.$$

Then the conditional density of Y given X' is at most one and depends on X' only through P. For a fixed level s, put $x_* = (s - R_0(X'))/(2(L + \beta))$. If $W_s(x, p)$ is the joint density of (x_*, P) under the nonnegative weight $1/(2|L + \beta|)$, then the supremum at s over all conditional hub densities $f(y \mid p) \le 1$ is

$$\Phi(s) = \int_{\mathbb{R}} \sup_{|A|=1} \int_A W_s(x, p)\, dx\, dp. \tag{166}$$

In particular, $p_Q(s) \le \Phi(s)$ for the independent-hub form. The same assertion can be formulated for window probabilities, without assuming that the weighted pushforward has a density.

Proof. The identity $2X_0(L + \beta) + 2\varepsilon LP = 2(L + \beta)(X_0 + \varepsilon P) - 2\varepsilon\beta P$ proves the first assertion. Its conditional density is $f_0(y - \varepsilon P)$. Given X', the map from Y to Q is affine with derivative $2(L + \beta)$; hence

$$p_Q(s) = \iint f(x \mid p) W_s(x, p)\, dx\, dp.$$

For each p, maximizing a nonnegative linear functional subject to $0 \le f(\cdot \mid p) \le 1$ and $\int f(\cdot \mid p) = 1$ gives the integral over a set of measure one on which W_s is largest. Measurable approximate maximizers can be chosen by thresholding W_s and filling a level set; monotone approximation then gives (166). The window version follows by applying the same conditional maximization to the nonnegative conditional window kernel before taking the density limit. □

Conjecture 13.28 (A sufficient adapted-hub estimate)**.** *Let X' have independent coordinate densities bounded by one, and let w, v, e be orthonormal with $w_i^2 + v_i^2 + e_i^2 \le 1/2$. Set $L = \langle w, X' \rangle$, $V = \langle v, X' \rangle$, and $P = \langle e, X' \rangle$. If the conditional density of Y given X' is $g(y \mid P) \le 1$ of unit mass, then $2YL + V^2$ has density bounded by an absolute constant. Equivalently, its functional (166) is bounded uniformly in the level.*

The conjecture is stronger than the estimate for a translated independent hub, because it permits an arbitrary probability density depending on P. This conditional estimate remains open. The preceding estimates also do not cover arbitrary quadratic twists on the orthogonal complement of $\operatorname{span}\{w, v\}$ or yield an exhaustive quantitative reduction of all three-eigenvalue matrices to the stated hub configurations.

13.9 Adapted hubs for Gaussian mixtures

The adapted-hub estimate can be proved for a class allowing arbitrarily small Gaussian component variances. The resulting theorem concerns the entangled core $2YL + \lambda V^2$; it does not assert an estimate for arbitrary light diagonal terms.

Lemma 13.29 (A conditional Gaussian bound)**.** *Suppose (L, V, P) is a Gaussian vector with covariance matrix at least hI_3, where $h > 0$. Let the conditional law of Y given (L, V, P) have density $g(y \mid P) \le 1$ and unit integral in y. For $|\lambda| \ge 1$ and arbitrary finite real measurable B, C, set*

$$Q = 2YL + \lambda V^2 + B(Y, P)V + C(Y, P).$$

Then

$$\|p_Q\|_\infty \le 4(2\pi)^{-3/4} h^{-3/4}. \tag{167}$$

If $h \ge (2\pi)^{-1}$, the sharper numerical bound is $\|p_Q\|_\infty \le 3$. The means of the Gaussian vector are arbitrary.

Proof. Condition on P. The conditional covariance of (L, V) is at least hI_2. Indeed, for the full covariance matrix $M \geq hI_3$, its Schur complement satisfies

$$a^T K a = \inf_b (a, b)^T M (a, b) \geq h|a|^2.$$

Regress L on V: $L = \mu + \alpha V + W$, where W is an independent Gaussian with variance at least h, and V also has variance at least h. Given P, the variable Y is independent of these Gaussian residuals. For fixed $Y = y$, the W factor and the one-dimensional quadratic Gaussian integral imply

$$|\mathbb{E}[e^{itQ} \mid P, Y = y]| \leq e^{-2hy^2t^2}(1 + 4h^2t^2)^{-1/4}.$$

The added terms change only a linear phase in V and a constant. Completing the Gaussian square shows that the modulus contributed by the linear phase is at most one, for every conditional mean. Using both $g \leq 1$ and $\int g = 1$ gives

$$|\mathbb{E}e^{itQ}| \leq (1 + 4h^2t^2)^{-1/4} \min\left\{1, \frac{\sqrt{\pi}}{\sqrt{2h}|t|}\right\}. \tag{168}$$

This integrable bound is independent of P. Set $a = \sqrt{2\pi h}$ and $u = 2h|t|$. Fourier inversion gives

$$\|p_Q\|_\infty \leq a^{-2} \int_0^\infty (1 + u^2)^{-1/4} \min(1, a/u)\, du.$$

Bound $(1 + u^2)^{-1/4}$ by $u^{-1/2}$ and split at $u = a$. The integral is at most $4\sqrt{a}$, proving (167). For $h = (2\pi)^{-1}$, $a = 1$; splitting at one and using one below that point and $u^{-1/2}$ above it gives the sharper value $1 + 2 = 3$. The envelope decreases with h, so this holds for larger h too. □

Corollary 13.30 (A Gaussian covariance floor)**.** *Let $X' = Z + G$, where Z is an arbitrary random vector and G is an independent Gaussian vector with covariance at least $(2\pi)^{-1}I$. Let L, V, P be orthonormal linear projections and let Y have a conditional density at most one depending on X' only through P. Then every Q in Lemma 13.29 has density at most three. No spread hypothesis on the frame is needed.*

Proof. Generate the conditional hub after X', using independent auxiliary randomness. Conditional on Z, the vector (L, V, P) is Gaussian with the required covariance floor and with arbitrary mean. Its hub kernel is still $g(y \mid P)$. Apply Lemma 13.29 and average. □

Lemma 13.31 (A uniform partition of fixed-dimensional frames)**.** *For each positive integer d there is $\eta_d > 0$ such that every finite family $r_i \in \mathbb{R}^d$ satisfying*

$$\sum_i r_i r_i^T = I_d, \qquad |r_i|^2 \leq \tfrac{1}{2}$$

can be partitioned into A, B with

$$\sum_{i \in A} r_i r_i^T \geq \eta_d I_d, \qquad \sum_{i \in B} r_i r_i^T \geq \eta_d I_d. \tag{169}$$

The constant does not depend on the number of vectors.

Proof. We first recall the finite vector-matroid base-packing criterion: a family of rank d contains two disjoint bases if and only if $|E \setminus F| \geq 2(d - \operatorname{rank} F)$ for every subfamily F (Edmonds, 1965). In a Parseval frame as above, the energy orthogonal to a subspace of codimension k is k. Since a single vector contributes at most $1/2$, at least $2k$ vectors lie outside that subspace. Thus two disjoint bases exist for each fixed frame.

Suppose the uniform assertion were false. Choose frames whose best lower bound in (169) tends to zero. Sort each frame in decreasing order of vector norm, pad with zero vectors, and take a diagonal subsequence so that its ith vector converges to r_i for each fixed i. Because the total squared norm is d, every tail vector beyond index N has squared norm at most $d/(N+1)$. The limiting atoms satisfy $\sum_i r_i r_i^T \le I_d$. Put

$$S = I_d - \sum_i r_i r_i^T \ge 0, \qquad K = \ker S.$$

The projections of the limiting atoms onto K form a Parseval frame for K, each with squared norm at most $1/2$. They contain two disjoint finite bases of K. To justify this for the possibly infinite family, choose a finite initial segment whose covariance on K is at least $(1-\epsilon)I_K$, with $\epsilon < 1/(2d)$. Every codimension-k subspace of K then has outside energy greater than $k - 1/2$, so the number of outside vectors is at least $2k$. The finite base-packing criterion applies to this initial segment.

Let A_0, B_0 be the indices of these disjoint bases, taking both empty if $K = \{0\}$. The matrices

$$H_A = \sum_{i \in A_0} r_i r_i^T + S/2, \qquad H_B = \sum_{i \in B_0} r_i r_i^T + S/2$$

are positive definite. A null vector would belong to K and be orthogonal to a basis of K. Let $\gamma > 0$ be the smaller of their least eigenvalues. Choose N containing both bases and large enough that $d/(2\sqrt{N+1}) < \gamma/4$.

In each approximating frame assign the indicated basis vectors to their prescribed sides, and assign the other first N vectors arbitrarily. Partition the tail independently by fair coin flips. If T is the tail covariance and T_A its contribution to side A,

$$\mathbb{E}\|T_A - T/2\|_F^2 = \tfrac{1}{4} \sum_{i>N} |r_i^{\mathrm{app}}|^4 \le \frac{d^2}{4(N+1)}.$$

There is consequently a tail partition with operator error at most $d/(2\sqrt{N+1})$ on both sides. For fixed N, the tail covariance converges to $I_d - \sum_{i \le N} r_i r_i^T \ge S$. The prescribed basis covariances converge as well. Thus sufficiently late approximating frames have both partition covariances at least $\gamma I_d/2$, a contradiction. □

Theorem 13.32 (An adapted hub for Gaussian component mixtures)**.** *Let*

$$X_i = M_i + \sqrt{T_i}\, G_i, \qquad T_i > 0,$$

where the pairs (M_i, T_i) are independent across coordinates and the standard Gaussian variables G_i are independent of all these pairs. Suppose

$$\mathbb{E} T_i^{-1/2} \le \sqrt{2\pi} \qquad \text{for every } i. \tag{170}$$

Let w, v, e be orthonormal with $w_i^2 + v_i^2 + e_i^2 \le 1/2$, put $(L, V, P) = (\langle w, X' \rangle, \langle v, X' \rangle, \langle e, X' \rangle)$, and let Y have conditional density $g(y \mid P) \le 1$ of unit mass. For $|\lambda| \ge 1$ and arbitrary finite real measurable B, C,

$$Q = 2YL + \lambda V^2 + B(Y,P)V + C(Y,P)$$

has density bounded by

$$4\, 2^{-3/4} 3^{3/2} \eta_3^{-3/4}, \tag{171}$$

where η_3 is the constant in Lemma 13.31. In particular this gives a uniform bound for

$$aX_0^2 + 2X_0(L+\beta) + \lambda V^2 + 2\varepsilon LP$$

for every independent cap-one hub law, every a, β, ε, and every light law satisfying (170).

Proof. Put $r_i = (w_i, v_i, e_i)^T$ and choose a partition A, B from Lemma 13.31. Its two covariance sums satisfy $S_A \geq \eta_3 I_3$ and $S_B \geq \eta_3 I_3$. Realize Y after X' using fresh auxiliary randomness and the prescribed kernel $g(y \mid P)$. This preserves the joint law relevant to Q and ensures that the same kernel remains valid after conditioning on the latent mixture variables. Conditional on the mixture variables, the covariance of (L, V, P) is

$$M = \sum_i T_i r_i r_i^T = M_A + M_B.$$

If $h = \lambda_{\min}(M)$, Lemma 13.29 bounds the conditional density by $4(2\pi)^{-3/4} h^{-3/4}$, uniformly in all conditional means and in P. It remains to average this bound.

For side A, set $R_A = (S_A^{-1/2} r_i)_{i \in A}$, so that $R_A R_A^T = I_3$. For any positive diagonal matrix D, compression satisfies

$$(R_A D R_A^T)^{-1/2} \leq R_A D^{-1/2} R_A^T. \tag{172}$$

Indeed the corresponding resolvent inequality follows from a Schur complement, and integrating it against $\pi^{-1} s^{-1/2} ds$ proves (172), using $x^{-1/2} = \pi^{-1} \int_0^\infty s^{-1/2}(x+s)^{-1}\, ds$. Since $M_A = S_A^{1/2}[R_A \operatorname{diag}(T_i) R_A^T] S_A^{1/2}$, taking traces and using (170) gives

$$\mathbb{E}\lambda_{\min}(M_A)^{-1/2} \leq \eta_3^{-1/2} \sum_{i \in A} |S_A^{-1/2} r_i|^2 \mathbb{E} T_i^{-1/2} \leq 3\sqrt{2\pi}\, \eta_3^{-1/2}.$$

The same bound holds for M_B. The positive random variables $a_0 = \lambda_{\min}(M_A)$ and $b_0 = \lambda_{\min}(M_B)$ are independent, and $h \geq a_0 + b_0 \geq 2\sqrt{a_0 b_0}$. Therefore, by independence and the monotonicity of positive moments,

$$\mathbb{E} h^{-3/4} \leq 2^{-3/4} (\mathbb{E} a_0^{-1/2})^{3/4} (\mathbb{E} b_0^{-1/2})^{3/4} \leq 2^{-3/4} (3\sqrt{2\pi}\, \eta_3^{-1/2})^{3/2}.$$

Combining the two inequalities yields (171).

For the final assertion, set $Y = X_0 + \varepsilon P$. Then its conditional density is $f_0(y - \varepsilon P) \leq 1$, and

$$a X_0^2 + 2X_0(L + \beta) + \lambda V^2 + 2\varepsilon L P = 2YL + \lambda V^2 + a(Y - \varepsilon P)^2 + 2\beta(Y - \varepsilon P),$$

which is in the stated class with $B = 0$. □

Remark 13.33 (Scope of the mixture condition). Condition (170) bounds the average peak of the Gaussian components and implies that each marginal density is at most one. For centered Gaussian scale mixtures, meaning that each M_i is deterministic, it is equivalent to the cap-one assumption: the maximum is attained at M_i and equals $(2\pi)^{-1/2} \mathbb{E} T_i^{-1/2}$. Thus Theorem 13.32 covers every cap-one centered Gaussian scale mixture, even when its component variances have no positive lower bound. A mixture of many narrow components separated in location can have a bounded final density while violating (170); such laws are not covered.

Corollary 13.34 (Laplace light coordinates)**.** *Let the light coordinates have independent translated Laplace densities $e^{-2|x - m_i|}$. For the adapted forms of Theorem 13.32, no spread condition is needed, and*

$$\|p_Q\|_\infty \leq 12\pi^{-3/4}\Gamma(1/4) < 19.$$

Proof. The Laplace law has the representation $m_i + \sqrt{T_i} G_i$ with T_i exponential of rate two: its characteristic function is $\mathbb{E} e^{-t^2 T_i/2} = 4/(4 + t^2)$. For the covariance $M = \sum_i T_i r_i r_i^T$, the same resolvent compression argument, now with exponent $3/4$, gives

$$\mathbb{E}\lambda_{\min}(M)^{-3/4} \leq \mathbb{E} \operatorname{tr} M^{-3/4} \leq \sum_i |r_i|^2 \mathbb{E} T_i^{-3/4} = 3\, 2^{3/4} \Gamma(1/4).$$

Average (167). No partition of the frame is needed because the full negative $3/4$ moment exists here. □

13.10 A limitation of density-cap arguments for adapted hubs

Proposition 13.35 (Conditional information is not controlled by marginal caps)**.** *For every sufficiently large integer m, there is a joint law (L, V, P) whose joint density and all its coordinate marginal densities are bounded by one, and a conditional hub density $g(y \mid P) \le 1$ of unit mass, such that*

$$\|p_{2YL+V^2}\|_\infty \ge \frac{1}{4}\log m - \frac{1}{2}\log 9.$$

Here coordinate marginals include all one- and two-coordinate projections of this three-dimensional vector. The constructed joint law is not asserted to be an orthogonal projection of a product law.

Proof. Let $\delta = 1/m$. Partition $[0,1] \times [1,2]$ into squares

$$C_{jk} = [j\delta, (j+1)\delta] \times [1 + k\delta, 1 + (k+1)\delta], \qquad 0 \le j, k < m.$$

Assign to each square a distinct disjoint unit interval I_{jk} on the P axis. Put joint density one on $\bigcup_{j,k} C_{jk} \times I_{jk}$ and zero elsewhere. Its total integral is $m^2\delta^2 = 1$. The (L, V) marginal is uniform on the rectangle; the (L, P) and (V, P) densities are δ on their supports; and the P density is δ^2. The one-dimensional L, V densities are one.

Write (ℓ_j, v_k) for the center of C_{jk}. When $P \in I_{jk}$, let Y be uniform on the unit interval centered at $-v_k^2/(2\ell_j)$. This kernel depends only on P. For cells whose lower L endpoint is at least $8\sqrt{\delta}$, the root $-v^2/(2\ell)$ stays in this unit interval throughout the cell. The mean-value theorem bounds its distance from the center root by

$$\frac{\delta}{\ell_{\min}} + \frac{\delta}{\ell_{\min}^2} \le \frac{\sqrt{\delta}}{8} + \frac{1}{64} < \frac{1}{2}.$$

The same remains true for the root $(s - v^2)/(2\ell)$ when $|s| \le \sqrt{\delta}$, since the added displacement is at most $1/16$. Let b_m be the first grid boundary at least $8\sqrt{\delta}$; then $b_m \le 9\sqrt{\delta}$ for large m. For every such level s, the conditional density formula in Y therefore gives

$$p_{2YL+V^2}(s) \ge \int_{b_m}^{1} \int_1^2 \frac{dv\, d\ell}{2\ell} = \frac{1}{2}\log(1/b_m) \ge \frac{1}{4}\log m - \frac{1}{2}\log 9.$$

This holds on a nonempty interval of levels, and hence proves an essential-supremum lower bound. □

This example shows why the adapted problem requires more than caps on the displayed joint and marginal densities: P can encode a fine (L, V) cell and allow the hub to choose a separate center for that cell. It is an obstruction to a reduced proof strategy, not a counterexample to the bounded-density conjecture for independent coordinates. The mixture theorem above uses both the conditional Gaussian structure and the independence of two frame blocks, and therefore retains information absent from this example.

13.11 Collisions on shells and a logarithmic support dependence

Lemma 13.36 (A shell collision estimate)**.** *Let w, v, b be orthonormal, put $L = w \cdot X'$, $V = v \cdot X'$, $B = b \cdot X'$, and let $Y = \lambda(V - v_0)^2 + \gamma B$, where $\lambda \ne 0$. For almost every ℓ, let h_ℓ be the unnormalized density of Y on the slice $L = \ell$, restricted to $r \le |V - v_0| \le K$, where $0 < r < K$. Then*

$$\int_{\mathbb{R}} h_\ell(y)^2\, dy \le \frac{8}{|\lambda|}\log(K/r).$$

No bound on the coordinate leverages is required.

Proof. Take an independent copy (L', V', B') and put $S = V + V' - 2v_0$, $D = V - V'$, $E = B - B'$. The square integral is the collision density at $(\ell, \ell, 0)$ of $(L, L', Y - Y')$ for the two restricted measures, and $Y - Y' = \lambda SD + \gamma E$. First restrict to pairs on the same side of the fold v_0. Then $2r \le |S| \le 2K$. For fixed $S = s$ put $n = (\lambda s, \gamma)/\sqrt{\lambda^2 s^2 + \gamma^2}$. On the product space of the two copies, the coefficient vectors of

$$L, \quad L', \quad (S + 2v_0)/\sqrt{2}, \quad (n_1 D + n_2 E)/\sqrt{2}$$

are orthonormal. Their joint density is at most four by (146). Undoing the two factors $\sqrt{2}$ bounds the density in the unscaled coordinates by two. Restrictions of the underlying measure can only decrease it. Thus the contribution of the same-side pairs is at most

$$\int_{2r \le |s| \le 2K} \frac{2\,ds}{\sqrt{\lambda^2 s^2 + \gamma^2}} \le \frac{4}{|\lambda|} \log(K/r).$$

For opposite-side pairs fix $D = d$ instead. Now $2r \le |D| \le 2K$, and the four orthogonal directions are $L, L', D/\sqrt{2}, (n_1 S + n_2 E)/\sqrt{2}$, with $n = (\lambda d, \gamma)/\sqrt{\lambda^2 d^2 + \gamma^2}$. The same calculation gives the remaining four units. These collision identities can first be applied to mollified densities and bounded restrictions; disintegration and monotone convergence then give the asserted almost-everywhere statement. □

Theorem 13.37 (Logarithmic dependence on bounded light supports)**.** *Suppose each light coordinate has density at most one and is supported in an interval of length at most K. In the normal form at the start of this section suppose $a = 0$ and the light diagonal coefficients vanish. Then, uniformly in all cross terms, affine terms, and the hub law,*

$$\|p_Q\|_\infty \le \frac{1}{2} + \frac{C}{\sqrt{|\lambda|}} \sqrt{1 + \log(1 + K^2|\lambda|)}$$

for an absolute constant C.

Proof. We have $K \ge 1$. On every frozen slice the light polynomial, after completing the square, has the form in Lemma 13.36, up to a constant; its direction b, coefficient γ, and fold location may depend on the frozen value. A missing b direction can be supplied by an extra, independent coordinate whose coefficient in Y is zero. The exponential-tilt proof of Proposition 13.18, applied just in direction v, gives

$$\rho_{L,V}(\ell, u) \le 2 \exp\{-2(|u - \mathbb{E}V| - 2K)_+^2 / K^2\}.$$

Consequently the slice mass on $|V - \mathbb{E}V| > 2K + t$ is at most $\sqrt{2\pi} K e^{-2t^2/K^2}$.

Put $W = 2K + t$ and $D_0 = |v_0 - \mathbb{E}V|$. On the remaining part of the slice and outside the fold slab of radius $r \le W$, the values of $|V - v_0|$ lie between $\max(r, D_0 - W)$ and $D_0 + W$. The ratio of these endpoints is at most $3W/r$: if $D_0 \le 2W$ use the upper bound $3W$, and otherwise use the ratio bound three. The fold slab has slice mass at most $4r$, since $\rho_{L,V} \le 2$. Cauchy–Schwarz and Lemma 13.36 therefore give, for every Borel set S,

$$\mu_{\ell_0, \ell}(S) \le 4r + \sqrt{\frac{8}{|\lambda|} \log(3W/r)} \sqrt{|S|} + \sqrt{2\pi} K e^{-2t^2/K^2}.$$

Here and below harmless changes in the polynomial's additive constant do not affect the estimate.

Write $x = |S|$ and $\mu = |\lambda|$. If $x < \mu$, take

$$r = \sqrt{x/\mu}, \qquad t = K \sqrt{\tfrac{1}{2} \log_+(\sqrt{2\pi} K \sqrt{\mu/x})}.$$

Then $r < 1 \le K \le W$, and the last term is at most $\sqrt{x/\mu}$. For an absolute C_0 this proves

$$\mu_{\ell_0,\ell}(S) \le \frac{C_0}{\sqrt{\mu}}\sqrt{x\log\Big(e + \frac{1+K^2\mu}{x}\Big)}.$$

Indeed $\log(3W/r) \le C[1+\log(K\sqrt{\mu/x})]$, since the extra square root of a logarithm in W contributes at most the same expression. If $x \ge \mu$, the displayed bound follows instead from the total slice mass $\rho_L(\ell) \le \sqrt{2}$, after enlarging C_0.

This last displayed expression defines an increasing concave profile for all $x > 0$, with value zero at zero. To check concavity, $x \mapsto x\log(e + A/x)$ is increasing and concave for every $A > 0$, as direct differentiation shows; composition with the increasing concave square root preserves both properties. Finally, substituting $x = 2e^{-u}$ shows

$$\int_0^2 \sqrt{x\log(e + A/x)}\,\frac{dx}{x} \le C_1\sqrt{1+\log(1+A)}.$$

For example use $\log(e + (A/2)e^u) \le u + \log(e + A/2)$ and integrate $e^{-u/2}\sqrt{u+\log(e+A/2)}$. Lemma 13.4 proves the theorem. □

Proposition 13.38 (A weighted collision criterion)**.** *In the notation of Lemma 13.36, set*

$$G = \sqrt{\lambda^2(|V-v_0| + |V'-v_0|)^2 + \gamma^2}, \qquad Z = (Y - Y')/G.$$

Suppose that, uniformly in the frozen parameters, the disintegrated collision density of (L, L', Z) at $(\ell, \ell, 0)$ is at most C_0 for almost every ℓ. More explicitly, in each of the two branch classes used in the proof of Lemma 13.36, assume that the nonnegative density $F_\ell(s)$ in the remaining shell coordinate at $Z = 0$ satisfies $\int F_\ell(s)\,ds \le C_0$. Then the density restricted to $|V - v_0| \ge r$ satisfies

$$\int h_\ell(y)^2\,dy \le \frac{8}{|\lambda|}\log\Big(1 + \frac{C_0}{8r}\Big).$$

In particular, if this hypothesis holds uniformly and $|\lambda| \ge 1$, the corresponding zero-diagonal one-hub forms have uniformly bounded densities, with a constant depending only on C_0.

Proof. In the same-side class use $s = S$ and in the opposite-side class use $s = D$. The orthogonal projection calculation above gives $0 \le F_\ell(s) \le 2$, even though its direction depends on s. The collision integral in either class is at most $|\lambda|^{-1}\int_{|s|\ge 2r} F_\ell(s)|s|^{-1}\,ds$. The maximum under the constraints $F \le 2$ and $\int F \le C_0$ is obtained by filling the intervals closest to the two endpoints $\pm 2r$. Its value is $4|\lambda|^{-1}\log(1 + C_0/(8r))$. Adding the two classes proves the first assertion. The fold slab again has mass at most $4r$. Hence

$$\mu_{\ell_0,\ell}(S) \le 4r + \sqrt{\frac{8|S|}{|\lambda|}\log(1 + C_0/(8r))}.$$

For $|\lambda| \ge 1$, choose $r = \sqrt{|S|}$ near zero and use the trivial slice mass bound away from zero. An increasing concave majorant is $C(C_0)\sqrt{x\log(e + (1+C_0)^2/x)}$. Its integral against dx/x on $(0, 2)$ is finite, so Lemma 13.4 applies. □

The weighted collision hypothesis is an additional nonlinear condition. The flat projection density bound proves the pointwise bound $F_\ell \le 2$ but does not prove its integral bound. Thus the proposition gives a sufficient condition whose validity for arbitrary product densities remains open.

13.12 Averaging the hub quadratic phase

Lemma 13.39 (A parabola extension estimate)**.** *If $0 \le f \le 1$ and $\int f = 1$, and $g(s,\tau) = \int f(x) e^{isx^2 + i\tau x}\,dx$, then*

$$\iint_{\mathbb{R}^2} |g(s,\tau)|^6\,ds\,d\tau \le \frac{4\pi^3}{\sqrt{3}}.$$

The same bound holds if f in the definition of g is replaced by a complex function dominated in absolute value by f.

Proof. The cube g^3 is the Fourier transform of the pushforward of $f(x_1)f(x_2)f(x_3)\,dx$ under $T(x) = (x_1 + x_2 + x_3, x_1^2 + x_2^2 + x_3^2)$. If r is the distance from x to the line spanned by $(1,1,1)$, the coarea Jacobian is $2\sqrt{3}\,r$. Each nonempty regular fiber is a circle of radius r. Its image density m is consequently at most $\pi/\sqrt{3}$, since the product of the three densities is at most one. It has integral one. Plancherel gives

$$\iint |g|^6 = (2\pi)^2 \int m^2 \le (2\pi)^2(\pi/\sqrt{3}) \int m.$$

For a complex amplitude, the absolute value of its pushforward is bounded by the same nonnegative density m, so the argument is unchanged. This coarea proof is the elementary one-dimensional parabola case of the Fourier extension estimates associated with Strichartz (1977); Foschi (2007); Hundertmark and Zharnitsky (2006). □

Theorem 13.40 (A degenerate hub with a quadratic partner)**.** *Let l, u be linearly independent, let $L = l \cdot X'$, $U = u \cdot X'$, and put $G = \operatorname{Gram}(l,u)$. Assume $(l_i, u_i)G^{-1}(l_i, u_i)^T \le 1/2$ for every i. For independent coordinates with density at most one, consider*

$$Q = aH^2 + 2H(L + \beta) + \sigma L^2 + \lambda U^2 + \sum_i \phi_i(X_i),$$

where the ϕ_i are arbitrary quadratic polynomials. If $\sigma\lambda \ne 0$ and $a\sigma \ne 1$, define

$$C_6 = (4\pi^3/\sqrt{3})^{1/6},$$
$$B_0 = (4\pi)^{5/6} C_6 |\lambda|^{-1/2} |\sigma|^{-1/6} |1 - a\sigma|^{-1/6} (\det G)^{-5/12} |u|^{-1/6}.$$

Then, for every $T > 0$,

$$\int_T^{2T} |\mathbb{E} e^{itQ}|\,dt \le B_0 T^{-1/6}, \tag{173}$$

and

$$\|p_Q\|_\infty \le \frac{7}{\pi} \left(\frac{B_0}{6(1 - 2^{-1/6})} \right)^{6/7}.$$

Proof. The leverage assumption also implies $u_i^2/|u|^2 \le 1/2$. Absorb all linear hub terms into β. Write $\phi_i(x) = d_i x^2 + b_i x + c_i$ and $\psi_i^{(t)}(\theta) = \int f_i(x) e^{itd_i x^2 + i\theta x}\,dx$. For a fixed hub value x use the exact identity

$$ax^2 + 2xL + \sigma L^2 = \sigma(L + x/\sigma)^2 + (a - 1/\sigma)x^2.$$

Two Fresnel linearizations, with Gaussian damping followed by dominated convergence, therefore give

$$|\varphi_Q(t)| \le \frac{1}{4\pi|t|\sqrt{|\sigma\lambda|}} \int_{\mathbb{R}} |g_H(t(a - 1/\sigma), 2t\beta + \zeta/\sigma)| K_t(\zeta)\,d\zeta,$$

where $K_t(\zeta) = \int \prod_i |\psi_i^{(t)}(\zeta l_i + \xi u_i + t b_i)|\,d\xi$. Lemma 13.1, first in the u direction and then in both directions, yields

$$\|K_t\|_\infty \le 4\pi/|u|, \quad \|K_t\|_1 \le (4\pi)^2 (\det G)^{-1/2}, \quad \|K_t\|_{6/5} \le (4\pi)^{11/6} (\det G)^{-5/12} |u|^{-1/6}.$$

Apply Hölder with exponents 6 and 6/5 on $[T, 2T] \times \mathbb{R}$. The map

$$(t, \zeta) \longmapsto (t(a - 1/\sigma),\ 2t\beta + \zeta/\sigma)$$

has Jacobian $|1-a\sigma|/\sigma^2$. Lemma 13.39 bounds the first factor by $C_6(\sigma^2/|1-a\sigma|)^{1/6}$. The second factor is at most $T^{5/6}(4\pi)^{11/6}(\det G)^{-5/12}|u|^{-1/6}$. Multiplying these factors and $1/(4\pi T\sqrt{|\sigma\lambda|})$ proves (173). Since Q is real, the negative-frequency bound is identical. The full Fourier integral is at most

$$2T_0 + \frac{2B_0T_0^{-1/6}}{1 - 2^{-1/6}}.$$

Its minimum occurs at $T_0 = [B_0/(6(1 - 2^{-1/6}))]^{6/7}$. Fourier inversion proves the density estimate. □

Theorem 13.41 (Uniformity through zero partner curvature)**.** *In Theorem 13.40, assume $a = 0$. There is a finite bound depending only on $|\lambda|$, $\det G$, and $|u|$, uniform in every $\sigma \in \mathbb{R}$, in the affine terms, and in the light diagonal terms. One explicit bound is*

$$\|p_Q\|_\infty \le 1 + \frac{13}{4}C_H + C_{\mathrm{av}}|\lambda|^{-1/2}(\det G)^{-5/12}|u|^{-1/6}, \qquad C_H = 3\,2^{4/3}(|\lambda| \det G)^{-1/3},$$

where

$$C_{\mathrm{av}} = \frac{(4\pi)^{5/6}C_6}{\pi}\left\{\frac{2}{1 - 2^{-1/6}} + \frac{2}{3(2^{11/6} - 1)}\right\}.$$

Proof. When $\sigma = 0$, Proposition 13.13 with empty localization set, after changing the two frame coordinates, gives the bound C_H. Suppose $\sigma \neq 0$. Set $Q_1 = Q - \sigma L^2$ and $Q_2 = Q - \sigma(L + \beta)^2$. Each has the preceding zero-curvature form and density at most C_H, since their difference consists only of affine light terms. Fix $\delta \ge |\sigma|$ and put $\Lambda = \sqrt{\delta/|\sigma|} \ge 1$. On $|L| \le \Lambda$, comparison with Q_1 bounds a window of radius δ by $4C_H\delta$. On $|L| > \Lambda$, $|L + \beta| \ge \Lambda/2$, conditioning on X' and using the affine hub slope gives probability at most $\delta/|L + \beta| \le 2\delta$. On the remaining event, $|Q - Q_2| \le |\sigma|(L + \beta)^2 < \delta/4$, so comparison with Q_2 gives at most $(5/2)C_H\delta$. Consequently

$$\sup_s \frac{\mathbb{P}(|Q - s| \le \delta)}{2\delta} \le 1 + \frac{13}{4}C_H.$$

To pass to smaller scales, put $h = |\sigma|$. The preceding theorem gives an integrable characteristic function. For the uniform probability density U_h on $[-h, h]$, Fourier inversion and $0 \le 1 - \sin z/z \le \min(2, z^2/6)$ imply

$$\|p_Q - p_Q * U_h\|_\infty \le \frac{1}{2\pi}\int |\varphi_Q(t)|(1 - \mathrm{sinc}(ht))\,dt.$$

Use (173) on blocks $T = 2^j/h$, $j \in \mathbb{Z}$. On such a block the last factor is at most $\min(2, (2/3)4^j)$. The geometric sums over $j \ge 0$ and $j \le -1$ are respectively bounded by $2/(1 - 2^{-1/6})$ and $2/[3(2^{11/6} - 1)]$. Thus the error is at most $B_0h^{1/6}$ times their sum divided by π. When $a = 0$, the factor $|\sigma|^{-1/6}$ in B_0 cancels $h^{1/6}$ exactly. The estimate for $p_Q * U_h$ already follows from the window bound at $\delta = h$. Adding the two bounds proves the result. □

Corollary 13.42 (A normalized degenerate hub)**.** *If w, v are orthonormal and $w_i^2 + v_i^2 \le 1/2$, then*

$$4H(w \cdot X') + 2(w \cdot X')^2 + 2(v \cdot X')^2 + \sum_i \phi_i(X_i) + b_H H$$

has density less than 18*, for every choice of the independent coordinate laws with density at most one.*

Proof. Apply Theorem 13.40 with $l = 2w$, $u = v$, $a = 0$, $\sigma = 1/2$, and $\lambda = 2$. Then $\det G = 4$, $|u| = 1$, and the exact expression for B_0 is less than 7.5. The resulting exact density bound is less than 18. □

For a quadratic polynomial confined to $\operatorname{span}(w, v)$, completing the square in V produces $\lambda'(V+\kappa L+\gamma)^2+\sigma L^2$ and affine terms. The replacement $u = v+\kappa w$ preserves $\det \operatorname{Gram}(w, u) = 1$ and the relative leverage $w_i^2 + v_i^2$. Hence the uniform theorem applies whenever the transverse coefficient $|\lambda'|$ stays bounded below. For $a \neq 0$, the bound retains its factors $|\sigma|^{-1/6}|1 - a\sigma|^{-1/6}$ and diverges as the corresponding curvatures vanish.

13.13 Hub diagonals and square-root profiles with random centers

Here H is independent of (t, R), has density $f_H \leq 1$, and

$$Q = aH^2 + 2Ht + R.$$

The random variable t may be a shifted linear projection of the light coordinates. We use the following explicitly joint hypothesis: there are a random center C coupled with t and a constant K such that for every nonnegative Borel function Φ,

$$\mathbb{E}\Phi(t, R) \leq K\mathbb{E}\int_{\mathbb{R}} \Phi(t, r)|r - C|^{-1/2}\, dr. \tag{174}$$

For example, if $R = \lambda(V - D_0)^2 + C$ and V has conditional density at most κ given (t, D_0, C), then (174) holds with $K = \kappa/\sqrt{|\lambda|}$, by the two-root change of variables. The auxiliary variables used to represent the right-hand side of (174) are sampled independently of H.

Lemma 13.43 (A scalar discriminant bound)**.** *For a probability density $0 \leq f \leq 1$ and $\Delta = t^2 + as \neq 0$,*

$$\int_{\mathbb{R}} f(x)|s - ax^2 - 2tx|^{-1/2}\, dx \leq 2^{7/4}|\Delta|^{-1/4}.$$

If $a = 0$ and $t \neq 0$, the sharper bound is $2|t|^{-1/2}$.

Proof. For $a \neq 0$, put $A = |a|$, $d = |\Delta|$, and $z = A(x + t/a)/\sqrt{d}$. The sublevel condition $|s - ax^2 - 2tx| < u$ becomes $|z^2 - \operatorname{sgn}\Delta| < Au/d$. If $\Delta > 0$, the length of this set in the z variable is

$$2\{\sqrt{1 + r} - \sqrt{(1 - r)_+}\}, \qquad r = Au/d,$$

which is at most $2\sqrt{2}\, r$. For $r \leq 1$ rationalize the difference and use $\sqrt{1 + r} + \sqrt{1 - r} \geq \sqrt{2}$; for $r \geq 1$ use $\sqrt{1 + r} \leq \sqrt{2}\, r$. If $\Delta < 0$, its length is $2\sqrt{(r - 1)_+} \leq r$, so the same bound holds. After returning to x,

$$|\{x : |s - ax^2 - 2tx| < u\}| \leq 2\sqrt{2}\, u/\sqrt{d}.$$

Layer cake and $f \leq 1$, $\int f = 1$ yield

$$\int f|s - ax^2 - 2tx|^{-1/2} \leq \int_0^\infty \min(1, 2\sqrt{2}\, v^{-2}/\sqrt{d})\, dv = 2^{7/4}d^{-1/4}.$$

For $a = 0$ the sublevel length is exactly $u/|t|$, and the same integral gives $2|t|^{-1/2}$. □

Theorem 13.44 (Composition with a random center)**.** *Under (174), for almost every s,*

$$p_Q(s) \leq 1 + 2^{7/4}K\,\mathbb{E}|t^2 + a(s - C)|^{-1/4}. \tag{175}$$

If $C = \sigma_c t^2 + \eta_1 t + \eta_0 + C'$ for deterministic σ_c, η_1, η_0, and the conditional density of t given C' is bounded by D, then, provided $D_H = 1 - a\sigma_c \neq 0$,

$$\|p_Q\|_\infty \leq 1 + 2^{13/4}K\sqrt{D}\,|D_H|^{-1/4}.$$

At $a = 0$ the first bound requires no information about the random center beyond (174).

Proof. For fixed (t, r) the roots of $ax^2 + 2tx + r = s$ have derivative $2|ax + t|$. Roots with $|ax + t| \geq 1$ contribute at most one in total. Let

$$\mathcal{K}_s(t, r) = \sum_{\substack{x:\, ax^2+2tx+r=s \\ |ax+t|<1}} \frac{f_H(x)}{2|ax + t|}.$$

Changing r at fixed t on each monotonicity branch gives

$$\int \mathcal{K}_s(t, r)\Psi(r)\, dr = \int f_H(x)\mathbf{1}_{\{|ax+t|<1\}}\Psi(s - ax^2 - 2tx)\, dx.$$

Apply (174) to $\mathcal{K}_s$, use this identity, and drop the indicator. Lemma 13.43, with its level parameter replaced by $s - C$, proves (175). The formulas also follow by smoothing in s, which avoids exceptional critical values; Tonelli then justifies all the nonnegative integrals.

Conditionally on C', the discriminant in the second assertion is a quadratic polynomial in t with leading coefficient D_H. For real roots, factor it and use Cauchy–Schwarz. For conjugate nonreal roots, its absolute value is at least $|D_H|(t - r)^2$, where r is their common real part. In either case

$$\mathbb{E}(|t^2 + a(s - C)|^{-1/4} \mid C') \leq |D_H|^{-1/4} \sup_r \mathbb{E}(|t - r|^{-1/2} \mid C') \leq 2\sqrt{2}\,\sqrt{D}\,|D_H|^{-1/4}.$$

The final bound is the bathtub integral over the interval of length $1/D$ centered at r. Multiplication of the constants proves the result. □

Proposition 13.45 (Randomness at the degenerate center)**.** *Assume* (174)*, the center representation, and the conditional density cap D in Theorem 13.44, now allowing $D_H = 0$. Suppose $\sigma_c \neq 0$. For a random variable Z write $\alpha_1(Z) = \sup_{s\in\mathbb{R},\, u>0} \mathbb{P}(|Z - s| \leq u)/\sqrt{u}$. With the convention that the second argument below is infinite when $D_H = 0$, put*

$$A_* = \min\left\{\alpha_1(C'), 2\sqrt{2}\sqrt{|\sigma_c|/|D_H|}\right\}.$$

If $A_* < \infty$, then

$$\|p_Q\|_\infty \leq 1 + KD|\sigma_c|^{-1/2}\{12 + 2\log_+ |\sigma_c| + 4\log_+ A_*\}.$$

In particular, center randomness with a finite $\alpha_1(C')$ gives a finite bound at $D_H = 0$.

Proof. Retain the indicator in the proof of Theorem 13.44, and condition on (H, C'). The density cap D reduces the remaining integral to

$$D\int_{|ax+\tau|<1} |s - c' - \eta_0 - ax^2 - (2x + \eta_1)\tau - \sigma_c\tau^2|^{-1/2}\, d\tau.$$

The interval has length two. A quadratic with leading coefficient $-\sigma_c$ and critical value h_c has integral over any such interval at most

$$|\sigma_c|^{-1/2}\{8 + 2\log_+(|\sigma_c|/|h_c|)\}. \tag{176}$$

To verify this bound, after translating and dividing by $|\sigma_c|$ write the integrand as $|z^2 - c|^{-1/2}$. On $|z| \leq 2\sqrt{|c|}$ its integral over the whole region is at most $\pi + 2\,\mathrm{arcosh}\,2 < 6$. Outside that region it is at most $2/(\sqrt{3}|z|)$. On a set of length at most two the integral of this last majorant is at most $(4/\sqrt{3})\log(1 + 1/(2\sqrt{|c|}))$. Since this is at most $(4/\sqrt{3})\log(3/2) + (2/\sqrt{3})\log_+(1/|c|)$, (176) follows. The case $c = 0$ is interpreted with an infinite right side.

The critical value here is $s - C' - q_c(H)$, where

$$q_c(x) = \eta_0 + ax^2 - (x + \eta_1/2)^2/\sigma_c,$$

whose quadratic coefficient is $-D_H/\sigma_c$. The variables C' and H are independent. Convolution does not increase the concentration function, so

$$\mathbb{P}(|C' + q_c(H) - s| \le u) \le \min(1, A_*\sqrt{u}).$$

For the bound from $q_c(H)$ use the elementary quadratic sublevel length $2\sqrt{2u/|D_H/\sigma_c|}$ and $f_H \le 1$. Layer cake now gives

$$\mathbb{E}\log_+ \frac{1}{|s - C' - q_c(H)|} \le \int_0^\infty \min(1, A_* e^{-v/2})\, dv \le 2 + 2\log_+ A_*.$$

Insert this and $\log_+(|\sigma_c|/|h_c|) \le \log_+ |\sigma_c| + \log_+(1/|h_c|)$ into (176). The near-root contribution is the asserted expression after multiplication by KD, and the far roots cost at most one. □

Corollary 13.46 (An independent transverse square)**.** *Suppose $V = v \cdot X_S$ and $t = w \cdot X_T + \beta$ use disjoint coordinate sets, and D_0, C are arbitrary functions of X_T. For $Q = aH^2 + 2Ht + \lambda(V - D_0)^2 + C$,*

$$\|p_Q\|_\infty \le \frac{2^{13/4}}{|v|\sqrt{|\lambda|}}\, \alpha_1(t^2 - aC)^{1/2}.$$

When $a = 0$ one has the unconditional bound

$$\|p_Q\|_\infty \le 2^{13/4} |\lambda|^{-1/2} |v|^{-1} |w|^{-1/2}.$$

Proof. Condition first on (H, X_T) and integrate the independent square. Since $\|p_V\|_\infty \le \sqrt{2}/|v|$, the resulting density is bounded by

$$\frac{\sqrt{2}}{|v|\sqrt{|\lambda|}}\, \mathbb{E}|s - aH^2 - 2Ht - C|^{-1/2}.$$

There is no far-root term in this argument. The scalar lemma bounds the expectation by $2^{7/4}\mathbb{E}|t^2 + a(s - C)|^{-1/4}$. For any Z with $\alpha_1(Z) < \infty$, layer cake gives $\mathbb{E}|Z - s|^{-1/4} \le 2\alpha_1(Z)^{1/2}$. This proves the first displayed bound. For $a = 0$, use instead the scalar estimate $2|t|^{-1/2}$ and the linear density cap $\sqrt{2}/|w|$. Its bathtub integral is $2\sqrt{2}(\sqrt{2}/|w|)^{1/2}$, which yields the second bound. □

The square-root hypothesis is a statement about the joint law with the hub partner. A concentration estimate for the marginal law of R alone cannot replace it. For example, if $R = 2Yt$ with an additional independent variable Y, then at $a = 0$ the complete form is $2(H + Y)t$; for suitable interval laws this product has a logarithmically unbounded density. Likewise, at $C = t^2/a$ and opposite signs of a and λ, the form $(aH + t)^2/a + \lambda V^2$ has a hyperbolic two-dimensional singularity. These examples have only two effective quadratic directions and explain the determinant and center-randomness requirements without contradicting the rank-three target.

14 Convolution bounds for independent coordinate blocks

The estimates in this section use disjoint groups of the original independent coordinates. A phase carried by one group may be arbitrary, but the independence required of a separate smoothing term is explicit.

14.1 Product densities and two-coordinate quadratic forms

Lemma 14.1 (A sharp product-density estimate)**.** *Let U, V be independent real random variables with probability densities $f \le M$ and $g \le N$. The density q of UV satisfies*

$$\|q\|_2^2 \le 4MN. \tag{177}$$

Equality holds for the centered uniform densities of heights M, N. Consequently, for independent cap-one variables U, V and $c \neq 0$,

$$\|p_{2cUV+\alpha U+\beta V+\gamma}\|_2 \le \sqrt{2}\,|c|^{-1/2} \tag{178}$$

for all real α, β, γ.

Proof. The change of variables $(x, y) \mapsto (x, xy)$, away from the null set $x = 0$, gives $q(z) = \int f(x)g(z/x)\,dx/|x|$ for almost every z. All integrands in the following calculation are nonnegative, so Tonelli's theorem applies even before finiteness is known. Expanding the square, putting $z = xy$, and then $x = rx'$ gives

$$\begin{aligned}\|q\|_2^2 &= \int_{\mathbb{R}^2} \frac{f(x)f(x')}{|xx'|} \int_{\mathbb{R}} g(z/x)g(z/x')\,dz\,dx\,dx' \\ &= \int_{\mathbb{R}} K_f(r)K_g(r)\,dr, \qquad K_f(r) = \int_{\mathbb{R}} f(x)f(rx)\,dx.\end{aligned}$$

The mass and height bounds imply $K_f(r) \le M\min(1, |r|^{-1})$ and $K_g(r) \le N\min(1, |r|^{-1})$ for $r \neq 0$. Since the integral of $\min(1, |r|^{-1})^2$ over the real line equals 4, this proves (177). For centered uniform densities both correlation inequalities are equalities, proving sharpness.

Finally,

$$2cUV + \alpha U + \beta V + \gamma = 2c\left(U + \frac{\beta}{2c}\right)\left(V + \frac{\alpha}{2c}\right) + \gamma - \frac{\alpha\beta}{2c}.$$

Translations preserve the caps and independence. For a scalar dilation by $a \neq 0$, the squared L^2 norm of a density is divided by $|a|$. Apply (177) with $M = N = 1$ and $a = 2c$. □

Theorem 14.2 (Two coordinates with an independent L^2 remainder)**.** *Let $H = (H_1, H_2)$ and $Y = (Y_1, \dots, Y_m)$ consist of independent cap-one original coordinates. Let Z be independent of (H, Y) and have a density in $L^2(\mathbb{R})$. Suppose*

$$Q = H^T C H + 2H^T(BY + d) + R(Y) + Z,$$

where C is real symmetric, B is a real 2-by-m matrix of rank two, $d \in \mathbb{R}^2$, and R is an arbitrary finite measurable function. Put

$$D = \det(BB^T), \qquad P = \|p_{BY+d}\|_2, \qquad R_2 = \|p_Z\|_2.$$

Then Q has a density in $C_0(\mathbb{R})$ and

$$\|p_Q\|_\infty \le 3\,2^{-2/3}(PR_2)^{2/3} \le 3\,2^{-1/3}D^{-1/6}R_2^{2/3}. \tag{179}$$

If $Z = 2cUV + \alpha U + \beta V + \gamma$, where U, V are independent cap-one coordinates independent of (H, Y) and $c \neq 0$, then

$$\|p_Q\|_\infty \le 3D^{-1/6}|c|^{-1/3}. \tag{180}$$

Proof. By Lemma 4.2, the density of $S = BY + d$ is at most $2D^{-1/2}$. Thus $P \le \sqrt{2}D^{-1/4}$. For fixed t, push the complex measure $e^{itR(Y)}\,d\mathbb{P}_Y$ forward under $Y \mapsto S$. Its total variation is dominated by the law of S, so it has a density a_t with $|a_t| \le p_S$ almost everywhere. In particular, $\|a_t\|_2 \le P$.

Let $V = Q - Z$. The joint density of H has L^2 norm at most one. Cauchy–Schwarz in the original H coordinates, followed by two-dimensional Plancherel, yields

$$\begin{aligned}|\varphi_V(t)| &\le \left(\int_{\mathbb{R}^2}\left|\int_{\mathbb{R}^2} e^{2ith^T s}a_t(s)\,ds\right|^2 dh\right)^{1/2}\\ &= (2|t|)^{-1}\|\widehat{a}_t\|_2 = \frac{\pi}{|t|}\|a_t\|_2 \le \frac{\pi P}{|t|} \qquad (t \ne 0).\end{aligned}$$

The factor $e^{ith^T Ch}$ has modulus one throughout this argument. In particular, no sign condition on C is needed.

Independence of Z gives $\varphi_Q = \varphi_V\varphi_Z$. For any $T > 0$, Plancherel and Cauchy–Schwarz give

$$\begin{aligned}\int_{\mathbb{R}} |\varphi_Q(t)|\,dt &\le 2T + \pi P\int_{|t|>T}|t|^{-1}|\varphi_Z(t)|\,dt\\ &\le 2T + \pi P(2/T)^{1/2}(2\pi)^{1/2}R_2\\ &= 2T + 2\pi^{3/2}PR_2T^{-1/2}.\end{aligned}$$

Take $T = (\pi^{3/2}PR_2/2)^{2/3}$. The last expression is $6T$. Fourier inversion therefore gives a C_0 density with bound $6T/(2\pi) = 3\,2^{-2/3}(PR_2)^{2/3}$. Substituting the bound for P proves (179). Equation (178) proves (180). □

Remark 14.3. The independence of Z is a hypothesis of Theorem 14.2. The function $R(Y)$ can depend on the entire slope bank Y, but the proof does not permit replacing Z by a variable coupled to that bank. For example, if $BB^T = \kappa^2 I_2$ and $|c| = \kappa^{-2}$, the bound in (180) is 3, independently of κ and of the measurable phase R.

14.2 A quadratic pair and an independent square

Lemma 14.4 (Fractional integrals of a capped density)**.** *For a probability density $f \le 1$, $a \in \mathbb{R}$, and $0 < \alpha < 1$,*

$$\int_{\mathbb{R}} f(x)|x-a|^{-\alpha}\,dx \le \frac{2^\alpha}{1-\alpha}. \tag{181}$$

Proof. Layer cake and the density cap bound the left side by

$$\int_0^\infty \min(1, 2t^{-1/\alpha})\,dt = \frac{2^\alpha}{1-\alpha}.$$

The bound is attained by the uniform density on $[a-1/2, a+1/2]$. □

Lemma 14.5 (A pair with one zero quadratic coefficient)**.** *Let X, Y be independent cap-one variables and let $b \ne 0$. For every $1 < p < \infty$, the variable*

$$T = aX^2 + 2bXY + 2uX + 2vY + c$$

has a density satisfying

$$\|p_T\|_p \le p\,|b|^{-1+1/p}. \tag{182}$$

Proof. For almost every x, conditioning on $X = x$ leaves an affine function of Y of nonzero slope $2(bx + v)$. Writing $\alpha = 1 - 1/p$, its density has L^p norm at most $|2(bx + v)|^{-\alpha}$, since $\|f_Y\|_p \le 1$. Minkowski's integral inequality and Lemma 14.4 give

$$\|p_T\|_p \le (2|b|)^{-\alpha} \int f_X(x)|x + v/b|^{-\alpha}\, dx \le p\,|b|^{-\alpha}.$$

The quadratic term aX^2 and the remaining terms only translate the conditional density. □

Lemma 14.6 (An L^p bound for a square)**.** *For a cap-one variable X, $\gamma \neq 0$, and $1 < p < 2$,*

$$\|p_{\gamma X^2+2wX}\|_p^p \le |\gamma|^{1-p}\frac{2^{p-1}}{2-p}. \tag{183}$$

In particular, $\|p_{\gamma X^2+2wX}\|_{3/2} \le 2|\gamma|^{-1/3}$.

Proof. Complete the square and put $m = -w/\gamma$. The two-root density formula and $ds = 2|\gamma|r\, dr$ give

$$\begin{aligned}\|p_{\gamma X^2+2wX}\|_p^p &= (2|\gamma|)^{1-p}\int_0^\infty (f_X(m + r) + f_X(m - r))^p r^{1-p}\, dr \\ &\le |\gamma|^{1-p}\int_{\mathbb{R}} f_X(x)|x - m|^{-(p-1)}\, dx.\end{aligned}$$

Here the sum of the two density values is at most two. Apply (181) with $\alpha = p - 1$. □

Theorem 14.7 (A physical pair and a square)**.** *Suppose A has an original-coordinate principal block*

$$\begin{pmatrix} a & b & 0 \\ b & 0 & 0 \\ 0 & 0 & \gamma \end{pmatrix}, \qquad b\gamma \neq 0.$$

All other blocks of A and all affine terms are arbitrary. Then $Q = X^T AX + b_0^T X + c_0$ has a density in $C_0(\mathbb{R})$ and

$$\|p_Q\|_\infty \le 6|b|^{-2/3}|\gamma|^{-1/3}. \tag{184}$$

If $b^2 \ge |a| + 1$ and $|\gamma| \ge 1$, this is at most 6; these conditions also guarantee three eigenvalues of modulus at least one in the displayed principal block.

Proof. Condition on all coordinates outside the displayed block. The remaining polynomial is a sum of two independent variables of the forms in Lemmas 14.5 and 14.6, with arbitrary induced affine coefficients. Young's inequality with exponents 3 and 3/2 gives

$$\|p_T * p_S\|_\infty \le \|p_T\|_3\|p_S\|_{3/2} \le 6|b|^{-2/3}|\gamma|^{-1/3}.$$

The convolution belongs to C_0: approximate its two factors in their respective L^p norms by continuous compactly supported functions and use Young's inequality to obtain uniform convergence of the convolutions. The bound is uniform in the conditioned coordinates. Measurable conditional versions may be obtained by taking limits of centered interval averages of the conditional laws. Dominated convergence then shows that their average is continuous and vanishes at infinity, and retains the bound.

The two eigenvalues of the upper-left pair are $(a \pm \sqrt{a^2 + 4b^2})/2$. Both have modulus at least one exactly when $b^2 \ge |a| + 1$. The third eigenvalue of the displayed block is γ. □

15 Qualitative regularity and variational formulations

Absolute continuity, continuity of a density, and a bound uniform over all dimensions and matrices are distinct conclusions. We first prove that every fixed quadratic matrix of rank at least three has a C_0 density for the product laws under consideration. The estimates used for this regularity statement may depend on the matrix.

15.1 Continuity for rough product densities

Lemma 15.1 (A quadratic integral near its critical point)**.** *Let B be an invertible real symmetric d-by-d matrix, where $d = 3$ or 4. Write $\Lambda = \|B\|$, $\sigma = s_d(B) > 0$, and $q(x) = x^T Bx$. The pushforward under q of Lebesgue measure restricted to $|x| \le r$ has a density bounded by*

$$C_d \Lambda^{(d-2)/2} |\det B|^{-1/2} r^{d-2}, \qquad C_3 = 2\pi, \quad C_4 = 4\pi. \tag{185}$$

If $F(x) = \prod_{i=1}^d f_i(x_i)$ with probability densities $f_i \le 1$, the pushforward of $F(x)\mathbf{1}_{\{|x|>R\}}\,dx$ has density at most

$$\frac{d^{3/2}}{\sigma R}. \tag{186}$$

Proof. Use the linear change of variables $z = |B|^{1/2}x$, followed by an orthogonal change in this deterministic integral. Its Jacobian is $|\det B|^{-1/2}$, the quadratic is a sum of positive squares minus a sum of negative squares, and the image of the ball lies in $|z| \le \rho = \sqrt{\Lambda} r$.

For a definite form, polar coordinates give the bound $|\mathbb{S}^{d-1}|\rho^{d-2}/2$. This is $2\pi\rho$ for $d = 3$ and $\pi^2\rho^2$ for $d = 4$. For signature (p, m) with $p, m > 0$ and $p + m = d$, put $u = |z_+|^2$ and $v = |z_-|^2$. The quadratic is $u - v$, while the radial measure is

$$\frac{|\mathbb{S}^{p-1}||\mathbb{S}^{m-1}|}{4} u^{p/2-1} v^{m/2-1}\,du\,dv, \qquad u, v \ge 0, \quad u + v \le \rho^2.$$

In dimension three, signature $(2, 1)$ gives a density at most $\pi \int_0^{\rho^2} v^{-1/2}\,dv = 2\pi\rho$; the opposite signature follows by reflection of the output. In dimension four, signature $(2, 2)$ gives at most $\pi^2\rho^2/2$, because the allowed radial interval on $u - v = t$ has length at most $\rho^2/2$. For signature $(3, 1)$, the density is at most $2\pi\rho \int_0^{\rho^2} v^{-1/2}\,dv = 4\pi\rho^2$. Reflection treats $(1, 3)$. These estimates prove (185).

For the second assertion, $|Bx| \ge \sigma R$ when $|x| > R$. Hence at least one coordinate satisfies $|\partial_i q(x)| = 2|(Bx)_i| \ge 2\sigma R/\sqrt{d}$. For fixed x_{-i}, a quadratic in x_i has at most two inverse branches. On the indicated set their inverse derivatives are each at most $\sqrt{d}/(2\sigma R)$. The cap on f_i and change of variables therefore bound the conditional pushforward density by $\sqrt{d}/(\sigma R)$. Integrating the other probability densities and summing over the possible i proves (186). □

Lemma 15.2 (The region with only one large derivative)**.** *In the notation of Lemma 15.1, fix $0 < r < R < \infty$. Let*

$$E_{r,R,\delta} = \{x : r \le |x| \le R, \text{ at most one } |\partial_i q(x)| \text{ exceeds } \delta\}.$$

For $0 < \delta \le 1$, the pushforward of $F\mathbf{1}_{E_{r,R,\delta}}\,dx$ has density at most

$$C(B, r, R)\delta^{d-2}, \tag{187}$$

uniformly over all products of cap-one probability densities.

Proof. Partition the set measurably according to an index i maximizing $|(Bx)_i|$, breaking ties by the smallest index. On this part,

$$|\partial_i q| \geq 2\sigma r/\sqrt{d}, \qquad |(Bx)_j| \leq \delta/2 \quad (j \neq i).$$

Write $Bx = se_i + v$, where $v_i = 0$. Then $|s| \leq \Lambda R$, $|v| \leq \sqrt{d-1}\delta/2$, and

$$x = sB^{-1}e_i + B^{-1}v.$$

The projection onto the $d-1$ coordinates other than i therefore lies within distance $\epsilon = \sqrt{d-1}\delta/(2\sigma)$ of a line segment of length at most $2\Lambda R/\sigma$. In $\mathbb{R}^{d-1}$ the volume of this neighborhood is at most

$$\frac{2\Lambda R}{\sigma}\,\omega_{d-2}\epsilon^{d-2} + \omega_{d-1}\epsilon^{d-1} \leq C(B,R)\delta^{d-2},$$

where ω_j is the volume of the unit ball in $\mathbb{R}^j$. The estimate also holds when the projected segment is a point.

For each fixed value of these $d-1$ coordinates, the inverse-branch argument from the preceding lemma bounds the one-dimensional pushforward density by $\sqrt{d}/(\sigma r)$. Here all density factors are bounded by one, so integration over the projected set costs at most its Lebesgue volume. Summing over i proves (187). □

Lemma 15.3 (Continuity in three or four invertible coordinates)**.** *Let $d = 3$ or 4, let B be invertible and symmetric, and let $X_1, \dots, X_d$ be independent with densities bounded by one. For every affine term, $X^TBX + b^TX + c$ has a density in $C_0(\mathbb{R})$. A height bound independent of the affine term and of the coordinate laws is*

$$M_B = C_d\|B\|^{(d-2)/2}|\det B|^{-1/2} + d^{3/2}/s_d(B). \tag{188}$$

For fixed B, b, c, convergence of all coordinate densities in L^1 implies uniform convergence of these continuous output densities.

Proof. Translation of the original independent coordinates removes the affine term, up to a constant shift of the output. It suffices to consider $q(x) = x^TBx$. Splitting into the unit ball and its complement in Lemma 15.1 immediately gives (188).

We give the regularity argument in detail to allow rough, disconnected, and unbounded coordinate supports. For a product density F and measurable set E, denote by $p_{F;E}$ the density of the restricted pushforward $q_*(F\mathbf{1}_E\,dx)$. These restrictions are absolutely continuous because they are dominated by the full pushforward, which is absolutely continuous by Lemma 3.1.

Let $F = \prod_i f_i$ and $G = \prod_i g_i$ be two products of cap-one probability densities. On the set

$$\mathcal{G}_\delta = \{x : \text{at least two } |\partial_i q(x)| \text{ exceed } \delta\},$$

the telescoping identity

$$F - G = \sum_{i=1}^{d}(f_i - g_i)\prod_{k<i} f_k \prod_{k>i} g_k$$

and coordinatewise change of variables give

$$\|p_{F;\mathcal{G}_\delta} - p_{G;\mathcal{G}_\delta}\|_\infty \leq \frac{2(d-1)}{\delta}\sum_{i=1}^{d}\|f_i - g_i\|_1. \tag{189}$$

Indeed, for the summand containing $f_i - g_i$, every point of $\mathcal{G}_\delta$ has some $j \neq i$ with $|\partial_j q| > \delta$. Sum over those $d-1$ choices. On each choice, solve for x_j along the at most two inverse branches.

The jth density factor is at most one and the inverse derivative is at most δ^{-1}. Integration of the remaining factors gives $\|f_i - g_i\|_1$, since all other factors have mass one. The same estimate holds with an arbitrary measurable subset of $\mathcal{G}_\delta$ in place of that set.

Fix a desired error $\eta > 0$. Choose r sufficiently small and R sufficiently large that (185) and (186) are each less than η. For these fixed r, R, choose δ sufficiently small that (187) is less than η. The complement of $\mathcal{G}_\delta$ is covered by these three regions. Thus both F and G contribute at most 3η in density there. Equation (189) gives the uniform estimate

$$\|p_F - p_G\|_\infty \le 6\eta + \frac{2(d-1)}{\delta} \sum_{i=1}^{d} \|f_i - g_i\|_1. \tag{190}$$

The parameters depend only on B, η, not on the coordinate densities.

Each probability density $f_i \le 1$ can be approximated in L^1 by smooth compactly supported probability densities with the same cap. One explicit construction first truncates to a large bounded interval, places the missing mass with height one on a disjoint interval, and then convolves with a nonnegative smooth compactly supported approximate identity. Truncation and the added mass cost at most twice the discarded mass in L^1; convolution preserves the total mass and the cap.

For a smooth compactly supported product density, its pushforward under q belongs to C_0. To see this without assuming regularity at the critical value, remove a ball of radius r about the origin by a smooth cutoff. On the compact support that remains, the gradient of q never vanishes. Cover that support by finitely many coordinate charts with one nonzero partial derivative and take a smooth partition of unity. In each chart the coordinates (q, x_{-i}) form a smooth change of variables. The transformed integrand is smooth with compact support, and integration in x_{-i} gives a smooth compactly supported function of q. The removed part has density bounded by a constant times r^{d-2}, by (185). Letting $r \downarrow 0$ shows that the full density is a uniform limit of functions in C_0, and hence is in C_0.

Apply (190) to the smooth product approximations. Their output densities are uniformly Cauchy and therefore converge uniformly to a function in C_0. The input products converge in L^1, so their pushforward measures converge in total variation. The limiting continuous function is consequently a density of $q(X)$. Finally, the same inequality, first choosing η and then the input L^1 error, proves the stability assertion. □

Theorem 15.4 (Every fixed rank-three-or-higher matrix)**.** *Let A be any real symmetric matrix of rank at least three. For independent cap-one coordinates and arbitrary affine terms, $Q = X^T AX + b^T X + c$ has a bounded density in $C_0(\mathbb{R})$.*

Proof. There is an invertible original-coordinate principal submatrix $B = A_{II}$ of order $d = 3$ or 4. For completeness, symmetric elimination proves this as follows. If a nonzero diagonal entry is available, use it as an invertible one-coordinate pivot. If all diagonal entries vanish but the remaining matrix is nonzero, a nonzero off-diagonal entry gives the invertible block $\left(\begin{smallmatrix} 0 & a \\ a & 0 \end{smallmatrix}\right)$. An invertible pivot reduces rank by its order, through its Schur complement. Continue until the total pivot order first reaches three. The rank assumption permits doing so, and the total is either three or four. The corresponding principal submatrix of the original matrix is invertible, since its determinant is the product of the successive nonzero pivot determinants.

Condition on $Y = X_{I^c}$. The conditional quadratic matrix in the coordinates X_I is the fixed invertible matrix B; all conditioning effects are affine. Lemma 15.3 supplies for every y a conditional density $p_y \in C_0$ bounded by M_B. There is a jointly measurable choice: for each (y, t) take the limit of

$$\frac{m}{2}\,\mathbb{P}\big(|X_I^T B X_I + b(y)^T X_I + c(y) - t| < 1/m\big) \quad \text{as } m \to \infty.$$

The limit exists at every t because the conditional continuous density exists. Its average $p(t) = \mathbb{E}p_Y(t)$ is a density of Q. The bound M_B permits dominated convergence both as t approaches a finite point and as $|t| \to \infty$. Thus p is continuous, vanishes at infinity, and is bounded. □

Remark 15.5. The constant M_B in this proof can deteriorate when the selected principal block becomes ill-conditioned. Theorem 15.4 therefore gives a qualitative regularity statement for every fixed matrix; the dimension-independent constant in Conjecture 1.1 remains a separate question.

15.2 Recovery from one fixed smoothing kernel

Let $\mathfrak{C}$ be the optimal constant in Conjecture 1.1, as an extended nonnegative real number:

$$\mathfrak{C} = \sup_{\substack{n\geq 3,\ A=A^T,\ s_3(A)\geq 1 \\ X \text{ independent},\ 0\leq f_i\leq 1 \\ b\in\mathbb{R}^n,\ c\in\mathbb{R}}} \|p_{X^TAX+b^TX+c}\|_\infty.$$

Every density in this definition exists by Lemma 3.1. The next identities are valid whether $\mathfrak{C}$ is finite or infinite.

Lemma 15.6 (Approximate-identity recovery)**.** *Let $K \geq 0$ be any probability density in $L^1(\mathbb{R})$, and set $K_r(x) = r^{-1}K(x/r)$. For every probability density p and every $r_0 > 0$,*

$$\sup_{0<r\leq r_0} \|K_r * p\|_\infty = \|p\|_\infty. \tag{191}$$

Proof. Young's inequality gives the upper bound. Translation continuity in L^1 and dominated convergence give

$$\|K_r * p - p\|_1 \leq \int_{\mathbb{R}} K(z)\, \|p(\,\cdot - rz) - p\|_1\, dz \longrightarrow 0.$$

If the left side of (191) were a finite number M smaller than $\|p\|_\infty$, take $r_j \downarrow 0$ and an almost-everywhere convergent subsequence of the L^1 approximations. The limit would satisfy $p \leq M$ almost everywhere, a contradiction. The same argument treats $\|p\|_\infty = \infty$. □

Theorem 15.7 (An exact fixed-kernel formulation)**.** *For every fixed probability density $K \in L^1(\mathbb{R})$,*

$$\mathfrak{C} = \sup_{\substack{A,X,b,c \text{ as above} \\ s_3(A)\geq 1}} s_3(A)\, \|K * p_Q\|_\infty. \tag{192}$$

In particular,

$$\mathfrak{C} = \sup_{\substack{A,X,b,c \text{ as above} \\ s_3(A)\geq 1,\ y\in\mathbb{R}}} \frac{s_3(A)}{2}\, \mathbb{P}(|Q - y| \leq 1). \tag{193}$$

Proof. First observe the equivalent weighted form

$$\mathfrak{C} = \sup s_3(A)\|p_Q\|_\infty.$$

The weighted supremum is at least the original one because $s_3(A) \geq 1$. For the reverse inequality, normalize each instance by $Q \mapsto Q/s_3(A)$, including its affine term. The new matrix has third singular value one and its density height is $s_3(A)\|p_Q\|_\infty$.

Let the right side of (192) be $\mathfrak{C}_K$. Young's inequality gives $\mathfrak{C}_K \le \mathfrak{C}$. Fix an admissible instance and write $s = s_3(A)$. For $0 < r \le s$, the form Q/r is still admissible, with matrix A/r and density $p_r(u) = rp_Q(ru)$. Direct substitution gives

$$(K * p_r)(y/r) = r(K_r * p_Q)(y), \qquad \frac{s}{r}\|K * p_r\|_\infty = s\|K_r * p_Q\|_\infty.$$

Taking the supremum over $0 < r \le s$ and applying Lemma 15.6 shows $\mathfrak{C}_K \ge s\|p_Q\|_\infty$. The supremum over the original instance proves equality.

Take $K = \frac{1}{2}\mathbf{1}_{[-1,1]}$ for (193). For an absolutely continuous law the endpoints carry no mass, and the box convolution is the continuous function $y \mapsto \frac{1}{2}\mathbb{P}(|Q-y| \le 1)$. Its essential supremum is therefore its pointwise supremum, as used in that identity. □

Lemma 15.8 (Stability at a fixed interval width)**.** *Suppose probability laws μ_j converge weakly to an atomless probability law μ. For every fixed $a > 0$,*

$$\sup_{y\in\mathbb{R}} |\mu_j([y-a, y+a]) - \mu([y-a, y+a])| \longrightarrow 0. \tag{194}$$

Proof. Let F_j, F be the distribution functions. Because μ is atomless, F is continuous. Weak convergence gives $F_j(t) \to F(t)$ at every t. A finite grid whose successive F values differ by at most a prescribed ϵ, together with two tail points of total μ mass at most 2ϵ, proves by monotonicity that $\sup_t |F_j(t) - F(t)| \to 0$. For clarity, one may choose a bounded interval containing all but ϵ of each tail of μ, subdivide it finely using uniform continuity of F, and bracket each t between adjacent grid points. The same uniform bound controls $F_j(t-)$, since $F(t-) = F(t)$. Now $\mu_j([y-a, y+a]) = F_j(y+a) - F_j((y-a)-)$. Subtract the corresponding expression for μ to obtain (194). □

15.3 A countable heat-kernel formulation

Let $\mathcal{U}$ be the countable class of finite unions of bounded half-open intervals with rational endpoints and total Lebesgue measure exactly one. For $n \ge 3$, let $\mathcal{A}_n$ consist of real symmetric matrices with rational entries such that every entry and every principal minor is nonzero, the spectrum is simple, and $s_3(A) > 1$. Put

$$\gamma_\epsilon(t) = \frac{1}{\sqrt{2\pi\epsilon}} \exp\left(-\frac{t^2}{2\epsilon^2}\right).$$

Theorem 15.9 (A countable exact formulation)**.** *The full optimal constant satisfies*

$$\mathfrak{C} = \sup_{\substack{n\ge 3,\ A\in\mathcal{A}_n,\ b\in\mathbb{Q}^n \\ E_1,\dots,E_n\in\mathcal{U} \\ \epsilon\in\mathbb{Q}_{>0},\ y\in\mathbb{Q}}} \int_{E_1\times\cdots\times E_n} \gamma_\epsilon(x^T Ax + b^T x - y)\, dx. \tag{195}$$

There is no bound here on n, the number of intervals, their locations, or their rational denominators.

Proof. Lemma 15.6, with a Gaussian kernel, shows that for each output law

$$\|p_Q\|_\infty = \sup_{\epsilon>0,\ y\in\mathbb{R}} \mathbb{E}\gamma_\epsilon(Q-y).$$

The convolution at every positive ϵ is continuous, so a pointwise supremum in y is appropriate. Constants in Q can be absorbed into y. Every value on the right side of (195) comes from an

admissible product law, and is at most $\mathfrak{C}$. We prove the reverse inequality by approximating an arbitrary fixed smoothed value.

First, for any bounded nonnegative measurable payoff h,

$$\sup_{\substack{0\le f\le 1\\ \int f=1}} \int hf = \sup_{|E|=1} \int_E h. \tag{196}$$

Here the suprema need not be attained. One elementary proof chooses $\tau = \inf\{t \ge 0 : |\{h > t\}| \le 1\}$. Monotone convergence from levels above τ gives $|\{h > \tau\}| \le 1$. For $\delta > 0$, choose a set E of measure one that contains $\{h > \tau\}$ and whose remaining points have $h \ge \tau - \delta$; if $\tau = 0$, fill with arbitrary points. Such a set exists by the definition of τ and the nonatomic nature of Lebesgue measure. In the identity $\int h(f - \mathbf{1}_E) = \int (h - \tau)(f - \mathbf{1}_E)$, the contribution from $\{h > \tau\}$ is nonpositive. The contribution outside E is also nonpositive, and the remaining contribution is at most δ. This proves (196) by letting $\delta \downarrow 0$.

Apply (196) successively to each original coordinate of the bounded payoff $\gamma_\epsilon(x^T Ax + b^T x - y)$, averaging over the others. At each step the smoothed value can be preserved up to an arbitrarily small prescribed error. After finitely many steps the input laws are uniform on measurable sets E_i of measure one.

Each such set can be approximated in symmetric difference by a member of $\mathcal{U}$. Indeed, Lebesgue regularity gives a finite bounded union of rational intervals F with $|E_i \triangle F|$ arbitrarily small. Its total length is rational. If that length exceeds one, remove or shorten interval pieces, retaining rational endpoints, until the length is one. If it is less than one, add a disjoint rational interval of the missing rational length. The additional change has measure $||F| - 1|$, which is at most $|E_i \triangle F|$. Thus the final symmetric difference can also be made arbitrarily small. The telescoping inequality

$$\left\| \prod_i f_i - \prod_i g_i \right\|_1 \le \sum_i \|f_i - g_i\|_1$$

shows that these replacements alter the fixed smoothed value by at most the last sum times $\|\gamma_\epsilon\|_\infty$.

The resulting product set is bounded. Next replace A by $(1 + \delta)A$ and then by a nearby rational symmetric matrix A_δ whose entries and principal minors are all nonzero and whose spectrum is simple. Such matrices are dense: each forbidden condition is the zero set of a nonzero polynomial in the independent matrix entries, including the spectral discriminant. Their finite union has empty interior, and its complement is open and contains rational points in every sufficiently small ball. Choose $\|A_\delta - (1 + \delta)A\| < \delta/2$. Singular-value stability gives $s_3(A_\delta) > 1$, while $A_\delta \to A$. Also approximate b by rational vectors. On the fixed bounded product set, the corresponding quadratic polynomials converge uniformly. Continuity of the Gaussian payoff therefore preserves the smoothed integral in the limit.

Finally approximate $\epsilon > 0$ and y by positive rational and rational parameters, respectively. The kernels are uniformly bounded in a neighborhood of the fixed positive ϵ, and depend continuously on all their parameters. Dominated convergence preserves the integral. All approximation errors can be chosen arbitrarily small. Every smoothed value in the full problem is therefore bounded by the right side of (195), proving equality. □

Remark 15.10. The fixed-kernel and countable formulations are exact identities for the optimal density bound. Neither identity by itself gives a finite bound for the supremum. The fixed-width stability in Lemma 15.8 also does not assert uniform control as the interval width tends to zero.

16 Further coordinate block estimates and spectral stability

The estimates in this section apply to blocks of original independent coordinates with densities bounded by one. Their conclusions are uniform in affine terms. The deterministic spectral inequalities control perturbations of coordinate submatrices.

16.1 A linear bank with two crossing directions

Lemma 16.1 (Negative moments from a density bound)**.** *If a probability law on $\mathbb{R}^d$ has density at most K, then for $0 < p < d$ and every $z \in \mathbb{R}^d$,*

$$\mathbb{E}|Y - z|^{-p} \le \frac{d}{d-p}(Kv_d)^{p/d},$$

where v_d is the volume of the unit ball.

Proof. For every $r > 0$, the probability of the ball of radius r is at most $\min\{1, Kv_d r^d\}$. Integrate this bound in

$$\mathbb{E}|Y - z|^{-p} = \int_0^\infty \mathbb{P}\{|Y - z| < t^{-1/p}\}\, dt$$

and split at $t = (Kv_d)^{p/d}$. The two contributions are $(Kv_d)^{p/d}$ and $p(Kv_d)^{p/d}/(d-p)$. □

Theorem 16.2 (An arbitrary-rank crossing block with a linear bank)**.** *Let X_I and X_J be disjoint original coordinate banks and let*

$$Q = F(X_I) + 2X_I^{\mathsf{T}} M X_J + b_J^{\mathsf{T}} X_J,$$

where F is any real measurable function. If $s_2(M) > 0$, then

$$\|p_Q\|_\infty \le \frac{2\sqrt{\pi}}{\sqrt{s_1(M)s_2(M)}}. \tag{197}$$

The rank and all smaller singular values of M are unrestricted.

Proof. Condition on $X_I = x$. The remaining polynomial is linear in X_J, with coefficient vector $c(x) = 2M^{\mathsf{T}}x + b_J$. The one-dimensional product-projection estimate of Livshyts et al. (2016) gives a conditional density bound $\sqrt{2}/|c(x)|$ whenever $c(x) \neq 0$. Let u_1, u_2 and v_1, v_2 be corresponding left and right singular vectors for the two largest singular values s_1, s_2 of M. The vector $U = (u_1^{\mathsf{T}} X_I, u_2^{\mathsf{T}} X_I)$ has density bounded by two. Therefore

$$Z = (2s_1U_1 + v_1^{\mathsf{T}} b_J, 2s_2U_2 + v_2^{\mathsf{T}} b_J)$$

has density at most $1/(2s_1s_2)$, and $|c(X_I)| \ge |Z|$. Lemma 16.1 with $d = 2, p = 1$ gives

$$\mathbb{E}|c(X_I)|^{-1} \le 2\sqrt{\frac{\pi}{2s_1s_2}}.$$

In particular, the exceptional set $c(X_I) = 0$ has probability zero. Averaging the conditional density inequality proves (197). This argument retains the full matrix M; the projection onto two singular directions is used only for a lower bound on the length of $c(X_I)$. □

The condition that the J bank enters linearly is essential to this proof. Theorem 16.2 includes a bilinear form with an arbitrary quadratic or measurable addition on one bank. Its conditioning argument is closely related to the independent-block density estimates in Banks et al. (2024).

16.2 A singular physical pair and a random slope

Theorem 16.3 (Rank-one physical pair)**.** *Let $X, Y, Z_1, \dots, Z_m$ be independent original coordinates. Suppose*

$$Q = aX^2 + 2bXY + cY^2 + 2X\langle u, Z\rangle + 2Y\langle v, Z\rangle + F(Z) + 2\alpha X + 2\beta Y,$$

where $a \neq 0$, $ac = b^2$, F is real measurable, and $w = v - (b/a)u \neq 0$. Then

$$\|p_Q\|_\infty \leq \frac{4\,2^{1/4}}{\sqrt{|a|\,|w|}}. \tag{198}$$

The dependence of F on all coordinates of Z is unrestricted.

Proof. Complete the square only in the original coordinate X. With $\eta = \beta - b\alpha/a$ this writes

$$Q = a\left(X + \frac{bY + \langle u, Z\rangle + \alpha}{a}\right)^2 + 2Y(\langle w, Z\rangle + \eta) + \widetilde{F}(Z).$$

Conditioning on (Y, Z) and using the scalar square density bound gives

$$p_Q(t) \leq |a|^{-1/2}\mathbb{E}k_a(t - 2Y(\langle w, Z\rangle + \eta) - \widetilde{F}(Z)), \quad k_a(u) = |u|^{-1/2}\mathbf{1}_{\{au>0\}},$$

for almost every t. For any probability density $f \leq M$,

$$\sup_t \int_0^\infty y^{-1/2} f(t \pm y)\, dy \leq 2\sqrt{M},$$

by filling the interval of length $1/M$ nearest zero. If $\ell \neq 0$, the variable $2\ell Y$ has density at most $1/(2|\ell|)$. Hence the expectation of the one-sided kernel over Y is at most $\sqrt{2}\,|\ell|^{-1/2}$, uniformly in its translate. The scalar projection $\langle w, Z\rangle + \eta$ has density at most $\sqrt{2}/|w|$ by Livshyts et al. (2016). Lemma 16.1 with $d = 1, p = 1/2$ gives

$$\mathbb{E}|\langle w, Z\rangle + \eta|^{-1/2} \leq 2\sqrt{2\sqrt{2}/|w|}.$$

Multiplication by the preceding bound for the slope gives $\sqrt{2} \cdot 2\sqrt{2\sqrt{2}} = 4\,2^{1/4}$, as asserted. All changes of variables can be integrated against nonnegative output test functions, so the almost-everywhere conditional formulas introduce no exceptional-level assumption. □

Corollary 16.4 (An aggregate bound for singular pairs)**.** *Under Theorem 16.3, put*

$$W = a^2 \sum_{i \neq j} w_i^2 w_j^2.$$

If $W > 0$, then

$$\|p_Q\|_\infty \leq 4\,2^{1/4}|a|^{-1/4}W^{-1/8}.$$

Proof. The inequality $|w|^4 \geq \sum_{i\neq j} w_i^2 w_j^2 = W/a^2$ inserted into (198) proves the statement. □

16.3 Spectral energy in a coordinate set

For a coordinate set H, let $P_H : \mathbb{R}^{|H|} \to \mathbb{R}^n$ be the coordinate inclusion. Compression is written $T_{HH} = P_H^{\mathsf{T}} T P_H$. Eigenvalues of a positive semidefinite matrix are listed in decreasing order.

Lemma 16.5 (Spectral-square spill)**.** *Let A be symmetric, $C = A_{HH}$, $B = A_{H,H^c}$, and $0 \le G \le A^2$. Suppose $\lambda_k(G_{HH}) \ge g^2$. For $0 \le r < k$ put $c = s_{r+1}(C)$. If $c < g$, then*

$$s_{k-r}(B) \ge \sqrt{g^2 - c^2}.$$

In particular, one may use $G = \Lambda^2 \mathbf{1}_{\{|A| \ge \Lambda\}}$ for any $\Lambda > 0$.

Proof. The exact block identity is $(A^2)_{HH} = C^2 + BB^{\mathsf{T}}$. Write $C = C_r + E$, where C_r consists of its r largest absolute eigenmodes. Then $C_r E = 0$, $\operatorname{rank} C_r \le r$, and $E^2 \le c^2 I$. Thus

$$BB^{\mathsf{T}} + C_r^2 = (A^2)_{HH} - E^2 \ge G_{HH} - c^2 I.$$

The left side has at least k eigenvalues at least $g^2 - c^2$. Adding a positive semidefinite matrix of rank at most r can increase that count by at most r. Consequently BB^{T} has at least $k - r$ eigenvalues of that size, which is the assertion. The stated spectral projection gives an admissible G by the spectral theorem. □

Lemma 16.6 (A pair in an approximate rank-two self block)**.** *Let $C = C^{\mathsf{T}}$ be m by m, $m \ge 3$, and write $N = \binom{m}{2}$. If $s_2(C) \ge b$ and $s_3(C) < a$, then some principal two-coordinate block K satisfies*

$$s_{\min}(K) > b/N - a.$$

Proof. Let $C_2 = \lambda_1 uu^{\mathsf{T}} + \lambda_2 vv^{\mathsf{T}}$ be the two largest absolute eigenmodes. The orthonormality of u, v gives $\sum_{i<j}(u_i v_j - u_j v_i)^2 = 1$. Choose a pair I whose summand is at least $1/N$. Then

$$|\det(C_2)_{II}| \ge s_1(C) s_2(C)/N, \qquad \|(C_2)_{II}\| \le s_1(C).$$

It follows that $s_{\min}((C_2)_{II}) \ge b/N$. The remaining matrix has norm $s_3(C) < a$, so the singular-value perturbation inequality proves the claim. □

Theorem 16.7 (Six spectral-square directions on a coordinate set)**.** *Let $s_3(A) \ge s > 0$ and let H have size $m \ge 6$. Suppose $0 \le G \le A^2$ and $\lambda_6(G_{HH}) \ge g^2$. For $N = \binom{m}{2}$, choose $0 < a < b/N$ and $0 < b < g$, and set $\kappa = b/N - a$. Then*

$$\|p_Q\|_\infty \le \max\left\{ \frac{2\sqrt{6\pi m}}{a}, \frac{9.2 + \log_+(10000\kappa/s)}{\kappa}, \frac{5\sqrt{2e}}{\sqrt{g^2 - b^2}} \right\}. \tag{199}$$

All entries of the actual matrix A outside H are retained.

Proof. Put $C = A_{HH}$ and $B = A_{H,H^c}$. If $s_3(C) \ge a$, condition on the complementary original coordinates and apply Corollary 9.8. This gives the first term. If $s_3(C) < a$ and $s_2(C) \ge b$, Lemma 16.6 gives a physical pair K with $s_{\min}(K) > \kappa$, hence $|\det K| > \kappa^2$. A definite pair gives π/κ by Lemma 3.3. An indefinite pair gives the estimate in Theorem 3.7. The function

$$x \longmapsto \frac{9.2 + \log_+(10000x/s)}{x}$$

is decreasing on $(0, \infty)$: on its logarithmic branch its derivative has numerator $1 - 9.2 - \log(10000x/s) < 0$, and elsewhere this is immediate. Substitute $x = \sqrt{|\det K|} > \kappa$; the second term in (199) also exceeds π/κ.

The remaining case has $s_2(C) < b$. Lemma 16.5 with $k = 6, r = 1$ gives $s_5(B) > \sqrt{g^2 - b^2}$. The five-volume case of Corollary 4.10 gives

$$\|p_Q\|_\infty \le 5\sqrt{2e}\, e_5(BB^{\mathsf{T}})^{-1/10} \le 5\sqrt{2e}/s_5(B),$$

which proves the third term. These three alternatives are exhaustive. □

The carrier hypothesis itself implies $s_6(A) \ge g$, so one may always take $s = g$ if no stronger global spectral lower bound is available. In particular, this theorem does not extract six directions from the three-direction premise of Conjecture 1.1.

16.4 Stability under additional spectrum

Proposition 16.8 (Localized anticommutator bound)**.** *Let $A = A_0 + R$ be symmetric and define*

$$\nu_H = \max\{0, -\lambda_{\min}((A_0R + RA_0)_{HH})\}.$$

For every $j \le |H|$,

$$\lambda_j((A^2)_{HH}) \ge \lambda_j((A_0^2)_{HH}) - \nu_H.$$

If $A_0R + RA_0 = 0$, all these inequalities hold with zero loss. This includes disjoint spectral packets, for which $A_0R = 0$.

Proof. The identity $A^2 = A_0^2 + A_0R + RA_0 + R^2$, together with $R^2 \ge 0$, gives $(A^2)_{HH} \ge (A_0^2)_{HH} - \nu_H I$. Apply monotonicity of ordered eigenvalues under this order. The zero-loss assertions follow from the same identity. □

Theorem 16.9 (Low-rank additions with an operator-norm remainder)**.** *Let $A = A_0 + S + R$ be symmetric, $\|R\| \le \varepsilon$, and $r_H = \operatorname{rank}(SP_H)$. If $k + r_H \le |H|$, then*

$$\sqrt{\lambda_k((A^2)_{HH})} \ge \left(\sqrt{\lambda_{k+r_H}((A_0^2)_{HH})} - \varepsilon\right)_+. \tag{200}$$

No restriction is imposed on the signs or magnitudes of the nonzero eigenvalues of S.

Proof. For any rectangular matrices T, E with $\operatorname{rank} E \le r$, the singular-value variational principle gives

$$s_k(T + E) \ge s_{k+r}(T). \tag{201}$$

Indeed, take a $(k + r)$-dimensional subspace on which $|Tx| \ge s_{k+r}(T)|x|$ and intersect it with $\ker E$. The intersection has dimension at least k, and $T + E = T$ there. Apply (201) to $T = A_0P_H$, $E = SP_H$, then use $\|RP_H\| \le \varepsilon$. Finally,

$$s_j(MP_H)^2 = \lambda_j((M^2)_{HH})$$

for symmetric M, which proves (200). □

Remark 16.10 (The amount of spectral reserve)*.* A different argument notes that

$$\operatorname{range}((A_0 + S)^2 - A_0^2) \subset \operatorname{range}(S) + A_0 \operatorname{range}(S).$$

Rank interlacing of the squares then gives a valid but weaker loss of at most $2 \operatorname{rank} S$. The fact that this difference can have rank $2 \operatorname{rank} S$ does not make that spectral reserve loss necessary. Theorem 16.9 uses the rectangular operators before squaring and loses at most $\operatorname{rank} S$, and often only r_H.

Corollary 16.11 (An adaptive split of a signed addition)**.** *Let $E = E^{\mathsf{T}}$, let S_h contain its eigenmodes with absolute eigenvalue greater than $h \geq 0$, and put $r_H(h) = \operatorname{rank}(S_h P_H)$. If $\|T\| \leq \delta$ and $A = A_0 + E + T$, then, whenever $k + r_H(h) \leq |H|$,*

$$\sqrt{\lambda_k((A^2)_{HH})} \geq \left(\sqrt{\lambda_{k+r_H(h)}((A_0^2)_{HH})} - h - \delta\right)_+ .$$

The threshold h may be chosen to maximize the bound.

Proof. The remaining part $E - S_h$ has operator norm at most h. Apply Theorem 16.9 with remainder $E - S_h + T$. □

Proposition 16.12 (Separate positive and negative reserves)**.** *Let $C = C_0 + P - N + T$ be symmetric m by m, where $P, N \geq 0$ and $\|T\| \leq \delta$. Write $p_j = \lambda_j(P)$ and $n_j = \lambda_j(N)$ in decreasing order, and set $p_{m+1} = n_{m+1} = 0$. For $1 \leq u, v \leq m$,*

$$\lambda_u(C) \geq \max_{0 \leq r \leq m-u} \{\lambda_{u+r}(C_0) - n_{r+1}\} - \delta,$$
$$\lambda_v(-C) \geq \max_{0 \leq r \leq m-v} \{\lambda_{v+r}(-C_0) - p_{r+1}\} - \delta.$$

If the active bounds are positive and supply a total of at least three positive and negative directions, their minimum is a lower bound for $s_3(C)$.

Proof. Write $N = N_r + N_{\mathrm{tail}}$, where N_r has rank at most r and $\|N_{\mathrm{tail}}\| = n_{r+1}$. The favorable term $P \geq 0$ can only increase ordered eigenvalues, so

$$\lambda_{u+r}(C_0 + P + T - N_{\mathrm{tail}}) \geq \lambda_{u+r}(C_0) - \delta - n_{r+1}.$$

Subtracting N_r loses at most r indices by rank interlacing, which proves the first inequality after maximizing in r. Apply the same argument to $-C$ for the second. Strictly positive positive and negative spectral directions are disjoint, proving the final assertion. □

16.5 A complementary Schur block protects three directions

Theorem 16.13 (Complementary Schur protection)**.** *On an original coordinate block H of size m, suppose*

$$A_{HH} = B + U\Theta U^{\mathsf{T}} + T, \qquad B = B^{\mathsf{T}}, \quad \Theta = \Theta^{\mathsf{T}}, \quad \|T\| \leq \delta.$$

Let the columns of W form an orthonormal basis of range U*, and let the columns of Z form an orthonormal basis of its orthogonal complement. Suppose $D = Z^{\mathsf{T}} B Z$ is invertible and $s_3(D) \geq \mu > 0$. Put*

$$R = (W^{\mathsf{T}} B Z) D^{-1}, \quad \rho = \|R\|, \quad \ell = \frac{\sqrt{\rho^2 + 4} - \rho}{2}.$$

If $\delta < \ell^2 \mu$, then for every Θ,

$$\|p_Q\|_\infty \leq \frac{2\sqrt{6\pi m}}{\ell^2 \mu - \delta}.$$

Proof. The orthogonal basis $[W \ Z]$ is used for matrix algebra only. In this basis the block without T equals

$$L \begin{pmatrix} S & 0 \\ 0 & D \end{pmatrix} L^{\mathsf{T}}, \quad L = \begin{pmatrix} I & R \\ 0 & I \end{pmatrix}, \quad S = W^{\mathsf{T}}(B + U\Theta U^{\mathsf{T}})W - RDR^{\mathsf{T}}.$$

No invertibility assumption on S is needed. A singular-value decomposition of R decomposes L into two-dimensional shears $\left(\begin{smallmatrix}1 & r\\ 0 & 1\end{smallmatrix}\right)$ and identity blocks. The smaller singular value of such a shear is $(\sqrt{r^2+4}-r)/2$, decreasing in $r \geq 0$, so $s_{\min}(L) = \ell$. Multiplication on each side by an invertible matrix reduces every singular value by at most its smallest singular value. Consequently

$$s_3(B + U\Theta U^{\mathsf{T}}) \geq \ell^2 s_3(\operatorname{diag}(S, D)) \geq \ell^2 \mu.$$

The perturbation T costs at most δ. Condition outside H and apply Corollary 9.8 to finish. □

Corollary 16.14 (Many large eigenvalues of one sign)**.** *If A has at least $n-1$ eigenvalues at least $s > 0$, or at least $n-1$ eigenvalues at most $-s$, and $n \geq 4$, then*

$$\|p_Q\|_\infty \leq \frac{2\sqrt{24\pi}}{s}.$$

The remaining eigenvalue may have any sign and magnitude.

Proof. Every principal four-coordinate block has at least three eigenvalues of the indicated sign and magnitude by interlacing. Apply Corollary 9.8 with $m = 4$. □

These results give bounds under additional spectral hypotheses. They do not establish that $s_3(A) \geq 1$ alone guarantees a bounded coordinate block, linear bank, or complementary Schur block satisfying their respective hypotheses.

16.6 Limitations of density interpolation and joint caps

Proposition 16.15 (An L^3 bound and a square-root interval bound)**.** *There are probability densities p_N, $N \geq 2$, such that*

$$\|p_N\|_3 \leq 2^{1/3}, \qquad \sup_z \int_{z-\varepsilon}^{z+\varepsilon} p_N(x)\,dx \leq 3\sqrt{\varepsilon} \quad (\varepsilon > 0),$$

while $\|p_N\|_\infty = N$.

Proof. Set

$$p_N = N\mathbf{1}_{[0,N^{-3}]} + (1 - N^{-2})\mathbf{1}_{[2,3]}.$$

Its mass is one, its essential supremum is N, and $\int p_N^3 = 1 + (1-N^{-2})^3 \leq 2$. An interval of length 2ε receives at most $\min\{N^{-2}, 2N\varepsilon\}$ from the spike. Dividing by $\sqrt{\varepsilon}$, the maximum over $\varepsilon > 0$ occurs at $\varepsilon = (2N^3)^{-1}$ and equals $\sqrt{2}N^{-1/2} \leq 1$. If $0 < \varepsilon \leq 1$, the other part has mass at most $\min\{1, 2\varepsilon\} \leq 2\sqrt{\varepsilon}$. For $\varepsilon \geq 1$, the total mass is already at most $1 \leq 3\sqrt{\varepsilon}$. This proves all assertions. □

Thus a uniform L^3 bound and a uniform square-root interval bound do not imply a uniform density bound. An additional hypothesis is needed to combine these estimates for quadratic forms.

Proposition 16.16 (A moving quadratic ridge under a joint density cap)**.** *For every $0 < \varepsilon < 1$ there are a uniform X on $[0,1]$ and a triple (Y, Z, W) independent of X, with joint probability density bounded by one such that*

$$T = 4XY + Z^2 - YW, \qquad \|p_T\|_\infty \geq \frac{1}{9\varepsilon}.$$

The quadratic matrix has spectrum $\{1, 0, \sqrt{17}/2, -\sqrt{17}/2\}$. The triple is not required to have independent coordinates.

Proof. For $L > 0$, define

$$S_{\varepsilon,L} = \{(y, z, w) : \varepsilon \le y \le 2\varepsilon,\ L \le w \le 2L,\ \sqrt{yw} \le z \le \sqrt{yw + \varepsilon}\}.$$

Its volume

$$V_\varepsilon(L) = \int_\varepsilon^{2\varepsilon} \int_L^{2L} (\sqrt{yw + \varepsilon} - \sqrt{yw})\, dw\, dy$$

is positive and continuous. It tends to zero as $L \downarrow 0$, because it is at most $\varepsilon^{3/2}L$. For $L \ge 1$, rationalization and $yw \le 4\varepsilon L$ give

$$V_\varepsilon(L) \ge \frac{\varepsilon^{3/2}\sqrt{L}}{2\sqrt{5}},$$

so it tends to infinity as $L \to \infty$. Choose L for which $V_\varepsilon(L) = 1$ and let the triple have density $\mathbf{1}_{S_{\varepsilon,L}}$. On this set, $0 \le Z^2 - YW \le \varepsilon$, while $0 \le 4XY \le 8\varepsilon$. Thus $T \in [0, 9\varepsilon]$ almost surely. Given the triple, $4XY$ is uniform on an interval of positive length; consequently T has a density. A probability density supported on an interval of length 9ε has essential supremum at least its reciprocal.

In coordinates (X, Y, Z, W) the matrix is

$$A_* = \begin{pmatrix} 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & -1/2 \\ 0 & 0 & 1 & 0 \\ 0 & -1/2 & 0 & 0 \end{pmatrix}.$$

The (X, Y, W) block has eigenvalues $0, \pm\sqrt{2^2 + (1/2)^2}$, and Z contributes eigenvalue one. This proves the stated spectrum. □

Proposition 16.16 concerns a weaker independence assumption than Conjecture 1.1. It shows that keeping one independent hub and a bounded joint density for the remaining three statistics does not preserve enough information for the desired conclusion. It is not a counterexample to that conjecture.

17 Gaussian limits and rank-three examples

This section gives a sharp Gaussian density bound and explicit examples for specified input classes or output levels. For related Gaussian quadratic-form estimates, see Götze et al. (2019); Bobkov et al. (2020); Choudhary and Kuchibhotla (2026).

17.1 The sharp Gaussian ceiling, including affine terms

Lemma 17.1 (Extracting a central square)**.** *For every $\lambda \ge 1$ and $m \in \mathbb{R}$, the law of $\lambda(Z + m)^2$, where Z is standard normal, is a convolution of the law of a central χ_1^2 variable and a probability measure on $[0, \infty)$.*

Proof. The elementary Gaussian integral gives, for $t \ge 0$,

$$\mathbb{E}e^{-t\lambda(Z+m)^2} = (1 + 2\lambda t)^{-1/2} \exp\left(-\frac{\lambda m^2 t}{1 + 2\lambda t}\right).$$

First consider the centered factor. For $\lambda > 1$, the finite nonnegative measure

$$\nu_\lambda(dx) = \frac{e^{-x/(2\lambda)} - e^{-x/2}}{2x} \mathbf{1}_{\{x>0\}}\, dx$$

has mass $\frac{1}{2}\log\lambda$. A compound Poisson sum with this intensity measure has Laplace transform

$$\exp\left(-\int_0^\infty (1-e^{-tx})\nu_\lambda(dx)\right) = \left(\frac{1+2t}{1+2\lambda t}\right)^{1/2}.$$

The identity follows by integrating $\int_0^\infty (e^{-ax}-e^{-bx})\,dx/x = \log(b/a)$; it may also be checked by differentiation in t and evaluation at zero. At $\lambda = 1$, use the constant zero variable. Multiplication by $(1+2t)^{-1/2}$ extracts the central square.

The remaining noncentral factor is the Laplace transform of an independent compound Poisson sum: take Poisson intensity $m^2/2$ and jumps distributed as $\lambda\chi_2^2$. Indeed its transform is

$$\exp\left[\frac{m^2}{2}\left((1+2\lambda t)^{-1}-1\right)\right] = \exp\left(-\frac{\lambda m^2 t}{1+2\lambda t}\right).$$

Uniqueness of Laplace transforms proves the stated convolution identity. All variables in this distributional construction can be chosen independently. □

Theorem 17.2 (Sharp Gaussian density ceiling)**.** *Let $G \sim N(\mu,\Sigma)$, where $\Sigma \succeq \sigma^2 I$ and $\sigma > 0$. If A is real symmetric and $s_3(A) \geq a > 0$, then, for arbitrary b, c, the quadratic polynomial $G^{\mathsf{T}}AG + b^{\mathsf{T}}G + c$ has a continuous density vanishing at infinity and*

$$\|p_{G^{\mathsf{T}}AG+b^{\mathsf{T}}G+c}\|_\infty \leq \frac{1}{2\sqrt{2}\,\sigma^2 a}. \tag{202}$$

The constant is sharp. In particular, $\Sigma \succeq I/12$ and $s_3(A) \geq 1$ give the bound $3\sqrt{2}$.

Proof. Write $G = \mu + \Sigma^{1/2}Z$. The standard Gaussian vector Z may be orthogonally rotated to diagonalize $B = \Sigma^{1/2}A\Sigma^{1/2}$. Its coordinates remain independent. The singular-value product inequality gives

$$s_3(B) \geq s_{min}(\Sigma^{1/2})^2 s_3(A) \geq \sigma^2 a.$$

For completeness, this inequality follows by writing $A = \Sigma^{-1/2}B\Sigma^{-1/2}$ and applying $s_j(UDV) \leq \|U\|\|V\|s_j(D)$, which itself follows from the best rank-$(j-1)$ approximation characterization of s_j.

Put $d = \sigma^2 a$, and divide the output by d. Select three eigenvalues of B/d with absolute values at least one. Completing the square in each corresponding coordinate and using Lemma 17.1 shows that the entire output is distributed as

$$\epsilon_1 Z_1^2 + \epsilon_2 Z_2^2 + \epsilon_3 Z_3^2 + W, \qquad \epsilon_j \in \{-1,1\},$$

where the three displayed standard normals are independent of W. The constant terms, the extracted Poisson terms with their signs, all other eigenvalue coordinates, and linear Gaussian terms from the kernel are absorbed into W. This argument uses independence after a Gaussian rotation only.

Convolution cannot increase a density supremum. If all three signs coincide, the density is a reflection of the χ_3^2 density, whose maximum is

$$\max_{x>0}\frac{x^{1/2}e^{-x/2}}{\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi e}} < \frac{1}{2\sqrt{2}}.$$

For the mixed signature, reflection reduces to $D = Z_1^2 + Z_2^2 - Z_3^2$. The first two squares sum to an exponential variable of rate $1/2$. Therefore, for $t \geq 0$,

$$p_D(t) = \frac{e^{-t/2}}{2\sqrt{2\pi}}\int_0^\infty v^{-1/2}e^{-v}\,dv = \frac{e^{-t/2}}{2\sqrt{2}}.$$

For $u > 0$, the same convolution gives

$$p_D(-u) = \frac{e^{-u/2}}{2\sqrt{2\pi}} \int_0^\infty (u+v)^{-1/2} e^{-v}\, dv \le \frac{e^{-u/2}}{2\sqrt{2}}.$$

These formulas give a continuous density, with maximum $p_D(0) = 1/(2\sqrt{2})$, vanishing at infinity. Convolution with a probability measure preserves these properties. Rescaling by d proves (202). Equality holds for a centered isotropic three-dimensional Gaussian and $A = a\,\mathrm{diag}(1, 1, -1)$. □

17.2 A global ceiling for diffuse Lindeberg arrays

Theorem 17.3 (Uniform convergence of diffuse quadratic densities)**.** *For each m, let R_m be an n_m-by-3 matrix with orthonormal columns and rows $r_{mi}^{\mathsf T}$, such that*

$$\tau_m := \max_i \|r_{mi}\|^2 \longrightarrow 0.$$

Fix $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$ with $|\lambda_j| \ge 1$. Let the physical coordinates X_{mi} be independent, with densities bounded by one, finite means μ_{mi}, and finite variances v_{mi}. Suppose

$$h_m := \sum_i r_{mi}\mu_{mi} \longrightarrow h, \qquad \Sigma_m := \sum_i v_{mi} r_{mi} r_{mi}^{\mathsf T} \longrightarrow \Sigma, \tag{203}$$

and, for every $\varepsilon > 0$,

$$\sum_i \mathbb{E}\left[\|r_{mi}(X_{mi} - \mu_{mi})\|^2 \mathbf{1}_{\{\|r_{mi}(X_{mi}-\mu_{mi})\| > \varepsilon\}}\right] \longrightarrow 0. \tag{204}$$

Then the forms

$$Q_m = X_m^{\mathsf T} R_m \Lambda R_m^{\mathsf T} X_m$$

have continuous bounded densities for all sufficiently large m. If $G \sim N(h, \Sigma)$ and $Q_\infty = G^{\mathsf T}\Lambda G$, then

$$\|p_{Q_m} - p_{Q_\infty}\|_\infty \longrightarrow 0, \qquad \limsup_m \|p_{Q_m}\|_\infty \le 3\sqrt{2}. \tag{205}$$

The last constant is sharp within this class.

Proof. Every cap-one density with finite variance satisfies $\mathrm{Var}(X) \ge 1/12$. One direct proof, writing $\mu = \mathbb{E}X$, is

$$\mathbb{P}(|X - \mu| > t) \ge 1 - 2t \quad (0 \le t \le 1/2), \qquad \mathbb{E}(X-\mu)^2 \ge \int_0^{1/2} 2t(1-2t)\, dt = \frac{1}{12}.$$

Consequently $\Sigma_m \succeq I_3/12$ and $\Sigma \succeq I_3/12$.

Here are details of the required vector central limit argument. Fix $u \in \mathbb{R}^3$, and put $\xi_{mi} = u^{\mathsf T} r_{mi}(X_{mi} - \mu_{mi})$. The scalar Lindeberg condition follows from (204), and $\sum_i \mathbb{E}\xi_{mi}^2 \to u^{\mathsf T}\Sigma u$. The expansion of e^{ix} to order two, splitting its remainder at $|x| = \delta$, gives

$$\sum_i \left|\mathbb{E}e^{i\xi_{mi}} - 1 + \tfrac{1}{2}\mathbb{E}\xi_{mi}^2\right| \longrightarrow 0.$$

Indeed, the small part is at most a constant times $\delta \sum_i \mathbb{E}\xi_{mi}^2$, and the large part tends to zero by Lindeberg; then let $\delta \downarrow 0$. The same splitting shows $\max_i \mathbb{E}\xi_{mi}^2 \to 0$. Taking logarithms of the product of the scalar characteristic functions therefore proves

$$R_m^{\mathsf T} X_m \implies N(h, \Sigma).$$

The continuity theorem for characteristic functions and the continuous mapping theorem now give pointwise convergence $\varphi_{Q_m}(t) \to \varphi_{Q_\infty}(t)$.

For large m, all row leverages are at most $1/2$, so Theorem 4.6 applies to the actual finite product laws. Since $|\det \Lambda| \geq 1$, it gives the common integrable bound

$$|\varphi_{Q_m}(t)| \leq g(t) := \min\{1, (\pi e)^{3/2}|t|^{-3/2}\}.$$

The pointwise limit satisfies the same bound. Dominated convergence and Fourier inversion give

$$\|p_{Q_m} - p_{Q_\infty}\|_\infty \leq \frac{1}{2\pi}\int_{\mathbb{R}} |\varphi_{Q_m}(t) - \varphi_{Q_\infty}(t)|\, dt \longrightarrow 0.$$

Theorem 17.2, with the covariance floor already proved, gives the final ceiling.

For sharpness, take $n_m = 3m$, with m rows equal to $e_j/\sqrt{m}$ for each $j = 1, 2, 3$, and use independent centered unit uniforms. The hypotheses hold with $h = 0$, $\Sigma = I_3/12$, and $\Lambda = \mathrm{diag}(1, 1, -1)$. The limiting density has maximum $3\sqrt{2}$, so uniform convergence gives $\|p_{Q_m}\|_\infty \to 3\sqrt{2}$. □

Remark 17.4 (Hypotheses of the limiting assertion). The nonzero spectrum of every finite matrix $R_m \Lambda R_m^{\mathsf{T}}$ is exactly the spectrum of Λ. The three projected coordinates need not be independent before the Gaussian limit. The proof also allows $Q_m = Y_m^{\mathsf{T}} \Lambda Y_m + \beta_m^{\mathsf{T}} Y_m + c_m$, where $Y_m = R_m^{\mathsf{T}} X_m$, $\beta_m \to \beta$, and $c_m \to c$, because the physical linear terms are separable phases in Theorem 4.6. Finite variance, convergence of the covariances and means, and the Lindeberg condition are hypotheses of this limiting assertion. Weak convergence alone would not supply (205); the integrable product-law Fourier bound is the step that excludes narrow surviving density peaks.

17.3 One heavy coordinate and two Gaussianized banks

Theorem 17.5 (A uniform gap at the cone tip)**.** *Let $0 < a \leq 1$, $d = \sqrt{1 - a^2}$, and*

$$Q_a = -aX^2 + 2dXB + aB^2 + C^2,$$

where B, C are independent $N(0, 1/12)$, independent of X. Over all probability densities $0 \leq f_X \leq 1$, the unique maximizer of $p_{Q_a}(0)$, up to null sets, is the centered unit interval density. Write the maximum as $C(a)$. Then

$$C(a) \leq \frac{\sqrt{6\pi}\ \mathrm{erf}(\sqrt{3/2}) + 3\sqrt{2}}{2} < 4.111374 < 3\sqrt{2}. \tag{206}$$

This is a bound at output zero.

Proof. Conditional on $X = x$, complete the square:

$$Q_a = a(B + dx/a)^2 + C^2 - x^2/a.$$

The density of $(\sqrt{a}(B + dx/a), C)$ is

$$\frac{6}{\pi\sqrt{a}} \exp\{-6[(u/\sqrt{a} - dx/a)^2 + v^2]\}.$$

Polar coordinates in (u, v) therefore give the conditional zero-density kernel, for almost every x,

$$K_a(x) = \frac{3}{\pi\sqrt{a}} \int_0^{2\pi} e^{-6F_a(\theta)x^2}\, d\theta, \qquad F_a(\theta) = \frac{(\cos\theta - d)^2}{a^2} + \frac{\sin^2\theta}{a}. \tag{207}$$

The conditional density at any level is at most $6/\sqrt{a}$. For each $x \neq 0$ it is continuous in the level near zero, so dominated convergence against f_X proves continuity of the unconditional density there and identifies its value with $\int f_X(x)K_a(x)\,dx$.

The function F_a is strictly positive. Thus K_a is even and strictly decreasing in $|x|$. Filling the interval closest to zero is consequently the unique cap-one maximizer: explicitly, with $I = [-1/2, 1/2]$,

$$\int (\mathbf{1}_I - f_X)(K_a - K_a(1/2))\,dx \geq 0,$$

and equality forces $f_X = \mathbf{1}_I$ almost everywhere. Put

$$L(s) = \int_{-1/2}^{1/2} e^{-6sx^2}\,dx, \qquad R(s) = \sqrt{1+s}\,L(s).$$

Then $C(a) = \frac{3}{\pi\sqrt{a}} \int L(F_a(\theta))\,d\theta$. We claim that $R(0) = 1$ and R is strictly decreasing on $(0, \infty)$. If $M(s) = \int_{-1/2}^{1/2} x^2 e^{-6sx^2}\,dx$, integration by parts gives $L(s) - 12sM(s) = e^{-3s/2}$, while $e^y > 1 + y$ gives

$$L(s) = e^{-3s/2} \int_{-1/2}^{1/2} e^{6s(1/4-x^2)}\,dx > (1+s)e^{-3s/2} \qquad (s > 0).$$

Since $L' = -6M$, these identities imply

$$\frac{d}{ds} \log R(s) = \frac{1}{2(1+s)} - \frac{L(s) - e^{-3s/2}}{2sL(s)} < 0.$$

Write $\alpha = \sqrt{1+a}$, $\beta = \sqrt{1-a}$. Direct expansion gives the perfect-square identity

$$a^2(1 + F_a(\theta)) = (\alpha - \beta\cos\theta)^2.$$

The tangent-half-angle substitution, or an elementary rational integral, gives

$$\int_0^{2\pi} \frac{d\theta}{\alpha - \beta\cos\theta} = \frac{2\pi}{\sqrt{\alpha^2 - \beta^2}}.$$

Consequently

$$d\nu_a(\theta) = \frac{(1 + F_a(\theta))^{-1/2}}{\pi\sqrt{2a}}\,d\theta$$

is a probability measure and

$$\frac{C(a)}{3\sqrt{2}} = \int R(F_a)\,d\nu_a. \tag{208}$$

We next show $\nu_a\{F_a \geq 1\} \geq 1/2$. At $a = 1$, $F_a \equiv 1$. For $0 < a < 1$, put

$$c_a = \frac{\alpha - a\sqrt{2}}{\beta}, \qquad \theta_0 = \arccos c_a, \qquad \kappa = \sqrt{\frac{\alpha+\beta}{\alpha-\beta}}.$$

Here $0 < c_a < 1$: the upper inequality follows from $\alpha + \beta > \sqrt{2}$, and the lower one also follows from the comparison below. The event $F_a < 1$ is exactly $\cos\theta > c_a$. The same substitution gives

$$\nu_a\{F_a < 1\} = \frac{2}{\pi} \arctan\left(\kappa \tan\frac{\theta_0}{2}\right).$$

But

$$c_a - \frac{\beta}{\alpha} = \frac{a(2 - \sqrt{2}\,\alpha)}{\alpha\beta} \geq 0.$$

Equivalently, $\kappa \tan(\theta_0/2) \le 1$, so the displayed probability is at most $1/2$. As $R \le 1$ everywhere and $R(s) \le R(1)$ for $s \ge 1$, (208) yields

$$C(a) \le \frac{3\sqrt{2}}{2}(1 + R(1)), \qquad R(1) = \sqrt{\frac{\pi}{3}}\,\mathrm{erf}(\sqrt{3/2}) < 1.$$

This proves the exact bound and its strict gap below $3\sqrt{2}$.

The decimal upper bound can also be certified without numerical quadrature. Expanding the exponential and integrating gives the alternating series

$$L(1) = \sum_{j=0}^{\infty} \frac{(-3/2)^j}{j!(2j+1)}.$$

Its term magnitudes decrease, so the even partial sum through $j = 12$ is an upper bound. Since $\sqrt{2} < 14142135623731/10^{13}$, exact rational arithmetic gives

$$\frac{3\sqrt{2}}{2} + 3L(1) < \frac{3}{2}\frac{14142135623731}{10^{13}} + 3\sum_{j=0}^{12} \frac{(-3/2)^j}{j!(2j+1)} < 4.111374.$$

The expression on the left equals the exact ceiling in (206). □

Remark 17.6 (Exact finite product realization). The matrix on the three displayed statistics is

$$M_a = \begin{pmatrix} -a & d & 0 \\ d & a & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad M_a^2 = I_3, \qquad \mathrm{spec}(M_a) = \{1, 1, -1\}.$$

Replace B, C by two independent normalized sums of m centered unit uniforms, retaining the independent heavy coordinate X. The corresponding matrix in the $1 + 2m$ original coordinates is $T_m M_a T_m^{\mathsf{T}}$, where T_m has one row e_1^{T}, m rows $e_2^{\mathsf{T}}/\sqrt{m}$, and m rows $e_3^{\mathsf{T}}/\sqrt{m}$. Its columns are orthonormal, so its nonzero spectrum is exactly $\{1, 1, -1\}$. For each fixed $a > 0$ and fixed heavy density, its cone-tip density converges to that of Q_a.

The bank characteristic function is $\psi_m(t) = [\sin(t/(2\sqrt{m}))/(t/(2\sqrt{m}))]^m$. For $|t| \le \sqrt{m}$, the elementary bound $|\sin u/u| \le e^{-cu^2}$ on $|u| \le 1/2$, for some $c > 0$, gives a common Gaussian majorant. For $\sqrt{m} \le |t| \le 4\sqrt{m}$, compactness gives the bound ρ^m, with a fixed $\rho < 1$. For $|t| \ge 4\sqrt{m}$, the bound $(2\sqrt{m}/|t|)^m$ has integral tending to zero. The central limit expansion on compact intervals and these three bounds prove L^1 convergence of ψ_m to the Gaussian characteristic function. Thus the bank densities converge uniformly, and are uniformly bounded, by Fourier inversion. In the conditional polar formula used for (207), replace the product of Gaussian bank densities by the product of these two bank densities. Uniform convergence of the products, the finite angular range, and $\int f_X = 1$ prove the asserted cone-tip limit. Neither this argument nor (206) gives a global bound over all output levels, or a convergence rate uniform as $a \downarrow 0$.

17.4 An exact dense Householder example

Proposition 17.7 (The Householder cube cone tip). *Let X_1, X_2, X_3 be independent centered unit uniforms, and put*

$$e = \frac{1}{\sqrt{3}}(1, 1, 1)^{\mathsf{T}}, \qquad A = I - 2ee^{\mathsf{T}} = \frac{1}{3}\begin{pmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}.$$

Then $\operatorname{spec}(A) = \{1, 1, -1\}$*, and the density of* $Q = X^{\mathsf{T}}AX$ *is continuous at zero, with*

$$p_Q(0) = 6\sqrt{3}\,\operatorname{artanh}\left(\frac{\sqrt{2}-1}{\sqrt{3}}\right) = 2.5343552034\ldots. \tag{209}$$

In particular, this is also the limit of $\mathbb{P}(|Q| \le \varepsilon)/(2\varepsilon)$*.*

Proof. The matrix acts by -1 on e and by $+1$ on $e^{\perp}$, proving the spectral statement. Choose an orthonormal basis u, v of $e^{\perp}$, and let $u_\theta = u\cos\theta + v\sin\theta$. For $\mathcal{C} = [-1/2, 1/2]^3$, polar coordinates in the positive two-plane give the following density version:

$$p_Q(s) = \frac{1}{2}\int_{\mathbb{R}}\int_0^{2\pi} \mathbf{1}_{\{s+z^2>0\}}\mathbf{1}_{\mathcal{C}}\big(\sqrt{s+z^2}\,u_\theta + ze\big)\,d\theta\,dz.$$

Indeed the radial measure is $r\,dr$, and changing from r to $s = r^2 - z^2$ contributes the factor $1/2$. Whenever the cube indicator is nonzero, $|z| \le \sqrt{3}/2$. As $s \to 0$, its integrand converges except at $z = 0$ or where the limiting ray meets the cube boundary. For each θ there are only finitely many such nonzero radii. Dominated convergence therefore proves continuity at zero.

Put $d_\pm(\theta) = u_\theta \pm e$ and $M_\pm(\theta) = \max_i |d_{\pm,i}(\theta)|$. At zero, each ray remains in the cube up to $r = 1/(2M_\pm(\theta))$, so

$$p_Q(0) = \frac{1}{4}\int_0^{2\pi}\left(\frac{1}{M_+(\theta)} + \frac{1}{M_-(\theta)}\right)d\theta = \frac{1}{2}\int_0^{2\pi}\frac{d\theta}{M_+(\theta)}.$$

The last equality uses $M_-(\theta) = M_+(\theta+\pi)$. With a suitable origin of the angular coordinate, the three coordinates of d_+ are phase shifts by $2\pi/3$ of

$$a_0 + b_0\cos\theta, \qquad a_0 = 1/\sqrt{3}, \quad b_0 = \sqrt{2/3}.$$

Every negative coordinate has absolute value at most $b_0 - a_0 < a_0$, whereas the largest positive coordinate is at least a_0, because their sum is $\sqrt{3}$. Thus the largest absolute coordinate is the largest positive one. Each coordinate is maximal on a sector of width $2\pi/3$, giving

$$p_Q(0) = \frac{3}{2}\int_{-\pi/3}^{\pi/3}\frac{d\theta}{a_0 + b_0\cos\theta}.$$

Finally $t = \tan(\theta/2)$ yields

$$\int_{-\pi/3}^{\pi/3}\frac{d\theta}{a_0 + b_0\cos\theta} = \frac{4}{\sqrt{b_0^2 - a_0^2}}\operatorname{artanh}\left(\frac{1}{\sqrt{3}}\sqrt{\frac{b_0 - a_0}{b_0 + a_0}}\right).$$

Since $\sqrt{b_0^2 - a_0^2} = 1/\sqrt{3}$ and $\sqrt{(b_0 - a_0)/(b_0 + a_0)} = \sqrt{2} - 1$, this proves (209). Continuity proves the stated small-ball limit. □

The Householder calculation concerns the cone tip and does not locate the global maximum of that example. The one-heavy calculation has the same output-level restriction. The diffuse Lindeberg theorem is global in the output level, under its stated array hypotheses. These distinctions leave the unrestricted indefinite product-law endpoint open.

18 Counterexamples to stronger assertions and open cases

The following counterexamples delimit several stronger intermediate estimates. None is a counterexample to Conjecture 1.1.

18.1 A bounded joint density cannot replace a product law

Proposition 18.1 (A thin quadratic shell)**.** *There is no bound for the density of a nondegenerate three-variable quadratic that depends only on an upper bound for the joint input density and a lower bound for the three eigenvalue moduli.*

Proof. Fix $0 < \delta < 1$ and put $R = e^{1/\delta}$. In the change of variables

$$x_1 = r, \quad x_3 = z, \quad x_2 = (t - z^2)/r, \qquad 1 < r < R, \quad 0 < z < 1, \quad 0 < t < \delta,$$

the Jacobian is $1/r$. The resulting region $E_\delta \subset \mathbb{R}^3$ has volume $\delta \log R = 1$. Give X the density $\mathbf{1}_{E_\delta} \le 1$. Then $x_1 x_2 + x_3^2 = t$ is uniform on $(0, \delta)$. The quadratic

$$Q = 2X_1X_2 + 2X_3^2$$

has matrix eigenvalues $1, -1, 2$, yet its density equals $1/(2\delta)$ on $(0, 2\delta)$. This tends to infinity. The coordinates of X are dependent. □

Example 18.2 (Conditional densities can be arbitrarily high)**.** Let X_1, X_2 be independent uniforms on $(0, 1)$, and set $L = (X_1 + X_2)/\sqrt{2}$, $V = (X_1 - X_2)/\sqrt{2}$. For $0 < \ell < 1/\sqrt{2}$, the conditional law of V given $L = \ell$ is uniform on $(-\ell, \ell)$ and has density $1/(2\ell)$. The marginal density of L there is 2ℓ. Thus the conditional density norm is unbounded, and its pth moment is infinite for every $p \ge 2$. Both coordinates before rotation have density one. An unconditional projection theorem therefore cannot be substituted for a conditional-density estimate.

18.2 Uniform pointwise Fourier decay is too strong

Proposition 18.3. *Even for $Q = X_1^2 + X_2^2 + X_3^2$, there is no function $b(t) \to 0$ as $t \to \infty$ such that $|\mathbb{E}e^{itQ}| \le b(t)$ uniformly over all independent coordinate densities bounded by one.*

Proof. Fix $t > 0$ and $0 < \eta < \pi/2$. The set

$$F = \{x > 0 : \operatorname{dist}(tx^2, 2\pi\mathbb{Z}) < \eta\}$$

has infinite measure. Indeed, its intervals near $\sqrt{2\pi j/t}$ have lengths comparable, for large j, to $\eta/(\sqrt{t}\sqrt{j})$, whose sum diverges. Choose a measurable subset $E \subset F$ of measure one, and take all three coordinate densities equal to $\mathbf{1}_E$. Each expectation $\mathbb{E}e^{itX_i^2}$ has real part at least $\cos\eta$, hence modulus at least $\cos\eta$. Independence gives $|\mathbb{E}e^{itQ}| \ge (\cos\eta)^3$. Since η is arbitrary, the supremum over admissible laws equals one for every fixed $t > 0$. The laws may depend on t. This is consistent with the Riemann–Lebesgue lemma for each fixed law and with the bounded density in Theorem 3.4. □

18.3 An arbitrarily small aligned quadratic phase can destroy an integral bound

Theorem 18.4 (Aligned-twist obstruction)**.** *Let $p = (1, 1)/\sqrt{2}$ and fix $\kappa \ne 0$. Among independent coordinate densities $f_1, f_2 \le 1$, the quantity*

$$J_\kappa = \int_{\mathbb{R}} \left| \mathbb{E}e^{iu\langle p, X\rangle + i\kappa\langle p, X\rangle^2} \right| du$$

is unbounded. In contrast, $J_0 \le 2\pi\sqrt{2}$ for every such pair. Thus small operator norm of a quadratic phase is insufficient for a law-uniform bound on this Fourier integral, even when both frame leverages equal $1/2$.

Proof. For an integer $N \geq 2$, set $\sigma = 1/N$. Choose a spacing $D > 0$ sufficiently large below, and take

$$f_1(x) = \frac{1}{N} \sum_{j=0}^{N-1} \gamma_\sigma(x - \sqrt{2}Dj), \qquad f_2(x) = \frac{1}{N} \sum_{k=0}^{N-1} \gamma_\sigma(x - \sqrt{2}DNk),$$

where γ_σ is the centered Gaussian density of variance σ^2. Each isolated bump has height $1/\sqrt{2\pi} < 1$. The tails of other bumps converge uniformly to zero as D grows. More explicitly, choose the nearest center: all other centers are at least $(m - 1/2)\sqrt{2}D$ away for integers $m \geq 1$, so their contribution is bounded by a convergent Gaussian tail series tending to zero. Thus $f_1, f_2 \leq 1$ for sufficiently large D.

The linear statistic $L = \langle p, X \rangle$ is an equally weighted mixture of N^2 Gaussians with variance σ^2 and distinct means $m_{jk} = D(j + Nk)$. For one component of mean m, completing the square in its Gaussian integral gives

$$|T_m(u)| = (1 + 4\kappa^2\sigma^4)^{-1/4} \exp\left[-\frac{\sigma^2(u + 2\kappa m)^2}{2(1 + 4\kappa^2\sigma^4)} \right].$$

Its L^1 norm is

$$A_N = \frac{\sqrt{2\pi}}{\sigma}(1 + 4\kappa^2\sigma^4)^{1/4} \geq \sqrt{2\pi}N.$$

The frequency centers are separated by $2|\kappa|D$. Choose D so large that the disjoint windows $U_m = \{|u + 2\kappa m| < |\kappa|D\}$ each contain at least $(1 - \eta)A_N$ of their own component, with a fixed $\eta < 1/4$. The reverse triangle inequality on these windows yields

$$J_\kappa \geq \frac{1}{N^2} \sum_m \left(\int_{U_m} |T_m| - \int_{U_m^c} |T_m| \right) \geq (1 - 2\eta)A_N.$$

To see the first inequality, sum the cross-component errors over disjoint windows: for each component, their union is contained in the complement of its own window. The lower bound grows linearly with N.

For $\kappa = 0$, independence and Cauchy–Schwarz give

$$J_0 = \sqrt{2} \int |\widehat{f}_1(v)\widehat{f}_2(v)|\, dv \leq \sqrt{2}\, \|\widehat{f}_1\|_2 \|\widehat{f}_2\|_2 \leq 2\pi\sqrt{2},$$

by Plancherel and $\int f_i^2 \leq \int f_i = 1$. □

The input laws in this construction consist of many narrow separated components. Their final density caps arise from separation of component locations. They need not obey the average component-peak condition in the Gaussian-mixture theorem of Section 13. That theorem and the present obstruction have different hypotheses.

Two further failures are proved in the supplement: Proposition 16.15 shows that a bounded L^3 density norm and a square-root interval profile do not imply a bounded density, and Proposition 16.16 shows that an independent hub together with a bounded joint density for a dependent triple can still leave an unbounded output density. Neither provides an admissible product-law counterexample.

18.4 An independent-block estimate for separated phases

Lemma 18.5 (Phase-weighted projection)**.** *Let $Z = TX$ be a rank-k linear image of a product vector, and let $\psi(X)$ be any real measurable phase. Then*

$$\int_{\mathbb{R}^k} |\mathbb{E}e^{i\langle u, Z\rangle + i\psi(X)}|^2\, du \leq (2\pi)^k \|p_Z\|_\infty \leq (2\pi)^k 2^{k/2} \det(TT^T)^{-1/2}.$$

Proof. The complex pushforward has density $m(z) = \mathbb{E}[e^{i\psi(X)} \mid Z = z]p_Z(z)$, so $|m| \le p_Z$ and $\int |m|^2 \le \|p_Z\|_\infty$. Apply Plancherel and the projection theorem of Livshyts et al. (2016). This uses a bound on the complex pushforward, not on a conditional input density. □

Theorem 18.6 (Two independent blocks)**.** *Let the coordinates split as $K \sqcup I \sqcup J$, and let P be an $n \times k$ matrix. Put $G_K = P_K^T P_K$ and $G_I = P_I^T P_I$, and assume both are positive definite. Let E be any symmetric matrix with $E_{KI} = 0$. Then, allowing also an arbitrary extra phase depending on X_J,*

$$\int_{\mathbb{R}^k} |\mathbb{E}e^{i\langle Pu,X\rangle + iX^T EX + i\psi(X_J)}|\, du \le (2\pi)^k 2^{k/2} (\det G_K \det G_I)^{-1/4}.$$

Proof. Condition on X_J. The phase splits into one function of X_K and one function of X_I, together with a constant. Independence factors the conditional expectation. Apply Cauchy–Schwarz in u and Lemma 18.5 to each factor. The bound is independent of the frozen coordinates, so the triangle inequality and Fubini finish the proof. □

Corollary 18.7. *Let U have three orthonormal columns, let $\Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$ be invertible, and let $A = U\Lambda U^T + B$. If a split $K \sqcup I \sqcup J$ has $B_{KI} = 0$ and positive definite $G_K = U_K^T U_K$, $G_I = U_I^T U_I$, then every affine perturbation satisfies*

$$\|p_Q\|_\infty \le 6|\lambda_1\lambda_2\lambda_3|^{-1/3} (\det G_K \det G_I)^{-1/6}.$$

Proof. Fresnel-linearize the three displayed squares, using the regularization justified in Theorem 4.5. The preceding theorem, with the phase $tX^T BX$ and the affine phase absorbed into its coordinate blocks, gives

$$|\varphi_Q(t)| \le (2\pi)^{3/2} |\det \Lambda|^{-1/2} (\det G_K \det G_I)^{-1/4} |t|^{-3/2}.$$

Integrating its minimum with one gives $(3/\pi)$ times the $2/3$ power of the displayed coefficient, which is the stated bound. Arbitrary within-block quadratic terms are permitted, but the cross-block zero condition is exact. □

18.5 The remaining uniform estimate

A normalized family illustrating the remaining difficulty is

$$Q_\varepsilon = 2X_0 L + V^2 + 2\varepsilon LP, \tag{210}$$

where X_0 is an independent physical coordinate and L, V, P are orthonormal linear statistics of the other coordinates with a spread frame. In the orthonormal basis formed by the hub coordinate and these three vectors, its matrix is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & \varepsilon \\ 0 & 0 & 1 & 0 \\ 0 & \varepsilon & 0 & 0 \end{pmatrix}.$$

The characteristic polynomial is $z(z-1)(z^2 - 1 - \varepsilon^2)$, so its three nonzero eigenvalues are $1, \sqrt{1+\varepsilon^2}, -\sqrt{1+\varepsilon^2}$. The third eigenvalue modulus does not deteriorate when $\varepsilon \downarrow 0$.

For $\varepsilon = 0$, the independent-hub/transverse-square argument gives a uniform bound. For every fixed nonzero ε, Fourier decay gives a bounded density. The bounds for arbitrary cap-one coordinate laws deteriorate as $\varepsilon \downarrow 0$, even after symmetric-window recovery. The Gaussian-mixture results give uniform bounds under the additional hypotheses in Section 13.

An estimate for (210) alone does not cover the affine, hub-diagonal, and light-block terms that arise in more general matrices. In addition, the sufficient conditions above do not imply that every no-pivot matrix has a coordinate decomposition satisfying one of them.

To deduce the unrestricted conjecture from the results above, one would need a bound for the uncovered analytic cases and a quantitative decomposition covering all admissible matrices. A uniform estimate for the adapted-hub functional, extended to the additional quadratic terms, could address the analytic cases. The constants must be independent of dimension and of a vanishing leakage parameter. No counterexample to the unrestricted conjecture follows from the results above.

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